



NAND Flash Memory

**MT29F128G08CBECB, MT29F256G08CEECB, MT29F512G08C[K/M]ECB
MT29F1T08CUECB, MT29F2T08CTECB**

Features

- Open NAND Flash Interface (ONFI) 3.2-compliant¹
- JEDEC NAND Flash Interface Interoperability (JESD230) compliant
- Multiple-level cell (MLC) technology
- Organization
 - Page size x8: 18,256 bytes (16,384 + 1872 bytes)
 - Block size: 512 pages (8192K + 936K bytes)
 - Plane size: 2 planes x 1048 blocks per plane
 - Device size: 128Gb: 2096 blocks;
256Gb: 4192 blocks;
512Gb: 8384 blocks;
1Tb: 16,768 blocks
2Tb: 33,536 blocks
- NV-DDR2 I/O performance
 - Up to NV-DDR2 timing mode 3²
 - Clock rate: 12ns (NV-DDR2)
 - Read/write throughput per pin: 166 MT/s
- NV-DDR I/O performance
 - Up to NV-DDR timing mode 4³
 - Clock rate: 12ns (NV-DDR)
 - Read/write throughput per pin: 166 MT/s
- Asynchronous I/O performance
 - Up to asynchronous timing mode 5
 - ^tRC/^tWC: 20ns (MIN)
 - Read/write throughput per pin: 50 MT/s
- Array performance
 - Read page: 115µs (MAX)
 - Program page: 1600µs (TYP)
 - Erase block: 3ms (TYP)
- Operating Voltage Range
 - V_{CC}: 2.7–3.6V
 - V_{CCQ}: 1.7–1.95V, 2.7–3.6V
- Command set: ONFI/JEDEC NAND Flash Protocol
- Advanced Command Set
 - Program cache
 - Read cache sequential
 - Read cache random
 - One-time programmable (OTP) mode
 - Multi-plane commands
 - Multi-LUN operations
 - Read Unique ID
 - Copyback
- Read Retry
- First block (block address 00h) is valid when shipped from factory. For minimum required ECC, see Error Management (page 152)⁴.
- RESET (FFh) required as first command after power-on
- Operation status byte provides software method for detecting
 - Operation completion
 - Pass/fail condition
 - Write-protect status
- Data strobe (DQS) signals provide a hardware method for synchronizing data DQ in the NV-DDR/NV-DDR2 interface
- Copyback operations supported within the plane from which data is read
- Quality and reliability⁴
 - Data retention: JESD47 compliant; see qualification report
 - Endurance: 3000 PROGRAM/ERASE cycles
- Operating temperature:
 - Commercial: 0°C to +70°C
 - Industrial (IT): -40°C to +85°C
- Package
 - 152-ball BGA

- Notes:
1. The ONFI 3.2 specification is available at www.onfi.org.
 2. 1.8V V_{CCQ} and 3.3V V_{CCQ} up to NV-DDR2 timing mode 3 for 166MT/s.
 3. 1.8V V_{CCQ} and 3.3V V_{CCQ} up to NV-DDR timing mode 4 for 166MT/s.
 4. Read Retry operations are required to achieve specified endurance and for general array data integrity.

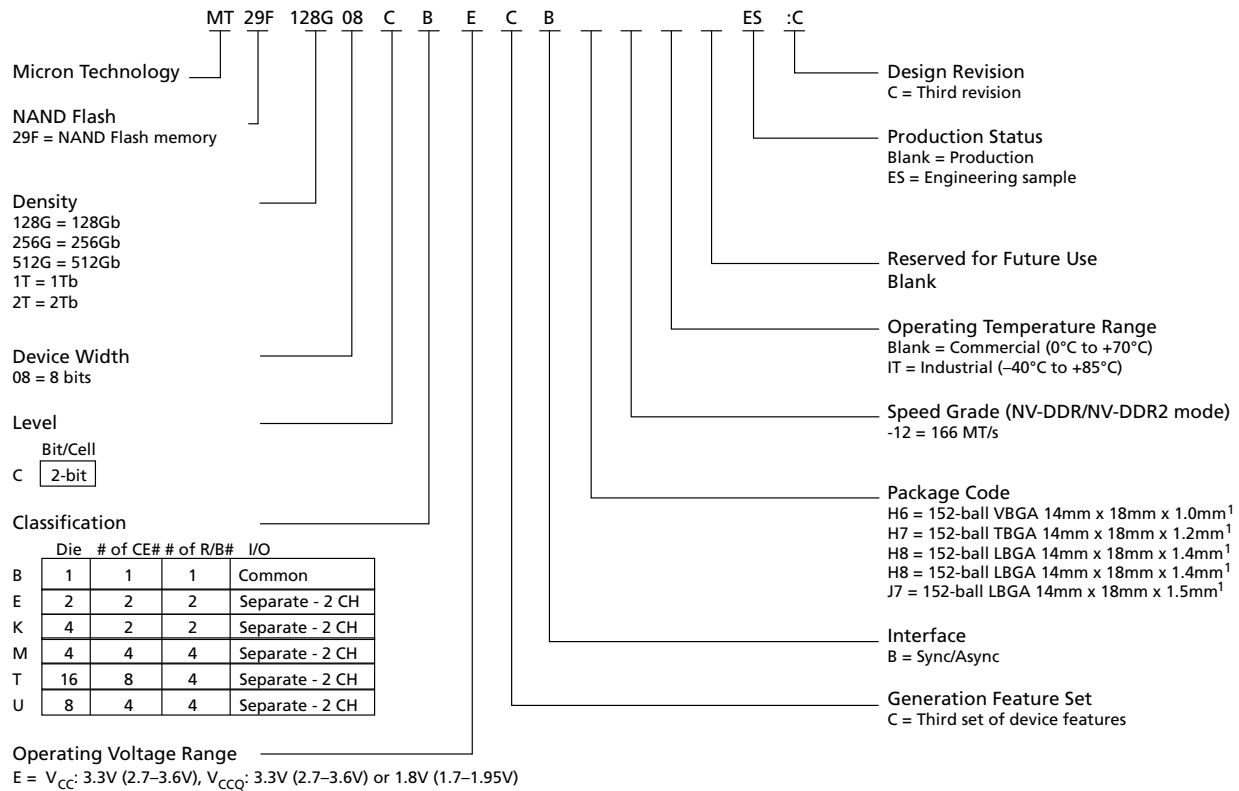


MLC 256Gb to 2Tb Async/Sync NAND Features

Part Numbering Information

Micron NAND Flash devices are available in different configurations and densities. Verify valid part numbers by using Micron's part catalog search at www.micron.com. To compare features and specifications by device type, visit www.micron.com/products. Contact the factory for devices not found.

Figure 1: Part Numbering



Note: 1. Pb-free package.



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Limited Warranty. In no event shall Micron be liable for any indirect, incidental, punitive, special or consequential damages (including without limitation lost profits, lost savings, business interruption, costs related to the removal or replacement of any products or rework charges) whether or not such damages are based on tort, warranty, breach of contract or other legal theory, unless explicitly stated in a written agreement executed by Micron's duly authorized representative.



General Description

Micron NAND Flash devices include an asynchronous data interface for high-performance I/O operations. These devices use a highly multiplexed 8-bit bus (DQx) to transfer commands, address, and data. There are five control signals used to implement the asynchronous data interface: CE#, CLE, ALE, WE#, and RE#. Additional signals control hardware write protection (WP#) and monitor device status (R/B#).

Some versions of this Micron NAND Flash device additionally includes a NV-DDR and/or a NV-DDR2 data interface for high-performance I/O operations. When the NV-DDR interface is active, WE# becomes CLK and RE# becomes W/R#. Data transfers include a bidirectional data strobe (DQS).

This hardware interface creates a low pin-count device with a standard pinout that remains the same from one density to another, enabling future upgrades to higher densities with no board redesign.

A target is the unit of memory accessed by a chip enable signal. A target contains one or more NAND Flash die. A NAND Flash die is the minimum unit that can independently execute commands and report status. A NAND Flash die, in the ONFI specification, is referred to as a logical unit (LUN). For further details, see Device and Array Organization.

Asynchronous, NV-DDR, and NV-DDR2 Signal Descriptions

Table 1: Asynchronous, NV-DDR, and NV-DDR2 Signal Definitions

Asynchronous Signal ¹	NV-DDR Signal ¹	NV-DDR2 Signal ¹	Type	Description ²
ALE	ALE	ALE	Input	Address latch enable: Loads an address from DQx into the address register.
CE#	CE#	CE#	Input	Chip enable: Enables or disables one or more die (LUNs) in a target.
CLE	CLE	CLE	Input	Command latch enable: Loads a command from DQx into the command register.
DQx	DQx	DQx	I/O	Data inputs/outputs: The bidirectional I/Os transfer address, data, and command information.
–	DQS	DQS, DQS_t	I/O	Data strobe: Provides a synchronous reference for data input and output.
–	–	DQS_c	I/O	Data strobe compliment: Provides a complementary signal to the data strobe signal optionally used in the NV-DDR2 interface for synchronous reference for data input and output.
ENi	ENi	ENi	Input	Enumerate input: Input to a NAND device (if first NAND device on the daisy chain have as NC) from ENo of a previous NAND device to support CE# pin reduction functionality.
ENo	ENo	ENo	Output	Enumerate output: Output from a NAND device to the ENi of the next NAND device in the daisy chain to support CE# pin reduction functionality.



MLC 256Gb to 2Tb Async/Sync NAND Asynchronous, NV-DDR, and NV-DDR2 Signal Descriptions

Table 1: Asynchronous, NV-DDR, and NV-DDR2 Signal Definitions (Continued)

Asynchronous Signal ¹	NV-DDR Signal ¹	NV-DDR2 Signal ¹	Type	Description ²
RE#	W/R#	RE#, RE_t	Input	Read enable and write/read: RE# transfers serial data from the NAND Flash to the host system when the asynchronous interface is active. When the NV-DDR interface is active, W/R# controls the direction of DQx and DQS.
–	–	RE_c	Input	Read enable complement: Provides a complementary signal to the read enable signal optionally used in the NV-DDR2 interface for synchronous reference for data output.
WE#	CLK	WE#	Input	Write enable and clock: WE# transfers commands, addresses when the asynchronous and NV-DDR2 interfaces are active, and serial data from the host system to the NAND Flash when the asynchronous interface is active. When the NV-DDR interface is active, CLK latches command and address cycles.
WP#	WP#	WP#	Input	Write protect: Enables or disables array PROGRAM and ERASE operations.
R/B#	R/B#	R/B#	Output	Ready/busy: An open-drain, active-low output that requires an external pull-up resistor. This signal indicates target array activity.
V _{CC}	V _{CC}	V _{CC}	Supply	V_{CC}: Core power supply
V _{CCQ}	V _{CCQ}	V _{CCQ}	Supply	V_{CCQ}: I/O power supply
V _{PP}	V _{PP}	V _{PP}	Supply	V_{PP}: The V _{PP} signal is an optional external high voltage power supply to the device. This high voltage power supply may be used to enhance operations (e.g., improved power efficiency).
V _{SS}	V _{SS}	V _{SS}	Supply	V_{SS}: Core ground connection
V _{SSQ}	V _{SSQ}	V _{SSQ}	Supply	V_{SSQ}: I/O ground connection
–	–	V _{REFQ}	Supply	V_{REFQ}: Reference voltage used with NV-DDR2 interface
NC	NC	NC	–	No connect: NCs are not internally connected. They can be driven or left unconnected.
DNU	DNU	DNU	–	Do not use: DNUs must be left unconnected.
RFU	RFU	RFU	–	Reserved for future use: RFUs must be left unconnected.

- Notes:
1. See Device and Array Organization for detailed signal connections.
 2. See Bus Operation - Asynchronous Interface, Bus Operation – NV-DDR Interface, and Bus Operation – NV-DDR2 Interface for detailed asynchronous, NV-DDR, and NV-DDR2 interface signal descriptions.



Architecture

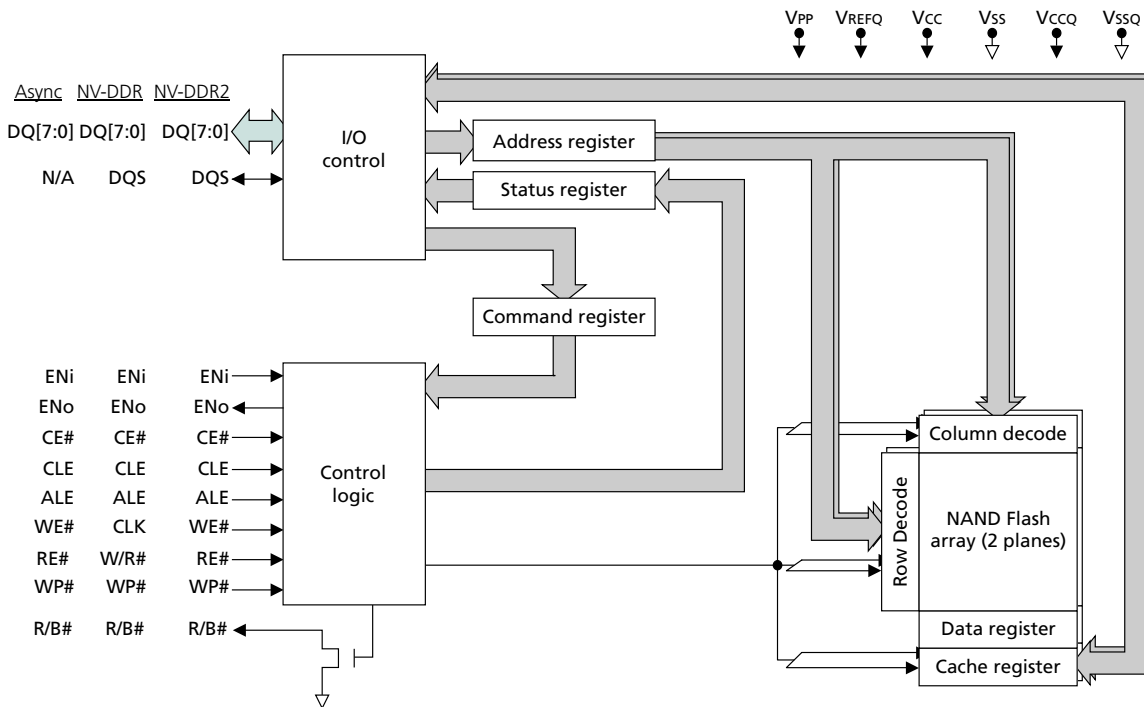
These devices use NAND Flash electrical and command interfaces. Data, commands, and addresses are multiplexed onto the same pins and received by I/O control circuits. The commands received at the I/O control circuits are latched by a command register and are transferred to control logic circuits for generating internal signals to control device operations. The addresses are latched by an address register and sent to a row decoder to select a row address, or to a column decoder to select a column address.

Data is transferred to or from the NAND Flash memory array, byte by byte, through a data register and a cache register.

The NAND Flash memory array is programmed and read using page-based operations and is erased using block-based operations. During normal page operations, the data and cache registers act as a single register. During cache operations, the data and cache registers operate independently to increase data throughput.

The status register reports the status of die (LUN) operations.

Figure 2: NAND Flash Die (LUN) Functional Block Diagram



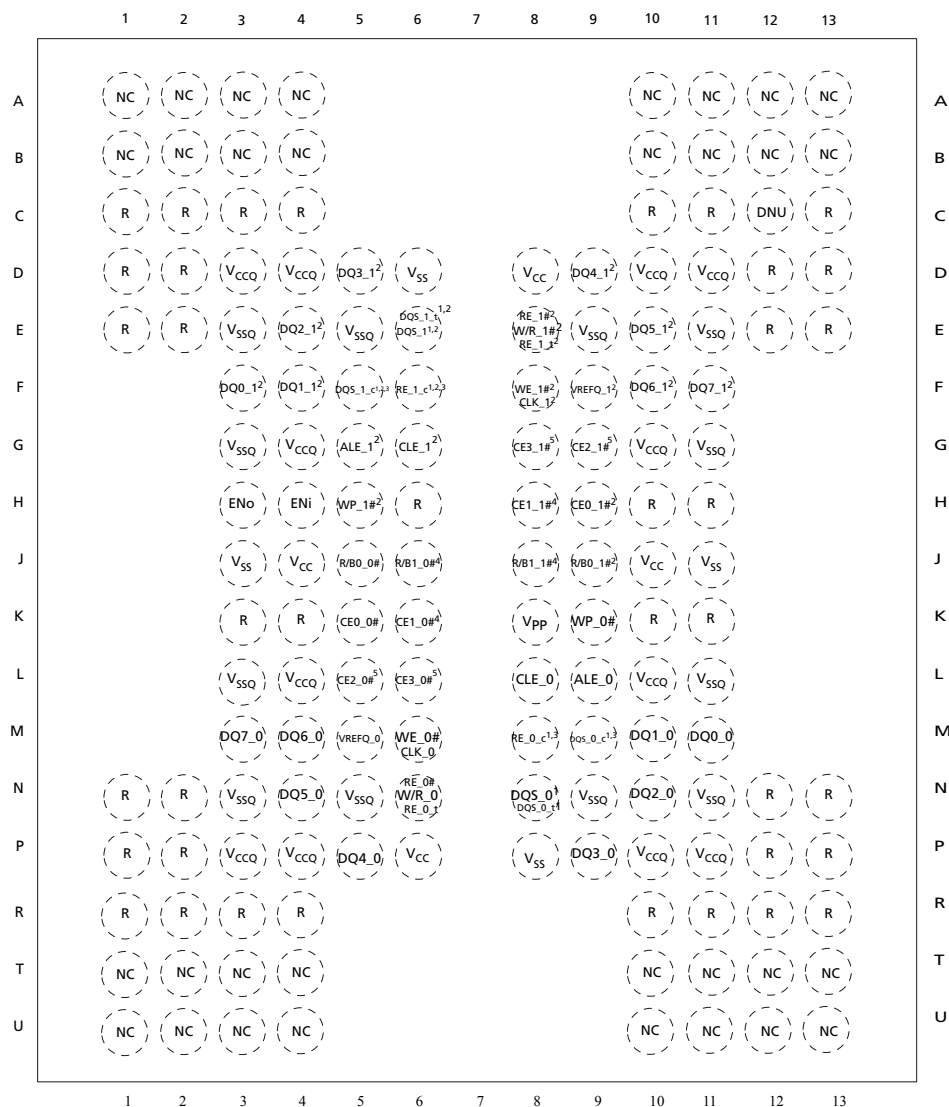
- Notes:
1. N/A: This signal is tri-stated when the asynchronous interface is active.
 2. Some devices do not include the NV-DDR or NV-DDR2 interface.
 3. Some devices do not include the V_{REFQ} or V_{PP} signal.



MLC 256Gb to 2Tb Async/Sync NAND Signal Assignments

Signal Assignments

Figure 3: 152-Ball BGA (Ball-Down, Top View)



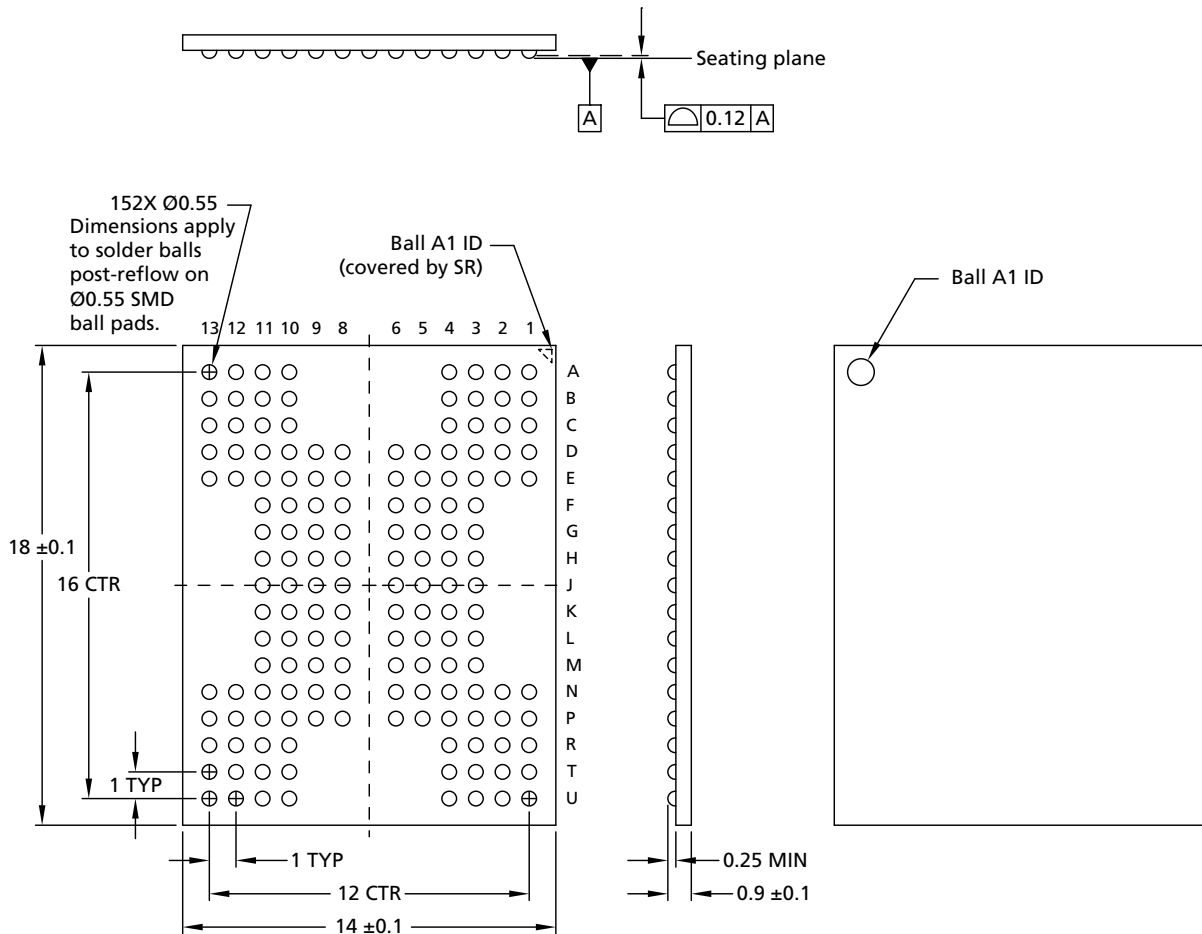
- Notes:
1. N/A: This signal is tri-stated when the asynchronous interface is active.
 2. These signals are available on dual, quad, and octal die packages. They are NC for other configurations.
 3. These signals are available when differential signaling is enabled.
 4. These signals are available on quad LUN four CE# or higher LUN stacked packages. They are NC for other configurations.
 5. These signals are available on 16 LUN stacked packages. They are NC for other configurations.



MLC 256Gb to 2Tb Async/Sync NAND Package Dimensions

Package Dimensions

Figure 4: 152-Ball VBGA – 14mm x 18mm (Package Code: H6)

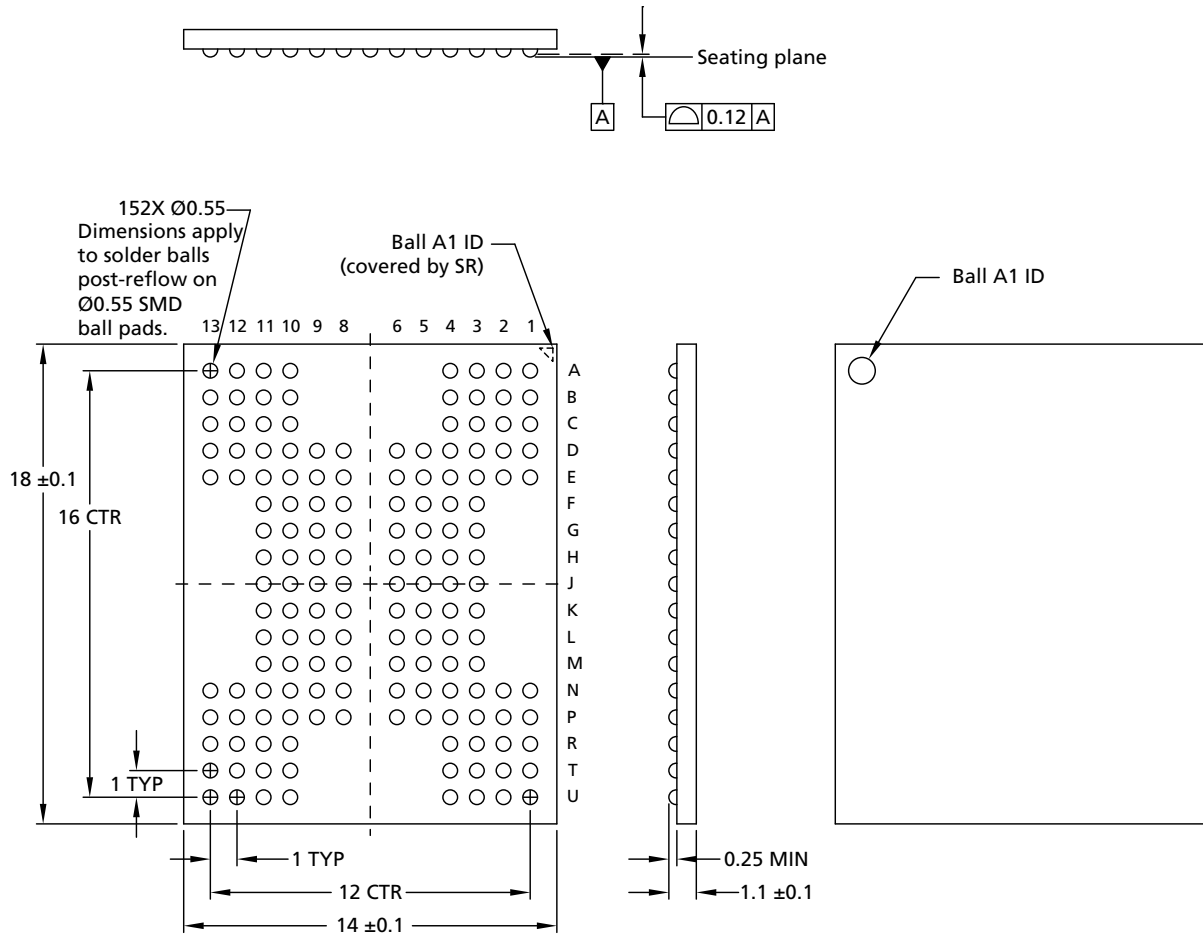


- Notes:
1. All dimensions are in millimeters.
 2. Solder ball material: SAC405 (95.5% Sn, 4% Ag, 0.5% Cu).



MLC 256Gb to 2Tb Async/Sync NAND Package Dimensions

Figure 5: 152-Ball TBGA – 14mm x 18mm (Package Code: H7)

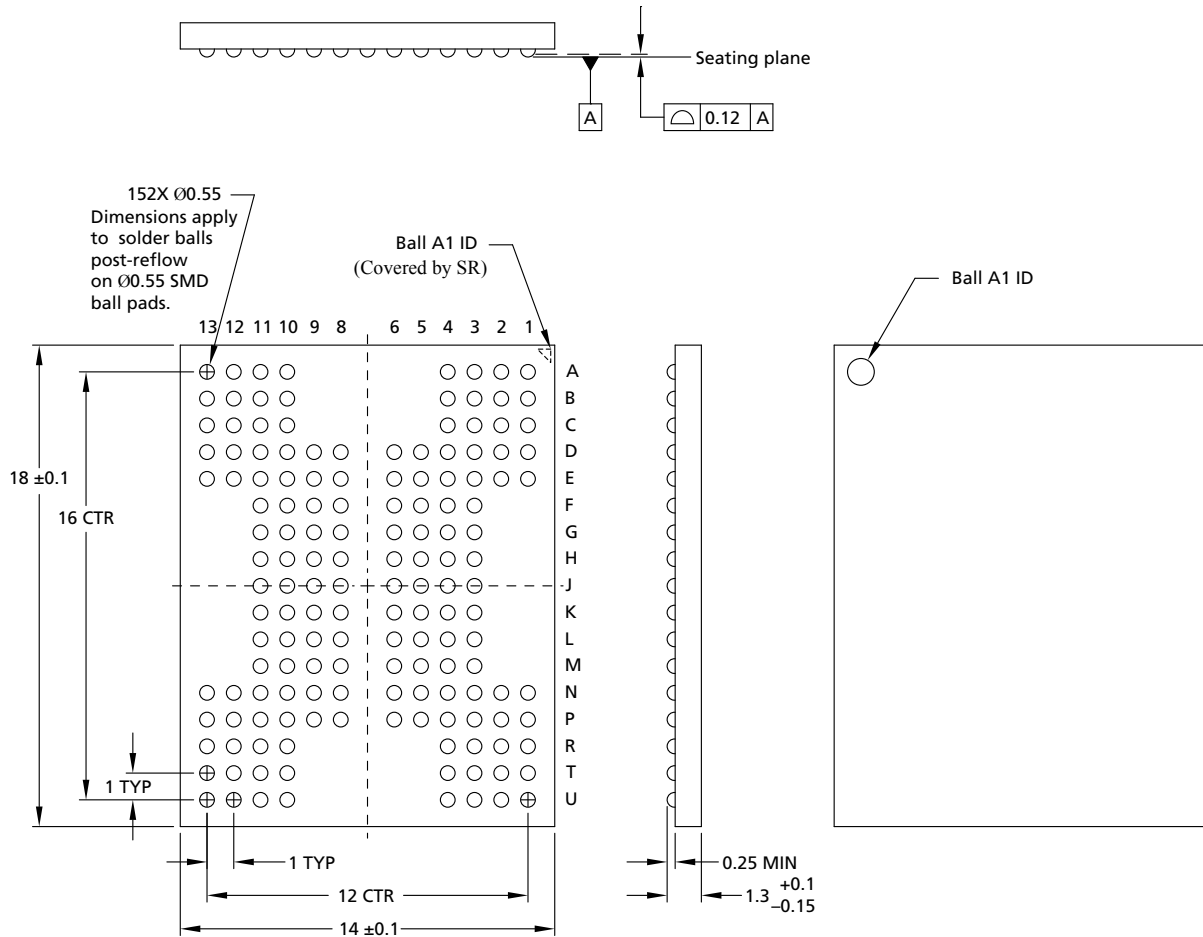


- Notes:
1. All dimensions are in millimeters.
 2. Solder ball material: SAC405 (95.5% Sn, 4% Ag, 0.5% Cu).



MLC 256Gb to 2Tb Async/Sync NAND Package Dimensions

Figure 6: 152-Ball LBGA – 14mm x 18mm (Package Code: H8)

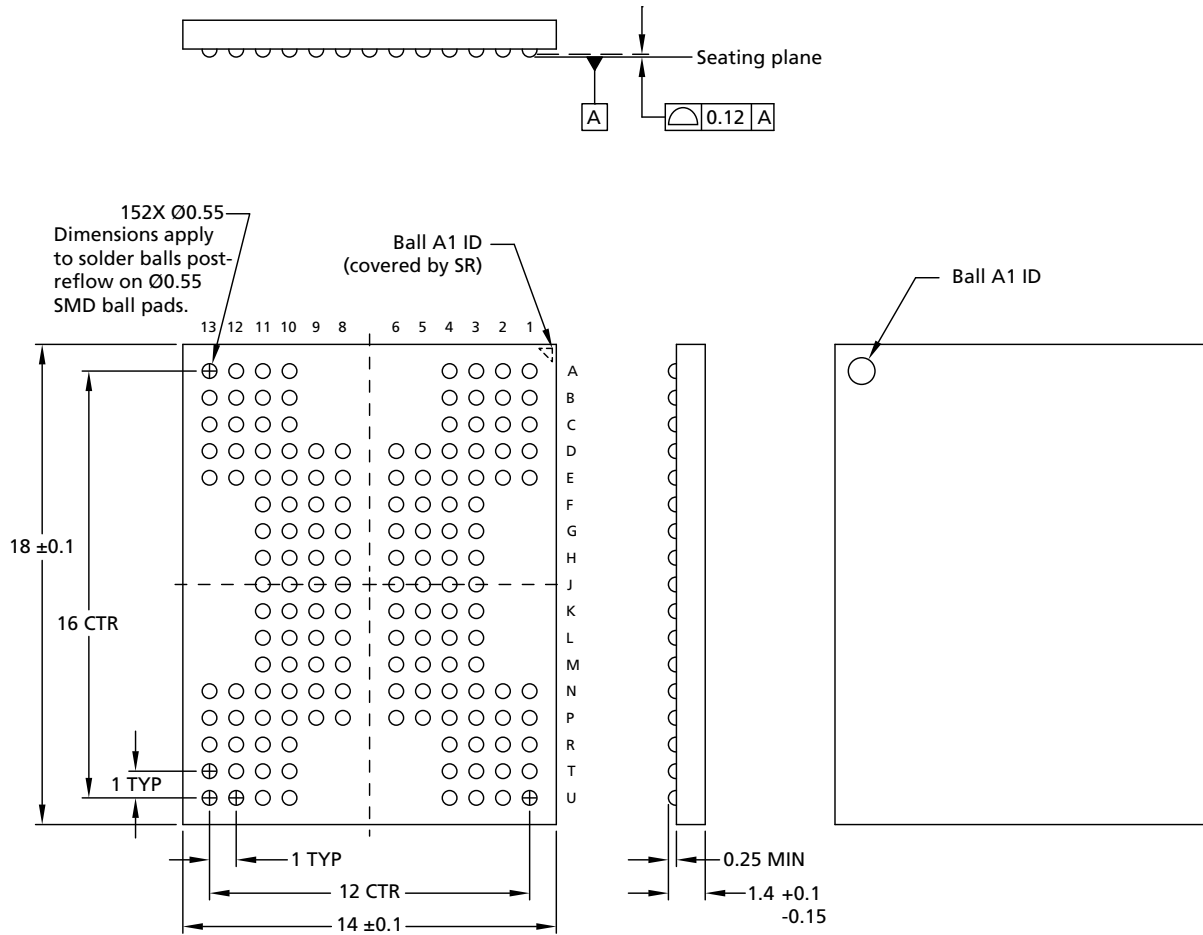


- Notes:
1. All dimensions are in millimeters.
 2. Solder ball material: SAC405 (95.5% Sn, 4% Ag, 0.5% Cu).



MLC 256Gb to 2Tb Async/Sync NAND Package Dimensions

Figure 7: 152-Ball LBGA – 14mm x 18mm (Package Code: J7)



- Notes:
1. All dimensions are in millimeters.
 2. Solder ball material: SAC405 (95.5% Sn, 4% Ag, 0.5% Cu).



MLC 256Gb to 2Tb Async/Async NAND Device and Array Organization

Device and Array Organization

Figure 8: Device Organization for Single-Die Package (152-ball BGA)

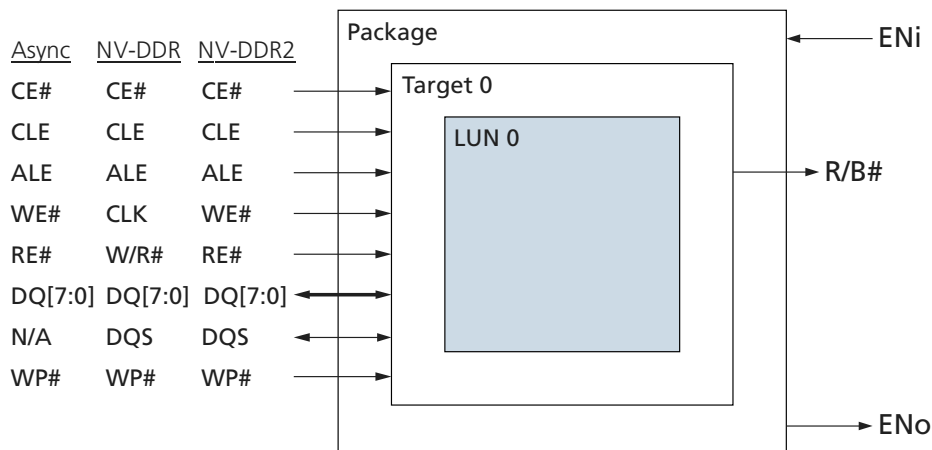
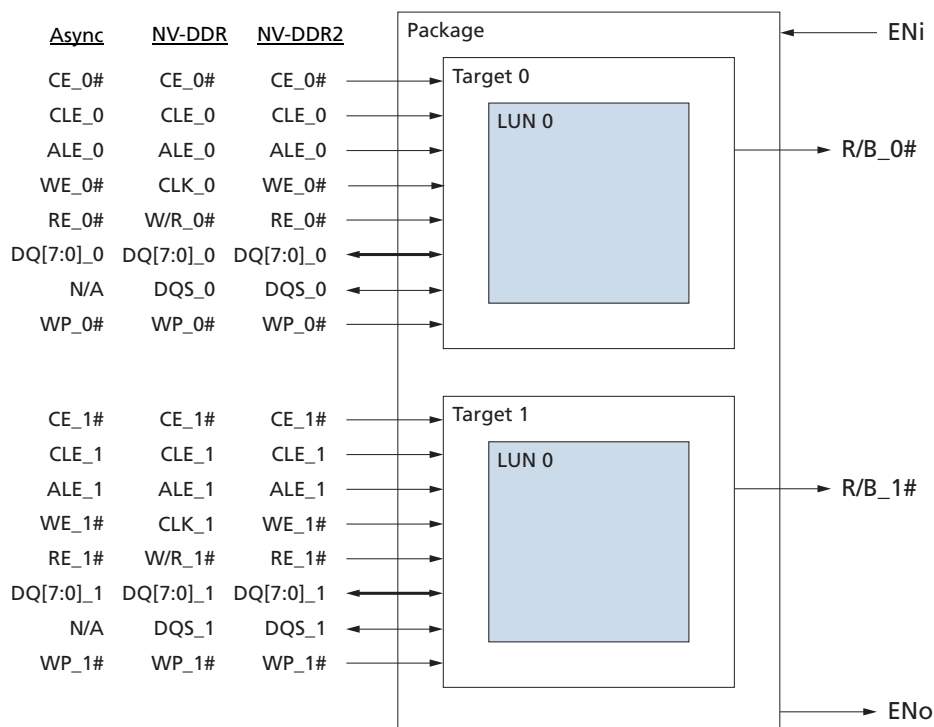


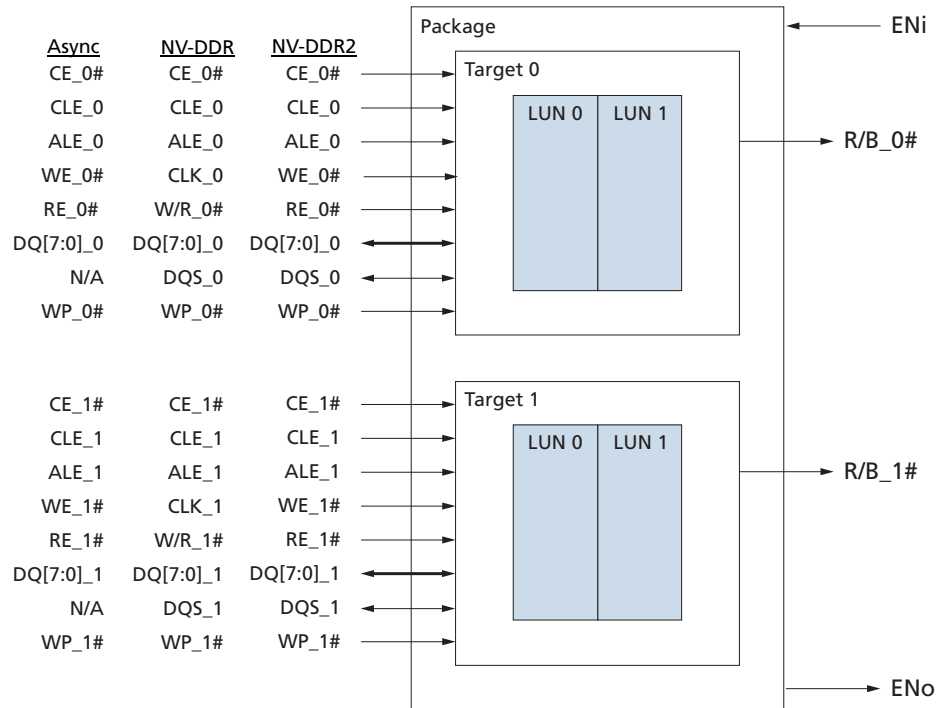
Figure 9: Device Organization for Two-Die Package (152-ball BGA)





MLC 256Gb to 2Tb Async/Sync NAND Device and Array Organization

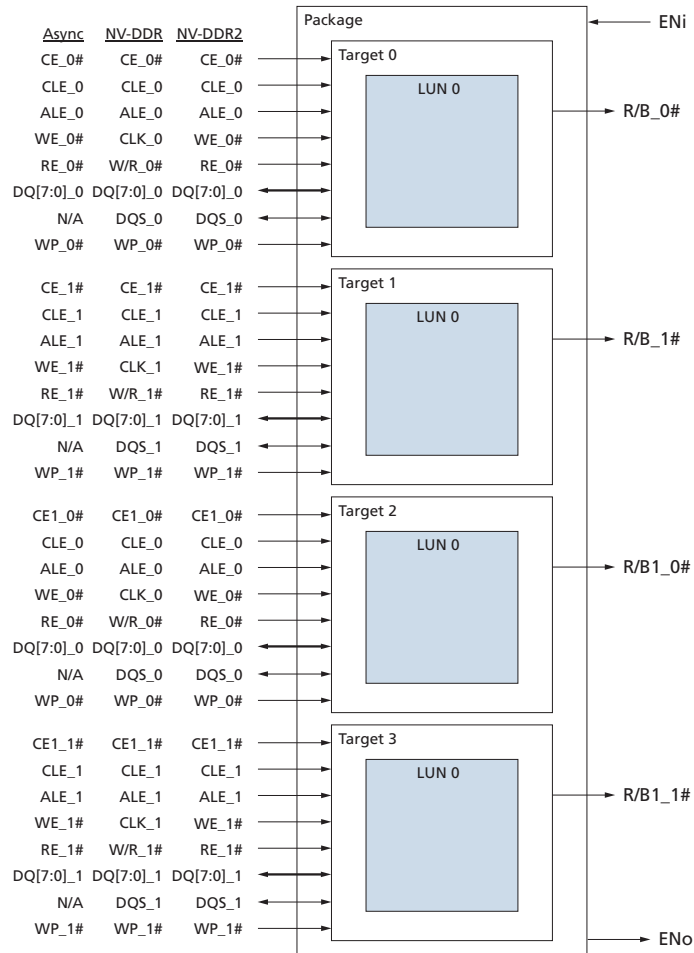
Figure 10: Device Organization for Four-Die Package with two CE# (152-ball BGA)





MLC 256Gb to 2Tb Async/Sync NAND Device and Array Organization

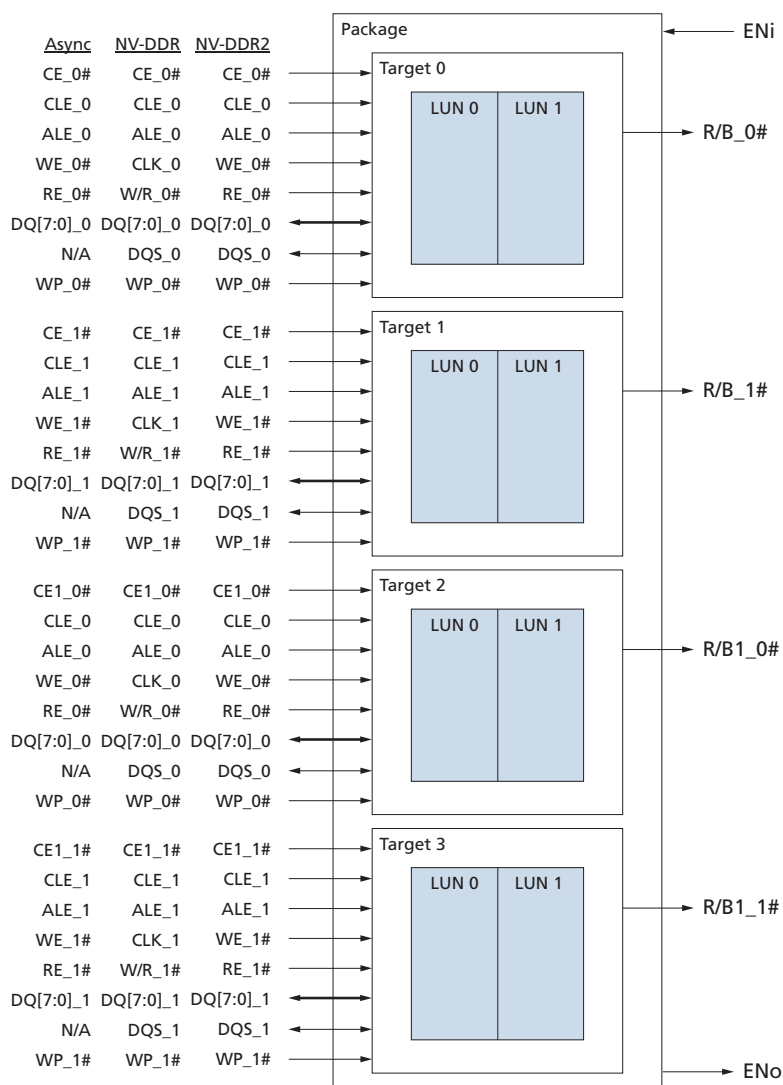
Figure 11: Device Organization for Four-Die Package with four CE# (152-ball BGA)





MLC 256Gb to 2Tb Async/Sync NAND Device and Array Organization

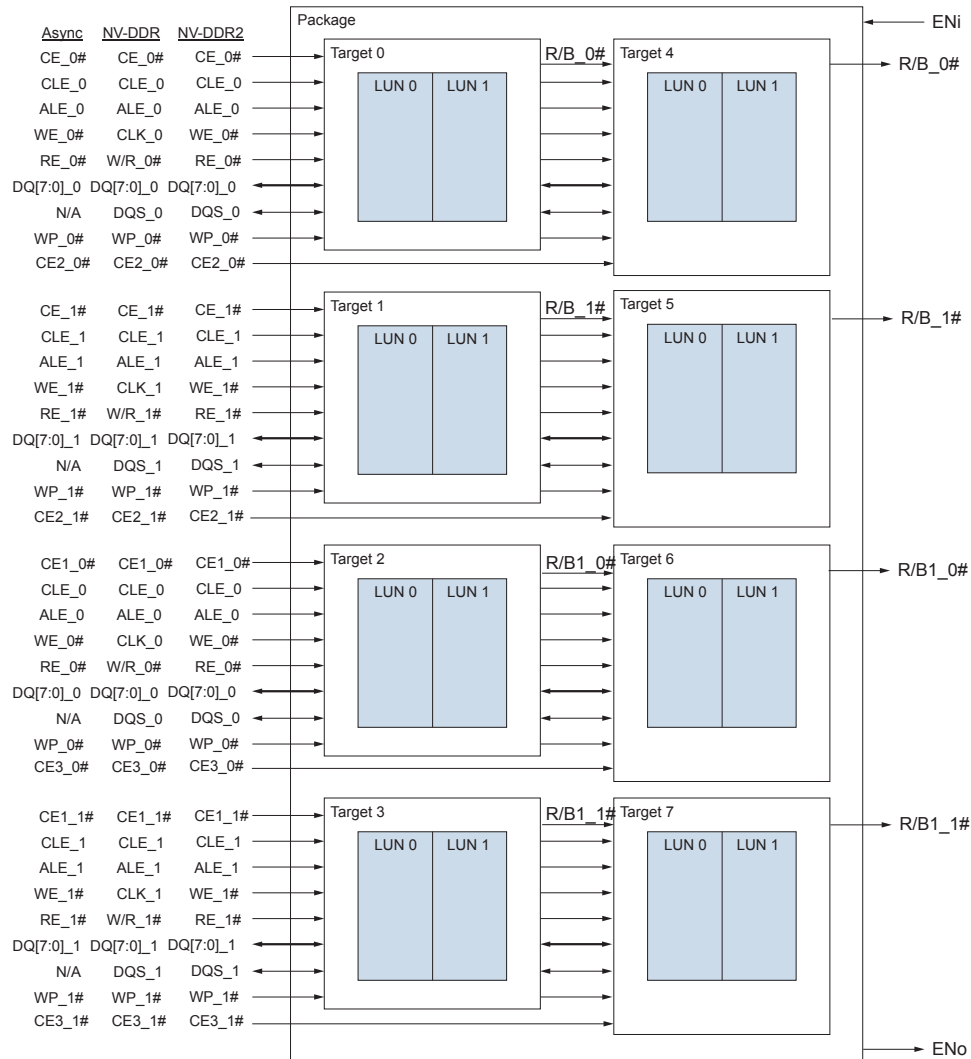
Figure 12: Device Organization for Eight-Die Package (152-ball BGA)





MLC 256Gb to 2Tb Async/Sync NAND Device and Array Organization

Figure 13: Device Organization for Sixteen-Die Package (152-ball BGA)





MLC 256Gb to 2Tb Async/Sync NAND Device and Array Organization

Figure 14: Array Organization per Logical Unit (LUN)

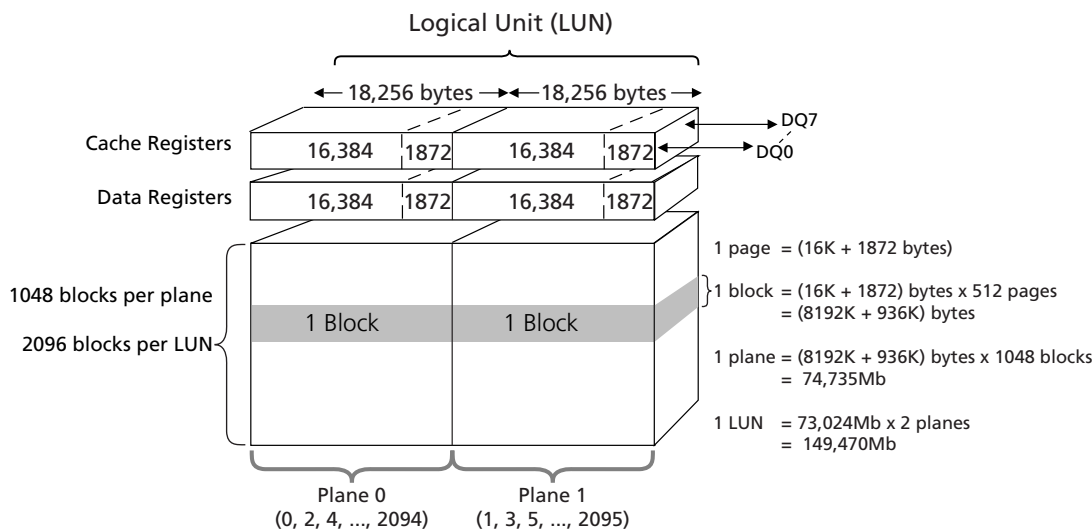


Table 2: Array Addressing for Logical Unit (LUN)

Cycle	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0
First	CA7	CA6	CA5	CA4	CA3	CA2	CA1	CA0 ²
Second	LOW	CA14 ³	CA13	CA12	CA11	CA10	CA9	CA8
Third	PA7	PA6	PA5	PA4	PA3	PA2	PA1	PA0
Fourth	BA15	BA14	BA13	BA12	BA11	BA10	BA9 ⁴	PA8
Fifth	LOW	LOW	LA0 ^{5,6}	BA20	BA19	BA18	BA17	BA16

- Notes:
1. CAx = column address, PAx = page address, BAx = block address, LAx = LUN address; the page address, block address, and LUN address are collectively called the row address.
 2. When using the synchronous interface, CA0 is forced to 0 internally; one data cycle always returns one even byte and one odd byte.
 3. Column addresses 18,255 (474Fh) through 32,767 (7FFFh) are invalid, out of bounds, do not exist in the device, and cannot be addressed.
 4. BA[9] is the plane-select bit:
Plane 0: BA[9] = 0
Plane 1: BA[9] = 1
 5. LA0 is the LUN-select bit. It is present only when two LUNs are shared on the target; otherwise, it should be held LOW.
LUN 0: LA0 = 0
LUN 1: LA0 = 1
 6. For single LUN Targets block addresses 2096 through 4095 are invalid, out of bounds, do not exist in the device, and cannot be addressed.
For two LUN Targets block addresses 2096 through 4095 and 6192 through 8191 are invalid, out of bounds, do not exist in the device, and cannot be addressed.



MLC 256Gb to 2Tb Async/Sync NAND Bus Operation – Asynchronous Interface

Bus Operation – Asynchronous Interface

The asynchronous interface is active when the NAND Flash device powers on. The I/O bus, DQ[7:0], is multiplexed sharing data I/O, addresses, and commands. The DQS signal, if present, is tri-stated when the asynchronous interface is active.

Asynchronous interface bus modes are summarized below.

Table 3: Asynchronous Interface Mode Selection

Mode	CE#	CLE	ALE	WE#	RE#	DQS	DQx	WP#	Notes
Standby	H	X	X	X	X	X	X	0V/V _{CCQ} ²	2
Bus idle	L	X	X	H	H	X	X	X	
Command input	L	H	L		H	X	input	H	
Address input	L	L	H		H	X	input	H	
Data input	L	L	L		H	X	input	H	
Data output	L	L	L	H		X	output	X	
Write protect	X	X	X	X	X	X	X	L	

- Notes:
1. DQS is tri-stated when the asynchronous interface is active.
 2. WP# should be biased to CMOS LOW or HIGH for standby.
 3. Mode selection settings for this table: H = Logic level HIGH; L = Logic level LOW; X = V_{IH} or V_{IL}.

Asynchronous Enable/Standby

A chip enable (CE#) signal is used to enable or disable a target. When CE# is driven LOW, all of the signals for that target are enabled. With CE# LOW, the target can accept commands, addresses, and data I/O. There may be more than one target in a NAND Flash package. Each target is controlled by its own chip enable; the first target (Target 0) is controlled by CE#; the second target (if present) is controlled by CE2#, etc.

A target is disabled when CE# is driven HIGH, even when the target is busy. When disabled, all of the target's signals are disabled except CE#, WP#, and R/B#. This functionality is also known as CE# "Don't Care". While the target is disabled, other devices can utilize the disabled NAND signals that are shared with the NAND Flash.

A target enters low-power standby when it is disabled and is not busy. If the target is busy when it is disabled, the target enters standby after all of the die (LUNs) complete their operations. Standby helps reduce power consumption.

Asynchronous Bus Idle

A target's bus is idle when CE# is LOW, WE# is HIGH, and RE# is HIGH.

During bus idle, all of the signals are enabled except DQS, which is not used when the asynchronous interface is active. No commands, addresses, and data are latched into the target; no data is output.



MLC 256Gb to 2Tb Async/Sync NAND Bus Operation – Asynchronous Interface

Asynchronous Pausing Data Input/Output

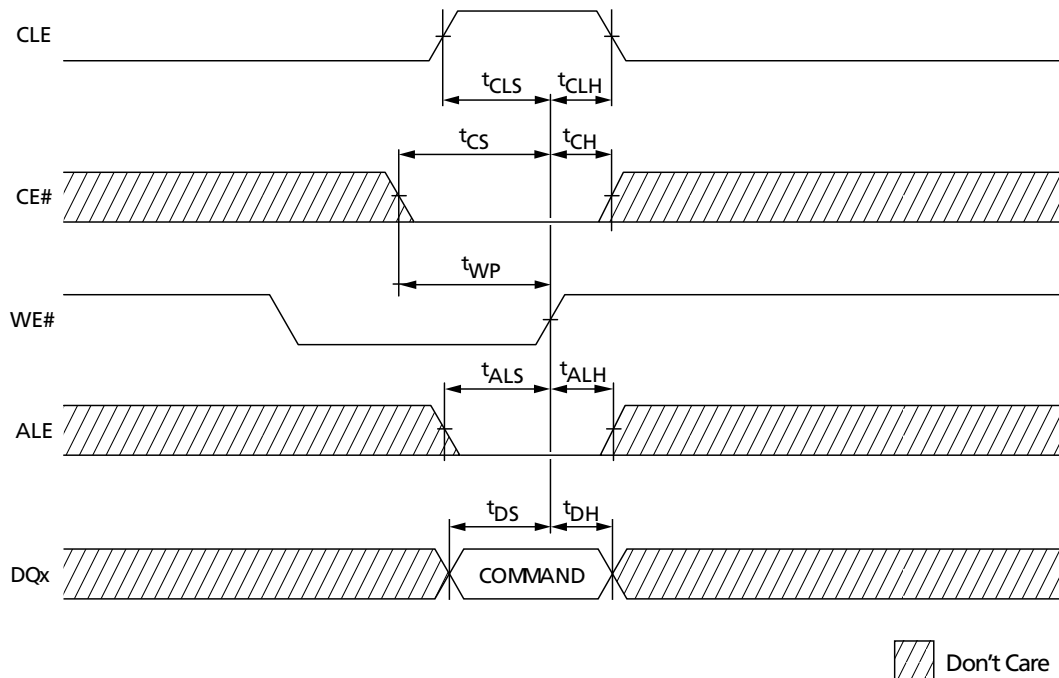
Pausing data input or data output is done by keeping WE# or RE# HIGH, respectively.

Asynchronous Commands

An asynchronous command is written from DQ[7:0] to the command register on the rising edge of WE# when CE# is LOW, ALE is LOW, CLE is HIGH, and RE# is HIGH.

Commands are typically ignored by die (LUNs) that are busy (RDY = 0); however, some commands, including READ STATUS (70h) and READ STATUS ENHANCED (78h), are accepted by die (LUNs) even when they are busy.

Figure 15: Asynchronous Command Latch Cycle





MLC 256Gb to 2Tb Async/Sync NAND Bus Operation – Asynchronous Interface

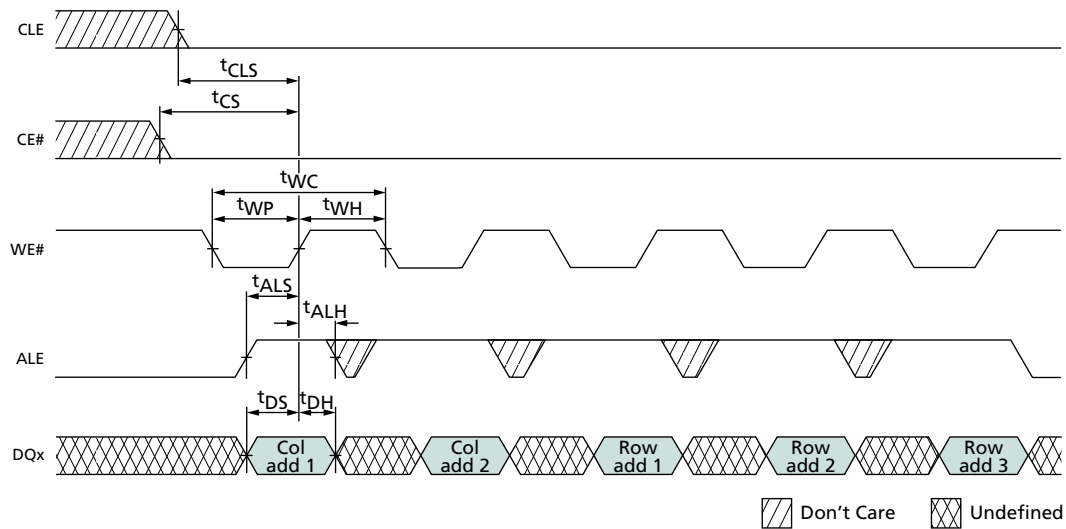
Asynchronous Addresses

An asynchronous address is written from DQ[7:0] to the address register on the rising edge of WE# when CE# is LOW, ALE is HIGH, CLE is LOW, and RE# is HIGH.

Bits that are not part of the address space must be LOW (see Device and Array Organization). The number of cycles required for each command varies. Refer to the command descriptions to determine addressing requirements (see Command Definitions).

Addresses are typically ignored by die (LUNs) that are busy (RDY = 0); however, some addresses are accepted by die (LUNs) even when they are busy; for example, address cycles that follow the READ STATUS ENHANCED (78h) command.

Figure 16: Asynchronous Address Latch Cycle





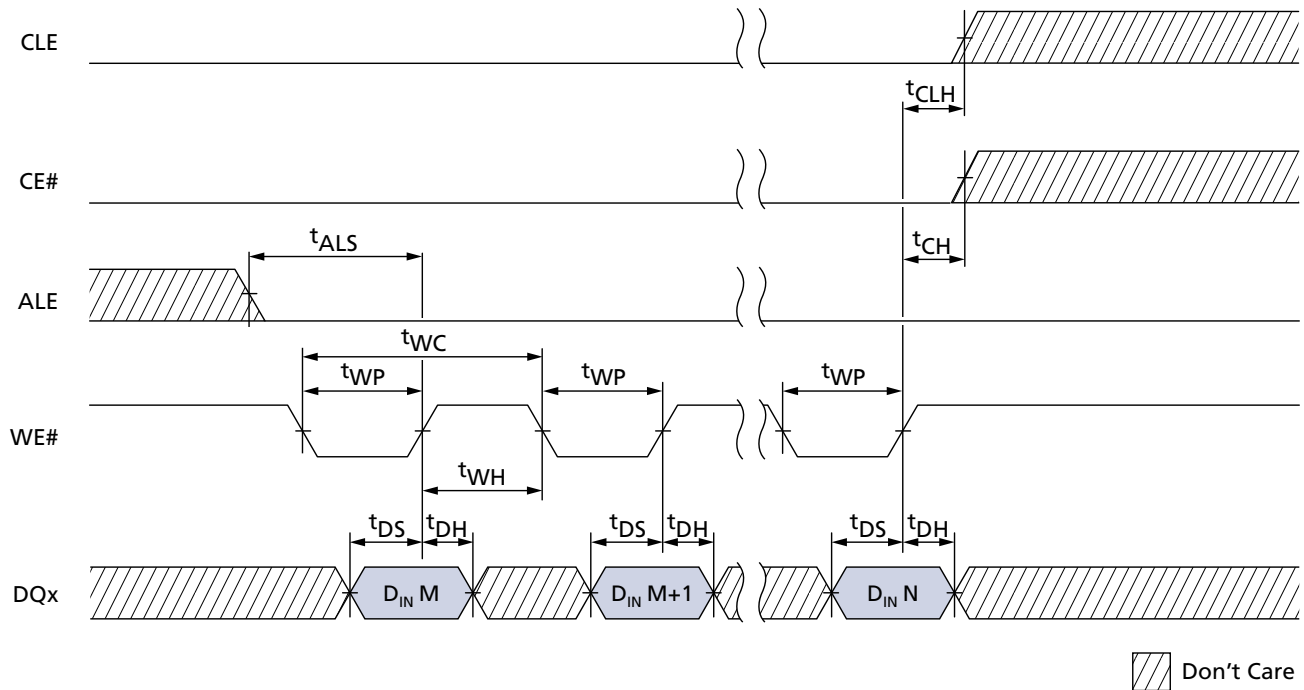
MLC 256Gb to 2Tb Async/Sync NAND Bus Operation – Asynchronous Interface

Asynchronous Data Input

Data is written from DQ[7:0] to the cache register of the selected die (LUN) on the rising edge of WE# when CE# is LOW, ALE is LOW, CLE is LOW, and RE# is HIGH.

Data input is ignored by die (LUNs) that are not selected or are busy (RDY = 0).

Figure 17: Asynchronous Data Input Cycles





MLC 256Gb to 2Tb Async/Sync NAND Bus Operation – Asynchronous Interface

Asynchronous Data Output

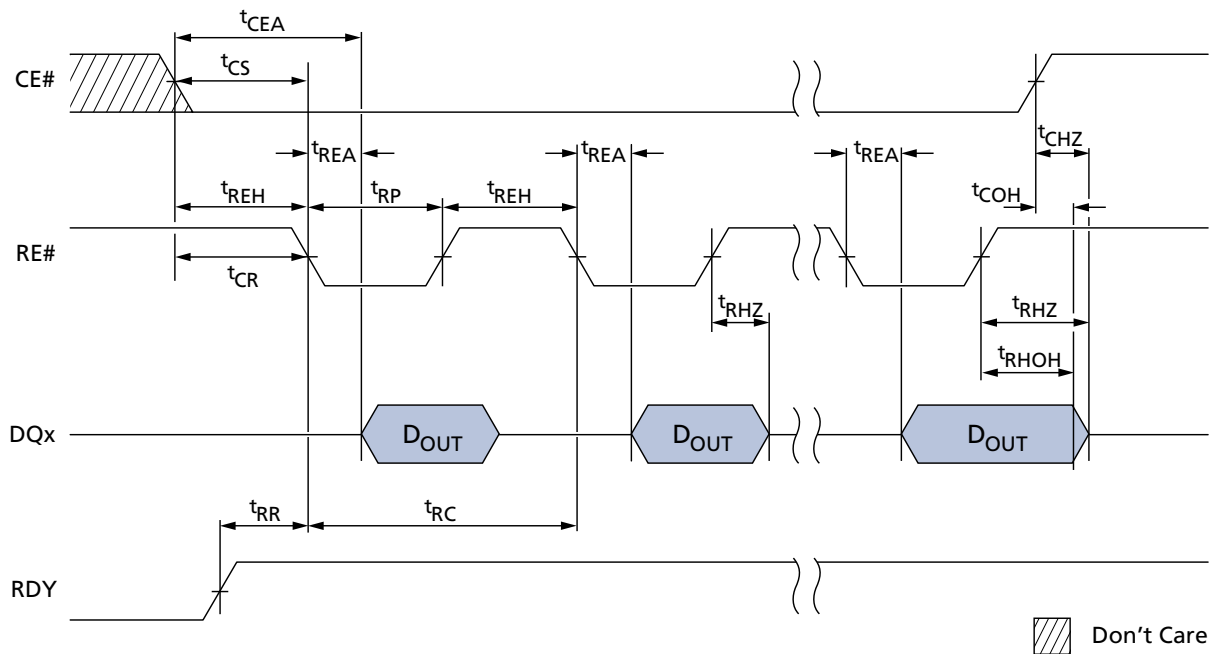
Data can be output from a die (LUN) if it is in a READY state. Data output is supported following a READ operation from the NAND Flash array. Data is output from the cache register of the selected die (LUN) to DQ[7:0] on the falling edge of RE# when CE# is LOW, ALE is LOW, CLE is LOW, and WE# is HIGH.

If the host controller is using a t_{RC} of 30ns or greater, the host can latch the data on the rising edge of RE# (see Figure 18 for proper timing). If the host controller is using a t_{RC} of less than 30ns, the host can latch the data on the next falling edge of RE# (see Figure 19 (page 32) for extended data output (EDO) timing).

Using the READ STATUS ENHANCED (78h) command prevents data contention following an interleaved die (multi-LUN) operation. After issuing the READ STATUS ENHANCED (78h) command, to enable data output, issue the READ MODE (00h) command.

Data output requests are typically ignored by a die (LUN) that is busy ($RDY = 0$); however, it is possible to output data from the status register even when a die (LUN) is busy by first issuing the READ STATUS (70h) or READ STATUS ENHANCED (78h) command.

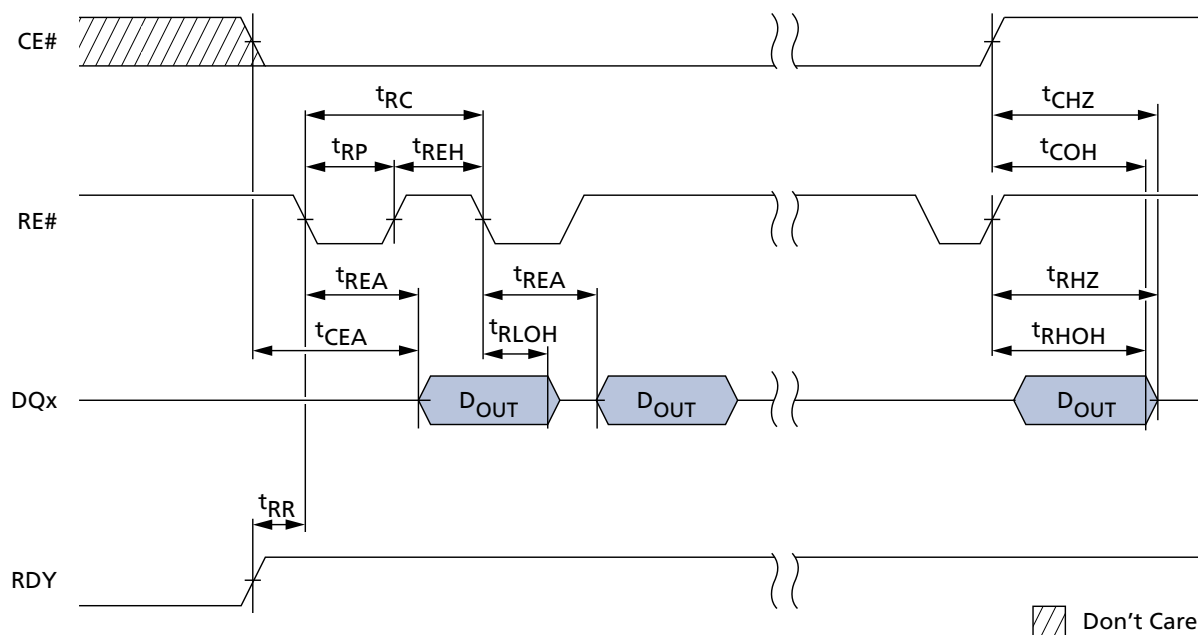
Figure 18: Asynchronous Data Output Cycles





MLC 256Gb to 2Tb Async/Sync NAND Bus Operation – Asynchronous Interface

Figure 19: Asynchronous Data Output Cycles (EDO Mode)



Write Protect

The write protect# (WP#) signal enables or disables PROGRAM and ERASE operations to a target. When WP# is LOW, PROGRAM and ERASE operations are disabled. When WP# is HIGH, PROGRAM and ERASE operations are enabled.

It is recommended that the host drive WP# LOW during power-on until V_{cc} and V_{ccq} are stable to prevent inadvertent PROGRAM and ERASE operations (see Device Initialization for additional details).

WP# must be transitioned only when the target is not busy and prior to beginning a command sequence. After a command sequence is complete and the target is ready, WP# can be transitioned. After WP# is transitioned, the host must wait t_{WW} before issuing a new command.

The WP# signal is always an active input, even when CE# is HIGH. This signal should not be multiplexed with other signals.

Ready/Busy#

The ready/busy# (R/B#) signal provides a hardware method of indicating whether a target is ready or busy. A target is busy when one or more of its die (LUNs) are busy (RDY = 0). A target is ready when all of its die (LUNs) are ready (RDY = 1). Because each die (LUN) contains a status register, it is possible to determine the independent status of each die (LUN) by polling its status register instead of using the R/B# signal (see Status Operations for details regarding die (LUN) status).

This signal requires a pull-up resistor, R_p, for proper operation. R/B# is HIGH when the target is ready, and transitions LOW when the target is busy. The signal's open-drain



MLC 256Gb to 2Tb Async/Sync NAND Bus Operation – Asynchronous Interface

driver enables multiple R/B# outputs to be OR-tied. Typically, R/B# is connected to an interrupt pin on the system controller (see Figure 20 (page 33)).

The combination of R_p and capacitive loading of the R/B# circuit determines the rise time of the R/B# signal. The actual value used for R_p depends on the system timing requirements. Large values of R_p cause R/B# to be delayed significantly. Between the 10- to 90-percent points on the R/B# waveform, the rise time is approximately two time constants (TC).

$$TC = R \times C$$

Where $R = R_p$ (resistance of pull-up resistor), and $C =$ total capacitive load.

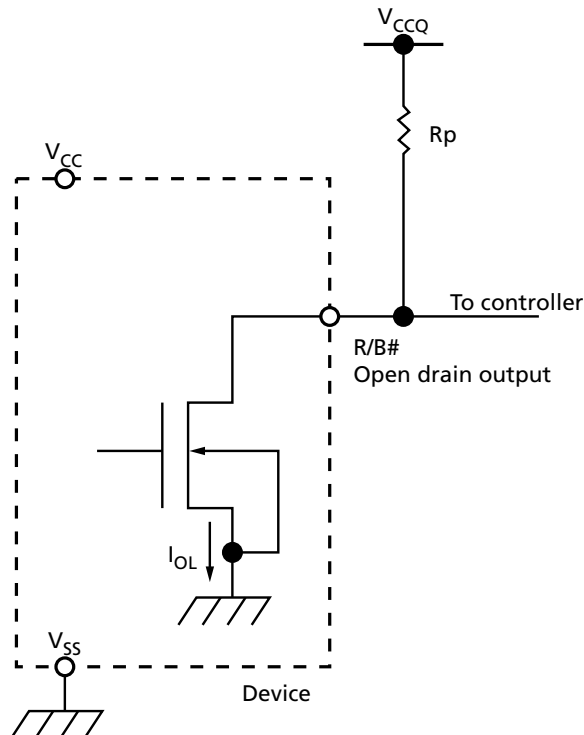
The fall time of the R/B# signal is determined mainly by the output impedance of the R/B# signal and the total load capacitance. Approximate R_p values using a circuit load of 100pF are provided in Figure 25 (page 36).

The minimum value for R_p is determined by the output drive capability of the R/B# signal, the output voltage swing, and V_{CCQ} .

$$R_p = \frac{V_{CCQ}(\text{MAX}) - V_{OL}(\text{MAX})}{I_{OL} + \Sigma i_l}$$

Where Σi_l is the sum of the input currents of all devices tied to the R/B# pin.

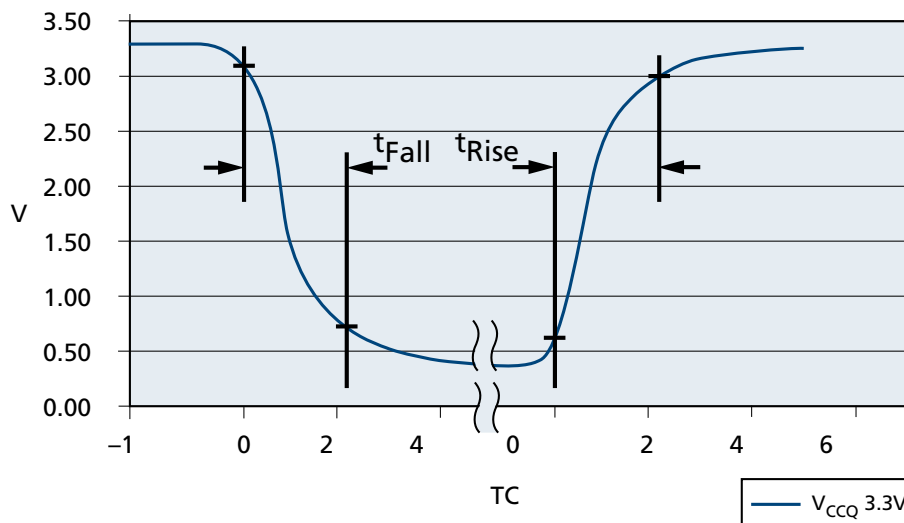
Figure 20: READ/BUSY# Open Drain





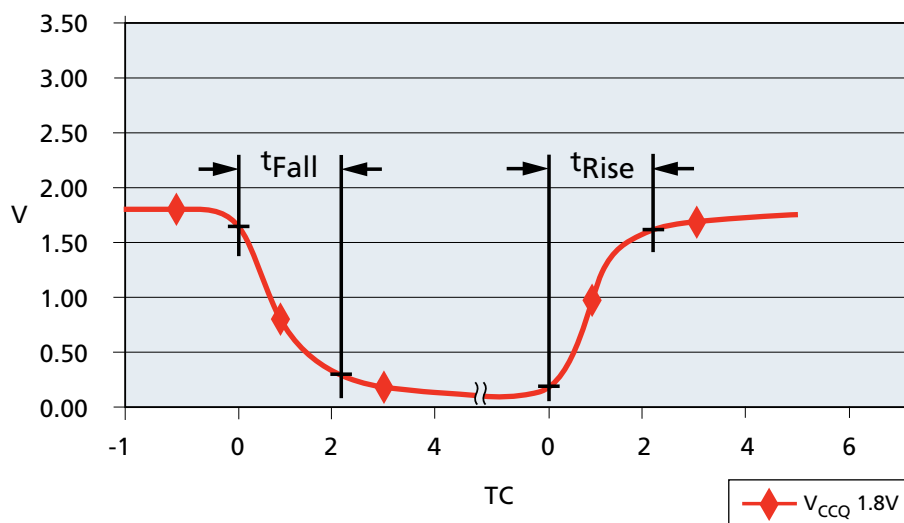
MLC 256Gb to 2Tb Async/Sync NAND Bus Operation – Asynchronous Interface

Figure 21: t_{Fall} and t_{Rise} ($V_{\text{CCQ}} = 2.7\text{-}3.6\text{V}$)



- Notes:
1. t_{FALL} is $V_{\text{OH(DC)}}$ to $V_{\text{OL(AC)}}$ and t_{RISE} is $V_{\text{OL(DC)}}$ to $V_{\text{OH(AC)}}$.
 2. t_{Rise} dependent on external capacitance and resistive loading and output transistor impedance.
 3. t_{Rise} primarily dependent on external pull-up resistor and external capacitive loading.
 4. $t_{\text{Fall}} = 10\text{ns}$ at 3.3V
 5. See TC values in Figure 25 (page 36) for approximate R_p value and TC.

Figure 22: t_{Fall} and t_{Rise} ($V_{\text{CCQ}} = 1.7\text{-}1.95\text{V}$)



- Notes:
1. t_{FALL} is $V_{\text{OH(DC)}}$ to $V_{\text{OL(AC)}}$ and t_{RISE} is $V_{\text{OL(DC)}}$ to $V_{\text{OH(AC)}}$.
 2. t_{Rise} is primarily dependent on external pull-up resistor and external capacitive loading.
 3. $t_{\text{Fall}} \approx 7\text{ns}$ at 1.8V.
 4. See TC values in Figure 25 (page 36) for TC and approximate R_p value.



MLC 256Gb to 2Tb Async/Sync NAND Bus Operation – Asynchronous Interface

Figure 23: IOL vs Rp ($V_{CCQ} = 2.7-3.6V$)

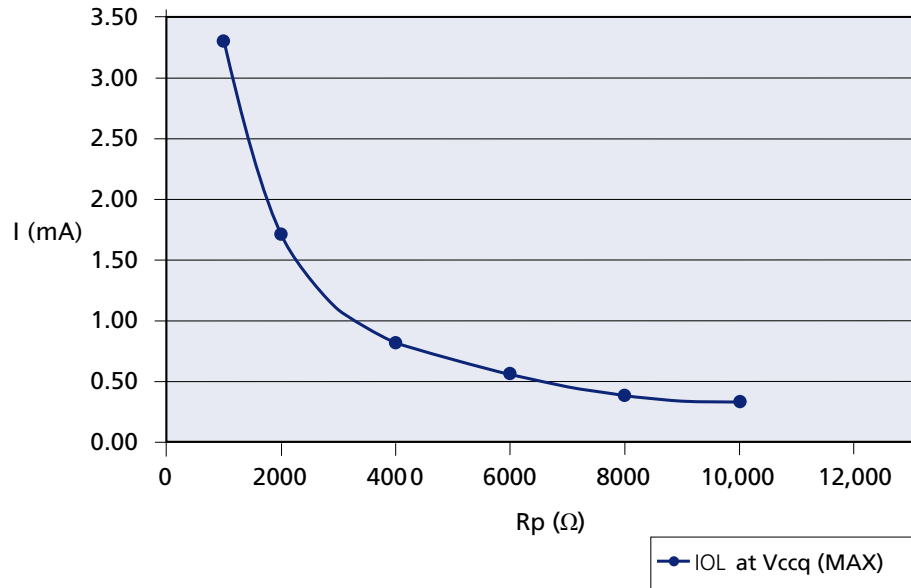
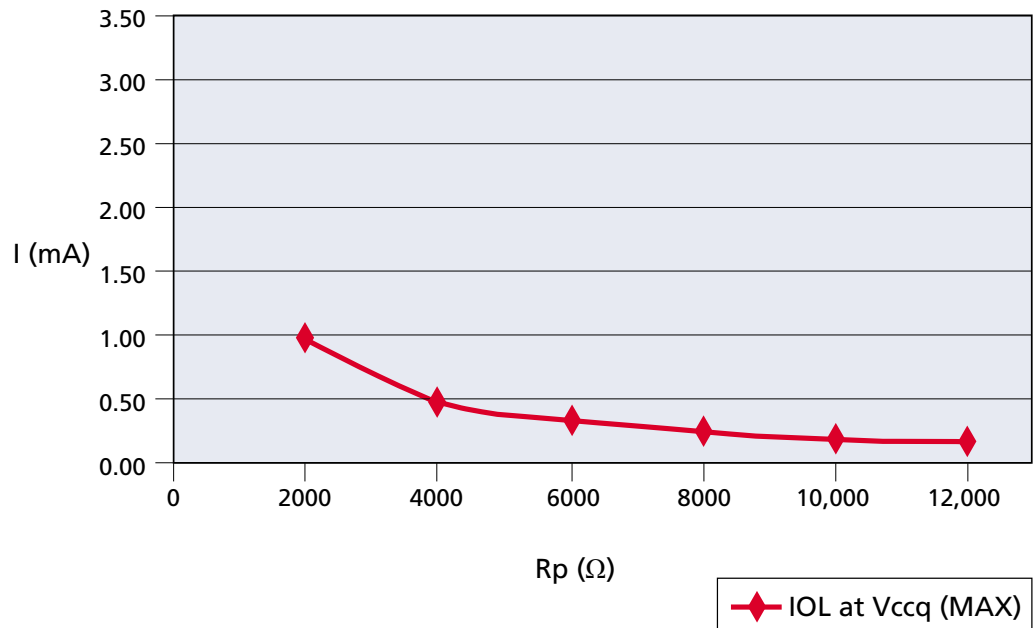


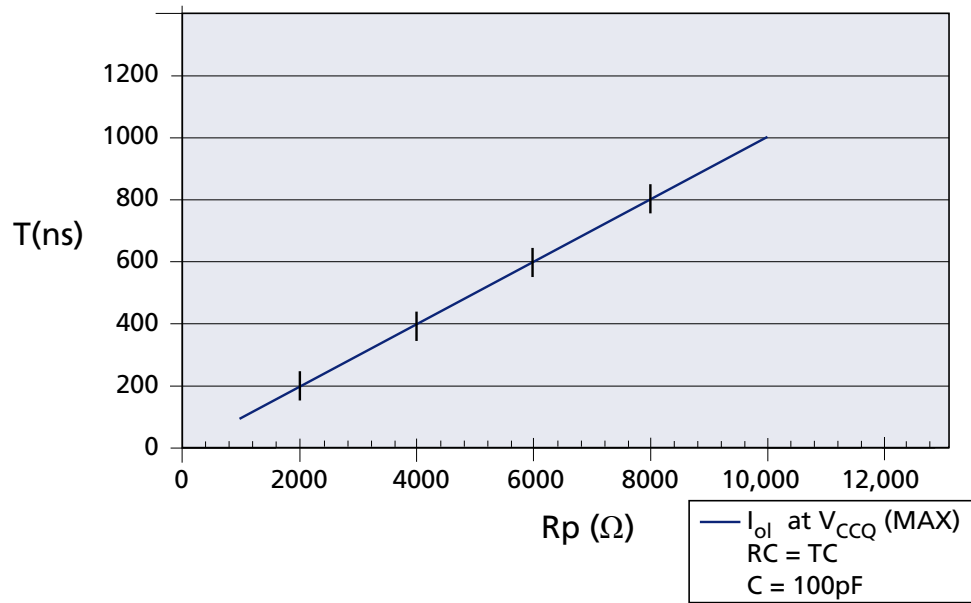
Figure 24: IOL vs Rp ($V_{CCQ} = 1.7-1.95V$)





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Figure 25: TC vs Rp





MLC 256Gb to 2Tb Async/Sync NAND Bus Operation – NV-DDR Interface

Bus Operation – NV-DDR Interface

These NAND Flash devices have two interfaces—a synchronous interface for fast data (NV-DDR) I/O transfer and an asynchronous interface that is backward compatible with existing NAND Flash devices.

The NAND Flash command protocol for both the asynchronous and NV-DDR interfaces is identical. However, there are some differences between the asynchronous and NV-DDR interfaces when issuing command, address, and data I/O cycles using the NAND Flash signals.

When the NV-DDR interface is activated on a target (see Activating Interfaces), the target is capable of high-speed NV-DDR data transfers. Existing signals are redefined for high-speed NV-DDR I/O. The WE# signal becomes CLK. DQS is enabled. The RE# signal becomes W/R#. CLK provides a clock reference to the NAND Flash device.

DQS is a bidirectional data strobe. During data output, DQS is driven by the NAND Flash device. During data input, DQS is controlled by the host controller while inputting data on DQ[7:0].

The direction of DQS and DQ[7:0] is controlled by the W/R# signal. When the W/R# signal is latched HIGH, the controller is driving the DQ bus and DQS. When the W/R# is latched LOW, the NAND Flash is driving the DQ bus and DQS.

Transition from the NV-DDR interface to the NV-DDR2 interface is not permitted.

The NV-DDR interface bus modes are summarized below.

Table 4: NV-DDR Interface Mode Selection

Mode	CE#	CLE	ALE	CLK	W/R#	DQS	DQ[7:0]	WP#	Notes
Standby	H	X	X	X	X	X	X	0V/V _{CCQ}	1, 2
Bus idle	L	L	L		H	X	X	X	
Bus driving	L	L	L		L	output	output	X	
Command input	L	H	L		H	X	input	H	3
Address input	L	L	H		H	X	input	H	3
Data input	L	H	H		H		input	H	4
Data output	L	H	H		L	See Note 5	output	X	5
Write protect	X	X	X	X	X	X	X	L	
Undefined	L	L	H		L	output	output	X	
Undefined	L	H	L		L	output	output	X	

- Notes:
1. CLK can be stopped when the target is disabled, even when R/B# is LOW.
 2. WP# should be biased to CMOS LOW or HIGH for standby.
 3. Commands and addresses are latched on the rising edge of CLK.



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4. During data input to the device, DQS is the “clock” that latches the data in the cache register.
5. During data output from the NAND Flash device, DQS is an output generated from CLK after t_{DQSCK} delay.
6. Mode selection settings for this table: H = Logic level HIGH; L = Logic level LOW; X = V_{IH} or V_{IL} .

NV-DDR Enable/Standby

In addition to the description found in Asynchronous Enable/Standby, the following requirements also apply when the NV-DDR interface is active.

Before enabling a target, CLK must be running and ALE and CLE must be LOW. When CE# is driven LOW, all of the signals for the selected target are enabled. The target is not enabled until t_{CS} completes; the target's bus is then idle.

Prior to disabling a target, the target's bus must be idle. A target is disabled when CE# is driven HIGH, even when it is busy. All of the target's signals are disabled except CE#, WP#, and R/B#. After the target is disabled, CLK can be stopped.

A target enters low-power standby when it is disabled and is not busy. If the target is busy when it is disabled, the target enters standby after all of the die (LUNs) complete their operations.

NV-DDR Bus Idle/Driving

A target's bus is idle or driving when CLK is running, CE# is LOW, ALE is LOW, and CLE is LOW.

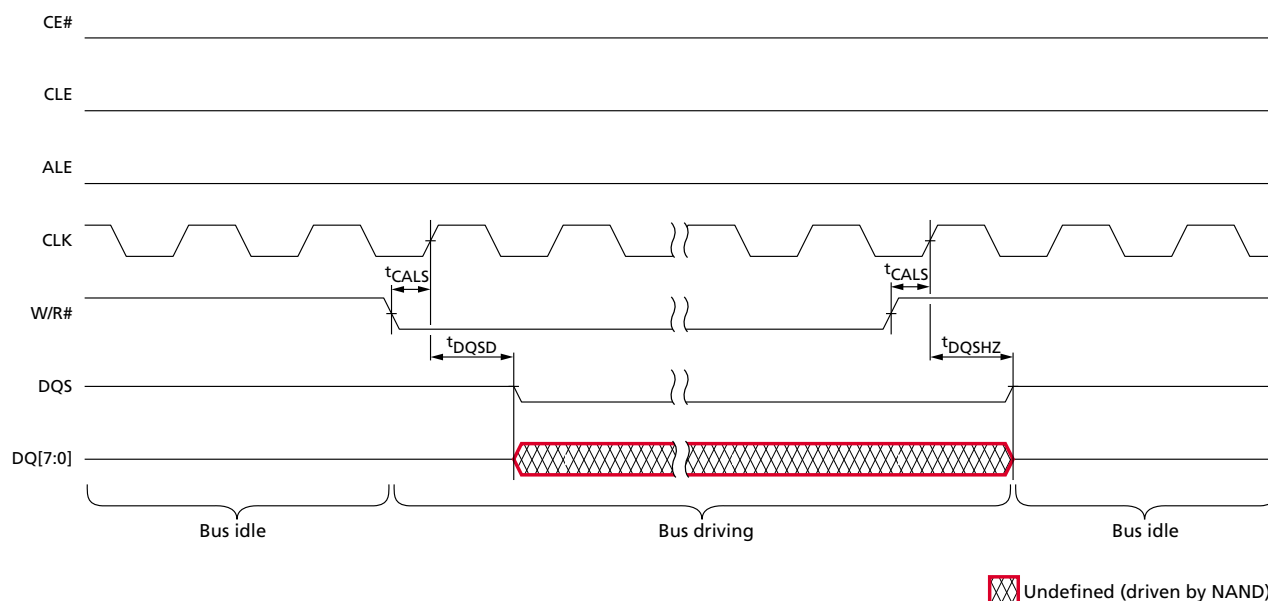
The bus is idle when W/R# transitions HIGH and is latched by CLK. During the bus idle mode, all signals are enabled; DQS and DQ[7:0] are inputs. No commands, addresses, or data are latched into the target; no data is output. When entering the bus idle mode, the host must wait a minimum of t_{CAD} before changing the bus mode. In the bus idle mode, the only valid bus modes supported are: bus driving, command, address, and NV-DDR data input.

The bus is driving when W/R# transitions LOW and is latched by CLK. During the bus driving mode, all signals are enabled; DQS is LOW and DQ[7:0] is driven LOW or HIGH, but no valid data is output. Following the bus driving mode, the only valid bus modes supported are bus idle and NV-DDR data output.



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Figure 26: NV-DDR Bus Idle/Driving Behavior



Note: 1. Only the selected die (LUN) drives DQS and DQ[7:0]. During an interleaved die (multi-LUN) operation, the host must use the READ STATUS ENHANCED (78h) to prevent data output contention.

NV-DDR Pausing Data Input/Output

Pausing data input or data output is done by setting ALE and CLE to LOW. The host may continue data transfer by setting ALE and CLE to HIGH after the applicable t_{CAD} time has passed.

NV-DDR Commands

A command is written from DQ[7:0] to the command register on the rising edge of CLK when CE# is LOW, ALE is LOW, CLE is HIGH, and W/R# is HIGH.

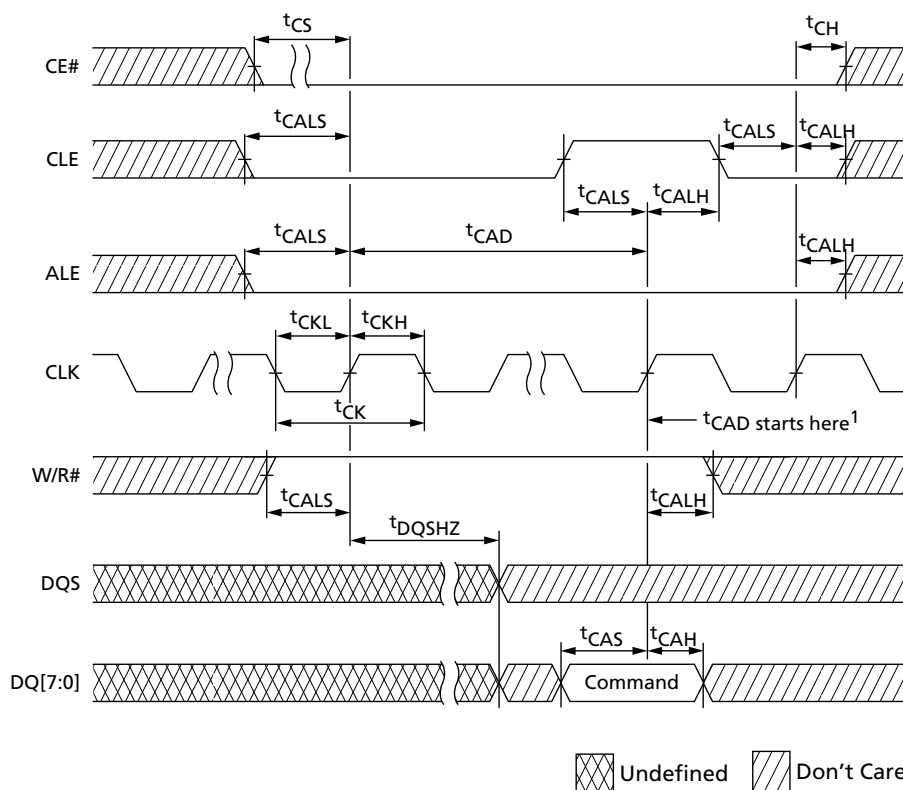
After a command is latched—and prior to issuing the next command, address, or data I/O—the bus must go to bus idle mode on the next rising edge of CLK, except when the clock period, t_{CK} , is greater than t_{CAD} .

Commands are typically ignored by die (LUNs) that are busy ($RDY = 0$); however, some commands, such as READ STATUS (70h) and READ STATUS ENHANCED (78h), are accepted by die (LUNs), even when they are busy.



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Figure 27: NV-DDR Command Cycle



Note: 1. When CE# remains LOW, t_{CAD} begins at the rising edge of the clock from which the command cycle is latched for subsequent command, address, data input, or data output cycle(s).

NV-DDR Addresses

A NV-DDR address is written from DQ[7:0] to the address register on the rising edge of CLK when CE# is LOW, ALE is HIGH, CLE is LOW, and W/R# is HIGH.

After an address is latched—and prior to issuing the next command, address, or data I/O—the bus must go to bus idle mode on the next rising edge of CLK, except when the clock period, t_{CK} , is greater than t_{CAD} .

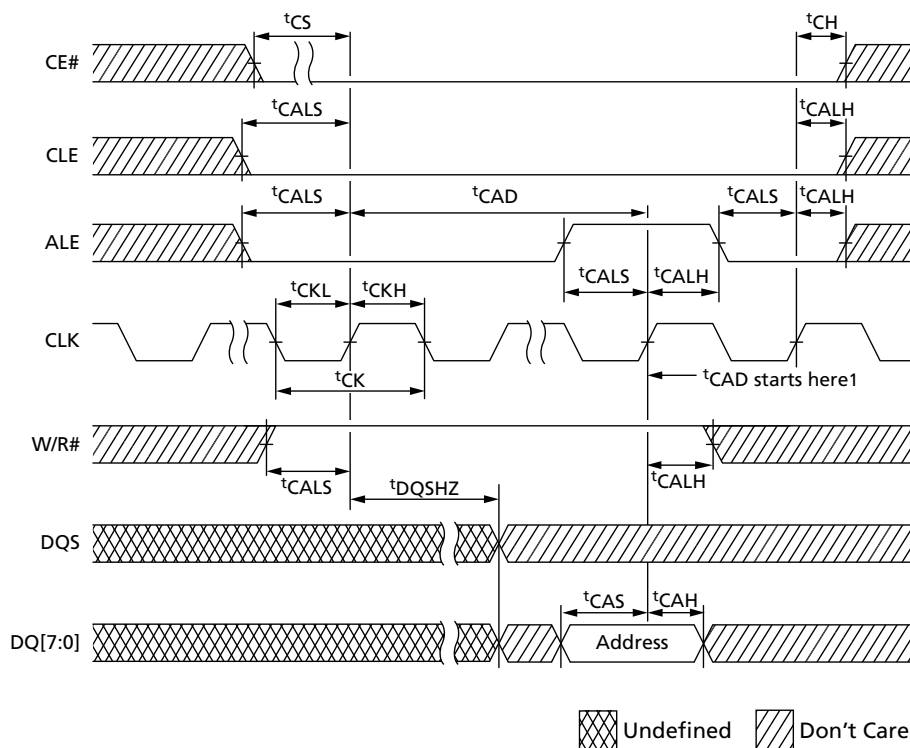
Bits not part of the address space must be LOW (see Device and Array Organization). The number of address cycles required for each command varies. Refer to the command descriptions to determine addressing requirements.

Addresses are typically ignored by die (LUNs) that are busy ($RDY = 0$); however, some addresses such as address cycles that follow the READ STATUS ENHANCED (78h) command, are accepted by die (LUNs), even when they are busy.



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Figure 28: NV-DDR Address Cycle



Note: 1. When CE# remains LOW, t_{CAD} begins at the rising edge of the clock from which the command cycle is latched for subsequent command, address, data input, or data output cycle(s).

NV-DDR DDR Data Input

To enter the NV-DDR data input mode, the following conditions must be met:

- CLK is running
- CE# is LOW
- W/R# is HIGH
- t_{CAD} is met
- DQS is LOW
- ALE and CLE are HIGH on the rising edge of CLK

Upon entering the NV-DDR data input mode after t_{DQSS} , data is written from DQ[7:0] to the cache register on each and every rising and falling edge of DQS (center-aligned) when CLK is running and the DQS to CLK skew meets t_{DSH} and t_{DSS} , CE# is LOW, W/R# is HIGH, and ALE and CLE are HIGH on the rising edge of CLK.

To exit NV-DDR data input mode, the following conditions must be met:

- CLK is running and the DQS to CLK skew meets t_{DSH} and t_{DSS}
- CE# is LOW
- W/R# is HIGH
- ALE and CLE are latched LOW on the rising edge of CLK



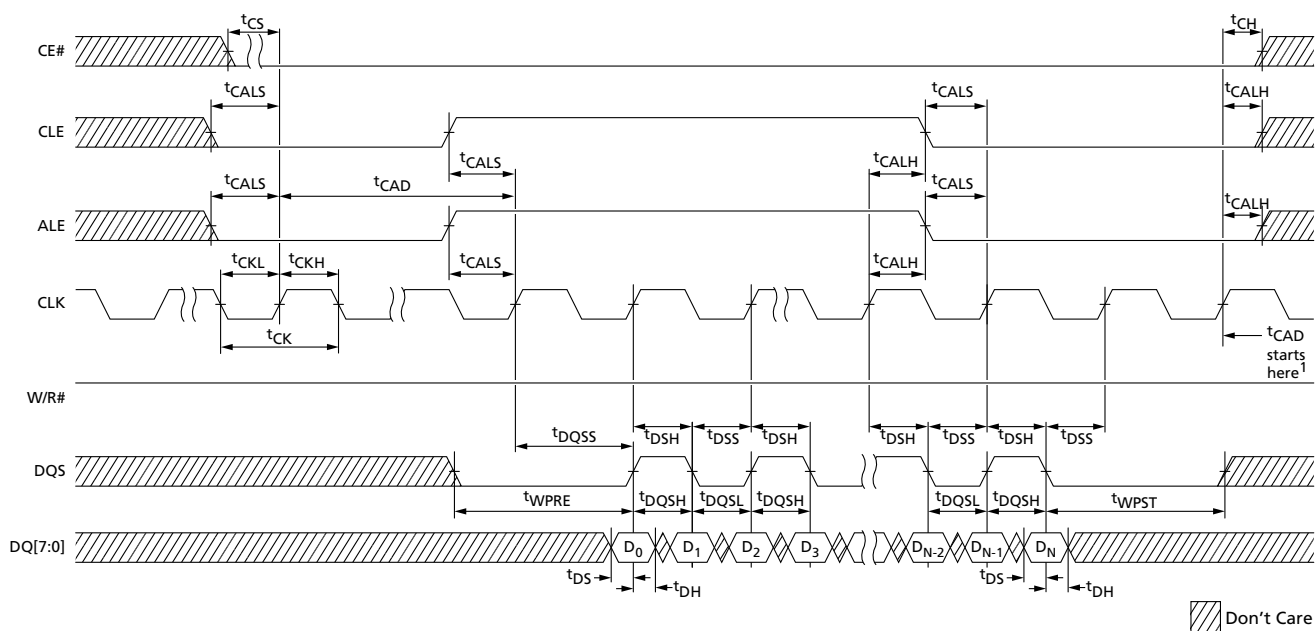
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- The final two data bytes of the data input sequence are written to DQ[7:0] to the cache register on the rising and falling edges of DQS after the last cycle in the data input sequence in which ALE and CLE are latched HIGH.
- DQS is held LOW for t_{WPST} (after the final falling edge of DQS)

Following t_{WPST} , the bus enters bus idle mode and t_{CAD} begins on the next rising edge of CLK. After t_{CAD} starts, the host can disable the target if desired.

Data input is ignored by die (LUNs) that are not selected or are busy.

Figure 29: NV-DDR Data Input Cycles



- Notes:
1. When CE# remains LOW, t_{CAD} begins at the first rising edge of the clock after t_{WPST} completes.
 2. t_{DSH} (MIN) generally occurs during t_{DQSS} (MIN).
 3. t_{DSS} (MIN) generally occurs during t_{DQSS} (MAX).

NV-DDR Data Output

Data can be output from a die (LUN) if it is ready. Data output is supported following a READ operation from the NAND Flash array.

To enter the NV-DDR data output mode, the following conditions must be met:

- CLK is running
- CE# is LOW
- The host has released the DQ[7:0] bus and DQS
- W/R# is latched LOW on the rising edge of CLK to enable the selected die (LUN) to take ownership of the DQ[7:0] bus and DQS within t_{WRCK}
- t_{CAD} is met
- ALE and CLE are HIGH on the rising edge of CLK



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Upon entering the NV-DDR data output mode, DQS will toggle HIGH and LOW with a delay of t_{DQSCK} from the respective rising and falling edges of CLK. DQ[7:0] will output data edge-aligned to the rising and falling edges of DQS, with the first transition delayed by no more than t_{AC} .

NV-DDR data output mode continues as long as CLK is running, CE# is LOW, W/R# is LOW, and ALE and CLE are HIGH on the rising edge of CLK.

To exit NV-DDR data output mode, the following conditions must be met:

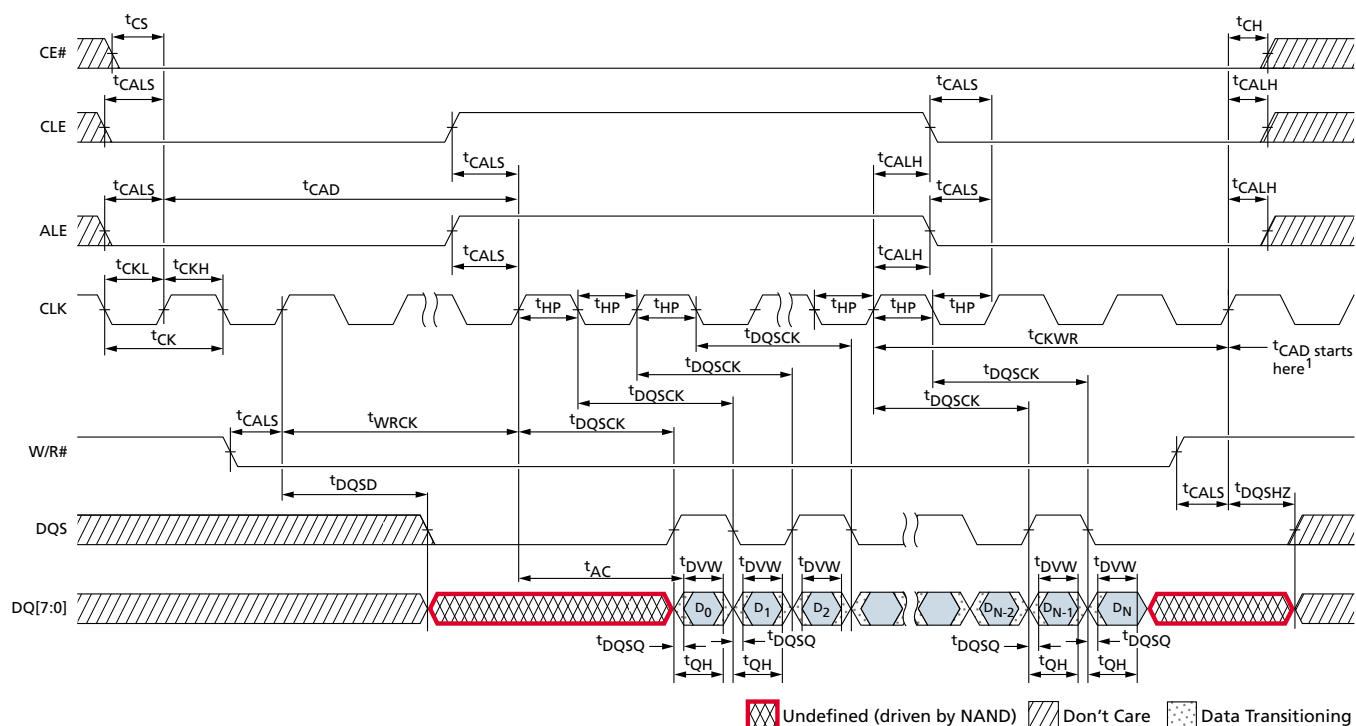
- CLK is running
- CE# is LOW
- W/R# is LOW
- ALE and CLE are latched LOW on the rising edge of CLK

The final two data bytes are output on DQ[7:0] on the final rising and falling edges of DQS. The final rising and falling edges of DQS occur t_{DQSCK} after the last cycle in the data output sequence in which ALE and CLE are latched HIGH. After t_{CKWR} , the bus enters bus idle mode and t_{CAD} begins on the next rising edge of CLK. Once t_{CAD} starts the host can disable the target if desired.

Data output requests are typically ignored by a die (LUN) that is busy ($RDY = 0$); however, it is possible to output data from the status register even when a die (LUN) is busy by issuing the READ STATUS (70h) or READ STATUS ENHANCED (78h) command.

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Figure 30: NV-DDR Data Output Cycles



- Notes:
1. When CE# remains LOW, t_{CAD} begins at the rising edge of the clock after t_{CKWR} for subsequent command or data output cycle(s).
 2. See Figure 27 (page 40) for details of W/R# behavior.
 3. t_{AC} is the DQ output window relative to CLK and is the long-term component of DQ skew.
 4. For W/R# transitioning HIGH, DQ[7:0] and DQS go to tri-state.
 5. For W/R# transitioning LOW, DQ[7:0] drives current state and DQS goes LOW.
 6. After final data output, DQ[7:0] is driven until W/R# goes HIGH, but is not valid.

Write Protect

See Write Protect under Bus Operations - Asynchronous Interface.

Ready/Busy#

See Ready/Busy# under Bus Operations - Asynchronous Interface.



Bus Operation – NV-DDR2 Interface

When the NV-DDR2 interface is activated on a target (see Activating Interfaces), the target is capable of high-speed DDR data transfers, and the DQS signal is enabled. DQS is a bidirectional data strobe. During data output, DQS is driven by the NAND Flash device. During data input, DQS is controlled by the host controller while inputting data on DQ[7:0].

For operations in NV-DDR2 mode, the NV-DDR2 interface must be selected (see Activating Interfaces). The capabilities that NV-DDR2 operations offers beyond NV-DDR operations include:

- Supported at 1.8V and 3.3V V_{CCQ}
- Support for differential signaling for the RE# and/or DQS signals (RE_c, DQS_c)
- Support for Warmup Cycles
- Support for On-die Termination (ODT)
- Support for V_{REFQ}

Use of differential signaling and external V_{REFQ} is optional for interface speeds 166MT/s or slower. If not using the differential signaling, statements about those signal types can be ignored.

Transition from the NV-DDR2 interface to the NV-DDR interface is not permitted.

The NV-DDR2 interface bus modes are summarized below:

Table 5: NV-DDR2 Interface Mode Selection

Mode	CE#	CLE	ALE	RE# (RE_t)	DQS (DQS_t)	DQ[7:0] ¹	WE#	WP#	Notes
Standby	H	X	X	X	X	X	X	0V/ V_{CCQ}	1, 2
Idle	L	L	L	H	H	X	H	X	6
Command input	L	H	L	H	H	input		H	3
Address input	L	L	H	H	X ⁴	input		H	3
Data input	L	L	L	H		input	H	H	2, 3
Data output	L	L	L			output	H	X	2, 3, 5
Write protect	X	X	X	X	X	X	X	L	

- Notes:
1. The current state of the device is data input, data output, or neither based on the commands issued.
 2. There are two data input/output cycles from the rising edge of DQS/RE# to the next rising edge of DQS/RE#.
 3. ODT may be enabled as part of the data input and data output cycles.
 4. If ODT is enabled, additional current consumption will occur if DQS is HIGH during the address input state.
 5. At the beginning of a data output burst, DQS shall be held HIGH for t_{DQSRH} after RE# transitions LOW to begin data output.



6. WE# is set HIGH during the Idle state.
7. Mode selection settings for this table: H = Logic level HIGH; L = Logic level LOW; X = V_{IH} or V_{IL} .

Differential Signaling

An enabler for higher speed operation is differential signaling for the RE# and DQS signals. A complementary RE# and complementary DQS signal may be optionally used to create differential signal pairs (RE_t/RE_c and DQS_t/DQS_c). When using differential signaling, RE# is referred to as RE_t and DQS is referred to as DQS_t, i.e., the "true" versions of the signals. Differential signaling may be used to improve signal integrity through enhanced noise immunity. Differential signaling shall only be enabled for use when the NV-DDR2 data interface is selected.

A device may support differential RE# and/or differential DQS signaling. The support for differential RE# and/or DQS is reported in the parameter page. Complementary RE# (i.e., RE_c) and complementary DQS (i.e., DQS_c) signals are individually configured/enabled. By default, differential signaling is disabled. The host may configure the device to use differential signaling using the NV-DDR2 Configuration feature address.

To begin using differential signaling, the host shall issue a SET FEATURES (EFh) command to the Timing Mode feature address that sets the Data Interface from asynchronous to NV-DDR2 operation. Then issue a SET FEATURES (EFh) command to the NV-DDR2 Configuration feature address to activate differential RE# and/or differential DQS signaling. The differential signaling is then enabled after CE# is brought HIGH.

To change from differential signaling to single-ended signaling, the host shall configure the device using the NV-DDR2 Configuration feature address to disable differential signaling. The differential signaling is disabled after CE# is brought HIGH.

A RESET (FFh) command will disable differential signaling. The SYNCHRONOUS RESET (FCh) and RESET LUN (FAh) commands have no effect on differential signaling.

Warmup Cycles

In order to support higher speed operation, warmup cycles for data output and data input may be provided. Warmup cycles shall only be enabled for use when the NV-DDR2 data interface is selected.

Warmup cycles for data output provides extra RE# and corresponding DQS transitions at the beginning of a data output burst. These extra RE#/DQS transitions do not have any data associated with them. The number of extra cycles is configured via the NV-DDR2 configuration feature address.

Warmup cycles for data input provides extra DQS transitions at the beginning of a data input burst. These extra DQS transitions do not have any data associated with them. The number of extra cycles is configured via the NV-DDR2 Configuration feature address. The number of cycles specified includes a full data input cycle (both rising and falling edge for DQS).

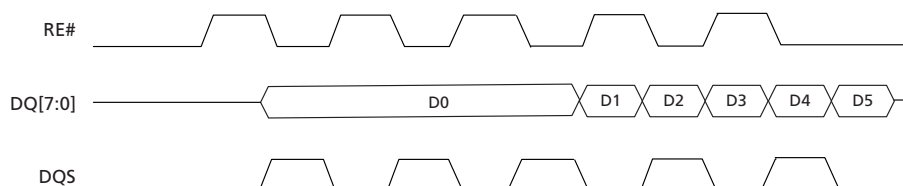
Warmup cycles are optional for both data output and data input, and if used, do not need to be configured to the same value. Warmup cycles apply to all commands. Warmup cycles are initiated at the start of each data burst when warmup cycles are enabled for that data transfer type. If the host pauses and then resumes a data transfer without exiting and re-entering the data burst, then the host shall not issue additional warmup cycles. Exiting and re-entering the data burst shall be performed by bringing ALE, CLE,



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or CE# HIGH without latching with WE#. In the case of not re-issuing warmup cycles, the host should take care to avoid signal integrity issues due to pausing the data transfer and resuming without warmup cycles.

Figure 31: Warmup Cycles for data output (2 warmup cycles)



On-die Termination (ODT)

On-die termination may be required at higher speeds depending on system topology. On-die termination applies to the DQ[7:0], DQS_t, DQS_c, RE_t, and RE_c signals. On-die termination is an optional capability that may be employed to meet higher speeds in particular topologies. If power needs to be optimized in a particular condition, then on-die termination may be disabled and the topology may be run at a slower timing mode. On-die termination shall only be enabled for use in the NV-DDR2 data interface.

On-die termination settings are configured during device initialization. The host may configure the ODT in a self-termination only configuration, or matrix termination which enables a combination of Target and non-Target termination to be specified.

For the more flexible matrix termination the host configures a matrix that defines the LUN(s) that terminate for a particular Volume. This matrix is configured using the ODT CONFIGURE (E2h) command. After the on-die termination matrix is defined, ODT is enabled and disabled based on the type of cycle (on for data input and output cycles, off for command, and address cycles). On-die termination applies for data input and output cycles for all command types, including both single data rate (SDR) and double data rate (DDR) transfers.

Volume addressing is required for use of non-Target on-die termination (ODT) functionality. See Configuration Operations for how to appoint Volume addresses. Once volume addresses have been appointed they can be selected with the VOLUME SELECT (E1h) command. For Target only termination, no ODT termination matrix is required.

The on-die termination configuration matrix and control mechanism utilize Volume addresses. Volume addressing is a required capability to utilize non-Target on-die termination.

When on-die termination is enabled via the NV-DDR2 Configuration feature address, the default is Target termination. For non-Target termination or termination topologies that use multiple terminators, the Volume address mechanism shall be used and the on-die termination configuration matrix shall be specified using the ODT CONFIGURE (E2h) command. As part of the ODT CONFIGURE (E2h) command, R_{TT} settings may be specified on a per LUN basis with individual values for:

- RE#
- DQ[7:0] and DQS for data output
- DQ[7:0] and DQS for data input



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On-die termination is enabled when ALE, CLE and CE# transition from HIGH to LOW.
On-die termination is disabled when ALE, CLE or CE# transitions from LOW to HIGH.

Table 6: On-die Termination DC Electrical Characteristics

Mode	Symbol	Min	Typ	Max	Units	Notes
R _{TT} effective impedance value for 50 Ohm setting	R _{TT2(EFF)}	32.5	50	67.5	Ohms	1
R _{TT} effective impedance value for 75 Ohm setting	R _{TT3(EFF)}	48.7	75	101.3	Ohms	1
R _{TT} effective impedance value for 100 Ohm setting	R _{TT4(EFF)}	65	100	135	Ohms	1
R _{TT} effective impedance value for 150 Ohm setting	R _{TT5(EFF)}	97.5	150	202.5	Ohms	1
Deviation of VM with respect to V _{CCQ} / 2	ΔVM	-7	-	7	Percent	2

- Notes: 1. R_{TT2} (EFF), R_{TT3} (EFF), R_{TT4} (EFF), and R_{TT5} (EFF) are determined by separately applying V_{IH(AC)} and V_{IL(AC)} to the signal being tested, and then measuring current I(V_{IH(AC)}) and I(V_{IL(AC)}), respectively. $R_{TT(EFF)} = (V_{IH[AC]} - V_{IL[AC]}) / (I(V_{IH[AC]}) - I(V_{IL[AC]}))$
2. Measure voltage (VM) at the tested signal with no load. $\Delta VM = [(2 \times VM) / V_{CCQ}] \times 100$.

Self-termination On-die Termination (ODT)

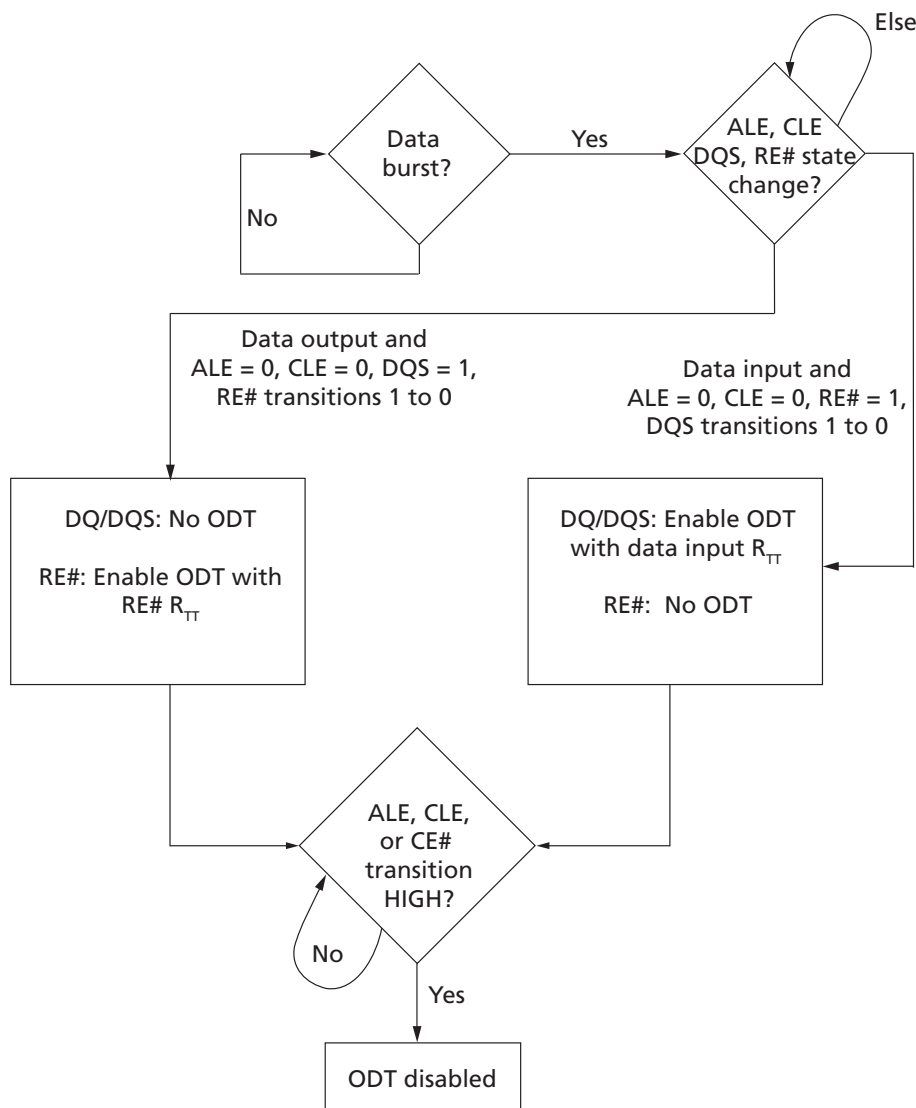
When self-termination is enabled, the LUN that is executing the command provides on-die termination. Figure 32 (page 49) defines the self-termination only ODT enable and disable requirements for the LUN that is executing the command when ODT is selected for use via SET FEATURES (EFh) command. If the ODT CONFIGURE (E2h) command is issued to a LUN on a Target, then the ODT mechanism used for that Target changes to matrix termination.

Self-termination is applied to DQS and DQ[7:0] signals during data input operations and RE# during data output operations.



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Figure 32: Self-termination only ODT behavioral flow



Matrix Termination

A LUN that is configured to act as a terminator using the configuration matrix (that is specified with the ODT CONFIGURE (E2h) command) may be located on the selected Volume as the Volume it is terminating for (Target termination) or a unselected Volume (non-Target termination). Based on the ODT configuration and the Volume a command is addressed to, LUNs enter different states which determine their ODT behavior; those states are listed in Table 7.

Table 7: LUN state for Matrix Termination

LUN is on selected Volume?	Terminator for selected Volume?	LUN state	ODT actions defined
Yes	N/A	Selected	Figure 33



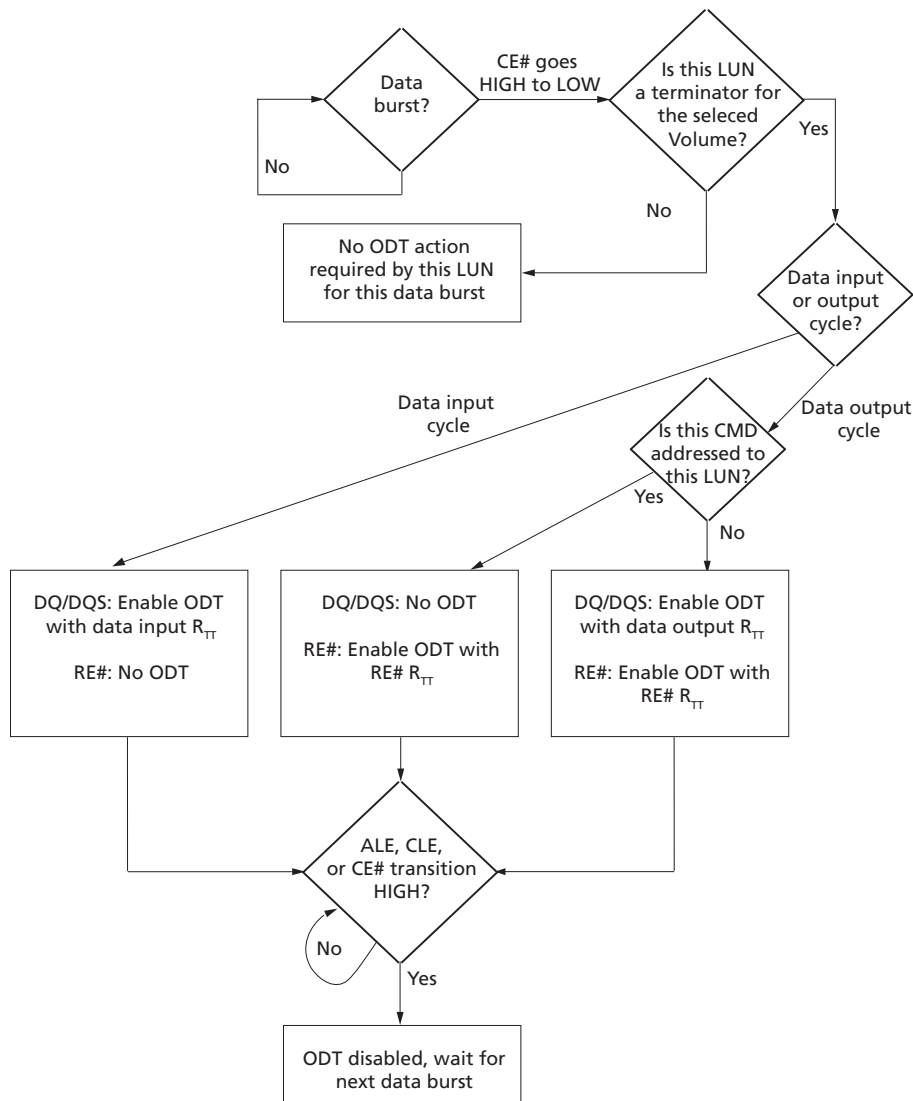
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Table 7: LUN state for Matrix Termination (Continued)

LUN is on selected Volume?	Terminator for selected Volume?	LUN state	ODT actions defined
No	Yes	Sniff	Figure 34
No	No	Deselected	No ODT actions

The LUN that a command is addressed to for execution may provide termination. Other LUNs on the selected Volume that are not responsible for execution of the command may also provide termination. Figure 33 (page 50) defines the ODT actions required for LUNs of each of these types on the selected Volume. LUNs on the selected Volume remain in an active state, and thus are aware of state information like whether there is a data burst currently and the type of cycle; these LUNs do not rely only on ALE, CLE, DQS and RE# signals.

Figure 33: ODT actions for LUNs on selected Volume

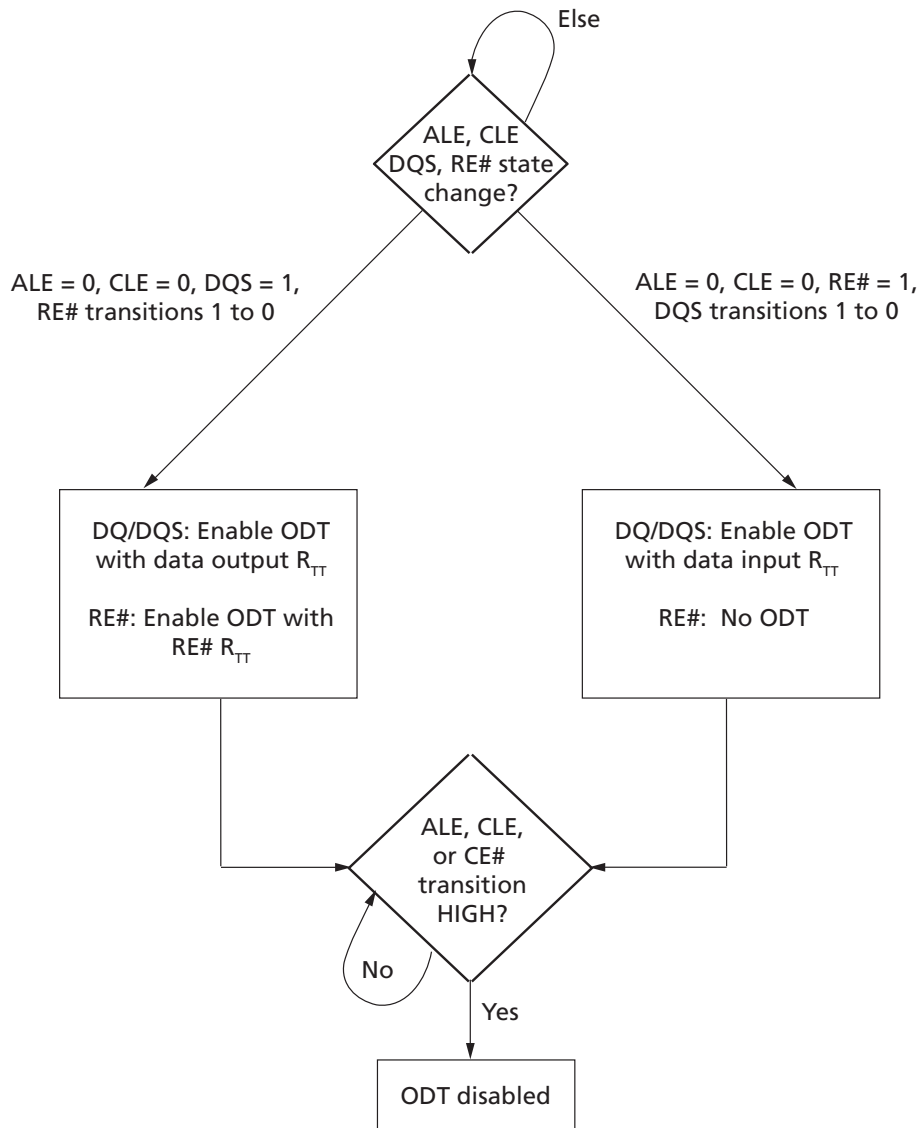




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The ODT configuration matrix also offers the flexibility of having LUNs on an unselected Volume provide termination for the selected Volume. When a LUN is placed in the Sniff state, it checks the ALE, CLE, DQS and RE# signals to determine when to enable or disable ODT. Figure 34 (page 51) defines the ODT actions for LUNs in the Sniff state on an unselected Volume.

Figure 34: ODT actions for LUNs in Sniff state on unselected Volume



Matrix Termination Examples

This section describes two examples of on-die termination configurations using matrix termination. In both examples, each Volume consists of two LUNs, referred to as H0N n -LUN0 and H0N n -LUN1. The following Volume addresses were appointed at initialization.



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Table 8: Volume appointment for Matrix Termination example

Volume	Appointed Volume Address
H0N0	0
H0N1	1
H0N2	2
H0N3	3

For optimal signal integrity and power consumption, the host may configure termination in a variety of ways. The host may configure a LUN to self terminate, perform non-Target termination for another Volume, or not perform any termination function. Using matrix termination, the termination R_{TT} values may be set differently for each LUN configured as a terminator, including the ability to specify different settings for data output operation and data input operation. The first example shows that a controller may configure the ODT matrix to perform stronger non-Target ODT for data output operations and weaker Target ODT for data input operations.

Table 9: Non-Target ODT for Data Output, Target ODT for Data Input settings configuration example

	Input values for ODT CONFIGURE (E2h)				
LUN	Byte M0	Byte M1	Byte R_{TT1}	Byte R_{TT2}	Notes
H0N0-LUN0	0Ch	00h	40h	00h	Terminates for Volumes 2 and 3 (non-Target) for data output with an R_{TT} value of 50 Ohms for DQ[7:0]/DQS
H0N0-LUN1	01h	00h	02h	03h	Terminates for Volume 0 (Target) for data input with an R_{TT} value of 100 Ohms for DQ[7:0]/DQS and 75 Ohms for RE#
H0N1-LUN0	02h	00h	02h	03h	Terminates for Volume 1 (Target) for data input with an R_{TT} value of 100 Ohms for DQ[7:0]/DQS and 75 Ohms for RE#
H0N1-LUN1	00h	00h	00h	00h	Does not act as terminator
H0N2-LUN0	00h	00h	00h	00h	Does not act as terminator
H0N2-LUN1	04h	00h	02h	03h	Terminates for Volume 2 (Target) for data input with an R_{TT} value of 100 Ohms for DQ[7:0]/DQS and 75 Ohms for RE#
H0N3-LUN0	08h	00h	02h	03h	Terminates for Volume 3 (Target) for data input with an R_{TT} value of 100 Ohms for DQ[7:0]/DQS and 75 Ohms for RE#



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Table 9: Non-Target ODT for Data Output, Target ODT for Data Input settings configuration example (Continued)

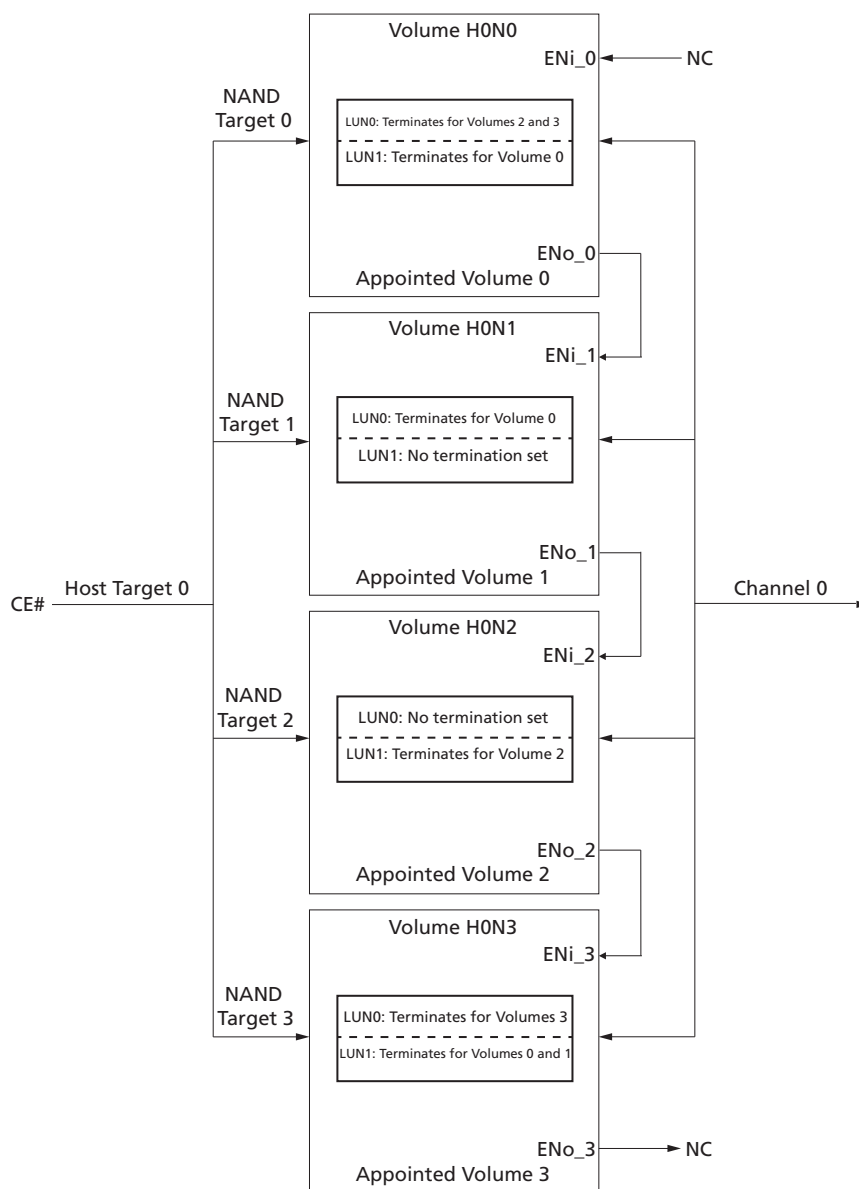
	Input values for ODT CONFIGURE (E2h)				
LUN	Byte M0	Byte M1	Byte R _{TT1}	Byte R _{TT2}	Notes
H0N3-LUN1	03h	00h	40h	00h	Terminates for Volumes 0 and 1 (non-Target) for data output with an R _{TT} value 50 Ohms for DQ[7:0]/DQS

Note: 1. See ODT CONFIGURE (E2h) (page 112) for details on input values for ODT CONFIGURE (E2h).



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Figure 35: Non-Target ODT for Data Output, Target ODT for Data Input configuration example



The second example uses parallel non-Target termination to achieve a stronger effective R_{TT} value for both data output and data input operations. For data output, two 50 Ohm terminators are used in parallel to achieve an effective 25 Ohms non-Target termination value. For data input, two 100 Ohm terminators are used in parallel to achieve an effective 50 Ohms non-Target termination value. This type of ODT matrix allows for stronger termination than may be available through a single device. It also allows for intermediate R_{TT} values with the use of different R_{TT} values for parallel LUNs. For example, if one terminator was configured for 75 Ohms and another terminator was configured for 100 Ohms for the same Volume then an effective R_{TT} value of 43 Ohms is achieved. In this example, parallel termination is used for data input and data output for



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DQ[7:0]/DQS, however, RE# is non-Target terminated with 100 Ohms using a single LUN.

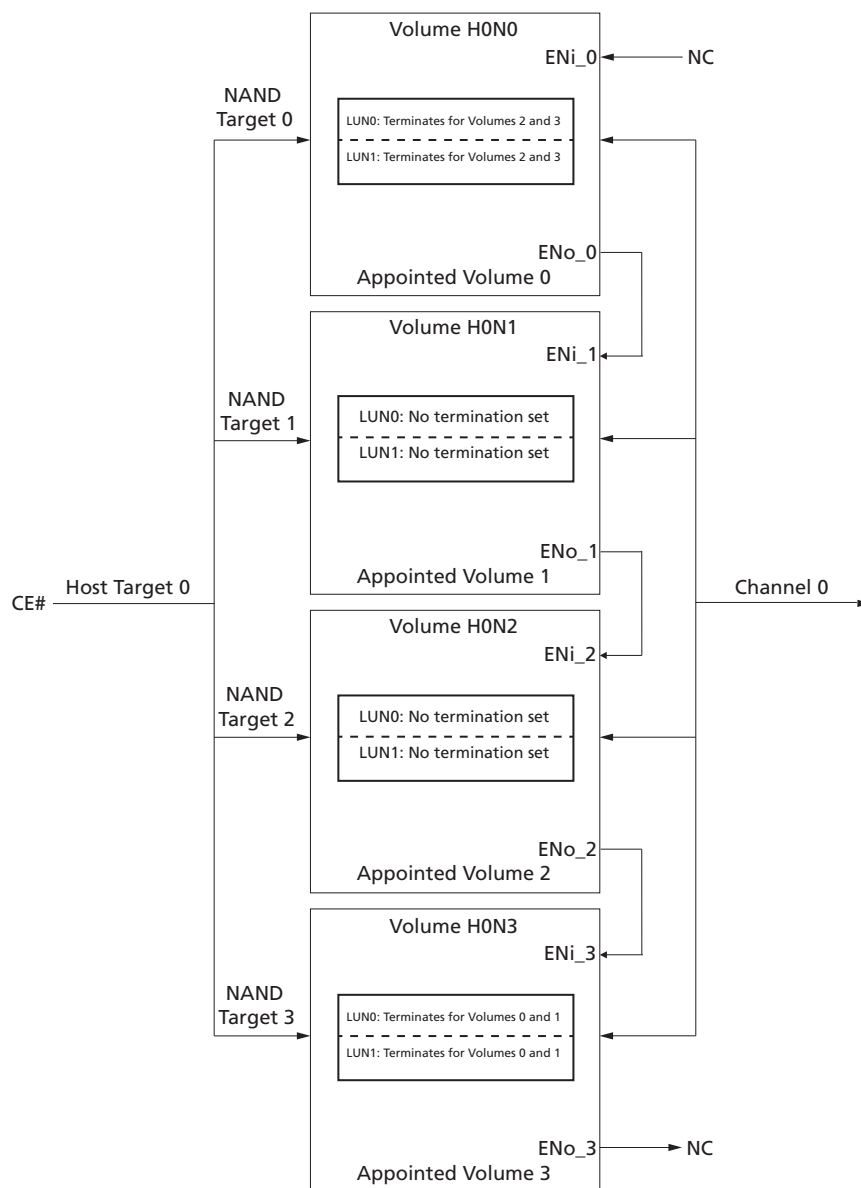
Table 10: Parallel Non-Target ODT settings configuration example

LUN	Input values for ODT CONFIGURE (E2h)				Notes
	Byte M0	Byte M1	Byte R _{TT1}	Byte R _{TT2}	
H0N0-LUN0	0Ch	00h	42h	00h	Terminates for Volumes 2 and 3 (non-Target) for data output with an R _{TT} value of 50 Ohms for DQ[7:0]/DQS. Terminates for Volumes 2 and 3 (non-Target) for data input with an R _{TT} value of 100 Ohms for DQ[7:0]/DQS.
H0N0-LUN1	0Ch	00h	42h	01h	Terminates for Volumes 2 and 3 (non-Target) for data output with an R _{TT} value of 50 Ohms for DQ[7:0]/DQS. Terminates for Volumes 0 and 1 (non-Target) for data input with an R _{TT} value of 100 Ohms for DQ[7:0]/DQS. Terminates for Volumes 2 and 3 (non-Target) with an R _{TT} value of 150 Ohms for RE#.
H0N1-LUN0	00h	00h	00h	00h	Does not act as terminator
H0N1-LUN1	00h	00h	00h	00h	Does not act as terminator
H0N2-LUN0	00h	00h	00h	00h	Does not act as terminator
H0N2-LUN1	00h	00h	00h	00h	Does not act as terminator
H0N3-LUN0	03h	00h	42h	01h	Terminates for Volumes 0 and 1 (non-Target) for data output with an R _{TT} value of 50 Ohms for DQ[7:0]/DQS. Terminates for Volumes 0 and 1 (non-Target) for data input with an R _{TT} value of 100 Ohms for DQ[7:0]/DQS. Terminates for Volumes 0 and 1 (non-Target) with an R _{TT} value of 150 Ohms for RE#.
H0N3-LUN1	03h	00h	42h	00h	Terminates for Volumes 0 and 1 (non-Target) for data output with an R _{TT} value of 50 Ohms for DQ[7:0]/DQS. Terminates for Volumes 0 and 1 (non-Target) for data input with an R _{TT} value of 100 Ohms for DQ[7:0]/DQS.



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Figure 36: Parallel Non-Target ODT configuration example



NV-DDR2 Standby

Prior to disabling a target, the target's bus must be idle. A target is disabled when CE# is driven HIGH, even when it is busy. All of the target's signals are disabled except CE#, WP#, and R/B#.

A target enters low-power standby when it is disabled and is not busy. If the target is busy when it is disabled, the target enters standby after all of the die (LUNs) complete their operations.



NV-DDR2 Idle

A target's bus is idle when CE# is LOW, ALE is LOW, CLE is LOW, RE# is HIGH, and DQS is HIGH and no internal LUN operations are ongoing or data being inputted or outputted from the target. DQS is driven HIGH to prevent the device from enabling ODT. If ODT is disabled, then DQS is a don't care during Idle states.

During the bus idle mode, all signals are enabled. No commands, addresses, or data are latched into the target; no data is output.

NV-DDR2 Pausing Data Input/Output

Pausing data input or data output may be done by placing the bus in an Idle state. The pausing of data output may also be done by pausing RE# and holding the signal(s) static HIGH or LOW until the data burst is resumed. The pausing of data input may also be done by pausing DQS and holding the signal(s) static HIGH or LOW until the data burst is resumed. WE# shall be held HIGH during data input and output burst pause time. ODT (if enabled) stays ON the entire pause time and warmup cycles (if enabled) are not re-issued when re-starting a data burst from pause. The host will be required to exit the data burst if it wishes to disable ODT or re-issue warmup cycles when re-starting. If the host wishes to end the data burst, after exiting the data burst, a new command is issued.

During the bus idle mode, all signals are enabled. No commands, addresses, or data are latched into the target; no data is output.

NV-DDR2 Commands

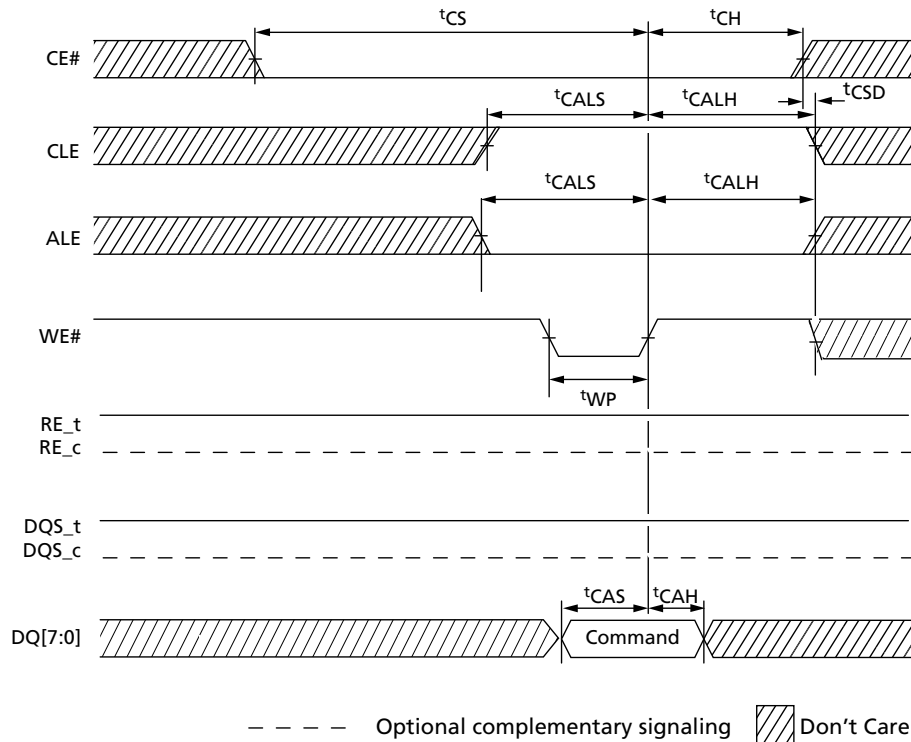
A command is written from DQ[7:0] to the command register on the rising edge of WE# when CE# is LOW, ALE is LOW, CLE is HIGH, and RE# is HIGH.

Commands are typically ignored by die (LUNs) that are busy (RDY = 0); however, some commands, such as READ STATUS (70h) and READ STATUS ENHANCED (78h), are accepted by die (LUNs), even when they are busy.



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Figure 37: NV-DDR2 Command Cycle



NV-DDR2 Addresses

A NV-DDR2 address is written from DQ[7:0] to the address register on the rising edge of WE# when CE# is LOW, ALE is HIGH, CLE is LOW, and RE# is HIGH.

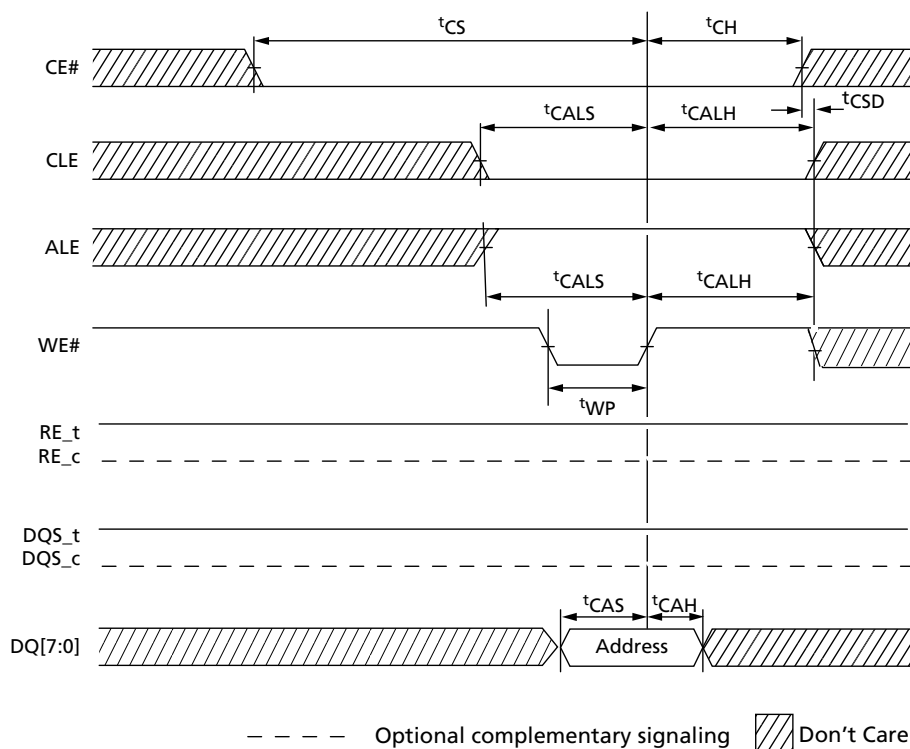
Bits not part of the address space must be LOW (see Device and Array Organization). The number of address cycles required for each command varies. Refer to the command descriptions to determine addressing requirements.

Addresses are typically ignored by die (LUNs) that are busy ($RDY = 0$); however, some addresses such as address cycles that follow the READ STATUS ENHANCED (78h) command, are accepted by die (LUNs), even when they are busy.



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Figure 38: NV-DDR2 Addresses Cycle



NV-DDR2 Data Input

To enter the NV-DDR2 data input mode, the following conditions must be met:

- CE# is LOW
- ALE and CLE are LOW
- RE# is HIGH
- t_{WPRE} is met
- DQS is LOW

Upon entering the NV-DDR2 data input mode after t_{WPRE} , data is written from DQ[7:0] to the cache register on each and every rising and falling edge of DQS (center-aligned) when CE# is LOW, RE# is HIGH, and ALE and CLE are LOW.

To exit NV-DDR2 data input mode, the following conditions must be met:

- CE# is LOW
- ALE and CLE are LOW
- RE# is HIGH
- The final two data bytes of the data input sequence are written to DQ[7:0] to the cache register on the rising and falling edges of DQS after the last cycle in the data input sequence
- DQS is held LOW for t_{WPST} (after the final falling edge of DQS)

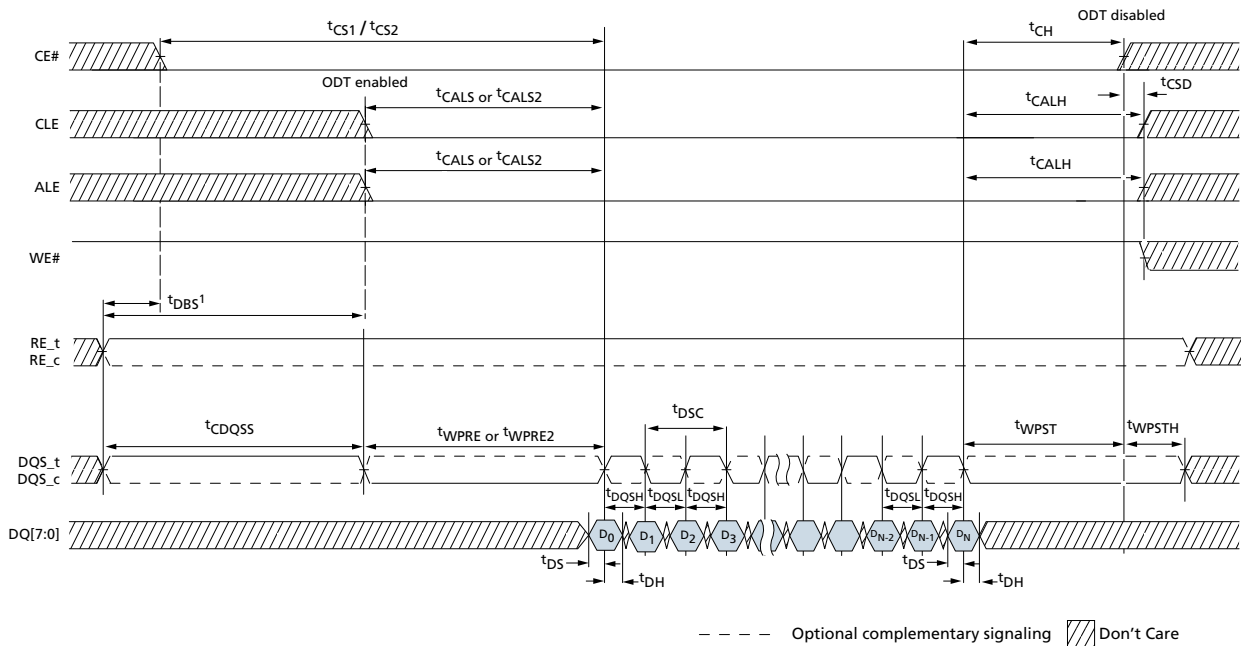
Following t_{WPST} , the bus enters bus idle mode.

Data input is ignored by die (LUNs) that are not selected or are busy.



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Figure 39: NV-DDR2 Data Input Cycles



Note: 1. ODT may not be required to be used for data input. If ODT is selected for use via SET FEATURES (EFh), then ODT is enabled and disabled during the points indicated.

NV-DDR2 Data Output

Data can be output from a die (LUN) if it is ready. Data output is supported following a READ operation from the NAND Flash array.

To enter the NV-DDR2 data output mode, the following conditions must be met:

- CE# is LOW
- The host has released the DQ[7:0] bus and DQS
- ALE and CLE are LOW
- t_{RPRE} is met

Upon entering the NV-DDR2 data output mode, DQS will toggle HIGH and LOW with a delay of t_{DQSRE} from the respective rising and falling edges of RE#. DQ[7:0] will output data edge-aligned to the rising and falling edges of DQS, with the first transition delayed by no more than t_{AC} .

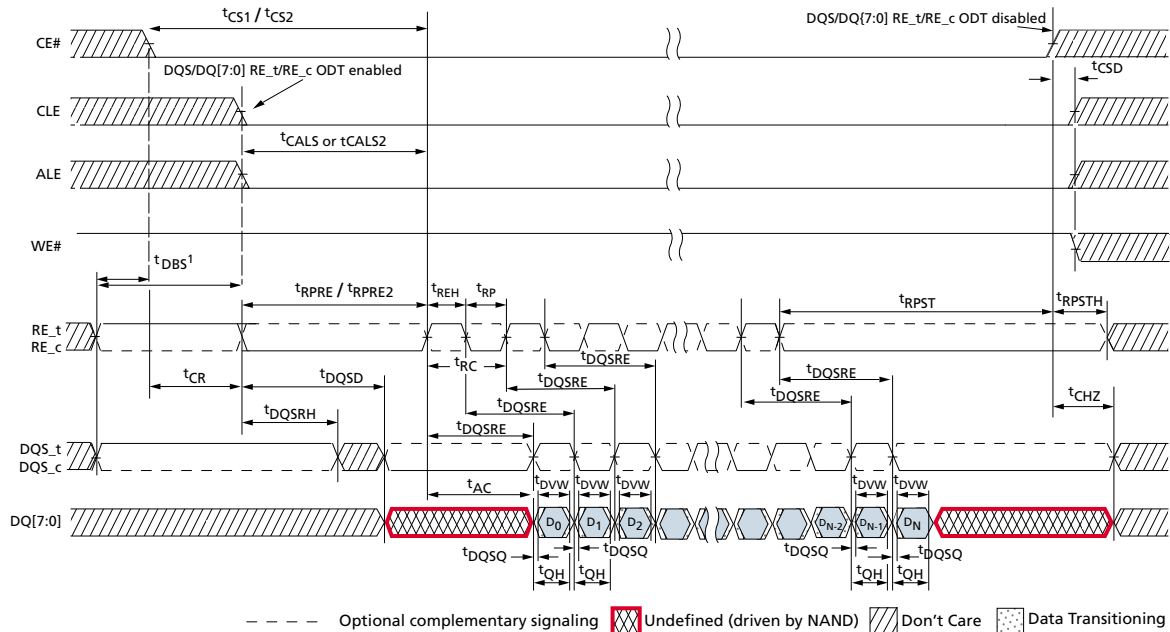
The final two data bytes are output on DQ[7:0] on the final rising and falling edges of DQS. The final rising and falling edges of DQS occur t_{DQSRE} after the last cycle in the data output sequence. The host must hold RE# for t_{RPST} after the last RE# falling edge for data output.

Data output requests are typically ignored by a die (LUN) that is busy (RDY = 0); however, it is possible to output data from the status register even when a die (LUN) is busy by issuing the READ STATUS (70h) or READ STATUS ENHANCED (78h) command.



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Figure 40: NV-DDR2 Data Output Cycles



Note: 1. ODT may not be required to be used for data output. If ODT is selected for use via SET FEATURES (EFh), then ODT is enabled and disabled during the points indicated.

Write Protect

See Write Protect under Bus Operations - Asynchronous Interface.

Ready/Busy#

See Ready/Busy# under Bus Operations - Asynchronous Interface.



Device Initialization

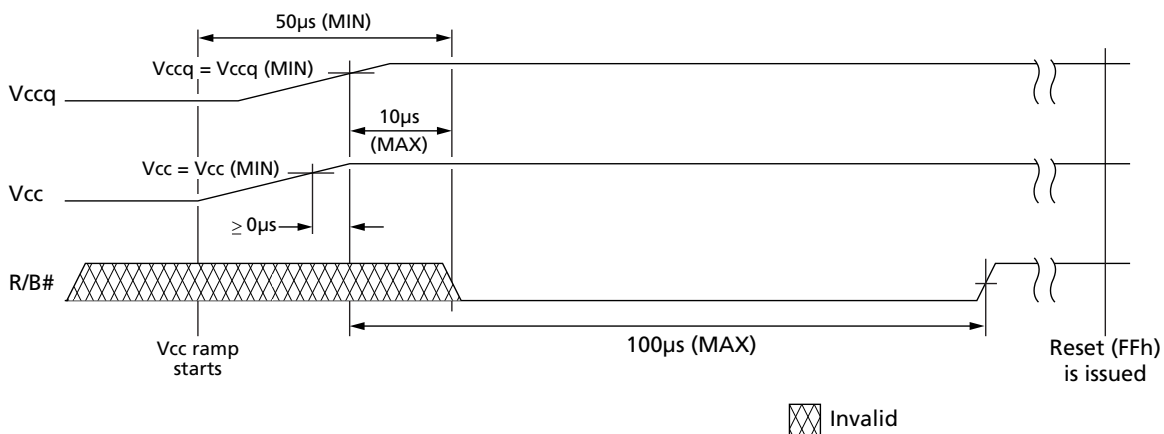
Some NAND Flash devices do not support V_{CCQ} . For these devices all references to V_{CCQ} are replaced with V_{CC} .

Micron NAND Flash devices are designed to prevent data corruption during power transitions. V_{CC} is internally monitored. (The $WP\#$ signal supports additional hardware protection during power transitions.) When ramping V_{CC} and V_{CCQ} , use the following procedure to initialize the device:

1. Ramp V_{CC} .
2. Ramp V_{CCQ} . V_{CCQ} may ramp before, simultaneously, or after V_{CC} .
3. The host must wait for $R/B\#$ to be valid and HIGH before issuing RESET (FFh) to any target (see Figure 41). The $R/B\#$ signal becomes valid when 50 μ s has elapsed since the beginning the V_{CC} ramp, and 10 μ s has elapsed since V_{CCQ} reaches V_{CCQ} (MIN) and V_{CC} reaches V_{CC} (MIN).
4. If not monitoring $R/B\#$, the host must wait at least 100 μ s after V_{CCQ} reaches V_{CCQ} (MIN) and V_{CC} reaches V_{CC} (MIN). If monitoring $R/B\#$, the host must wait until $R/B\#$ is HIGH.
5. The asynchronous interface is active by default for each target. Each LUN draws less than an average of I_{ST} measured over intervals of 1ms until the RESET (FFh) command is issued.
6. The RESET (FFh) command must be the first command issued to all targets (CE#s) after the NAND Flash device is powered on. Each target will be busy for t_{POR} after a RESET command is issued. The RESET busy time can be monitored by polling $R/B\#$ or issuing the READ STATUS (70h) command to poll the status register. The host must not issue an additional RESET (FFh) command during t_{POR} .
7. The device is now initialized and ready for normal operation.

At power-down, V_{CCQ} may go LOW, either before, simultaneously, or after V_{CC} going LOW.

Figure 41: R/B# Power-On Behavior



Note: 1. Disregard V_{CCQ} for devices that use only V_{CC} .

To initialize a discovered target, the following steps shall be taken. The initialization process should be followed for each connected CE# signal, including performing the READ PARAMETER PAGE (ECh) command for each target. Each chip enable corre-



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sponds to a unique target with its own independent properties that the host shall observe and subsequently use.

The host should issue the READ PARAMETER PAGE (ECh) command. This command returns information that includes the capabilities, features, and operating parameters of the device. When the information is read from the device, the host shall check the CRC to ensure that the data was received correctly and without error prior to taking action on that data.

If the CRC of the first parameter page read is not valid, the host should read redundant parameter page copies. The host can determine whether a redundant parameter page is present or not by checking if the first four bytes contain at least two bytes of the parameter page signature. If the parameter page signature is present, then the host should read the entirety of that redundant parameter page. The host should then check the CRC of that redundant parameter page. If the CRC is correct, the host may take action based on the contents of that redundant parameter page. If the CRC is incorrect, then the host should attempt to read the next redundant parameter page by the same procedure.

The host should continue reading redundant parameter pages until the host is able to accurately reconstruct the parameter page contents. The host may use bit-wise majority or other ECC techniques to recover the contents of the parameter page from the parameter page copies present. When the host determines that a parameter page signature is not present, then all parameter pages have been read.

After successfully retrieving the parameter page, the host has all information necessary to successfully communicate with that target. If the host has not previously mapped defective block information for this target, the host should next map out all defective blocks in the target. The host may then proceed to utilize the target, including erase and program operations.



V_{PP} Initialization

Some NAND Flash devices do not support V_{PP}.

Micron NAND Flash devices support the V_{PP} signal as a way for the host system to enhance NAND array operations by improving power efficiency. When ramping V_{PP}, use the following procedure to initialize the V_{PP} functionality:

1. V_{PP} can be ramped HIGH before, simultaneously, or after V_{CC,min} is reached.
2. V_{PP} must be within its valid voltage range prior to the SET FEATURES (EFh) command to enable the V_{PP} functionality.
3. After V_{PP} is successfully ramped and before V_{PP} functionality can be carried out, the V_{PP} feature must be activated via a SET FEATURES (EFh) command. See the Configurations Operations section for more details.
4. To deactivate V_{PP} functionality without a device power-down, a SET FEATURES (EFh) command must be issued to disable the V_{PP} functionality while V_{PP} is within its valid voltage range.

At power-down, V_{PP} can ramp LOW before, simultaneously, or after V_{CC,min} has reached.



Activating Interfaces

After performing the steps under Device Initialization (page 62), the asynchronous interface is active for all targets on the device.

Each target's interface is independent of other targets, so the host is responsible for changing the interface for each target.

If the host and NAND Flash device, through error, are no longer using the same interface, then steps under Activating the Asynchronous Interface are performed to re-synchronize the interfaces.

Activating the Asynchronous Interface

To activate the asynchronous NAND interface, once the synchronous interface is active, the following steps are repeated for each target:

1. The host pulls CE# HIGH, disables its input to CLK, and enables its asynchronous interface.
2. The host pulls CE# LOW and issues the RESET (FFh) command, using an asynchronous command cycle.
3. R/B# goes LOW for ^tRST.
4. After ^tITC, and during ^tRST, the device enters the asynchronous NAND interface. READ STATUS (70h) and READ STATUS ENHANCED (78h) are the only commands that can be issued.
5. After ^tRST, R/B# goes HIGH. Timing mode feature address (01h), subfeature parameter P1 is set to 00h, indicating that the asynchronous NAND interface is active and that the device is set to timing mode 0.

For further details, see Reset Operations.

Activating the NV-DDR Interface

To activate the NV-DDR NAND Flash interface, the following steps are repeated for each target:

1. Issue the SET FEATURES (EFh) command.
2. Write address 01h, which selects the Timing mode feature address.
3. Write P1 with 1Xh, where "X" is the timing mode used in the NV-DDR interface (see Configuration Operations).
4. Write P2–P4 as 00h-00h-00h.
5. R/B# goes LOW for ^tITC. The host should pull CE# HIGH. During ^tITC, the host should not issue any type of command, including status commands, to the NAND Flash device.
6. After ^tITC, R/B# goes HIGH and the synchronous interface is enabled. Before pulling CE# LOW, the host should enable the clock.

Activating the NV-DDR2 Interface

Transitions from NV-DDR directly to NV-DDR2 (or vice versa) is not supported. In this case, the host should transition to the asynchronous interface and then select the desired synchronous interface.

Prior to selecting the NV-DDR2 interface, it is recommended that settings for the NV-DDR2 interface be configured. Specifically:



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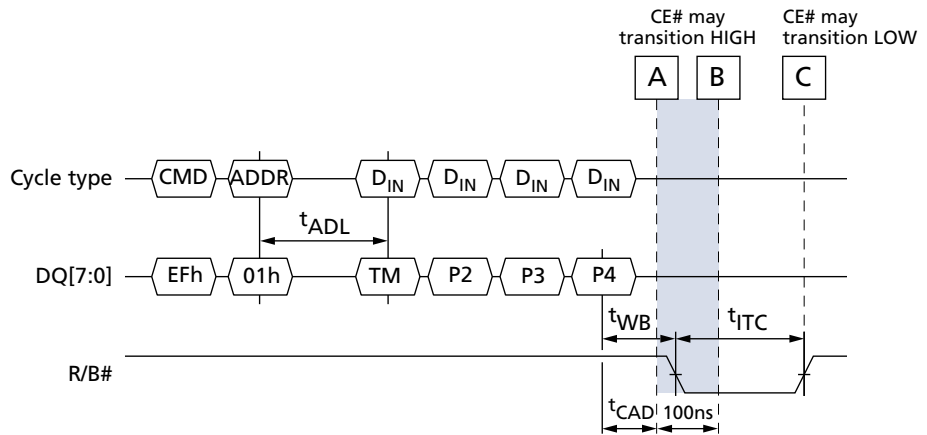
- SET FEATURES (EFh) command should be used to configure the NV-DDR2 Configuration feature address.
- If on-die termination (ODT) is used, the appropriate ODT CONFIGURE (E2) commands should be issued.

These actions should be completed prior to selecting the NV-DDR2 interface. If these settings are modified when the NV-DDR2 interface is already selected, the host should take care to ensure appropriate settings are applied in a manner that avoids signal integrity issues.

To activate the NV-DDR2 NAND Flash interface, the following steps are repeated for each target:

1. Issue the SET FEATURES (EFh) command.
2. Write address 01h, which selects the Timing mode feature address.
3. Write P1 with 2Xh, where "X" is the timing mode used in the NV-DDR2 interface (see Configuration Operations).
4. Write P2–P4 as 00h–00h–00h.
5. R/B# goes LOW for t_{ITC} . The host should pull CE# HIGH. During t_{ITC} , the host should not issue any type of command, including status commands, to the NAND Flash device.
6. After t_{ITC} , R/B# goes HIGH and the NV-DDR2 interface is enabled.

Figure 42: Activating Interfaces



Note: 1. TM = Timing mode.



CE# Pin Reduction and Volume Addressing

In higher density capacity implementations there may be a significant number of CE# pins required for a host to support where there are many NAND packages with two to four CE# pins per package. The CE# pin reduction mechanism enables a single CE# pin from the host to be shared by multiple NAND targets, thus enabling a significant reduction in the number of CE# pins required by the host system. The CE# pin reduction mechanism may be utilized with any data interface (asynchronous, NV-DDR, NV-DDR2).

In the CE# pin reduction mechanism, each NAND package is appointed a volume address during the initialization sequence. After initialization is complete, the host may address a particular volume (NAND target) by using the VOLUME SELECT (E1h) command. See VOLUME SELECT (E1h) for more details.

ENi and ENo pins are added to each NAND package and a daisy chain is created between NAND packages. The first NAND package in the chain has the ENi pin as not connected. All other NAND packages have their ENi pin connected to the previous package's ENo pin in a daisy chain configuration.

At power-on, the ENo pins are driven LOW by the NAND device. ENo shall be High-Z when all CE# signals on the NAND package are HIGH. When a NAND target has had a volume address appointed with the SET FEATURES (EFh) command, then the ENo pin shall be pulled HIGH by the NAND target when the corresponding CE# is LOW. This enables the next NAND target on a subsequent package in the daisy chain to accept commands because the ENi pin pulls HIGH when ENo is no longer pulling LOW. After a volume address has been appointed to a volume (i.e., NAND target), that volume shall become deselected and ignores the ENi pin until the next power cycle.

The state of ENi determines whether the NAND package is able to accept commands. If the ENi pin is HIGH and CE# is LOW for the NAND target, then the NAND target shall accept commands. If the ENi pin is LOW or CE# is HIGH for the NAND Target, then the NAND Target shall not accept commands.

To be selected to process a command, the VOLUME SELECT command shall be issued to the host target using the volume address that was previously appointed for a particular NAND target. After the CE# signal is pulled HIGH for t_{CEH} time, all LUNs on a volume revert to their previous states.



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Figure 43: CE# Pin Reduction Topology with Single Channel

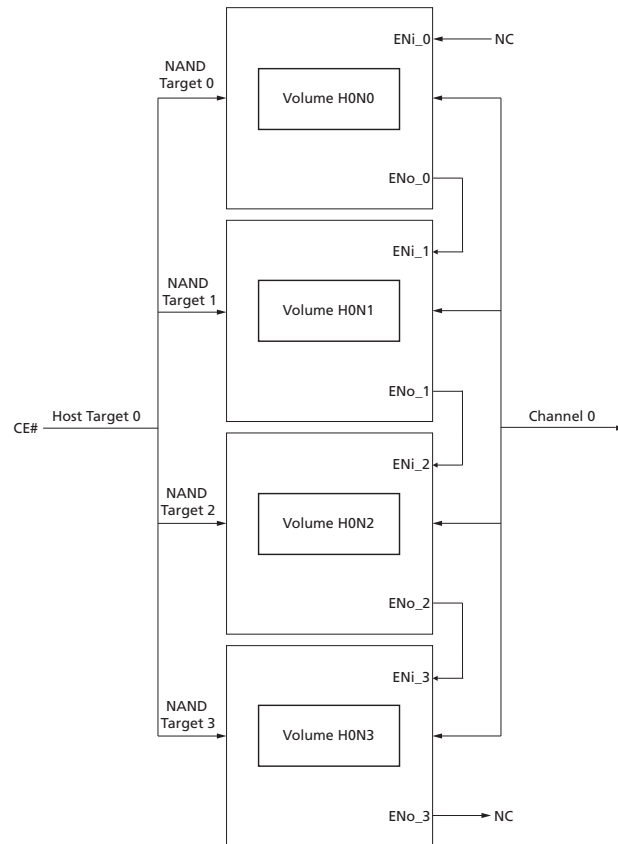
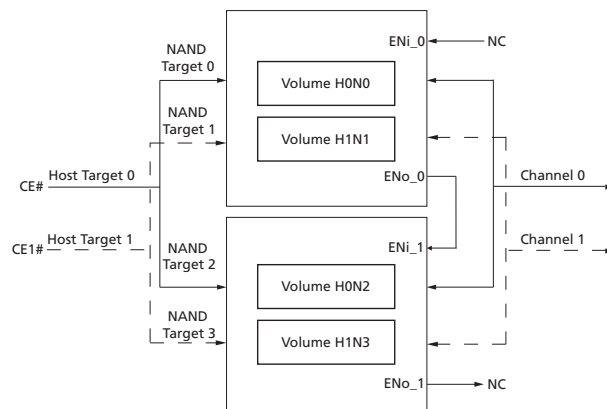


Figure 44: CE# Pin Reduction and Volume Addressing Topology with Dual Channel





Initialization Sequence

The host may issue a RESET (FFh) to all targets in parallel on the selected host target, or the host may sequentially issue RESET (FFh) to each target. The methodology chosen depends on host requirements for maximum current draw. To reset all targets in parallel, the host issues a RESET (FFh) as the first command issued to the NAND device(s). To reset targets sequentially, the host issues a READ STATUS (70h) command as the first command issued to all NAND targets on the selected host target.

During initialization when addressing a newly selected NAND target with an initial command, the host shall wait for the ENo signal to be propagated to the ENi of the subsequent target. Before addressing a new target, the host shall wait ($t_{\text{ENo}} + t_{\text{ENi}}$) and should also include signal propagation time.

In cases where there are multiple NAND targets within a package (multiple CE#), those NAND targets share the same ENo signal, and the host shall not stagger SET FEATURES (EFh) commands that appoint the volume addresses. If the SET FEATURES (EFh) commands are not issued simultaneously, then the host shall wait until volume appointment for previous NAND target(s) is complete before issuing the next SET FEATURES (EFh) command to appoint the volume address for the next NAND target that shares the ENo within a package.

After issuing the SET FEATURE (EFh) command to appoint the volume address, the host shall not issue another command to any NAND target on the associated host target (including status commands) until after the t_{FEAT} time has elapsed. This is to ensure that the proper NAND target responds to the next command, allowing for the proper ENo/ENi signal levels to be reflected.

The initialization sequence when utilizing the CE# reduction functionality is as follows:

1. The host powers on the NAND device(s).
2. The host pulls CE# LOW.
3. If resetting all NAND targets in parallel, then the host issues the RESET (FFh) command. This command is accepted by all NAND targets connected to the CE# (host target).
4. If resetting each NAND target sequentially, then:
 - a. The host issues the READ STATUS (70h) command to check for ready status. Issuing the READ STATUS (70h) command prior to any other command indicates sequential RESET (FFh) of each NAND target. The host then issues the RESET (FFh) command, which is only accepted by NAND devices with ENi set HIGH.
5. R/B# goes LOW for t_{RST} and can be monitored by the host by issuing READ STATUS (70h) commands and monitoring
6. The host configures the NAND target via commands necessary to perform that function (READ PARAMETER PAGE (ECh) command, SET FEATURES (EFh) command, etc).
7. The host issues SET FEATURES (EFh) command to the volume configuration feature address to appoint the volume address for the NAND target. The volume address specified shall be unique amongst all NAND targets. After the SET FEATURE (EFh) command completes, ENo is set to one and the volume is deselected until a VOLUME SELECT (E1h) command is issued that selects the volume. The host shall not issue another command to a NAND target connected to the associated host target until after t_{FEAT} time has elapsed.
8. For each NAND target connected to a host target, steps 4–7 are repeated for sequential initialization, and steps 5–7 are repeated for parallel initialization.



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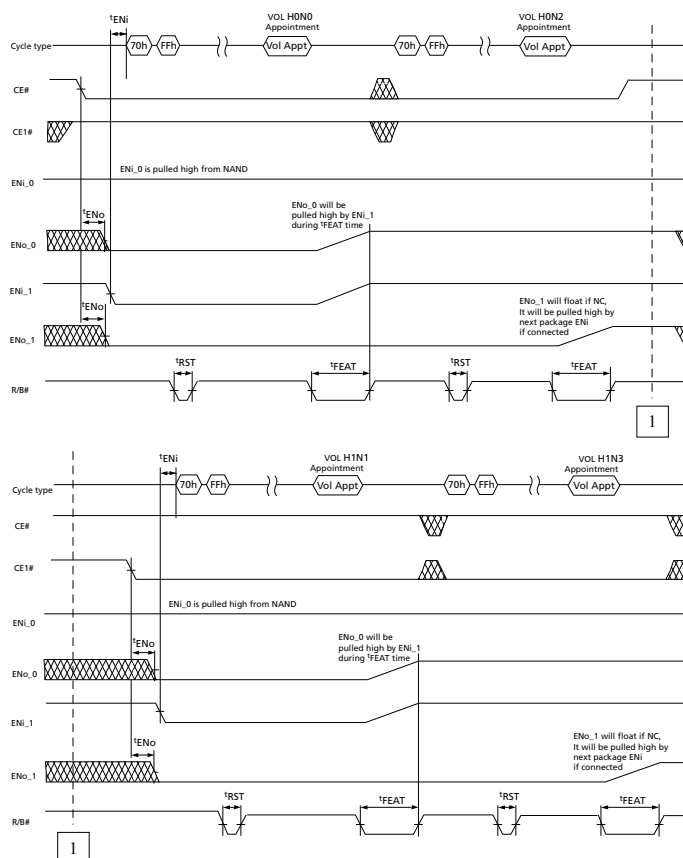
9. When no further NAND targets are found connected to the host target, then repeat steps 2–8 for the next host target (i.e., host CE# pin).
10. To complete the initialization process, a VOLUME SELECT (E1h) command is issued to select the next volume that is going to execute a command.

After volume addresses have been appointed to all NAND targets, the host may complete any additional initialization tasks (for example, configure on-die termination (ODT) for NV-DDR2 mode) and then proceed with normal operation. Prior to issuing a command to a volume, the VOLUME SELECT (E1h) command shall be issued.

The host CE# signal shall be kept LOW for steps 2–7. If the host CE# signal that is LOW for steps 2–7 is brought HIGH anytime after step 7 but before the initialization process is complete, then t_{CS} for Asynchronous timing mode 0 shall be used.

Figure 45 (page 70) is an example of sequential reset initialization for the CE# pin reduction topology of Figure 44 (page 68).

Figure 45: Example of Sequential Reset Initialization



Volume Appointment Without CE# Pin Reduction

If CE# pin reduction is not used (for example, if ENi and ENo are not connected) and the host desires to have the terminator on a package that does not share a CE# with the selected NAND target, then each package that shares the CE# must have a volume appointed at initialization using the SET FEATURE (EFh) command using the volume configuration feature.

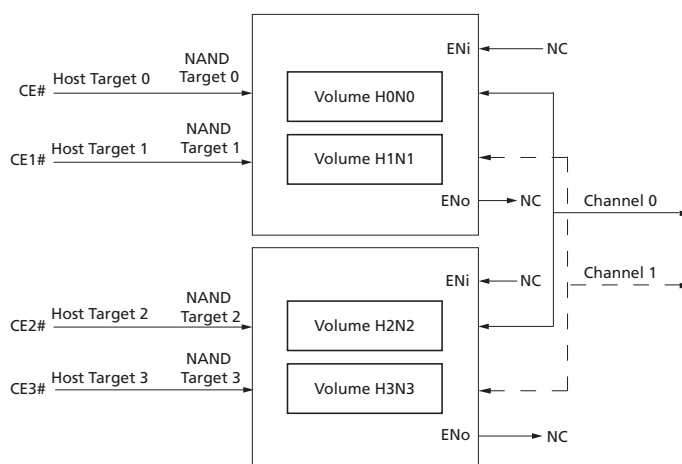


MLC 256Gb to 2Tb Async/Sync NAND CE# Pin Reduction and Volume Addressing

Each CE# must have a unique volume address appointed. Once all NAND targets have volume addresses appointed, the appointed volume addresses may be used for termination selection. Volume addressing is not required for operation due to discrete CE# signals; however, the VOLUME SELECT (E1h) command is required for terminator selection when using non-target termination schemes.

During operation, the CE# signal for the selected Volume and for any NAND targets assigned as a terminator for the selected volume need to be brought LOW. When CE# is brought LOW for an unselected volume, all LUNs that are not assigned as terminators for the selected volume are deselected.

Figure 46: Volume Addressing Without CE# Pin Reduction Topology with Dual Channel



Appointing Volume Addresses

To appoint a volume address, the SET FEATURE command is issued with a feature address of volume configuration. The volume address is not retained across power cycles; thus, if volume addressing is going to be used it needs to be appointed after each power-on prior to use of the NAND device(s).

Selecting a Volume

After volume addresses have been appointed, every NAND target (and associated LUN) is selected when the associated CE# is pulled LOW. The host issues a VOLUME SELECT (E1h) command to indicate the volume (NAND target) that shall execute the next command issued.

Multiple Volume Operation Restrictions

Volumes are independent entities. A multiple volume operation is when two or more Volumes are simultaneously processing commands. Before issuing a command to an unselected volume, CE# shall be pulled HIGH for a minimum of t_{CEH} , and the VOLUME SELECT (E1h) command shall then be issued to select the volume to issue a command to next. While commands (including multi-LUN operations) are being performed on the selected volume, a VOLUME SELECT (E1h) command is not required.

Issuing the same command to multiple volumes at the same time is not supported.



MLC 256Gb to 2Tb Async/Sync NAND CE# Pin Reduction and Volume Addressing

For a LUN level command (e.g., READ, PROGRAM), the host may select a different volume during a data input or data output operation and then resume the data transfer operation at a later time for a LUN level command. When re-selecting a Volume and associated LUN to complete the data input or data output operation, the following actions are required:

- Data input: The host shall issue a CHANGE ROW ADDRESS (85h) command prior to resuming data input.
- Data output: The host shall issue a CHANGE READ COLUMN ENHANCED (06h-E0h) or RANDOM DATA OUT (00h-05h-E0h) command prior to resuming data output.

For a target level command (GET FEATURES (EEh), SET FEATURES (EFh)), the host shall complete all data input or data output operations associated with that command prior to selecting a new volume.

A VOLUME SELECT command shall not be issued during the following atomic portions of the COPYBACK, READ, PROGRAM, and ERASE operations:

- READ operations:
 - READ PAGE (00h-30h)
 - COPYBACK READ (00h-35h)
 - READ PAGE MULTI-PLANE (00h-32h)
 - READ PAGE CACHE RANDOM (00h-31h)
- PROGRAM operations, note: The VOLUME SELECT (E1h) command may be issued prior to the 10h, 11h, or 15h command if the next command to this volume is CHANGE ROW ADDRESS (85h):
 - PROGRAM PAGE (80h-10h)
 - PROGRAM PAGE MULTI-PLANE (80h/81h-11h)
 - PROGRAM PAGE CACHE (80h-15h)
 - COPYBACK PROGRAM (85h-10h)
 - COPYBACK PROGRAM MULTI-PLANE (85h-11h)
- ERASE operations:
 - ERASE BLOCK (60h-D0h)
 - ERASE BLOCK MULTI-PLANE (60h-D1h or 60h-60h-D0h)

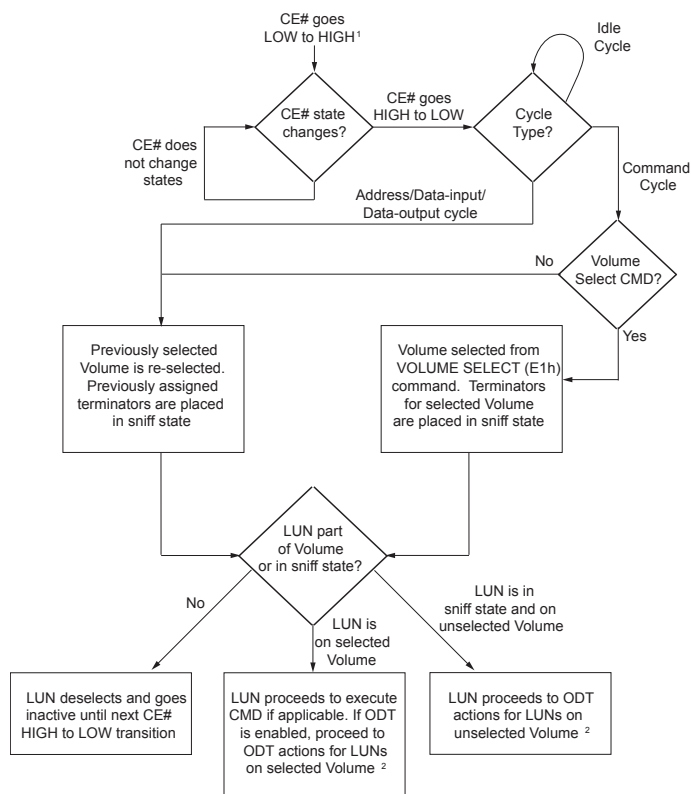
Volume Reversion

When using volume addressing, the LUNs shall support Volume reversion. Specifically, if CE# is transitioned from HIGH to LOW and a Volume Select is not the first command, then the LUN shall revert to the previously Selected, Sniff, and Deselected states based on the last specified volume address. If on-die termination is enabled when using the NV-DDR2 data interface there are additional actions described within On Die Termination (ODT).



MLC 256Gb to 2Tb Async/Sync NAND CE# Pin Reduction and Volume Addressing

Figure 47: Volume Reversion behavioral flow



- Notes:
1. This state is entered asynchronously when CE# transitions from LOW to HIGH.
 2. ODT actions for LUNs on a selected Volume are specified in ODT actions for LUNs on selected Volume. ODT actions for LUNs on an unselected Volume are specified in ODT actions for LUNs in Sniff state on unselected Volume.



MLC 256Gb to 2Tb Async/Sync NAND Command Definitions

Command Definitions

Table 11: Command Set

Command	Com- mand Cycle #1	Number of Valid Address Cycles	Data Input Cycles	Com- mand Cycle #2	Number of Valid Address Cycles #2	Com- mand Cycle #3	Valid While Se- lected LUN is Busy ¹	Valid While Oth- er LUNs are Busy ²	Note s
Reset Operations									
RESET	FFh	0	–	–	–	–	Yes	Yes	
SYNCHRONOUS RE- SET	FCh	0	–	–	–	–	Yes	Yes	
RESET LUN	FAh	3	–	–	–	–	Yes	Yes	
Identification Operations									
READ ID	90h	1	–	–	–	–			3
READ PARAMETER PAGE	ECh	1	–	–	–	–			
READ UNIQUE ID	EDh	1	–	–	–	–			
Configuration Operations									
VOLUME SELECT	E1h	1	–	–	–	–			
ODT CONFIGURE	E2h	1	4	–	–	–			
GET FEATURES	EEh	1	–	–	–	–			3
SET FEATURES	EFh	1	4	–	–	–			4
GET FEATURES by LUN	D4h	2	–	–	–	–		Yes	3
SET FEATURES by LUN	D5h	2	4	–	–	–		Yes	4
Status Operations									
READ STATUS	70h	0	–	–	–	–	Yes		
READ STATUS EN- HANCED	78h	3	–	–	–	–	Yes	Yes	
Column Address Operations									
CHANGE READ COL- UMN	05h	2	–	E0h	–	–		Yes	
CHANGE READ COL- UMN ENHANCED (ONFI)	06h	5	–	E0h	–	–		Yes	
CHANGE READ COL- UMN ENHANCED (JEDEC)	00h	5	–	05h	2	E0h		Yes	
CHANGE WRITE COLUMN	85h	2	Optional	–	–	–		Yes	
CHANGE ROW AD- DRESS	85h	5	Optional	11h (Op- tional)	–	–		Yes	5
Read Operations									



MLC 256Gb to 2Tb Async/Sync NAND Command Definitions

Table 11: Command Set (Continued)

Command	Com- mand Cycle #1	Number of Valid Address Cycles	Data Input Cycles	Com- mand Cycle #2	Number of Valid Address Cycles #2	Com- mand Cycle #3	Valid While Se- lected LUN is Busy ¹	Valid While Oth- er LUNs are Busy ²	Note s
READ MODE	00h	0	–	–	–	–		Yes	
READ PAGE	00h	5	–	30h	–	–		Yes	6
READ PAGE MULTI- PLANE	00h	5	–	32h	–	–		Yes	
READ PAGE CACHE SEQUENTIAL	31h	0	–	–	–	–		Yes	7
READ PAGE CACHE RANDOM	00h	5	–	31h	–	–		Yes	6,7
READ PAGE CACHE LAST	3Fh	0	–	–	–	–		Yes	7
Program Operations									
PROGRAM PAGE	80h	5	Yes	10h	–	–		Yes	
PROGRAM PAGE MULTI-PLANE	80h or 81h	5	Yes	11h	–	–		Yes	
PROGRAM PAGE CACHE	80h	5	Yes	15h	–	–		Yes	8
Erase Operations									
ERASE BLOCK	60h	3	–	D0h	–	–		Yes	
ERASE BLOCK MULTI-PLANE (ON- FI)	60h	3	–	D1h	–	–		Yes	
ERASE BLOCK MUL- TI-PLANE (JEDEC)	60h	3	–	60h	3	D0h		Yes	
Copyback Operations									
COPYBACK READ	00h	5	–	35h	–	–		Yes	6
COPYBACK PRO- GRAM	85h	5	Optional	10h	–	–		Yes	
COPYBACK PRO- GRAM MULTI- PLANE	85h	5	Optional	11h	–	–		Yes	

- Notes:
1. Busy means RDY = 0.
 2. These commands can be used for interleaved die (multi-LUN) operations.
 3. The READ ID (90h), GET FEATURES (EEh), and GET FEATURES by LUN (D4h) output identical data on rising and falling DQS edges.
 4. The SET FEATURES (EFh) and SET FEATURES by LUN (D5h) commands requires data transition prior to the rising edge of CLK or DQS, with identical data for the rising and falling edges.
 5. Command cycle #2 of 11h is conditional. See CHANGE ROW ADDRESS (85h) (page 120) for more details.



MLC 256Gb to 2Tb Async/Sync NAND Command Definitions

6. This command can be preceded by READ PAGE MULTI-PLANE (00h-32h) command to accommodate a maximum simultaneous multi-plane array operation.
7. Issuing a READ PAGE CACHE-series (31h, 00h-31h, 00h-32h, 3Fh) command when the array is busy (RDY = 1, ARDY = 0) is supported if the previous command was a READ PAGE (00h-30h) or READ PAGE CACHE-series command; otherwise, it is prohibited.
8. Issuing a PROGRAM PAGE CACHE (80h-15h) command when the array is busy (RDY = 1, ARDY = 0) is supported if the previous command was a PROGRAM PAGE CACHE (80h-15h) command; otherwise, it is prohibited.



Reset Operations

RESET (FFh)

The RESET (FFh) command is used to put a target into a known condition and to abort command sequences in progress. This command is accepted by all die (LUNs), even when they are busy.

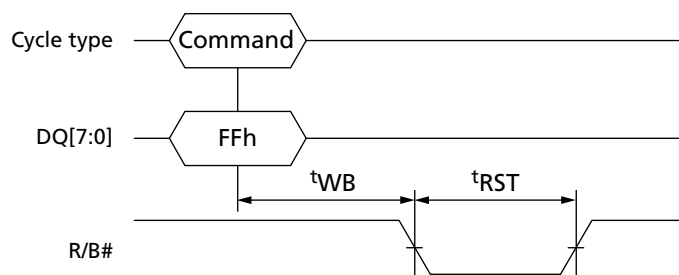
When FFh is written to the command register, the target goes busy for t_{RST} . During t_{RST} , the selected target (CE#) discontinues all array operations on all die (LUNs). All pending single- and multi-plane operations are cancelled. If this command is issued while a PROGRAM or ERASE operation is occurring on one or more die (LUNs), the data may be partially programmed or erased and is invalid. The command register is cleared and ready for the next command. The data register and cache register contents are invalid.

RESET must be issued as the first command to each target following power-up (see Device Initialization). Use of the READ STATUS ENHANCED (78h) command is prohibited during the power-on RESET. To determine when the target is ready, use READ STATUS (70h).

If the RESET (FFh) command is issued when the synchronous interface is enabled, the target's interface is changed to the asynchronous interface and the timing mode is set to 0. The RESET (FFh) command can be issued asynchronously when the synchronous interface is active, meaning that CLK does not need to be continuously running when CE# is transitioned LOW and FFh is latched on the rising edge of CLK. After this command is latched, the host should not issue any commands during t_{TTC} . After t_{TTC} , and during or after t_{RST} , the host can poll each LUN's status register.

If the RESET (FFh) command is issued when the asynchronous interface is active, the target's asynchronous timing mode remains unchanged. During or after t_{RST} , the host can poll each LUN's status register.

Figure 48: RESET (FFh) Operation





SYNCHRONOUS RESET (FCh)

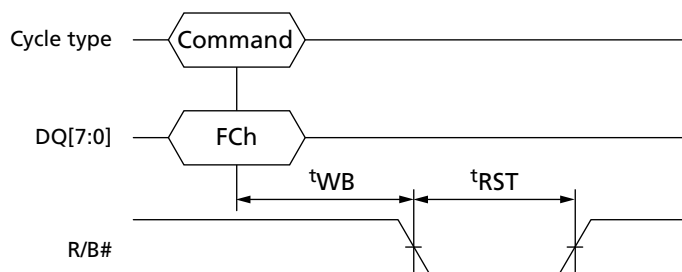
When the synchronous interface is active, the SYNCHRONOUS RESET (FCh) command is used to put a target into a known condition and to abort command sequences in progress. This command is accepted by all die (LUNs), even when they are BUSY.

When FCh is written to the command register, the target goes busy for t_{RST} . During t_{RST} , the selected target (CE#) discontinues all array operations on all die (LUNs). All pending single- and multi-plane operations are cancelled. If this command is issued while a PROGRAM or ERASE operation is occurring on one or more die (LUNs), the data may be partially programmed or erased and is invalid. The command register is cleared and ready for the next command. The data register and cache register contents are invalid and the synchronous interface remains active.

During or after t_{RST} , the host can poll each LUN's status register.

SYNCHRONOUS RESET is only accepted while the synchronous interface is active. Its use is prohibited when the asynchronous interface is active.

Figure 49: SYNCHRONOUS RESET (FCh) Operation





RESET LUN (FAh)

The RESET LUN (FAh) command is used to put a particular LUN on a target into a known condition and to abort command sequences in progress. This command is accepted by only the LUN addressed by the RESET LUN (FAh) command, even when that LUN is busy.

When FAh is written to the command register, the addressed LUN goes busy for t_{RST} . During t_{RST} , the selected LUN discontinues all array operations. All pending single- and multi-plane operations are canceled. If this command is issued while a PROGRAM or ERASE operation is occurring on the addressed LUN, the data may be partially programmed or erased and is invalid. The command register is cleared and ready for the next command. The data register and cache register contents are invalid.

If the RESET LUN (FAh) command is issued when the synchronous interface is enabled, the target's interface remains in synchronous mode.

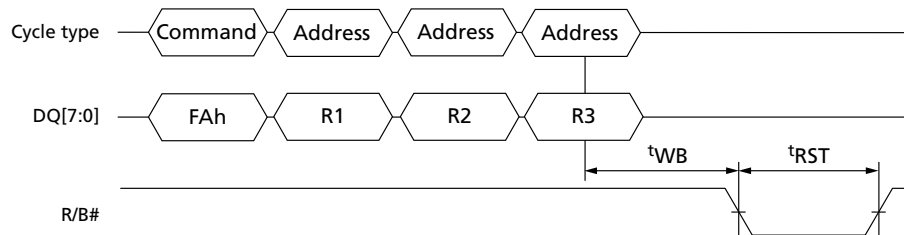
If the RESET LUN (FAh) command is issued when the asynchronous interface is enabled, the target's interface remains in asynchronous mode.

During or after t_{RST} , the host can poll each LUN's status register.

The RESET LUN (FAh) command is prohibited when not in the default array operation mode.

The RESET LUN (FAh) command can only be issued to a target (CE#) after the RESET (FFh) command has been issued as the first command to a target following power-up.

Figure 50: RESET LUN (FAh) Operation





Identification Operations

READ ID (90h)

The READ ID (90h) command is used to read identifier codes programmed into the target. This command is accepted by the target only when all die (LUNs) on the target are idle.

Writing 90h to the command register puts the target in read ID mode. The target stays in this mode until another valid command is issued.

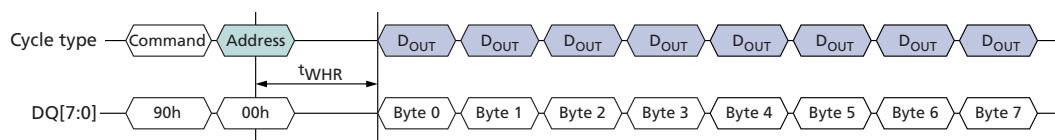
When the 90h command is followed by a 00h address cycle, the target returns a 5-byte identifier code that includes the manufacturer ID, device configuration, and part-specific information.

When the 90h command is followed by a 20h address cycle, the target returns the 4-byte ONFI identifier code.

When the 90h command is followed by a 40h address cycle, the target returns the 5-byte JEDEC identifier code.

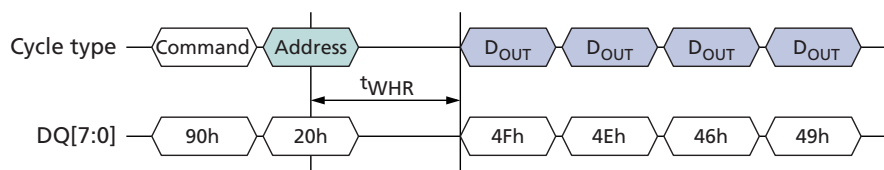
After the 90h and address cycle are written to the target, the host enables data output mode to read the identifier information. When the asynchronous interface is active, one data byte is output per RE# toggle. When the synchronous interface is active, one data byte is output per rising edge of DQS when ALE and CLE are HIGH; the data byte on the falling edge of DQS is identical to the data byte output on the previous rising edge of DQS.

Figure 51: READ ID (90h) with 00h Address Operation



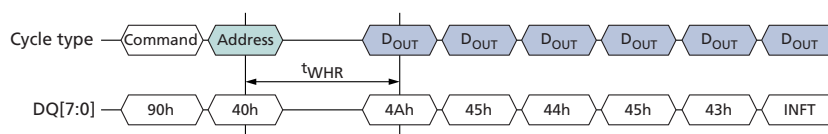
Note: 1. See the READ ID Parameter tables for byte definitions.

Figure 52: READ ID (90h) with 20h Address Operation



Note: 1. See the READ ID Parameter tables for byte definitions.

Figure 53: READ ID (90h) with 40h Address Operation



Note: 1. See the READ ID Parameter tables for byte definitions.



MLC 256Gb to 2Tb Async/Sync NAND Identification Operations

READ ID Parameter Tables

Table 12: Read ID Parameters for Address 00h

Device	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
MT29F128G08CBECB	2Ch	84h	64h	54h	A9h	00h	00h	00h
MT29F256G08CEECB	2Ch	84h	64h	54h	A9h	00h	00h	00h
MT29F512G08CKECB	2Ch	A4h	E5h	54h	A9h	00h	00h	00h
MT29F512G08CMECB	2Ch	84h	64h	54h	A9h	00h	00h	00h
MT29F1T08CUECB	2Ch	A4h	E5h	54h	A9h	00h	00h	00h
MT29F2T08CTECB	2Ch	A4h	E5h	54h	A9h	00h	00h	00h

Note: 1. h = hexadecimal.

Table 13: Read ID Parameters for Address 20h

Device	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4
All	4Fh	4Eh	46h	49h	XXh

Notes: 1. h = hexadecimal.
2. XXh = Undefined.

Table 14: Read ID Parameters for Address 40h

Device	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
Asynchronous or Synchronous	4Ah	45h	44h	45h	43h	05h

Note: 1. h = hexadecimal.



READ PARAMETER PAGE (ECh)

The READ PARAMETER PAGE (ECh) command is used to read the ONFI or JEDEC parameter page programmed into the target. This command is accepted by the target only when all die (LUNs) on the target are idle.

Writing ECh to the command register puts the target in read parameter page mode. The target stays in this mode until another valid command is issued.

When the ECh command is followed by an 00h or 40h address cycle, the target goes busy for t_R . If the READ STATUS (70h) command is used to monitor for command completion, the READ MODE (00h) command must be used to re-enable data output mode. Use of the READ STATUS ENHANCED (78h) command is prohibited while the target is busy and during data output.

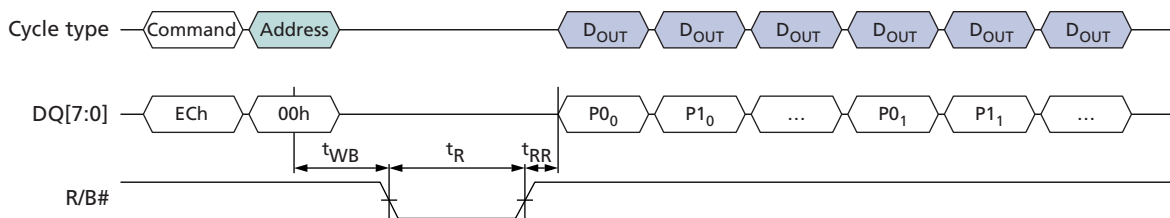
After t_R completes, the host enables data output mode to read the parameter page. When the asynchronous interface is active, one data byte is output per RE# toggle. When the synchronous interface is active, one data byte is output for each rising or falling edge of DQS.

A minimum of three copies of the parameter page are stored in the device. Each parameter page is 256 bytes. If desired, the CHANGE READ COLUMN (05h-E0h) command can be used to change the location of data output. Use of the CHANGE READ COLUMN ENHANCED (06h-E0h or 00h-05h-E0h) command is prohibited.

The READ PARAMETER PAGE (ECh) output data can be used by the host to configure its internal settings to properly use the NAND Flash device. Parameter page data is static per part, however the value can be changed through the product cycle of NAND Flash. The host should interpret the data and configure itself accordingly.

The host should continue reading redundant parameter pages until the host is able to accurately reconstruct the parameter page contents. Parameter pages returned by the target may have invalid CRC values; however, bit-wise majority may be used to recover the contents of the parameter page. The host may use bit-wise majority or other techniques to recover the contents of the parameter page from the parameter page copies present.

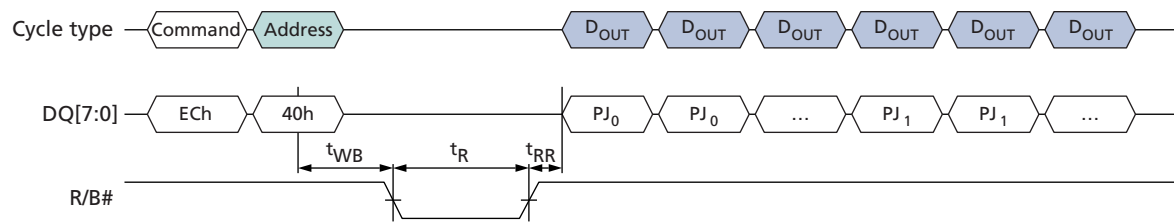
Figure 54: READ PARAMETER (ECh) with 00h Address Operation for ONFI





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Figure 55: READ PARAMETER (ECh) with 40h Address Operation for JEDEC





MLC 256Gb to 2Tb Async/Sync NAND Identification Operations

Parameter Page Data Structure Tables

Table 15: ONFI Parameter Page Data Structure

Byte	Description	Device	Values
Revision information and features block			
0–3	Parameter page signature Byte 0: 4Fh, "O" Byte 1: 4Eh, "N" Byte 2: 46h, "F" Byte 3: 49h, "I"	–	4Fh, 4Eh, 46h, 49h
4–5	Revision number Bit[15:7]: Reserved (0) Bit 8: 1 = supports ONFI version 3.2 Bit 7: 1 = supports ONFI version 3.1 Bit 6: 1 = supports ONFI version 3.0 Bit 5: 1 = supports ONFI version 2.3 Bit 4: 1 = supports ONFI version 2.2 Bit 3: 1 = supports ONFI version 2.1 Bit 2: 1 = supports ONFI version 2.0 Bit 1: 1 = supports ONFI version 1.0 Bit 0: Reserved (0)	–	FEh, 01h
6–7	Features supported Bit[15:13]: Reserved (0) Bit 12: 1 = supports external V _{pp} Bit 11: 1 = supports Volume addressing Bit 10: 1 = supports NV-DDR2 interface Bit 9: 0 = Reserved Bit 8: 1 = supports program page register clear enhancement Bit 7: 1 = supports extended parameter page Bit 6: 1 = supports interleaved (multi-plane) read operations Bit 5: 1 = supports NV-DDR interface Bit 4: 1 = supports odd-to-even page copyback Bit 3: 1 = supports interleaved (multi-plane) program and erase operations Bit 2: 1 = supports non-sequential page programming Bit 1: 1 = supports multiple LUN operations Bit 0: 1 = supports 16-bit data bus width	MT29F128G08CBECB	F8h, 1Dh
		MT29F256G08CEEBCB	
		MT29F512G08CMECB	
		MT29F512G08CKECB	FAh, 1Dh
		MT29F1T08CUECB	
		MT29F2T08CTECB	



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Table 15: ONFI Parameter Page Data Structure (Continued)

Byte	Description	Device	Values
8–9	Optional commands supported Bit[15:13]: Reserved (0) Bit 12: 1 = supports GET/SET FEATURES by LUN Bit 11: 1 = supports ODT CONFIGURE Bit 10: 1 = supports VOLUME SELECT Bit 9: 1 = supports RESET LUN Bit 8: 1 = supports small data move Bit 7: 1 = supports CHANGE ROW ADDRESS Bit 6: 1 = supports CHANGE READ COLUMN ENHANCED Bit 5: 1 = supports READ UNIQUE ID Bit 4: 1 = supports COPYBACK Bit 3: 1 = supports READ STATUS ENHANCED Bit 2: 1 = supports GET FEATURES and SET FEATURES Bit 1: 1 = supports read cache commands Bit 0: 1 = supports PROGRAM PAGE CACHE	–	FFh, 1Fh
10	ONFI-JEDEC JTG primary advanced command support Bit[7:4]: Reserved (0) Bit 3: 1 = supports ERASE BLOCK MULTI-PLANE Bit 2: 1 = supports COPYBACK PROGRAM MULTI-PLANE Bit 1: 1 = supports PROGRAM PAGE MULTI-PLANE Bit 0: 1 = supports CHANGE READ COLUMN	–	0Fh
11	Reserved (0)	–	All 00h
12–13	Extended parameter page length	–	03h, 00h
14	Number of parameter pages	–	3Ch
15–31	Reserved (0)	–	All 00h
Manufacturer information block			
32–43	Device manufacturer (12 ASCII characters) Micron	–	4Dh, 49h, 43h, 52h, 4Fh, 4Eh, 20h, 20h, 20h, 20h, 20h, 20h



MLC 256Gb to 2Tb Async/Sync NAND Identification Operations

Table 15: ONFI Parameter Page Data Structure (Continued)

Byte	Description	Device	Values
44–63	Device model (20 ASCII characters)	MT29F128G08CBECBH6	4Dh, 54h, 32h, 39h, 46h, 31h, 32h, 38h, 47h, 30h, 38h, 43h, 42h, 45h, 43h, 42h, 48h, 36h, 20h, 20h
		MT29F256G08CEECBH6	4Dh, 54h, 32h, 39h, 46h, 32h, 35h, 36h, 47h, 30h, 38h, 43h, 45h, 45h, 43h, 42h, 48h, 36h, 20h, 20h
		MT29F512G08CKECBH7	4Dh, 54h, 32h, 39h, 46h, 35h, 31h, 32h, 47h, 30h, 38h, 43h, 48h, 45h, 43h, 42h, 48h, 37h, 20h, 20h
		MT29F512G08CMECBH7	4Dh, 54h, 32h, 39h, 46h, 35h, 31h, 32h, 47h, 30h, 38h, 43h, 4Dh, 45h, 43h, 42h, 48h, 37h, 20h, 20h
		MT29F1T08CUECBH8	4Dh, 54h, 32h, 39h, 46h, 31h, 54h, 30h, 38h, 43h, 55h, 45h, 43h, 42h, 48h, 38h, 20h, 20h, 20h, 20h
		MT29F2T08CTECBJ7	4Dh, 54h, 32h, 39h, 46h, 32h, 54h, 30h, 38h, 43h, 54h, 45h, 43h, 42h, 4Ah, 37h, 20h, 20h, 20h, 20h
64	JEDEC manufacturer ID	–	2Ch
65–66	Date code	–	00h, 00h
67–79	Reserved (0)	–	All 00h
Memory organization block			
80–83	Number of data bytes per page	–	00h, 40h, 00h, 00h
84–85	Number of spare bytes per page	–	50h, 07h
86–91	Reserved (0)	–	All 00h
92–95	Number of pages per block	–	00h, 02h, 00h, 00h
96–99	Number of blocks per LUN	–	30h, 08h, 00h, 00h



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Table 15: ONFI Parameter Page Data Structure (Continued)

Byte	Description	Device	Values
100	Number of LUNs per chip enable	MT29F128G08CBECB	01h
		MT29F256G08CEECB	
		MT29F512G08CMECB	
		MT29F512G08CKECB	02h
		MT29F1T08CUECB	
		MT29F2T08CTECB	
101	Number of address cycles Bit[7:4]: Column address cycles Bit[3:0]: Row address cycles	–	23h
102	Number of bits per cell	–	02h
103–104	Bad blocks maximum per LUN	–	62h, 00h
105–106	Block endurance	–	03h, 03h
107	Guaranteed valid blocks at beginning of target	–	01h
108–109	Block endurance for guaranteed valid blocks	–	00h, 00h
110	Number of programs per page	–	01h
111	Reserved (0)	–	00h
112	Number of bits ECC correctability	–	FFh
113	Number of interleaved address bits Bit[7:4]: Reserved (0) Bit[3:0]: Number of interleaved address bits	–	01h
114	Interleaved operation attributes Bit[7:6]: Reserved (0) Bit 5: Reserved Bit 4: 1 = supports read cache Bit 3: Address restrictions for cache operations Bit 2: 1 = supports program cache Bit 1: 1 = no block address restrictions Bit 0: Overlapped/concurrent interleaving support	–	1Eh
115–127	Reserved (0)	–	All 00h
Electrical parameters block			
128	I/O pin capacitance per chip enable	MT29F128G08CBECB	06h
		MT29F256G08CEECB	06h
		MT29F512G08CKECB	0Ah
		MT29F512G08CMECB	05h
		MT29F1T08CUECB	09h
		MT29F2T08CTECB	09h



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Table 15: ONFI Parameter Page Data Structure (Continued)

Byte	Description	Device	Values
129–130	Asynchronous timing mode support Bit[15:6]: Reserved (0) Bit 5: 1 = supports timing mode 5 Bit 4: 1 = supports timing mode 4 Bit 3: 1 = supports timing mode 3 Bit 2: 1 = supports timing mode 2 Bit 1: 1 = supports timing mode 1 Bit 0: 1 = supports timing mode 0, shall be 1	–	3Fh, 00h
131–132	Reserved (0)	–	All 00h
133–134	^t PROG Maximum PROGRAM PAGE time (μs)	–	B8h, 0Bh
135–136	^t BERS Maximum BLOCK ERASE time (μs)	–	E0h, 2Eh
137–138	^t R Maximum PAGE READ time (μs)	–	73h, 00h
139–140	^t CCS Minimum change column setup time (ns)	–	2Ch, 01h
141	NV-DDR timing mode support Bit[7:6]: Reserved (0) Bit 5: 1 = supports timing mode 5 Bit 4: 1 = supports timing mode 4 Bit 3: 1 = supports timing mode 3 Bit 2: 1 = supports timing mode 2 Bit 1: 1 = supports timing mode 1 Bit 0: 1 = supports timing mode 0	MT29F128G08CBECB	1Fh
		MT29F256G08CEECB	
		MT29F512G08CKECB	
		MT29F512G08CMECB	
		MT29F1T08CUECB	
		MT29F2T08CTECB	
142	NV-DDR2 timing mode support Bit 7: 1 = supports timing mode 7 Bit 6: 1 = supports timing mode 6 Bit 5: 1 = supports timing mode 5 Bit 4: 1 = supports timing mode 4 Bit 3: 1 = supports timing mode 3 Bit 2: 1 = supports timing mode 2 Bit 1: 1 = supports timing mode 1 Bit 0: 1 = supports timing mode 0	MT29F128G08CBECB	0Fh
		MT29F256G08CEECB	
		MT29F512G08CKECB	
		MT29F512G08CMECB	
		MT29F1T08CUECB	
		MT29F2T08CTECB	
143	NV-DDR/NV-DDR2 features Bit[7:3]: Reserved (0) Bit 2: 1 = devices support CLK stopped for data input Bit 1: 1 = typical capacitance values present Bit 0: 0 = use ^t CAD MIN value	–	02h
144–145	CLK input pin capacitance, typical	MT29F128G08CBECB	28h, 00h
		MT29F256G08CEECB	
		MT29F512G08CKECB	40h, 00h
		MT29F512G08CMECB	20h, 00h
		MT29F1T08CUECB	39h, 00h
		MT29F2T08CTECB	30h, 00h



MLC 256Gb to 2Tb Async/Sync NAND Identification Operations

Table 15: ONFI Parameter Page Data Structure (Continued)

Byte	Description	Device	Values
146–147	I/O pin capacitance, typical	MT29F128G08CBECB	2Fh, 00h
		MT29F256G08CEECB	
		MT29F512G08CKECB	51h, 00h
		MT29F512G08CMECB	29h, 00h
		MT29F1T08CUECB	4Bh, 00h
		MT29F2T08CTECB	47h, 00h
148–149	Input capacitance, typical		
		MT29F128G08CBECB	28h, 00h
		MT29F256G08CEECB	
		MT29F512G08CKECB	40h, 00h
		MT29F512G08CMECB	20h, 00h
		MT29F1T08CUECB	39h, 00h
150	Input pin capacitance, maximum	MT29F2T08CTECB	30h, 00h
		MT29F128G08CBECB	05h
		MT29F256G08CEECB	
		MT29F512G08CKECB	08h
		MT29F512G08CMECB	04h
		MT29F1T08CUECB	07h
151	Driver strength support Bit[7:3]: Reserved (0) Bit 2: 1 = Supports 18 Ohm drive strength Bit 1: 1 = Supports 25 Ohm drive strength Bit 0: 1 = Supports driver strength settings	MT29F2T08CTECB	06h
			07h
		–	
152–153	^t R maximum interleaved (multi-plane) page read time (μs)	–	73h, 00h
154–155	^t ADL program page register clear enhancement value (ns)	–	46h, 00h
156–157	Reserved (0)	–	All 00h
158	NV-DDR2 features Bit[7:6]: Reserved (0) Bit 5: 0 = external V _{REFQ} not required for >= 200MT/s Bit 4: 1 = supports differential signaling for DQS Bit 3: 1 = supports differential signaling for RE# Bit 2: 1 = supports ODT value of 30 ohms Bit 1: 1 = supports matrix termination ODT Bit 0: 1 = supports self-termination ODT	–	1Fh
159	NV-DDR2 warmup cycles Bit[7:4]: Data input warmup cycles support Bit[3:0]: Data output warmup cycles support	–	44h
160–163	Reserved (0)	–	All 00h



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Table 15: ONFI Parameter Page Data Structure (Continued)

Byte	Description	Device	Values
Vendor block			
164–165	Vendor-specific revision number	–	01h, 00h
166	TWO-PLANE PAGE READ support Bit[7:1]: Reserved (0) Bit 0: 1 = Support for TWO-PLANE PAGE READ	–	01h
167	Read cache support Bit[7:1]: Reserved (0) Bit 0: 0 = Does not support Micron-specific read cache function	–	00h
168	READ UNIQUE ID support Bit[7:1]: Reserved (0) Bit 0: 0 = Does not support Micron-specific READ UNIQUE ID	–	00h
169	Programmable DQ output impedance support Bit[7:1]: Reserved (0) Bit 0: 0 = No support for programmable DQ output impedance by B8h command	–	00h
170	Number of programmable DQ output impedance settings Bit[7:3]: Reserved (0) Bit [2:0] = Number of programmable DQ output impedance settings	–	04h
171	Programmable DQ output impedance feature address Bit[7:0] = Programmable DQ output impedance feature address	–	10h
172	Programmable R/B# pull-down strength support Bit[7:1]: Reserved (0) Bit 0: 1 = Support programmable R/B# pull-down strength	–	01h
173	Programmable R/B# pull-down strength feature address Bit[7:0] = Feature address used with programmable R/B# pull-down strength	–	81h
174	Number of programmable R/B# pull-down strength settings Bit[7:3]: Reserved (0) Bit[2:0] = Number of programmable R/B# pull-down strength settings	–	04h
175	OTP mode support Bit[7:2]: Reserved (0) Bit 1: 1 = Supports Get/Set Features command set Bit 0: 0 = Does not support A5h/A0h/AFh OTP command set	–	02h
176	OTP page start Bit[7:0] = Page where OTP page space begins	–	02h



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Table 15: ONFI Parameter Page Data Structure (Continued)

Byte	Description	Device	Values
177	OTP DATA PROTECT address Bit[7:0] = Page address to use when issuing OTP DATA PROTECT command	–	01h
178	Number of OTP pages Bit[15:5]: Reserved (0) Bit[4:0] = Number of OTP pages	–	1Eh
179	OTP Feature Address	–	90h
180	Read Retry Options Bit[7:4]: Reserved (0) Bit[3:0] = Number of Read Retry options supported	–	0Ah
181–184	Read Retry Options available. A value of '1' in a bit position shows that Read Retry option is available for use. A value of '0' in a bit position shows that Read Retry option is not available for use. For example, the binary value of '37h, 00h, 00h, 00h' would represent that Read Retry options 0, 1, 2, 4, and 5 are available which would correspond to Read Retry input option selections of 00h, 01h, 02h, 04h, and 05h. It furthermore would show that Read Retry options 3, 6, and 7 to 32 which would correspond to Read Retry input option selections 03h, 06h, 07h to 20h, are not available.	–	FFh, 11h, 00h, 00h
185–252	Reserved (0)	–	All 00h
253	Parameter page revision	MT29F128G08CBECBH6	03h
		MT29F256G08CEECBH6	
		MT29F512G08CKECBH7	
		MT29F512G08CMECBH7	
		MT29F1T08CUECBH8	
		MT29F2T08CTECBJ7	02h
254–255	Integrity CRC	MT29F128G08CBECBH6	35h, 89h
		MT29F256G08CEECBH6	D3h, 23h
		MT29F512G08CKECBH7	3Ch, A7h
		MT29F512G08CMECBH7	CCh, 28h
		MT29F1T08CUECBH8	31h, 9Bh
		MT29F2T08CTECBJ7	44h, 17h
Redundant parameter pages			
256–511	Value of bytes 0–255	–	See bytes 0–255
512–767	Value of bytes 0–255	–	See bytes 0–255
...	...	–	...
14,848–15,103	Value of bytes 0–255	–	See bytes 0–255



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Table 15: ONFI Parameter Page Data Structure (Continued)

Byte	Description	Device	Values
15,104–15,359	Value of bytes 0–255	–	See bytes 0–255
Extended parameter pages			
15,360–15,361	Extended parameter page Integrity CRC	–	62h, AFh
15,362-15,365	Extended parameter page signature Byte 0: 45h, "E" Byte 1: 50h, "P" Byte 2: 50h, "P" Byte 3: 53h, "S"	–	45h, 50h, 50h, 53h
15,366-15,375	Reserved (0)	–	All 00h
15,376	Section 0 type	–	02h
15,377	Section 0 length	–	01h
15,378-15,391	Reserved (0)	–	All 00h
15,392	Number of bits ECC correctability	–	3Ch
15,393	ECC codeword size	–	0Ah
15,394–15,395	Bad blocks maximum per LUN	–	62h, 00h
15,396–15,397	Block endurance	–	03h, 03h
15,398-15,407	Reserved (0)	–	All 00h
Redundant extended parameter pages			
15,408-15,455	Value of bytes 15,360–15,407	–	See bytes 15,360–15,407
15,456-15,503	Value of bytes 15,360–15,407	–	See bytes 15,360–15,407
...	...	–	...
18,144–18,191	Value of bytes 15,360–15,407	–	See bytes 15,360–15,407
18,192–18,239	Value of bytes 15,360–15,407	–	See bytes 15,360–15,407
18,240 to end of page	Reserved (FFh)	–	All FFh

Table 16: JEDEC Parameter Page Definition

Byte	Description	Device	Values
Revision Information and Features Block			



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Table 16: JEDEC Parameter Page Definition (Continued)

Byte	Description	Device	Values
0–3	Parameter page signature Byte 0: 4Ah, "J" Byte 1: 45h, "E" Byte 2: 53h, "S" Byte 3: 44h, "D"	–	4Ah, 45h, 53h, 44h
4–5	Revision number Bit[15:3]: Reserved (0) Bit 2: 1 = supports JEDEC version 1.0 Bit 1: 1 = supports vendor specific parameter page Bit 0: Reserved (0)	–	06h, 00h
6–7	Features supported Bit[15:9]: Reserved (0) Bit 8: 1 = supports program page register clear enhancement Bit 7: 1 = supports external V _{pp} Bit 6: 1 = supports Toggle Mode DDR Bit 5: 1 = supports Synchronous DDR Bit 4: 1 = supports multi-plane read operations Bit 3: 1 = supports multi-plane program and erase operations Bit 2: 1 = supports non-sequential page programming Bit 1: 1 = supports multiple LUN operations Bit 0: 1 = supports 16-bit data bus width	MT29F128G08CBECB	B8h, 01h
		MT29F256G08CEEBCB	
		MT29F512G08CMECB	
		MT29F512G08CKECB	BAh, 01h
		MT29F1T08CUECB	
		MT29F2T08CTECB	
8–10	Features supported Bit[23:11]: Reserved (0) Bit 10: 1 = supports Synchronous Reset Bit 9: 1 = supports Reset LUN (Primary) Bit 8: 1 = supports Small Data Move Bit 7: 1 = supports Multi-plane Copyback Program (Primary) Bit 6: 1 = supports Random Data Out (Primary) Bit 5: 1 = supports Read Unique ID Bit 4: 1 = supports Copyback Bit 3: 1 = supports Read Status Enhanced (Primary) Bit 2: 1 = supports Get Features and Set Features Bit 1: 1 = supports Read Cache commands Bit 0: 1 = supports Page Cache Program command	–	FFh, 07h, 00h



MLC 256Gb to 2Tb Async/Sync NAND Identification Operations

Table 16: JEDEC Parameter Page Definition (Continued)

Byte	Description	Device	Values
11–12	Secondary commands supported Bit[15:8]: Reserved (0) Bit 7: 1 = supports secondary Read Status Enhanced Bit 6: 1 = supports secondary Multi-plane Block Erase Bit 5: 1 = supports secondary Multi-plane Copyback Program Bit 4: 1 = supports secondary Multi-plane Program Bit 3: 1 = supports secondary Random Data Out Bit 2: 1 = supports secondary Multi-plane Copyback Read Bit 1: 1 = supports secondary Multi-plane Read Cache Random Bit 0: 1 = supports secondary Multi-plane Read	–	58h, 00h
13	Number of Parameter Pages	–	23h
14–31	Reserved (0)	–	All 00h
Manufacturer information block			
32–43	Device manufacturer (12 ASCII characters) Micron	–	4Dh, 49h, 43h, 52h, 4Fh, 4Eh, 20h, 20h, 20h, 20h, 20h, 20h
44–63	Device model (20 ASCII characters)	MT29F128G08CBECBH6	4Dh, 54h, 32h, 39h, 46h, 31h, 32h, 38h, 47h, 30h, 38h, 43h, 42h, 45h, 43h, 42h, 48h, 36h, 20h, 20h
		MT29F256G08CEECBH6	4Dh, 54h, 32h, 39h, 46h, 32h, 35h, 36h, 47h, 30h, 38h, 43h, 45h, 45h, 43h, 42h, 48h, 36h, 20h, 20h
		MT29F512G08CKECBH7	4Dh, 54h, 32h, 39h, 46h, 35h, 31h, 32h, 47h, 30h, 38h, 43h, 48h, 45h, 43h, 42h, 48h, 37h, 20h, 20h
		MT29F512G08CMECBH7	4Dh, 54h, 32h, 39h, 46h, 35h, 31h, 32h, 47h, 30h, 38h, 43h, 4Dh, 45h, 43h, 42h, 48h, 37h, 20h, 20h
		MT29F1T08CUECBH8	4Dh, 54h, 32h, 39h, 46h, 31h, 54h, 30h, 38h, 43h, 55h, 45h, 43h, 42h, 48h, 38h, 20h, 20h, 20h, 20h
		MT29F2T08CTECBJ7	4Dh, 54h, 32h, 39h, 46h, 32h, 54h, 30h, 38h, 43h, 54h, 45h, 43h, 42h, 4Ah, 37h, 20h, 20h, 20h, 20h
64–69	JEDEC manufacturer ID	–	2Ch, 00h, 00h, 00h, 00h, 00h



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Table 16: JEDEC Parameter Page Definition (Continued)

Byte	Description	Device	Values
70–71	Reserved (0)	–	All 00h
72–79	Reserved (0)	–	All 00h
Memory organization block			
80–83	Number of data bytes per page	–	00h, 40h, 00h, 00h
84–85	Number of spare bytes per page	–	50h, 07h
86–89	Number of data bytes per partial page	–	00h, 04h, 00h, 00h
90–91	Number of spare bytes per partial page	–	75h, 00h
92–95	Number of pages per block	–	00h, 02h, 00h, 00h
96–99	Number of blocks per LUN	–	30h, 08h, 00h, 00h
100	Number of LUNs per chip enable	MT29F128G08CBECB	01h
		MT29F256G08CEEBCB	
		MT29F512G08CMECB	
		MT29F512G08CKECB	02h
		MT29F1T08CUECB	
		MT29F2T08CTECB	
101	Number of address cycles Bit[7:4]: Column address cycles Bit[3:0]: Row address cycles	–	23h
102	Number of bits per cell	–	02h
103	Number of programs per page	–	01h
104	Multi-plane operation addressing Bit[7:4]: Reserved (0) Bit[3:0]: Number of plane address bits	–	01h
105	Multi-plane operation attributes Bit[7:3]: Reserved (0) Bit 2: 1 = Read cache operations supported Bit 1: 1 = Program cache operations supported Bit 0: 1 = Address restrictions for cache operations	–	07h
106–143	Reserved (0)	–	All 00h
Electrical parameters block			
144–145	Asynchronous SDR speed grade support Bit[15:6]: Reserved (0) Bit 5: 1 = supports timing mode 5 Bit 4: 1 = supports timing mode 4 Bit 3: 1 = supports timing mode 3 Bit 2: 1 = supports timing mode 2 Bit 1: 1 = supports timing mode 1 Bit 0: 1 = supports timing mode 0, shall be 1	–	3Fh, 00h



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Table 16: JEDEC Parameter Page Definition (Continued)

Byte	Description	Device	Values
146–147	Synchronous DDR2 speed grade support Bit[15:8]: Reserved (0) Bit 7: 1 = supports timing mode 7 Bit 6: 1 = supports timing mode 6 Bit 5: 1 = supports timing mode 5 Bit 4: 1 = supports timing mode 4 Bit 3: 1 = supports timing mode 3 Bit 2: 1 = supports timing mode 2 Bit 1: 1 = supports timing mode 1 Bit 0: 1 = supports timing mode 0, shall be 1	MT29F128G08CBECB	0Fh, 00h
		MT29F256G08CEECB	
		MT29F512G08CKECB	
		MT29F512G08CMECB	
		MT29F1T08CUECB	
		MT29F2T08CTECB	
148–149	Synchronous DDR speed grade support Bit[15:6]: Reserved (0) Bit 5: 1 = supports timing mode 5 Bit 4: 1 = supports timing mode 4 Bit 3: 1 = supports timing mode 3 Bit 2: 1 = supports timing mode 2 Bit 1: 1 = supports timing mode 1 Bit 0: 1 = supports timing mode 0, shall be 1	MT29F128G08CBECB	1Fh, 00h
		MT29F256G08CEECB	
		MT29F512G08CKECB	
		MT29F512G08CMECB	
		MT29F1T08CUECB	
		MT29F2T08CTECB	
150	Asynchronous SDR features Bit[7:0]: Reserved (0)	–	00h
151	Reserved (0)	–	00h
152	Synchronous DDR features Bit[7:2]: Reserved (0) Bit 1: 1 = devices support CLK stopped for data input Bit 0: 0 = use ^t CAD MIN value	–	00h
153–154	^t PROG Maximum PROGRAM PAGE time (μs)	–	B8h, 0Bh
155–156	^t BERS Maximum BLOCK ERASE time (μs)	–	E0h, 2Eh
157–158	^t R Maximum Multi-PLANE PAGE READ time (μs)	–	73h, 00h
159–160	^t R Maximum PAGE READ time (μs)	–	73h, 00h
161–162	^t CCS Minimum change column setup time (ns)	–	2Ch, 01h
163–164	I/O pin capacitance, typical	MT29F128G08CBECB	2Fh, 00h
		MT29F256G08CEECB	
		MT29F512G08CKECB	51h, 00h
		MT29F512G08CMECB	29h, 00h
		MT29F1T08CUECB	4Bh, 00h
		MT29F2T08CTECB	47h, 00h
165–166	Input capacitance, typical	MT29F128G08CBECB	28h, 00h
		MT29F256G08CEECB	
		MT29F512G08CKECB	40h, 00h
		MT29F512G08CMECB	20h, 00h
		MT29F1T08CUECB	39h, 00h
		MT29F2T08CTECB	30h, 00h



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Table 16: JEDEC Parameter Page Definition (Continued)

Byte	Description	Device	Values
167–168	CLK input pin capacitance, typical	MT29F128G08CBECB	28h, 00h
		MT29F256G08CEECB	
		MT29F512G08CKECB	40h, 00h
		MT29F512G08CMECB	20h, 00h
		MT29F1T08CUECB	39h, 00h
		MT29F2T08CTECB	30h, 00h
169	Driver strength support Bit[7:3]: Reserved (0) Bit 2: 1 = Supports 18 Ohm drive strength Bit 1: 1 = Supports 25 Ohm drive strength Bit 0: 1 = Supports driver strength settings	–	07h
170–171	^t ADL program page register clear enhancement value (ns)	–	46h, 00h
172–207	Reserved (0)	–	00h
ECC and endurance block			
208	Guaranteed valid blocks at beginning of target	–	01h
209–210	Block endurance for guaranteed valid blocks	–	00h, 00h
211	Number of bits ECC correctability	–	3Ch
212	ECC codeword size		0Ah
213–214	Bad blocks maximum per LUN	–	62h, 00h
215–216	Block endurance	–	03h, 03h
217–218	Reserved (0)	–	All 00h
219–271	Reserved (0)	–	All 00h
Reserved			
272–419	Reserved (0)	–	All 00h
Vendor specific block			
420–421	Vendor-specific revision number	MT29F128G08CBECB	03h, 00h
		MT29F256G08CEECB	
		MT29F512G08CKECB	
		MT29F512G08CMECB	
		MT29F1T08CUECB	
		MT29F2T08CTECB	02h, 00h
422	Read Retry Options Bit[7:4]: Reserved (0) Bit[3:0] = Number of Read Retry options supported	–	0Ah



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Table 16: JEDEC Parameter Page Definition (Continued)

Byte	Description	Device	Values
423–426	Read Retry Options available. A value of '1' in a bit position shows that Read Retry option is available for use. A value of '0' in a bit position shows that Read Retry option is not available for use. For example, the binary value of '37h, 00h, 00h, 00h' would represent that Read Retry options 0, 1, 2, 4, and 5 are available which would correspond to Read Retry input option selections of 00h, 01h, 02h, 04h, and 05h. It furthermore would show that Read Retry options 3, 6, and 7 to 32 which would correspond to Read Retry input option selections 03h, 06h, 07h to 20h, are not available.	–	FFh, 11h, 00h, 00h
427–509	Reserved (0)	–	All 00h
CRC for Parameter Page			
510–511	Integrity CRC	MT29F128G08CBECBH6	24h, A0h
		MT29F256G08CEECBH6	13h, C0h
		MT29F512G08CKECBH7	C6h, B5h
		MT29F512G08CMECBH7	22h, FFh
		MT29F1T08CUECBH8	6Ch, 17h
		MT29F2T08CTECBJ7	28h, 31h
Redundant JEDEC Parameter Pages			
512–1023	See byte values 0-511	–	See bytes 0-511
1024–1535	See byte values 0-511	–	See bytes 0-511
...	...	–	...
17,408–17,919	See byte values 0-511	–	See bytes 0-511
17,920–18,431	See byte values 0-511	–	See bytes 0-511
18,432 to end of page	Reserved (FFh)	–	All FFh

Note: 1. h = hexadecimal.

READ UNIQUE ID (EDh)

The READ UNIQUE ID (EDh) command is used to read a unique identifier programmed into the target. This command is accepted by the target only when all die (LUNs) on the target are idle.

Writing EDh to the command register puts the target in read unique ID mode. The target stays in this mode until another valid command is issued.



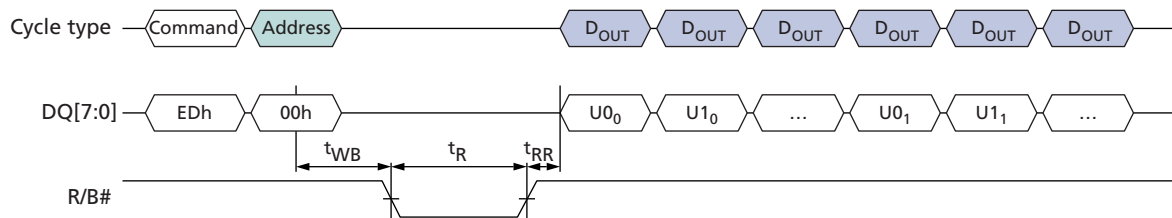
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When the EDh command is followed by a 00h address cycle, the target goes busy for t_R . If the READ STATUS (70h) command is used to monitor for command completion, the READ MODE (00h) command must be used to re-enable data output mode.

After t_R completes, the host enables data output mode to read the unique ID. When the asynchronous interface is active, one data byte is output per RE# toggle. When the synchronous interface is active, two data bytes are output, one byte for each rising or falling edge of DQS.

Sixteen copies of the unique ID data are stored in the device. Each copy is 32 bytes. The first 16 bytes of a 32-byte copy are unique data, and the second 16 bytes are the complement of the first 16 bytes. The host should XOR the first 16 bytes with the second 16 bytes. If the result is 16 bytes of FFh, then that copy of the unique ID data is correct. In the event that a non-FFh result is returned, the host can repeat the XOR operation on a subsequent copy of the unique ID data. If desired, the CHANGE READ COLUMN (05h-E0h) command can be used to change the data output location. Use of the CHANGE READ COLUMN ENHANCED (06h-E0h) command is prohibited.

Figure 56: READ UNIQUE ID (EDh) Operation





Configuration Operations

The SET FEATURES (EFh), GET FEATURES (EEh), SET FEATURES by LUN (D5h), and GET FEATURES by LUN (D4h) commands are used to modify the target's default power-on behavior. These commands use a one-byte feature address to determine which subfeature parameters will be read or modified. Each feature address (in the 00h to FFh range) is defined in Table 17. The SET FEATURES (EFh) and SET FEATURES by LUN (D5h) commands writes subfeature parameters (P1-P4) to the specified feature address. The GET FEATURES (EEh) and GET FEATURES by LUN (D4h) commands reads the subfeature parameters (P1-P4) at the specified feature address.

Unless otherwise specified, the values of the feature addresses do not change when RE-SET (FAh, FFh, FCh) is issued by the host.

Table 17: Feature Address Definitions

Feature Address	Definition
00h	Reserved
01h	Timing mode
02h	NV-DDR2 configuration
03h–0Fh	Reserved
10h	Programmable output drive strength
11h–2Fh	Reserved
30h	V _{PP} configuration
31h–57h	Reserved
58h	Volume configuration
59h–7Fh	Reserved
80h	Programmable output drive strength
81h	Programmable R/B# pull-down strength
82h–88h	Reserved
89h	Read Retry
8Ah–8Fh	Reserved
90h	Array operation mode
91h–FFh	Reserved

SET FEATURES (EFh)

The SET FEATURES (EFh) command writes the subfeature parameters (P1-P4) to the specified feature address to enable or disable target-specific features. This command is accepted by the target only when all die (LUNs) on the target are idle.

Writing EFh to the command register puts the target in the set features mode. The target stays in this mode until another command is issued.

The EFh command is followed by a valid feature address as specified in Table 17. The host waits for ^tADL before the subfeature parameters are input. When the asynchronous interface is active, one subfeature parameter is latched per rising edge of WE#. When the NV-DDR or NV-DDR2 interface is active, one subfeature parameter is latched per rising edge of DQS_t. The data on the falling edge of DQS_t should be identical to the



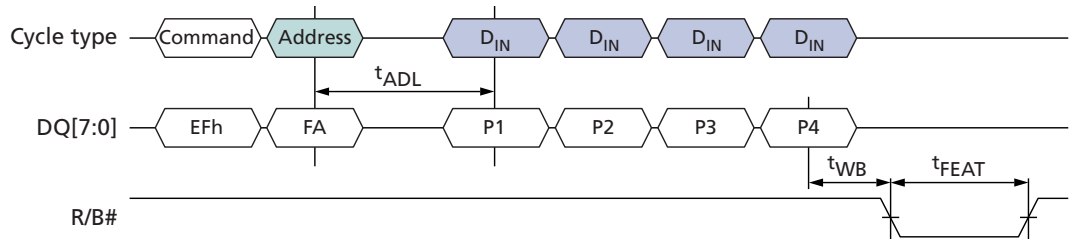
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subfeature parameter input on the previous rising edge of DQS_t. The device is not required to wait for the repeated data byte before beginning internal actions.

After all four subfeature parameters are input, the target goes busy for ^tFEAT. The READ STATUS (70h) command can be used to monitor for command completion.

Feature address 01h (Timing mode) operation is unique. If SET FEATURES is used to modify the interface type, the target will be busy for ^tITC. See Activating Interfaces (page 65) for details.

Figure 57: SET FEATURES (EFh) Operation



GET FEATURES (EEh)

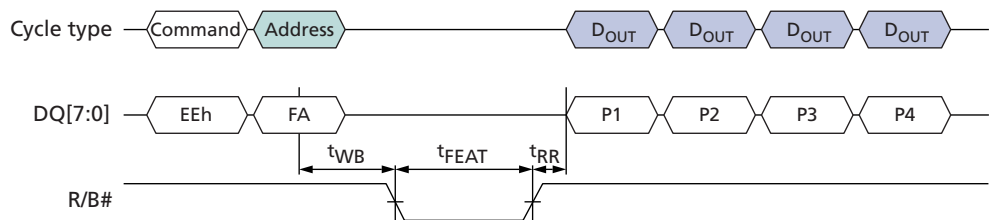
The GET FEATURES (EEh) command reads the subfeature parameters (P1-P4) from the specified feature address. This command is accepted by the target only when all die (LUNs) on the target are idle.

Writing EEh to the command register puts the target in get features mode. The target stays in this mode until another valid command is issued.

When the EEh command is followed by a valid feature address as specified in Table 17, the target goes busy for ^tFEAT. If the READ STATUS (70h) command is used to monitor for command completion, the READ MODE (00h) command must be used to re-enable data output mode. During and prior to data output, use of the READ STATUS ENHANCED (78h) command is prohibited.

After ^tFEAT completes, the host enables data output mode to read the subfeature parameters. When the asynchronous interface is active, one data byte is output per RE_# toggle. When the NV-DDR or NV-DDR2 interface is active, one subfeature parameter is output per DQS_t toggle on rising or falling edge of DQS_t.

Figure 58: GET FEATURES (EEh) Operation



GET/SET FEATURES by LUN (D4h/D5h)

The original GET FEATURES (EEh) and SET FEATURES (EFh) commands were target (CE#) based operations that did not take into account LUN addressing. By that, setting a feature address or getting a value for a feature address for a given target (CE#) applied to



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all LUNs on that target (CE#). GET FEATURES by LUN (D4h) and SET FEATURES by LUN (D5h) commands have the ability to also select a LUN address gives added flexibility to set the same feature address to different values for each LUN on a target (CE#) for more complex and versatile system solutions.

GET FEATURES by LUN (D4h) and SET FEATURES by LUN (D5h) operations work in the same manner as the non-LUN versions of those commands that did not take LUN addressing into account. The GET FEATURES by LUN (D4h) and SET FEATURES by LUN (D5h) commands both have an added LUN address cycle that is before the address cycle of the feature address which denotes the LUN to select for the operation. See Table 18 (page 102) for the decode of the LUN address cycle.

Table 18: GET/SET FEATURES by LUN Operation LUN address cycle decoding

Description	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
LUN											
LUN selection	LUN0								0	0b	
	LUN1								1	1b	
Reserved	—	—	—	—	—	—	—	—			

GET FEATURES by LUN (D4h) and SET FEATURES by LUN (D5h) can be issued to any addressable LUN that is not busy (RDY = 1, ARDY = 1). Target LUN status can be determined by using the READ STATUS ENHANCED (78h) command. When doing multiple LUN operations, refer to the Interleaved Die (Mutli-LUN) Operation section for details on proper operations.

Figure 59: GET FEATURES by LUN (D4h) Operation

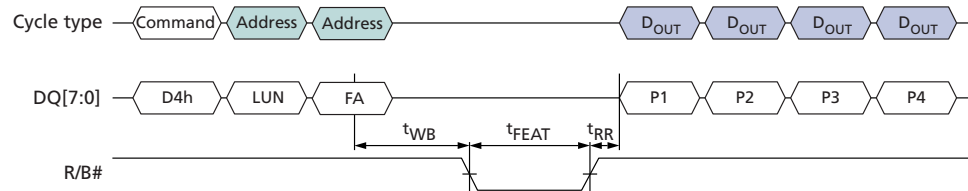
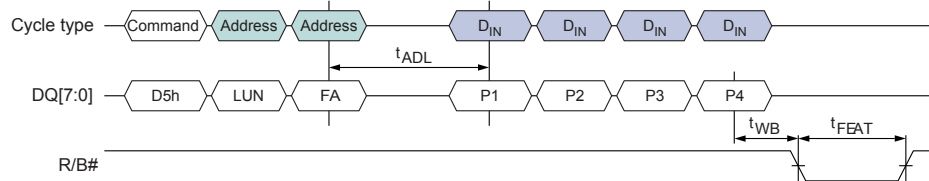


Figure 60: SET FEATURES by LUN (D5h) Operation





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Table 19: Feature Address 01h: Timing mode

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
P1											
Timing mode	Mode 0 (default)					0	0	0	0	x0h	1, 2
	Mode 1					0	0	0	1	x1h	
	Mode 2					0	0	1	0	x2h	
	Mode 3					0	0	1	1	x3h	
	Mode 4					0	1	0	0	x4h	
	Mode 5					0	1	0	1	x5h	
	Mode 6					0	1	1	0	x6h	
	Mode 7					0	1	1	1	x7h	
	Reserved					1	x	x	x	-	
Data interface	Asynchronous (default)			0	0					0xh	2
	NV-DDR			0	1					1xh	
	NV-DDR2			1	0					2xh	
	Reserved			1	1					3xh	
Program clear	Program command clears all cache registers on a target (default)		0							0b	
	Program command clears only the cache register of the selected plane of addressed LUN on a target		1							1b	
Reserved		0								0b	
P2											
Reserved		0	0	0	0	0	0	0	0	00h	
P3											
Reserved		0	0	0	0	0	0	0	0	00h	
P4											
Reserved		0	0	0	0	0	0	0	0	00h	

- Notes:
- Asynchronous timing mode 0 is the default, power-on value.
 - If the NV-DDR or NV-DDR2 interface is active, a RESET (FFh) command will change the timing mode and data interface bits of feature address 01h to their default values. If the asynchronous interface is active, a RESET (FFh) command will not change the values of the timing mode or data interface bits to their default values. Transition from the NV-DDR interface to the NV-DDR2 interface or vice versa is not permitted.



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Table 20: Feature Address 02h: NV-DDR2 configuration

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes	
P1												
Voltage Reference (VEN)	External V _{REFQ} is disabled and internal voltage is used as reference for DQ signals(default)								0	0b		
	External V _{REFQ} is enabled and used as reference for DQ signals								1	1b		
Complementary DQS (CMPD)	DQS_c signal disabled (default)							0		0b		
	DQS_c signal enabled							1		1b		
Complementary RE# (CMPR)	RE_c signal disabled (default)						0			0b		
	RE_c signal enabled						1			1b		
Reserved	-					0				0b		
DQ[7:0],DQS_t, DQS_c, RE_t, and RE_c ODT enable	ODT disabled (default)	0	0	0	0					0h	1	
	ODT enabled with R _{TT} of 150 Ohms	0	0	0	1					1h		
	ODT enabled with R _{TT} of 100 Ohms	0	0	1	0					2h		
	ODT enabled with R _{TT} of 75 Ohms	0	0	1	1					3h		
	ODT enabled with R _{TT} of 50 Ohms	0	1	0	0					4h		
	Reserved		0	1	0	1					5h	
			0	1	1	0					6h	
			0	1	1	1					7h	
			1	X	X	X					8h-Fh	
P2												



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Table 20: Feature Address 02h: NV-DDR2 configuration (Continued)

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
Warmup RE_t/ RE_c and DQS cycles for data output	0 cycles (default)					0	0	0	0	0h	3
	1 warmup cycle					0	0	0	1	1h	
	2 warmup cycles					0	0	1	0	2h	
	4 warmup cycles					0	0	1	1	3h	
	4 warmup cycles					0	1	0	0	4h	
	Reserved					0	1	0	1	5h	
						0	1	1	0	6h	
						0	1	1	1	7h	
						1	X	X	X	8h-Fh	
Warmup DQS cycles for data input	0 cycles (default)	0	0	0	0					0h	3
	1 warmup cycle	0	0	0	1					1h	
	2 warmup cycles	0	0	1	0					2h	
	4 warmup cycles	0	0	1	1					3h	
	4 warmup cycles	0	1	0	0					4h	
	Reserved	0	1	0	1					5h	
		0	1	1	0					6h	
		0	1	1	1					7h	
		1	X	X	X					8h-Fh	
P3											
Reserved		0	0	0	0	0	0	0	0	00h	
P4											
Reserved		0	0	0	0	0	0	0	0	00h	

- Notes:
1. R_{TT} settings may be specified separately for DQ[7:0], DQS, RE_t, and RE_c signals. The DQ[7:0], DQS signals may be specified separate for data input versus data output operation.
 2. If the NV-DDR2 interface is active, a RESET (FFh) command will change the bit values feature address 02h to their default values.
 3. Number of warmup cycles for DQS during data input and RE_t/RE_c and DQS_t/DQS_c for data output. The number of initial "dummy" RE_t/RE_c cycles at the start of data output operations. There are corresponding "dummy" DQS_t/DQS_c cycles to the "dummy" RE_t/RE_c cycles that the host shall ignore during data output.

Table 21: Feature Address 30h: V_{pp}

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
P1											
External V _{pp} configuration	Disabled (default)								0	00h	1
	Enabled								1	01h	



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Table 21: Feature Address 30h: V_{pp} (Continued)

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
Reserved		0	0	0	0	0	0	0		00h	
P2											
Reserved		0	0	0	0	0	0	0	0	00h	
P3											
Reserved		0	0	0	0	0	0	0	0	00h	
P4											
Reserved		0	0	0	0	0	0	0	0	00h	

Note: 1. This setting controls whether external V_{pp} is enabled. This setting is retained across RE-SET (FAh, FCh, FFh) commands.

Table 22: Feature Address 58h: Volume configuration

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
P1											
Volume Address	Field used to assign a value for a given Volume Address					X	X	X	X		1
Reserved	-	0	0	0	0					0h	
P2											
Reserved		0	0	0	0	0	0	0	0	00h	
P3											
Reserved		0	0	0	0	0	0	0	0	00h	
P4											
Reserved		0	0	0	0	0	0	0	0	00h	

Note: 1. After the Volume Address is appointed, the EN₀ pin for that Volume is set to one, the EN_i pin for that Volume is ignored until the next power cycle, and the Volume is deselected until a VOLUME SELECT (E1h) command is issued that selects the associated Volume. The host shall only set this feature once per power cycle for each Volume. The address specified is then used in VOLUME SELECT (E1h) command for accessing this NAND Target. This setting is retained across RESET (FAh, FCh, FFh) commands.

Table 23: Feature Addresses 10h and 80h: Programmable Output Drive Strength

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
P1											



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Table 23: Feature Addresses 10h and 80h: Programmable Output Drive Strength (Continued)

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
Output drive strength	18 Ohms							0	0	00h	1
	25 Ohms							0	1	01h	
	35 Ohms (default)							1	0	02h	
	50 Ohms							1	1	03h	
Reserved		0	0	0	0	0	0			00h	
P2											
Reserved		0	0	0	0	0	0	0	0	00h	
P3											
Reserved		0	0	0	0	0	0	0	0	00h	
P4											
Reserved		0	0	0	0	0	0	0	0	00h	

Note: 1. See Output Drive Impedance for details.

Table 24: Feature Address 81h: Programmable R/B# Pull-Down Strength

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
P1											
R/B# pull-down strength	Full (default)							0	0	00h	1
	Three-quarter							0	1	01h	
	One-half							1	0	02h	
	One-quarter							1	1	03h	
Reserved		0	0	0	0	0	0			00h	
P2											
Reserved		0	0	0	0	0	0	0	0	00h	
P3											
Reserved		0	0	0	0	0	0	0	0	00h	
P4											
Reserved		0	0	0	0	0	0	0	0	00h	

Note: 1. This feature address is used to change the default R/B# pull-down strength. Its strength should be selected based on the expected loading of R/B#. Full strength is the default, power-on value.

Table 25: Feature Addresses 89h: Read Retry

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
P1											



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Table 25: Feature Addresses 89h: Read Retry (Continued)

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
Read Retry	Disable (default)					0	0	0	0	00h	
	Option 1					0	0	0	1	01h	
	Option 2					0	0	1	0	02h	
	Option 3					0	0	1	1	03h	
	Option 4					0	1	0	0	04h	2
	Option 5					0	1	0	1	05h	
	Option 6					0	1	1	0	06h	
	Option 7					0	1	1	1	07h	
	Option 8					1	0	0	0	08h	
	Reserved					1	0	0	1	09h	
						1	0	1	0	0Ah	
						1	0	1	1	0Bh	
	Option 9					1	1	0	0	0Ch	2
	Reserved					1	1	0	1	0Dh	
						1	1	1	0	0Eh	
						1	1	1	1	0Fh	
Reserved		0	0	0	0					00h	
P2											
Reserved		0	0	0	0	0	0	0	0	00h	
P3											
Reserved		0	0	0	0	0	0	0	0	00h	
P4											
Reserved		0	0	0	0	0	0	0	0	00h	

Notes: 1. See Read Retry Operation for details.

2. ^tR will be longer with selected option. See Electrical Specifications - Array Characteristics section for details. Cache Read functions are not supported with these selected options.

Table 26: Feature Address 90h: Array Operation Mode

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
P1											
Array Operation Mode	Normal (default)								0	00h	2
	OTP Block								1	01h	
Reserved		0	0	0	0	0	0	0		00h	
P2											
Reserved		0	0	0	0	0	0	0	0	00h	



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Table 26: Feature Address 90h: Array Operation Mode (Continued)

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
P3											
Reserved		0	0	0	0	0	0	0	0	00h	
P4											
Reserved		0	0	0	0	0	0	0	0	00h	

- Notes:
1. See One-Time Programmable (OTP) Operations for details.
 2. A RESET (FFh) command will cause the bits of the array operation mode to change to their default values. If in NV-DDR/NV-DDR2 mode, a SYNCHRONOUS RESET (FCh) command will cause the bits of the array operation mode to change to their default values.



VOLUME SELECT (E1h)

The Volume Select function is used to select a particular Volume based on the address specified. VOLUME SELECT (E1h) command is required to be used when CE# pin reduction is used or when matrix termination is used.

This command is accepted by all Targets that share a particular CE# pin. The command may be executed with any LUN on the Volume in any state. The VOLUME SELECT (E1h) command may only be issued as the first command after CE# is pulled LOW; CE# shall have remained HIGH for t_{CEH} in order for the VOLUME SELECT (E1h) command to be properly received by all NAND Targets connected to the Host Target.

If Volumes that share a Host Target are configured to use different data interfaces, then the host shall issue the VOLUME SELECT (E1h) command using the Asynchronous data interface.

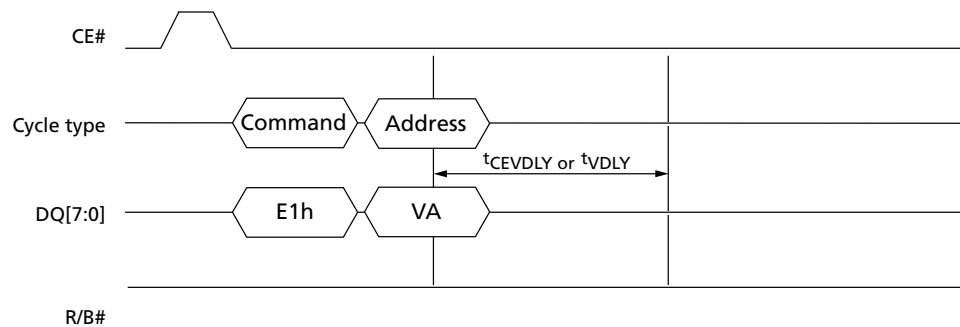
When the VOLUME SELECT (E1h) command is issued, all NAND Targets that have a Volume address that does not match the address specified shall be deselected to save power (equivalent behavior to CE# pulled HIGH). If one of the LUNs in an unselected Volume is the assigned terminator for the Volume addressed, then that LUN will enter the sniff state.

If the Volume address specified does not correspond to any appointed Volume address, then all NAND Targets shall be deselected until a subsequent VOLUME SELECT (E1h) command is issued. If the VOLUME SELECT (E1h) command is not the first command issued after CE# is pulled LOW, then the NAND Targets revert to their previous selected, deselected, or sniff states.

The host shall not issue new commands to any LUN on any Volume until after t_{VDLY} . This delay is required to ensure the appropriate Volume is selected for the next command issued. Also the host shall not bring CE# HIGH on any Volume until after t_{CEVDLY} . This delay is required to ensure the appropriate Volume is selected based on the previously issued VOLUME SELECT (E1h) command.

The Volume address is retained across all RESET (FAh, FCh, FFh) commands.

Figure 61: VOLUME SELECT (E1h) Operation





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Table 27: Volume Address

	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
Volume Address											
VA						Volume Address				-	
Reserved		0	0	0	0					-	

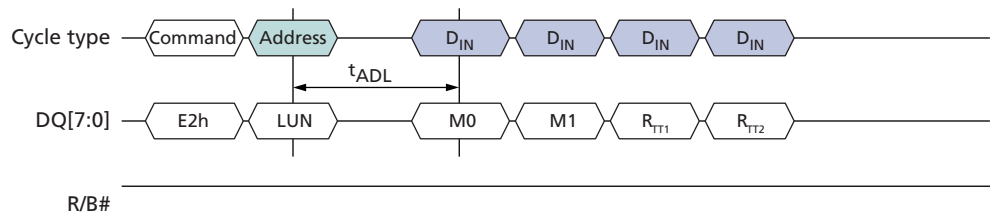


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ODT CONFIGURE (E2h)

The ODT CONFIGURE (E2h) command is used to configure on-die termination. Specifically, ODT CONFIGURE (E2h) specifies whether a particular LUN is a terminator for a Volume(s) and the R_{TT} settings. If the LUN is specified as a terminator for one or more Volumes, then the LUN shall enable on-die termination when either data input or data output cycles are executed on the Volume(s) it is acting as a terminator for depending on the settings of Table 28. The on-die termination settings are retained across all RE-SET (FAh, FCh, FFh) commands.

Figure 62: ODT CONFIGURE (E2h) Operation



The LUN address correspond to the same structure as the last address cycle for the NAND device which determines which LUN will act as the terminator.

The ODT Configuration Matrix structure is defined in Table 28. For the Volume Address fields M0 and M1, if a bit is set to one then the LUN shall act as the terminator for the corresponding Volume(s) (V_n) where n corresponds to the Volume address.

The ODT CONFIGURE (E2h) command is only available while in the NV-DDR2 interface operation mode and within the $1.8V_{CCQ}$ operational range.

Table 28: ODT Configuration Matrix

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
M0											
Volume Address	Volumes that will be terminated by selected LUN	V7	V6	V5	V4	V3	V2	V1	V0	-	
M1											
Volume Address	Volumes that will be terminated by selected LUN	V15	V14	V13	V12	V11	V10	V9	V8	-	
R_{TT0}											



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Table 28: ODT Configuration Matrix (Continued)

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
DQ[7:0]/DQS_t/ DQS_c R _{TT} and ODT enable for data input	ODT disabled (default)					0	0	0	0	0h	
	ODT enabled with R _{TT} of 150 Ohms					0	0	0	1	1h	
	ODT enabled with R _{TT} of 100 Ohms					0	0	1	0	2h	
	ODT enabled with R _{TT} of 75 Ohms					0	0	1	1	3h	
	ODT enabled with R _{TT} of 50 Ohms					0	1	0	0	4h	
	Reserved					0	1	0	1	5h	
						0	1	1	0	6h	
						0	1	1	1	7h	
						1	X	X	X	8h-Fh	
DQ[7:0]/DQS_t/ DQS_c R _{TT} and ODT enable for data output	ODT disabled (default)	0	0	0	0					0h	
	ODT enabled with R _{TT} of 150 Ohms	0	0	0	1					1h	
	ODT enabled with R _{TT} of 100 Ohms	0	0	1	0					2h	
	ODT enabled with R _{TT} of 75 Ohms	0	0	1	1					3h	
	ODT enabled with R _{TT} of 50 Ohms	0	1	0	0					4h	
	Reserved	0	1	0	1					5h	
		0	1	1	0					6h	
		0	1	1	1					7h	
		1	X	X	X					8h-Fh	
R _{TT1}											



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Table 28: ODT Configuration Matrix (Continued)

Subfeature Parameter	Options	DQ7	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0	Value	Notes
RE_t and RE_c R _{TT} ODT enable	ODT disabled (default)					0	0	0	0	0h	
	ODT enabled with R _{TT} of 150 Ohms					0	0	0	1	1h	
	ODT enabled with R _{TT} of 100 Ohms					0	0	1	0	2h	
	ODT enabled with R _{TT} of 75 Ohms					0	0	1	1	3h	
	ODT enabled with R _{TT} of 50 Ohms					0	1	0	0	4h	
Reserved	-					0	1	0	1	5h	
						0	1	1	0	6h	
						0	1	1	1	7h	
						1	X	X	X	8h-Fh	
Reserved	-	0	0	0	0					0h	



Status Operations

Each die (LUN) provides its status independently of other die (LUNs) on the same target through its 8-bit status register.

After the READ STATUS (70h) or READ STATUS ENHANCED (78h) command is issued, status register output is enabled. The contents of the status register are returned on DQ[7:0] for each data output request.

When the asynchronous or NV-DDR2 interfaces are active and status register output is enabled, changes in the status register are seen on DQ[7:0] as long as CE# and RE# are LOW; it is not necessary to toggle RE# to see the status register update.

When the NV-DDR interface is active and status register output is enabled, changes in the status register are seen on DQ[7:0] as long as CE# and W/R# are LOW and ALE and CLE are HIGH. DQS also toggles (with ^tDQSK delay) while ALE and CLE are captured HIGH. If status register output is enabled and CE# and W/R# are LOW and ALE and CLE are also captured LOW, changes in the status register are still seen asynchronously on DQ[7:0] but DQS does not toggle.

While monitoring the status register to determine when a data transfer from the Flash array to the data register (^tR) is complete, the host must issue the READ MODE (00h) command to disable the status register and enable data output (see READ MODE (00h) (page 124)).

The READ STATUS (70h) command returns the status of the most recently selected die (LUN). To prevent data contention during or following an interleaved die (multi-LUN) operation, the host must enable only one die (LUN) for status output by using the READ STATUS ENHANCED (78h) command (see Interleaved Die (Multi-LUN) Operations (page 150)).

Table 29: Status Register Definition

SR Bit	Definition	Independent per Plane ¹	Description
7	WP#	–	Write Protect: 0 = Protected 1 = Not protected In the normal array mode, this bit indicates the value of the WP# signal. In OTP mode this bit is set to 0 if a PROGRAM OTP PAGE operation is attempted and the OTP area is protected.
6	RDY	–	Ready/Busy I/O: 0 = Busy 1 = Ready This bit indicates that the selected die (LUN) is not available to accept new commands, address, or data I/O cycles with the exception of RESET (FFh), SYNCHRONOUS RESET (FCh), READ STATUS (70h), and READ STATUS ENHANCED (78h). This bit applies only to the selected die (LUN).


Table 29: Status Register Definition (Continued)

SR Bit	Definition	Independent per Plane ¹	Description
5	ARDY	–	Ready/Busy Array: 0 = Busy 1 = Ready This bit goes LOW (busy) when an array operation is occurring on any plane of the selected die (LUN). It goes HIGH when all array operations on the selected die (LUN) finish. This bit applies only to the selected die (LUN).
4	–	–	Reserved (0)
3	–	–	Reserved (0)
2	–	–	Reserved (0)
1	FAILC	Yes	Pass/Fail (N–1): 0 = Pass 1 = Fail This bit is set if the previous operation on the selected die (LUN) failed. This bit is valid only when RDY (SR bit 6) is 1. It applies to PROGRAM-, and COPYBACK PROGRAM-series operations (80h-10h, 80h-15h, 85h-10h). This bit is not valid following an ERASE-series or READ-series operation.
0	FAIL	Yes	Pass/Fail (N): 0 = Pass 1 = Fail This bit is set if the most recently finished operation on the selected die (LUN) failed. This bit is valid only when ARDY (SR bit 5) is 1. It applies to PROGRAM-, ERASE-, and COPYBACK PROGRAM-series operations (80h-10h, 80h-15h, 60h-D0h, 85h-10h). This bit is not valid following a READ-series operation.

Note: 1. After a multi-plane operation begins, the FAILC and FAIL bits are ORed together for the active planes when the READ STATUS (70h) command is issued. After the READ STATUS ENHANCED (78h) command is issued, the FAILC and FAIL bits reflect the status of the plane selected.

READ STATUS (70h)

The READ STATUS (70h) command returns the status of the last-selected die (LUN) on a target. This command is accepted by the last-selected die (LUN) even when it is busy (RDY = 0).

If there is only one die (LUN) per target, the READ STATUS (70h) command can be used to return status following any NAND command.

In devices that have more than one die (LUN) per target, during and following interleaved die (multi-LUN) operations, the READ STATUS ENHANCED (78h) command must be used to select the die (LUN) that should report status. In this situation, using the READ STATUS (70h) command will result in bus contention, as two or more die (LUNs) could respond until the next operation is issued. The READ STATUS (70h) command can be used following all single die (LUN) operations.

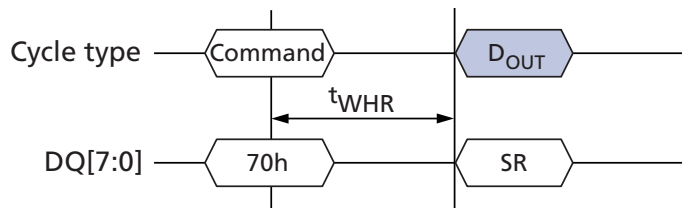
If following a multi-plane operation, regardless of the number of LUNs per target, the READ STATUS (70h) command indicates an error occurred (FAIL = 1), use the READ



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STATUS ENHANCED (78h) command—once for each plane—to determine which plane operation failed.

Figure 63: READ STATUS (70h) Operation



READ STATUS ENHANCED (78h)

The READ STATUS ENHANCED (78h) command returns the status of the addressed die (LUN) on a target even when it is busy (RDY = 0). This command is accepted by all die (LUNs), even when they are BUSY (RDY = 0).

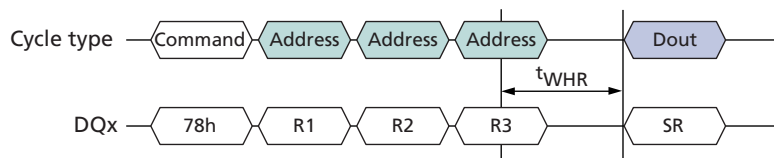
Writing 78h to the command register, followed by three row address cycles containing the page, block, and LUN addresses, puts the selected die (LUN) into read status mode. The selected die (LUN) stays in this mode until another valid command is issued. Die (LUNs) that are not addressed are deselected to avoid bus contention.

The selected LUN's status is returned when the host requests data output. The RDY and ARDY bits of the status register are shared for all of the planes of the selected die (LUN). The FAILC and FAIL bits are specific to the plane specified in the row address.

The READ STATUS ENHANCED (78h) command also enables the selected die (LUN) for data output. To begin data output following a READ-series operation after the selected die (LUN) is ready (RDY = 1), issue the READ MODE (00h) command, then begin data output. If the host needs to change the cache register that will output data, use the CHANGE READ COLUMN ENHANCED (06h-E0h) command after the die (LUN) is ready (see CHANGE READ COLUMN ENHANCED (06h-E0h)).

Use of the READ STATUS ENHANCED (78h) command is prohibited during the power-on RESET (FFh) command and when OTP mode is enabled. It is also prohibited following some of the other reset, identification, and configuration operations. See individual operations for specific details.

Figure 64: READ STATUS ENHANCED (78h) Operation



Column Address Operations

The column address operations affect how data is input to and output from the cache registers within the selected die (LUNs). These features provide host flexibility for managing data, especially when the host internal buffer is smaller than the number of data bytes or words in the cache register.



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When the asynchronous interface is active, column address operations can address any byte in the selected cache register.

When the synchronous interface is active, column address operations are aligned to word boundaries (CA0 is forced to 0), because as data is transferred on DQ[7:0] in two-byte units.

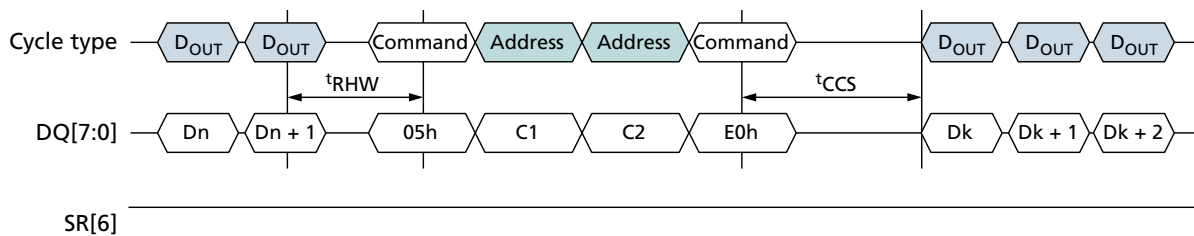
CHANGE READ COLUMN (05h-E0h)

The CHANGE READ COLUMN (05h-E0h) command changes the column address of the selected cache register and enables data output from the last selected die (LUN). This command is accepted by the selected die (LUN) when it is ready (RDY = 1; ARDY = 1). It is also accepted by the selected die (LUN) during CACHE READ operations (RDY = 1; ARDY = 0).

Writing 05h to the command register, followed by two column address cycles containing the column address, followed by the E0h command, puts the selected die (LUN) into data output mode. After the E0h command cycle is issued, the host must wait at least t_{CCS} before requesting data output. The selected die (LUN) stays in data output mode until another valid command is issued.

In devices with more than one die (LUN) per target, during and following interleaved die (multi-LUN) operations, the READ STATUS ENHANCED (78h) command must be issued prior to issuing the CHANGE READ COLUMN (05h-E0h). In this situation, using the CHANGE READ COLUMN (05h-E0h) command without the READ STATUS ENHANCED (78h) command will result in bus contention, as two or more die (LUNs) could output data.

Figure 65: CHANGE READ COLUMN (05h-E0h) Operation



CHANGE READ COLUMN ENHANCED (06h-E0h)

The CHANGE READ COLUMN ENHANCED (06h-E0h) command enables data output on the addressed die's (LUN's) cache register at the specified column address. This command is accepted by a die (LUN) when it is ready (RDY = 1; ARDY = 1).

Writing 06h to the command register, followed by two column address cycles and three row address cycles, followed by E0h, enables data output mode on the address LUN's cache register at the specified column address. Only the column, plane and LUN addresses are valid; the page and block addresses are ignored. After the E0h command cycle is issued, the host must wait at least t_{CCS} before requesting data output. The selected die (LUN) stays in data output mode until another valid command is issued.

Following a MULTI-PLANE READ PAGE operation, the CHANGE READ COLUMN ENHANCED (06h-E0h) command is used to select the cache register to be enabled for data



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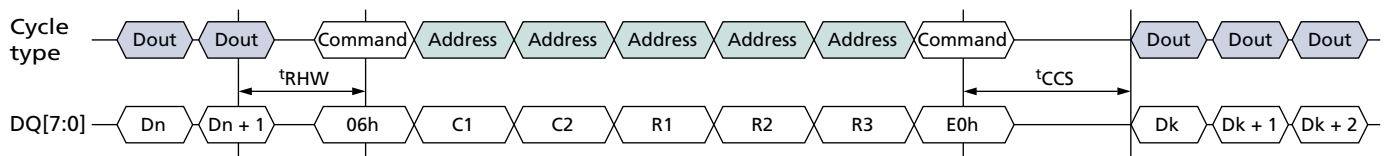
output. After data output is complete on the selected plane, the command can be issued again to begin data output on another plane.

In devices with more than one die (LUN) per target, after all of the die (LUNs) on the target are ready (RDY = 1), the CHANGE READ COLUMN ENHANCED (06h-E0h) command can be used following an interleaved die (multi-LUN) READ operation. Die (LUNs) that are not addressed are deselected to avoid bus contention.

In devices with more than one die (LUN) per target, during interleaved die (multi-LUN) operations where more than one or more die (LUNs) are busy (RDY = 1; ARDY = 0 or RDY = 0; ARDY = 0), the READ STATUS ENHANCED (78h) command must be issued to the die (LUN) to be selected prior to issuing the CHANGE READ COLUMN ENHANCED (06h-E0h). In this situation, using the CHANGE READ COLUMN ENHANCED (06h-E0h) command without the READ STATUS ENHANCED (78h) command will result in bus contention, as two or more die (LUNs) could output data.

If there is a need to update the column address without selecting a new cache register or LUN, the CHANGE READ COLUMN (05h-E0h) command can be used instead.

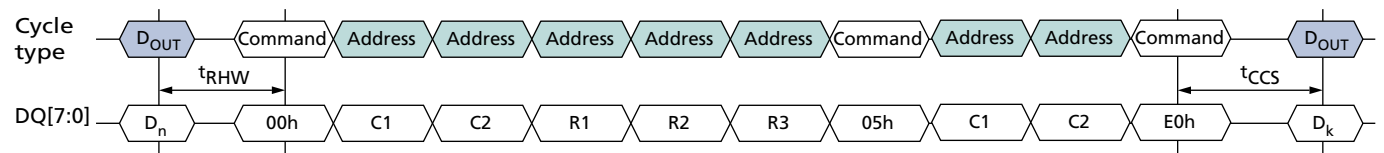
Figure 66: CHANGE READ COLUMN ENHANCED (06h-E0h) Operation



CHANGE READ COLUMN ENHANCED (00h-05h-E0h) Operation

This operation behaves the same as the CHANGE READ COLUMN ENHANCED (06h-E0h) command.

Figure 67: CHANGE READ COLUMN ENHANCED (00h-05h-E0h) Operation



CHANGE WRITE COLUMN (85h)

The CHANGE WRITE COLUMN (85h) command changes the column address of the selected cache register and enables data input on the last-selected die (LUN). This command is accepted by the selected die (LUN) when it is ready (RDY = 1; ARDY = 1). It is also accepted by the selected die (LUN) during CACHE PROGRAM operations (RDY = 1; ARDY = 0).

Writing 85h to the command register, followed by two column address cycles containing the column address, puts the selected die (LUN) into data input mode. After the second address cycle is issued, the host must wait at least t_{CCS} before inputting data. The selected die (LUN) stays in data input mode until another valid command is issued. Though



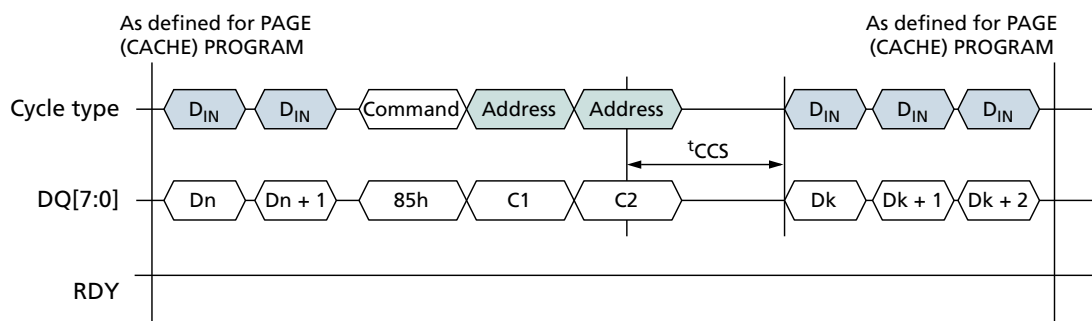
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data input mode is enabled, data input from the host is optional. Data input begins at the column address specified.

The CHANGE WRITE COLUMN (85h) command is allowed after the required address cycles are specified, but prior to the final command cycle (10h, 11h, 15h) of the following commands while data input is permitted: PROGRAM PAGE (80h-10h), PROGRAM PAGE MULTI-PLANE (80h-11h), PROGRAM PAGE CACHE (80h-15h), COPYBACK PROGRAM (85h-10h), and COPYBACK PROGRAM MULTI-PLANE (85h-11h).

In devices that have more than one die (LUN) per target, the CHANGE WRITE COLUMN (85h) command can be used with other commands that support interleaved die (multi-LUN) operations.

Figure 68: CHANGE WRITE COLUMN (85h) Operation



CHANGE ROW ADDRESS (85h)

The CHANGE ROW ADDRESS (85h) command changes the row address (block and page) where the cache register contents will be programmed in the NAND Flash array. It also changes the column address of the selected cache register and enables data input on the specified die (LUN). This command is accepted by the selected die (LUN) when it is ready (RDY = 1; ARDY = 1). It is also accepted by the selected die (LUN) during CACHE PROGRAMMING operations (RDY = 1; ARDY = 0).

Write 85h to the command register. Then write two column address cycles and three row address cycles. This updates the page and block destination of the selected plane for the addressed LUN and puts the cache register into data input mode. After the fifth address cycle is issued the host must wait at least t_{CCS} before inputting data. The selected LUN stays in data input mode until another valid command is issued. Though data input mode is enabled, data input from the host is optional. Data input begins at the column address specified.

The CHANGE ROW ADDRESS (85h) command is allowed after the required address cycles are specified, but prior to the final command cycle (10h, 11h, 15h) of the following commands while data input is permitted: PROGRAM PAGE (80h-10h), PROGRAM PAGE MULTI-PLANE (80h-11h), PROGRAM PAGE CACHE (80h-15h), COPYBACK PROGRAM (85h-10h), and COPYBACK PROGRAM MULTI-PLANE (85h-11h). When used with these commands, the LUN address and plane select bits are required to be identical to the LUN address and plane select bits originally specified.



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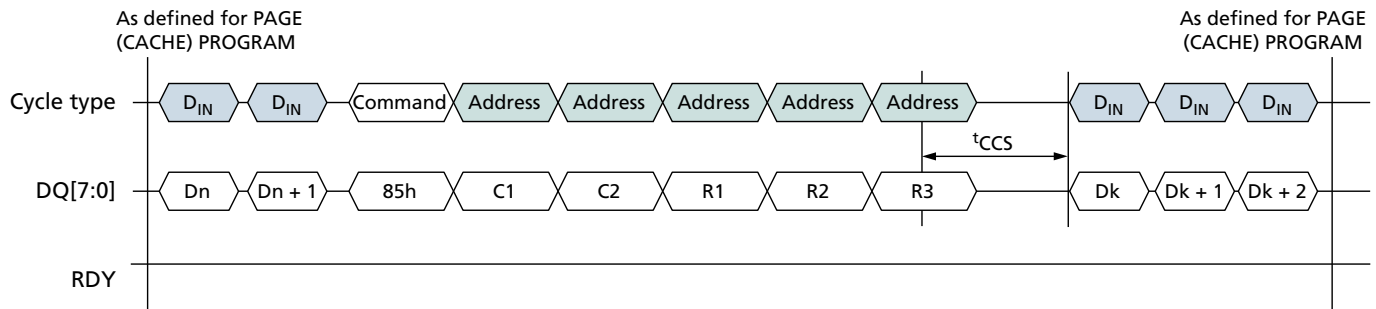
The CHANGE ROW ADDRESS (85h) command enables the host to modify the original page and block address for the data in the cache register to a new page and block address.

In devices that have more than one die (LUN) per target, the CHANGE ROW ADDRESS (85h) command can be used with other commands that support interleaved die (multi-LUN) operations.

The CHANGE ROW ADDRESS (85h) command can be used with the CHANGE READ COLUMN (05h-E0h) or CHANGE READ COLUMN ENHANCED (06h-E0h) commands to read and modify cache register contents in small sections prior to programming cache register contents to the NAND Flash array. This capability can reduce the amount of buffer memory used in the host controller.

To modify the cache register contents in small sections, first issue a PAGE READ (00h-30h) or COPYBACK READ (00h-35h) operation. When data output is enabled, the host can output a portion of the cache register contents. To modify the cache register contents, issue the 85h command, the column and row addresses, and input the new data. The host can re-enable data output by issuing the 11h command, waiting t_{DBSY} , and then issuing the CHANGE READ COLUMN (05h-E0h) or CHANGE READ COLUMN ENHANCED (06h-E0h) command. It is possible to toggle between data output and data input multiple times. After the final CHANGE ROW ADDRESS (85h) operation is complete, issue the 10h command to program the cache register to the NAND Flash array.

Figure 69: CHANGE ROW ADDRESS (85h) Operation





Read Operations

Read operations are used to copy data from the NAND Flash array of one or more of the planes to their respective cache registers and to enable data output from the cache registers to the host through the DQ bus.

Read Operations

The READ PAGE (00h-30h) command, when issued by itself, reads one page from the NAND Flash array to its cache register and enables data output for that cache register.

During data output the following commands can be used to read and modify the data in the cache registers: CHANGE READ COLUMN (05h-E0h) and CHANGE ROW ADDRESS (85h).

Read Cache Operations

To increase data throughput, the READ PAGE CACHE-series (31h, 00h-31h) commands can be used to output data from the cache register while concurrently copying a page from the NAND Flash array to the data register.

To begin a read page cache sequence, begin by reading a page from the NAND Flash array to its corresponding cache register using the READ PAGE (00h-30h) command. R/B# goes LOW during ^tR and the selected die (LUN) is busy (RDY = 0, ARDY = 0). After ^tR (R/B# is HIGH and RDY = 1, ARDY = 1), issue either of these commands:

- READ PAGE CACHE SEQUENTIAL (31h)—copies the next sequential page from the NAND Flash array to the data register
- READ PAGE CACHE RANDOM (00h-31h)—copies the page specified in this command from the NAND Flash array (any plane) to its corresponding data register

After the READ PAGE CACHE-series (31h, 00h-31h) command has been issued, R/B# goes LOW on the target, and RDY = 0 and ARDY = 0 on the die (LUN) for ^tRCBSY while the next page begins copying data from the array to the data register. After ^tRCBSY, R/B# goes HIGH and the die's (LUN's) status register bits indicate the device is busy with a cache operation (RDY = 1, ARDY = 0). The cache register becomes available and the page requested in the READ PAGE CACHE operation is transferred to the data register. At this point, data can be output from the cache register, beginning at column address 0. The CHANGE READ COLUMN (05h-E0h) command can be used to change the column address of the data output by the die (LUN).

After outputting the desired number of bytes from the cache register, either an additional READ PAGE CACHE-series (31h, 00h-31h) operation can be started or the READ PAGE CACHE LAST (3Fh) command can be issued.

If the READ PAGE CACHE LAST (3Fh) command is issued, R/B# goes LOW on the target, and RDY = 0 and ARDY = 0 on the die (LUN) for ^tRCBSY while the data register is copied into the cache register. After ^tRCBSY, R/B# goes HIGH and RDY = 1 and ARDY = 1, indicating that the cache register is available and that the die (LUN) is ready. Data can then be output from the cache register, beginning at column address 0. The CHANGE READ COLUMN (05h-E0h) command can be used to change the column address of the data being output.

For READ PAGE CACHE-series (31h, 00h-31h, 3Fh), during the die (LUN) busy time, ^tRCBSY, when RDY = 0 and ARDY = 0, the only valid commands are status operations (70h, 78h) and RESET (FFh, FCh). When RDY = 1 and ARDY = 0, the only valid commands during READ PAGE CACHE-series (31h, 00h-31h) operations are status opera-



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tions (70h, 78h), READ MODE (00h), READ PAGE CACHE-series (31h, 00h-31h), CHANGE READ COLUMN (05h-E0h), and RESET (FFh, FCh).

Multi-Plane Read Operations

Multi-plane read page operations improve data throughput by copying data from more than one plane simultaneously to the specified cache registers. This is done by prepending one or more READ PAGE MULTI-PLANE (00h-32h) commands in front of the READ PAGE (00h-30h) command.

When the die (LUN) is ready, the CHANGE READ COLUMN ENHANCED (06h-E0h) command determines which plane outputs data. During data output, the following commands can be used to read and modify the data in the cache registers: CHANGE READ COLUMN (05h-E0h) and CHANGE ROW ADDRESS (85h). See Multi-Plane Operations for details.

Multi-Plane Read Cache Operations

Multi-plane read cache operations can be used to output data from more than one cache register while concurrently copying one or more pages from the NAND Flash array to the data register. This is done by prepending READ PAGE MULTI-PLANE (00h-32h) commands in front of the PAGE READ CACHE RANDOM (00h-31h) command.

To begin a multi-plane read page cache sequence, begin by issuing a MULTI-PLANE READ PAGE operation using the READ PAGE MULTI-PLANE (00h-32h) and READ PAGE (00h-30h) commands. R/B# goes LOW during 'R and the selected die (LUN) is busy (RDY = 0, ARDY = 0). After 'R (R/B# is HIGH and RDY = 1, ARDY = 1), issue either of these commands:

- READ PAGE CACHE SEQUENTIAL (31h)—copies the next sequential page from the previously addressed planes from the NAND Flash array to the data registers.
- READ PAGE MULTI-PLANE (00h-32h) commands, if desired, followed by the READ PAGE CACHE RANDOM (00h-31h) command—copies the pages specified from the NAND Flash array to the corresponding data registers.

After the READ PAGE CACHE-series (31h, 00h-31h) command has been issued, R/B# goes LOW on the target, and RDY = 0 and ARDY = 0 on the die (LUN) for 'RCBSY while the next pages begin copying data from the array to the data registers. After 'RCBSY, R/B# goes HIGH and the LUN's status register bits indicate the device is busy with a cache operation (RDY = 1, ARDY = 0). The cache registers become available and the pages requested in the READ PAGE CACHE operation are transferred to the data registers. Issue the CHANGE READ COLUMN ENHANCED (06h-E0h) command to determine which cache register will output data. After data is output, the CHANGE READ COLUMN ENHANCED (06h-E0h) command can be used to output data from other cache registers. After a cache register has been selected, the CHANGE READ COLUMN (05h-E0h) command can be used to change the column address of the data output.

After outputting data from the cache registers, either an additional MULTI-PLANE READ CACHE-series (31h, 00h-31h) operation can be started or the READ PAGE CACHE LAST (3Fh) command can be issued.

If the READ PAGE CACHE LAST (3Fh) command is issued, R/B# goes LOW on the target, and RDY = 0 and ARDY = 0 on the die (LUN) for 'RCBSY while the data registers are copied into the cache registers. After 'RCBSY, R/B# goes HIGH and RDY = 1 and ARDY = 1, indicating that the cache registers are available and that the die (LUN) is ready. Issue the



CHANGE READ COLUMN ENHANCED (06h-E0h) command to determine which cache register will output data. After data is output, the CHANGE READ COLUMN ENHANCED (06h-E0h) command can be used to output data from other cache registers. After a cache register has been selected, the CHANGE READ COLUMN (05h-E0h) command can be used to change the column address of the data output.

For READ PAGE CACHE-series (31h, 00h-31h, 3Fh), during the die (LUN) busy time, t_{RCBSY} , when $RDY = 0$ and $ARDY = 0$, the only valid commands are status operations (70h, 78h) and RESET (FFh, FCh). When $RDY = 1$ and $ARDY = 0$, the only valid commands during READ PAGE CACHE-series (31h, 00h-31h) operations are status operations (70h, 78h), READ MODE (00h), multi-plane read cache-series (31h, 00h-32h, 00h-31h), CHANGE READ COLUMN (05h-E0h, 06h-E0h), and RESET (FFh, FCh).

See Multi-Plane Operations for additional multi-plane addressing requirements.

READ MODE (00h)

The READ MODE (00h) command disables status output and enables data output for the last-selected die (LUN) and cache register after a READ operation (00h-30h, 00h-35h) has been monitored with a status operation (70h, 78h). This command is accepted by the die (LUN) when it is ready ($RDY = 1$, $ARDY = 1$). It is also accepted by the die (LUN) during READ PAGE CACHE (31h, 3Fh, 00h-31h) operations ($RDY = 1$ and $ARDY = 0$).

In devices that have more than one die (LUN) per target, during and following interleaved die (multi-LUN) operations, the READ STATUS ENHANCED (78h) command must be used to select only one die (LUN) prior to issuing the READ MODE (00h) command. This prevents bus contention.



READ PAGE (00h-30h)

The READ PAGE (00h-30h) command copies a page from the NAND Flash array to its respective cache register and enables data output. This command is accepted by the die (LUN) when it is ready (RDY = 1, ARDY = 1).

To read a page from the NAND Flash array, write the 00h command to the command register, the write five address cycles to the address registers, and conclude with the 30h command. The selected die (LUN) will go busy (RDY = 0, ARDY = 0) for t_R as data is transferred.

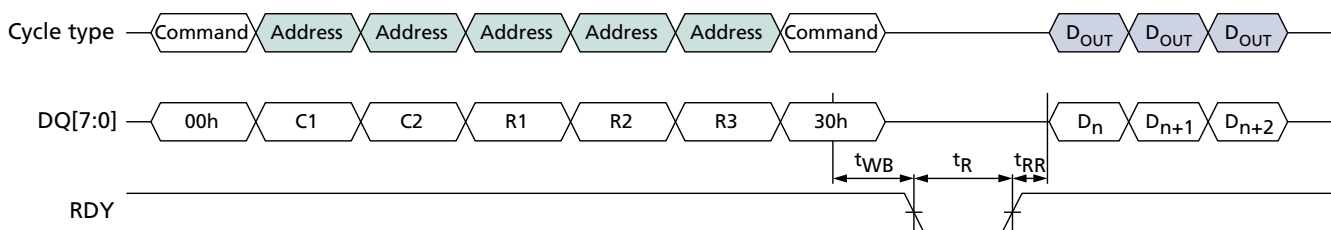
To determine the progress of the data transfer, the host can monitor the target's R/B# signal or, alternatively, the status operations (70h, 78h) can be used. If the status operations are used to monitor the LUN's status, when the die (LUN) is ready (RDY = 1, ARDY = 1), the host disables status output and enables data output by issuing the READ MODE (00h) command. When the host requests data output, output begins at the column address specified.

During data output the CHANGE READ COLUMN (05h-E0h) command can be issued.

In devices that have more than one die (LUN) per target, during and following interleaved die (multi-LUN) operations the READ STATUS ENHANCED (78h) command must be used to select only one die (LUN) prior to the issue of the READ MODE (00h) command. This prevents bus contention.

The READ PAGE (00h-30h) command is used as the final command of a multi-plane read operation. It is preceded by one or more READ PAGE MULTI-PLANE (00h-32h) commands. Data is transferred from the NAND Flash array for all of the addressed planes to their respective cache registers. When the die (LUN) is ready (RDY = 1, ARDY = 1), data output is enabled for the cache register linked to the last even plane addressed. When the host requests data output, output begins at the column address last specified in the READ PAGE (00h-30h) command. The CHANGE READ COLUMN ENHANCED (06h-E0h) command is used to enable data output in the other cache registers. See Multi-Plane Operations for additional multi-plane addressing requirements.

Figure 70: READ PAGE (00h-30h) Operation





READ PAGE CACHE SEQUENTIAL (31h)

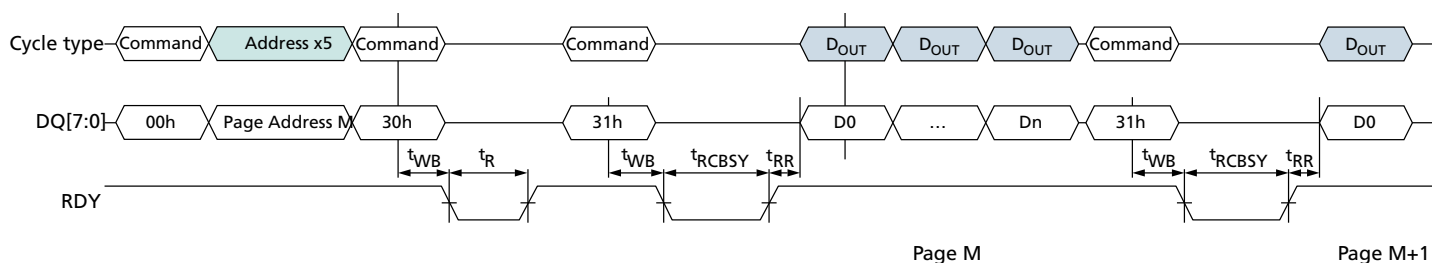
The READ PAGE CACHE SEQUENTIAL (31h) command reads the next sequential page within a block into the data register while the previous page is output from the cache register. This command is accepted by the die (LUN) when it is ready (RDY = 1, ARDY = 1). It is also accepted by the die (LUN) during READ PAGE CACHE (31h, 00h-31h) operations (RDY = 1 and ARDY = 0).

To issue this command, write 31h to the command register. After this command is issued, R/B# goes LOW and the die (LUN) is busy (RDY = 0, ARDY = 0) for 'RCBSY. After 'RCBSY, R/B# goes HIGH and the die (LUN) is busy with a cache operation (RDY = 1, ARDY = 0), indicating that the cache register is available and that the specified page is copying from the NAND Flash array to the data register. At this point, data can be output from the cache register beginning at column address 0. The CHANGE READ COLUMN (05h-E0h) command can be used to change the column address of the data being output from the cache register.

The READ PAGE CACHE SEQUENTIAL (31h) command can be used to cross block boundaries. If the READ PAGE CACHE SEQUENTIAL (31h) command is issued after the last page of a block is read into the data register, the next page read will be the next logical block in the plane which the 31h command was issued. Do not issue the READ PAGE CACHE SEQUENTIAL (31h) to cross die (LUN) boundaries. Instead, issue the READ PAGE CACHE LAST (3Fh) command.

If the READ PAGE CACHE SEQUENTIAL (31h) command is issued after a MULTI-PLANE READ PAGE operation (00h-32h, 00h-30h), the next sequential pages are read into the data registers while the previous pages can be output from the cache registers. After the die (LUN) is ready (RDY = 1, ARDY = 0), the CHANGE READ COLUMN ENHANCED (06h-E0h) command is used to select which cache register outputs data.

Figure 71: READ PAGE CACHE SEQUENTIAL (31h) Operation





READ PAGE CACHE RANDOM (00h-31h)

The READ PAGE CACHE RANDOM (00h-31h) command reads the specified block and page into the data register while the previous page is output from the cache register. This command is accepted by the die (LUN) when it is ready (RDY = 1, ARDY = 1). It is also accepted by the die (LUN) during READ PAGE CACHE (31h, 00h-31h) operations (RDY = 1 and ARDY = 0).

To issue this command, write 00h to the command register, then write five address cycles to the address register, and conclude by writing 31h to the command register. The column address in the address specified is ignored. The die (LUN) address must match the same die (LUN) address as the previous READ PAGE (00h-30h) command or, if applicable, the previous READ PAGE CACHE RANDOM (00h-31h) command. There is no restriction on the plane address.

After this command is issued, R/B# goes LOW and the die (LUN) is busy (RDY = 0, ARDY = 0) for 'RCBSY. After 'RCBSY, R/B# goes HIGH and the die (LUN) is busy with a cache operation (RDY = 1, ARDY = 0), indicating that the cache register is available and that the specified page is copying from the NAND Flash array to the data register. At this point, data can be output from the cache register beginning at column address 0. The CHANGE READ COLUMN (05h-E0h) command can be used to change the column address of the data being output from the cache register.

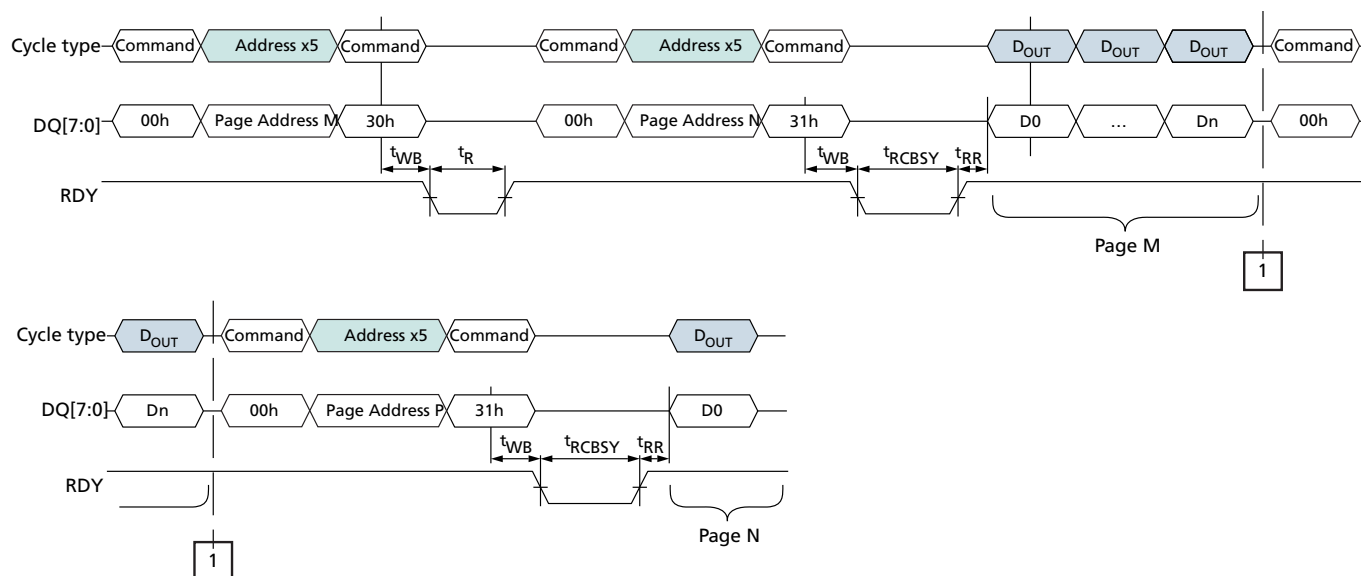
In devices that have more than one die (LUN) per target, during and following interleaved die (multi-LUN) operations the READ STATUS ENHANCED (78h) command followed by the READ MODE (00h) command must be used to select only one die (LUN) and prevent bus contention.

If a MULTI-PLANE CACHE RANDOM (00h-32h, 00h-31h) command is issued after a MULTI-PLANE READ PAGE operation (00h-32h, 00h-30h), then the addressed pages are read into the data registers while the previous pages can be output from the cache registers. After the die (LUN) is ready (RDY = 1, ARDY = 0), the CHANGE READ COLUMN ENHANCED (06h-E0h) command is used to select which cache register outputs data.



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Figure 72: READ PAGE CACHE RANDOM (00h-31h) Operation





READ PAGE CACHE LAST (3Fh)

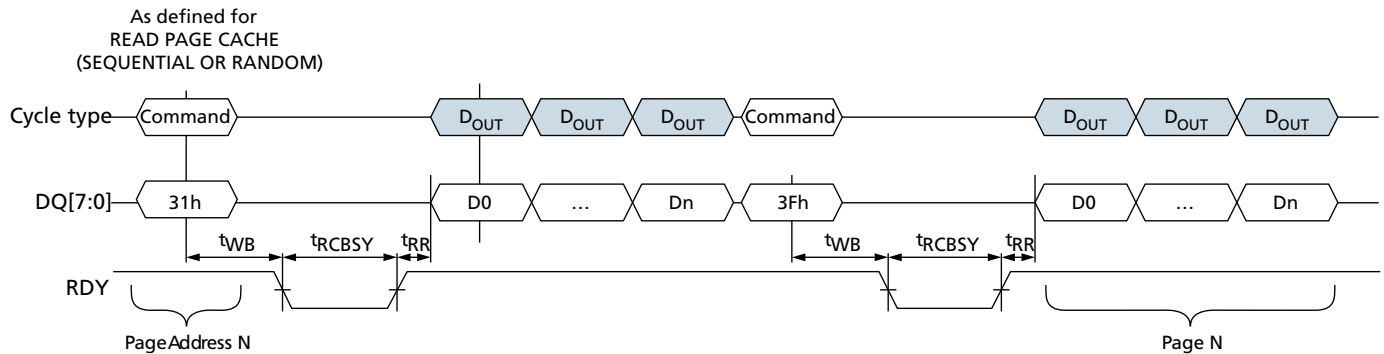
The READ PAGE CACHE LAST (3Fh) command ends the read page cache sequence and copies a page from the data register to the cache register. This command is accepted by the die (LUN) when it is ready (RDY = 1, ARDY = 1). It is also accepted by the die (LUN) during READ PAGE CACHE (31h, 00h-31h) operations (RDY = 1 and ARDY = 0).

To issue the READ PAGE CACHE LAST (3Fh) command, write 3Fh to the command register. After this command is issued, R/B# goes LOW and the die (LUN) is busy (RDY = 0, ARDY = 0) for t_{RCBSY} . After t_{RCBSY} , R/B# goes HIGH and the die (LUN) is ready (RDY = 1, ARDY = 1). At this point, data can be output from the cache register, beginning at column address 0. The CHANGE READ COLUMN (05h-E0h) command can be used to change the column address of the data being output from the cache register.

In devices that have more than one LUN per target, during and following interleaved die (multi-LUN) operations the READ STATUS ENHANCED (78h) command followed by the READ MODE (00h) command must be used to select only one die (LUN) and prevent bus contention.

If the READ PAGE CACHE LAST (3Fh) command is issued after a MULTI-PLANE READ PAGE CACHE operation (31h; 00h-32h, 00h-30h), the die (LUN) goes busy until the pages are copied from the data registers to the cache registers. After the die (LUN) is ready (RDY = 1, ARDY = 1), the CHANGE READ COLUMN ENHANCED (06h-E0h) command is used to select which cache register outputs data.

Figure 73: READ PAGE CACHE LAST (3Fh) Operation





READ PAGE MULTI-PLANE (00h-32h)

The READ PAGE MULTI-PLANE (00h-32h) command queues a plane to transfer data from the NAND flash array to its cache register. This command can be issued one or more times. Each time a new plane address is specified, that plane is also queued for data transfer. The READ PAGE (00h-30h) command is issued to select the final plane and to begin the read operation for all previously queued planes. All queued planes will transfer data from the NAND Flash array to their cache registers.

To issue the READ PAGE MULTI-PLANE (00h-32h) command, write 00h to the command register, then write five address cycles to the address register, and conclude by writing 32h to the command register. The column address in the address specified is ignored.

After this command is issued, R/B# goes LOW and the die (LUN) is busy (RDY = 0, ARDY = 0) for t_{DBSY}. After t_{DBSY}, R/B# goes HIGH and the die (LUN) is ready (RDY = 1, ARDY = 1). At this point, the die (LUN) and block are queued for data transfer from the array to the cache register for the addressed plane. During t_{DBSY}, the only valid commands are status operations (70h, 78h) and reset commands (FFh, FCh). Following t_{DBSY}, to continue the MULTI-PLANE READ operation, the only valid commands are status operations (70h, 78h), READ PAGE MULTI-PLANE (00h-32h), READ PAGE (00h-30h), and READ PAGE CACHE RANDOM (00h-31h).

Additional READ PAGE MULTI-PLANE (00h-32h) commands can be issued to queue additional planes for data transfer.

If the READ PAGE (00h-30h) command is used as the final command of a MULTI-PLANE READ operation, data is transferred from the NAND Flash array for all of the addressed planes to their respective cache registers. When the die (LUN) is ready (RDY = 1, ARDY = 1), data output is enabled for the cache register linked to the last even plane addressed. When the host requests data output, it begins at the column address specified in the READ PAGE (00h-30h) command. To enable data output in the other cache registers, use the CHANGE READ COLUMN ENHANCED (06h-E0h) command. Additionally, the CHANGE READ COLUMN (05h-E0h) command can be used to change the column address within the currently selected plane.

If the READ PAGE CACHE SEQUENTIAL (31h) is used as the final command of a MULTI-PLANE READ CACHE operation, data is copied from the previously read operation from each plane to each cache register and then data is transferred from the NAND Flash array for all previously addressed planes to their respective data registers. When the die (LUN) is ready (RDY = 1, ARDY = 0), data output is enabled. The CHANGE READ COLUMN ENHANCED (06h-E0h) command is used to determine which cache register outputs data first. To enable data output in the other cache registers, use the CHANGE READ COLUMN ENHANCED (06h-E0h) command. Additionally, the CHANGE READ COLUMN (05h-E0h) command can be used to change the column address within the currently selected plane.

If the READ PAGE CACHE RANDOM (00h-31h) command is used as the final command of a MULTI-PLANE READ CACHE operation, data is copied from the previously read operation from the data register to the cache register and then data is transferred from the NAND Flash array for all of the addressed planes to their respective data registers. When the die (LUN) is ready (RDY = 1, ARDY = 0), data output is enabled. The CHANGE READ COLUMN ENHANCED (06h-E0h) command is used to determine which cache register outputs data first. To enable data output in the other cache registers, use the CHANGE READ COLUMN ENHANCED (06h-E0h) command. Additionally, the CHANGE READ

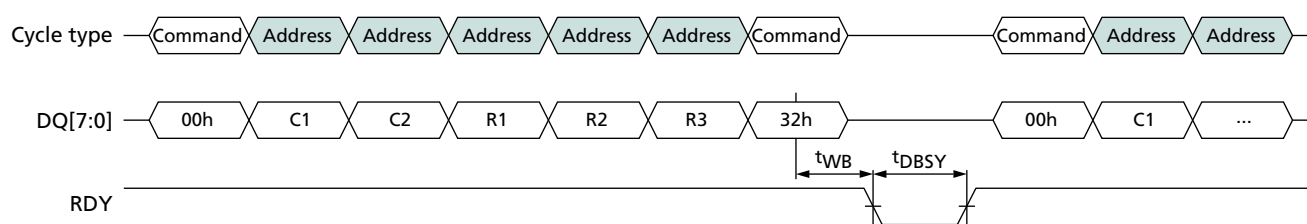


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COLUMN (05h-E0h) command can be used to change the column address within the currently selected plane.

See Multi-Plane Operations for additional multi-plane addressing requirements.

Figure 74: READ PAGE MULTI-PLANE (00h-32h) Operation





Read Retry Operations

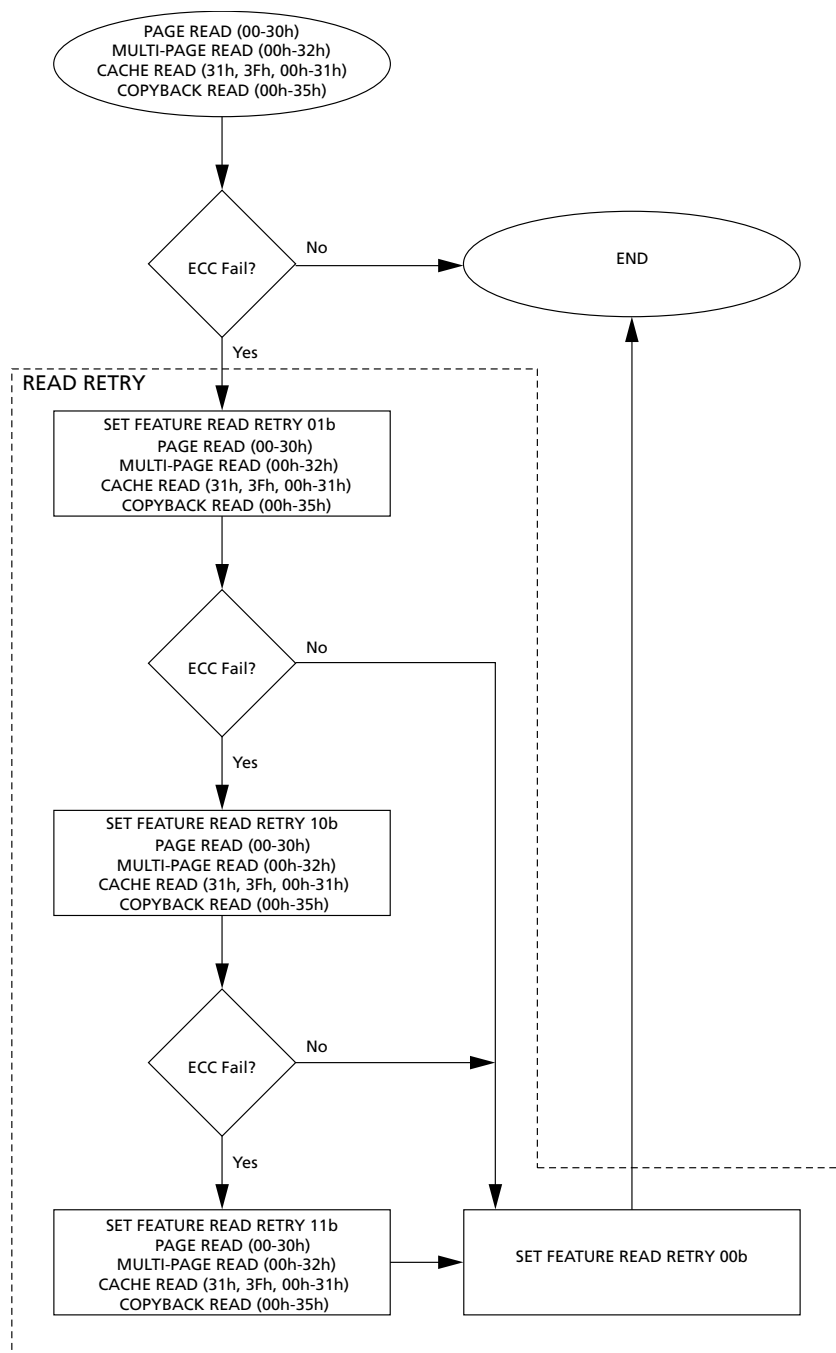
Read retry operations are used as an additional method to recover from bit errors beyond the ECC correction threshold.

Read Retry Operations

The read retry operations are a coordination of read operations and SET FEATURES (EFh) with feature address 89h, selecting different internal read settings in attempt to recover data that is beyond the ECC correction threshold. Using the read retry operations with any array read operation commands is allowed. See Configuration Operations and Feature Address 89h: Read Retry for details.

If reading a page has failed for bit errors beyond the ECC correction threshold, the host issues the SET FEATURE (EFh) command to feature address 89h with P1 subfeature set to a read retry option, as defined in that feature address. A new NAND array READ operation can now be performed. If the read still fails for bit errors beyond the ECC correction threshold, issue the SET FEATURES (EFh) command with the read retry (89h) feature address to select the next consecutive read retry option and repeat read retry operations until the data is correctable or the last option has been attempted. If the re-read is now correctable within the ECC threshold limits, the next read retry option should be set to its default value before the next NAND array READ operation. See Figure 75 (page 133) for flow diagram of read retry operations.

When the user writes to the read retry feature address, all subsequent reads use the internal NAND settings associated with that value until either the read retry feature address is rewritten or the device is powered down. This feature should not be used with the following commands: READ PARAMETER PAGE (ECh), READ UNIQUE ID (EDh), READ OTP PAGE (00h-30h).


Figure 75: Read Retry Flow Chart


Note: 1. There may be more read retry options than are shown in the figure for this NAND device. See Configuration Operations and Read Retry (89h) feature address for details.



Program Operations

Program operations are used to move data from the cache or data registers to the NAND array of one or more planes. During a program operation the contents of the cache and/or data registers are modified by the internal control logic.

Within a block, pages must be programmed sequentially from the least significant page address to the most significant page address (i.e. 0, 1, 2, 3, ...). Programming pages out of order within a block is prohibited.

Program Operations

The PROGRAM PAGE (80h-10h) command, when not preceded by the PROGRAM PAGE MULTI-PLANE (80h-11h) command, programs one page from the cache register to the NAND Flash array. When the die (LUN) is ready (RDY = 1, ARDY = 1), the host should check the FAIL bit to verify that the operation has completed successfully.

Program Cache Operations

The PROGRAM PAGE CACHE (80h-15h) command can be used to improve program operation system performance. When this command is issued, the die (LUN) goes busy (RDY = 0, ARDY = 0) while the cache register contents are copied to the data register, and the die (LUN) is busy with a program cache operation (RDY = 1, ARDY = 0). While the contents of the data register are moved to the NAND Flash array, the cache register is available for an additional PROGRAM PAGE CACHE (80h-15h) or PROGRAM PAGE (80h-10h) command.

For PROGRAM PAGE CACHE-series (80h-15h) operations, during the die (LUN) busy times, ^tCBSY and ^tLPROG, when RDY = 0 and ARDY = 0, the only valid commands are status operations (70h, 78h) and reset (FFh, FCh). When RDY = 1 and ARDY = 0, the only valid commands during PROGRAM PAGE CACHE-series (80h-15h) operations are status operations (70h, 78h), PROGRAM PAGE CACHE (80h-15h), PROGRAM PAGE (80h-10h), CHANGE WRITE COLUMN (85h), CHANGE ROW ADDRESS (85h), and reset (FFh, FCh).

Multi-Plane Program Operations

The PROGRAM PAGE MULTI-PLANE (80h-11h) command can be used to improve program operation system performance by enabling multiple pages to be moved from the cache registers to different planes of the NAND Flash array. This is done by prepending one or more PROGRAM PAGE MULTI-PLANE (80h-11h) commands in front of the PROGRAM PAGE (80h-10h) command. See Multi-Plane Operations for details.

Multi-Plane Program Cache Operations

The PROGRAM PAGE MULTI-PLANE (80h-11h) command can be used to improve program cache operation system performance by enabling multiple pages to be moved from the cache registers to the data registers and, while the pages are being transferred from the data registers to different planes of the NAND Flash array, free the cache registers to receive data input from the host. This is done by prepending one or more PROGRAM PAGE MULTI-PLANE (80h-11h) commands in front of the PROGRAM PAGE CACHE (80h-15h) command. See Multi-Plane Operations for details.

PROGRAM PAGE (80h-10h)

The PROGRAM PAGE (80h-10h) command enables the host to input data to a cache register, and moves the data from the cache register to the specified block and page address in the array of the selected die (LUN). This command is accepted by the die (LUN)



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when it is ready (RDY = 1, ARDY = 1). It is also accepted by the die (LUN) when it is busy with a PROGRAM PAGE CACHE (80h-15h) operation (RDY = 1, ARDY = 0).

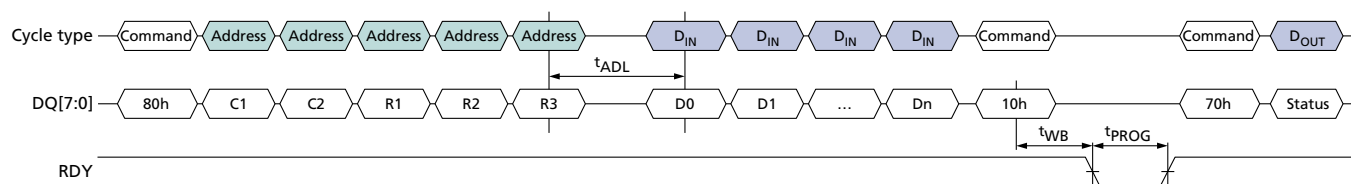
To input a page to the cache register and move it to the NAND array at the block and page address specified, write 80h to the command register. Unless this command has been preceded by a PROGRAM PAGE MULTI-PLANE (80h-11h) command, issuing the 80h to the command register clears all of the cache registers' contents on the selected target. Then write five address cycles containing the column address and row address. Data input cycles follow. Serial data is input beginning at the column address specified. At any time during the data input cycle the CHANGE WRITE COLUMN (85h) and CHANGE ROW ADDRESS (85h) commands may be issued. When data input is complete, write 10h to the command register. The selected LUN will go busy (RDY = 0, ARDY = 0) for t_{PROG} as data is transferred.

To determine the progress of the data transfer, the host can monitor the target's R/B# signal or, alternatively, the status operations (70h, 78h) may be used. When the die (LUN) is ready (RDY = 1, ARDY = 1), the host should check the status of the FAIL bit.

In devices that have more than one die (LUN) per target, during and following interleaved die (multi-LUN) operations, the READ STATUS ENHANCED (78h) command must be used to select only one die (LUN) for status output. Use of the READ STATUS (70h) command could cause more than one die (LUN) to respond, resulting in bus contention.

The PROGRAM PAGE (80h-10h) command is used as the final command of a multi-plane program operation. It is preceded by one or more PROGRAM PAGE MULTI-PLANE (80h-11h) commands. Data is transferred from the cache registers for all of the addressed planes to the NAND array. The host should check the status of the operation by using the status operations (70h, 78h). See Multi-Plane Operations for multi-plane addressing requirements.

Figure 76: PROGRAM PAGE (80h-10h) Operation





PROGRAM PAGE CACHE (80h-15h)

The PROGRAM PAGE CACHE (80h-15h) command enables the host to input data to a cache register; copies the data from the cache register to the data register; then moves the data register contents to the specified block and page address in the array of the selected die (LUN). After the data is copied to the data register, the cache register is available for additional PROGRAM PAGE CACHE (80h-15h) or PROGRAM PAGE (80h-10h) commands. The PROGRAM PAGE CACHE (80h-15h) command is accepted by the die (LUN) when it is ready (RDY = 1, ARDY = 1). It is also accepted by the die (LUN) when busy with a PROGRAM PAGE CACHE (80h-15h) operation (RDY = 1, ARDY = 0).

To input a page to the cache register to move it to the NAND array at the block and page address specified, write 80h to the command register. Unless this command has been preceded by a PROGRAM PAGE MULTI-PLANE (80h-11h) command, issuing the 80h to the command register clears all of the cache registers' contents on the selected target. Then write five address cycles containing the column address and row address. Data input cycles follow. Serial data is input beginning at the column address specified. At any time during the data input cycle the CHANGE WRITE COLUMN (85h) and CHANGE ROW ADDRESS (85h) commands may be issued. When data input is complete, write 15h to the command register. The selected LUN will go busy (RDY = 0, ARDY = 0) for tCBSY to allow the data register to become available from a previous program cache operation, to copy data from the cache register to the data register, and then to begin moving the data register contents to the specified page and block address.

To determine the progress of tCBSY, the host can monitor the target's R/B# signal or, alternatively, the status operations (70h, 78h) can be used. When the LUN's status shows that it is busy with a PROGRAM CACHE operation (RDY = 1, ARDY = 0), the host should check the status of the FAILC bit to see if a previous cache operation was successful.

If, after tCBSY, the host wants to wait for the program cache operation to complete, without issuing the PROGRAM PAGE (80h-10h) command, the host should monitor ARDY until it is 1. The host should then check the status of the FAIL and FAILC bits.

In devices with more than one die (LUN) per target, during and following interleaved die (multi-LUN) operations, the READ STATUS ENHANCED (78h) command must be used to select only one die (LUN) for status output. Use of the READ STATUS (70h) command could cause more than one die (LUN) to respond, resulting in bus contention.

The PROGRAM PAGE CACHE (80h-15h) command is used as the final command of a multi-plane program cache operation. It is preceded by one or more PROGRAM PAGE MULTI-PLANE (80h-11h) commands. Data for all of the addressed planes is transferred from the cache registers to the corresponding data registers, then moved to the NAND Flash array. The host should check the status of the operation by using the status operations (70h, 78h). See Multi-Plane Operations for multi-plane addressing requirements.



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Figure 77: PROGRAM PAGE CACHE (80h–15h) Operation (Start)

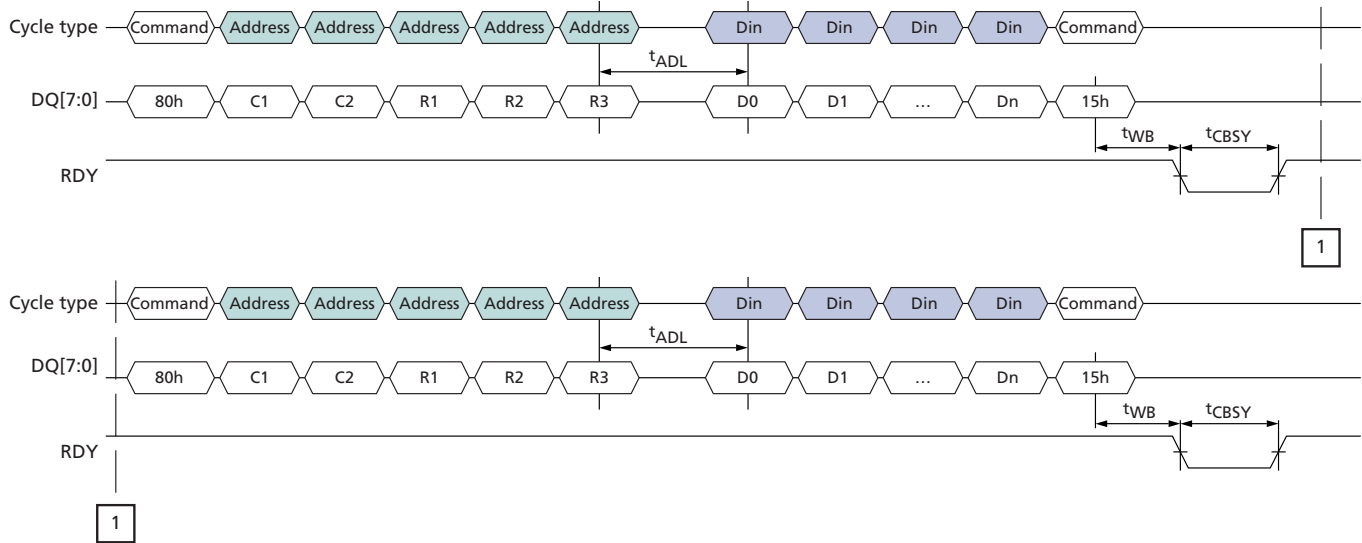
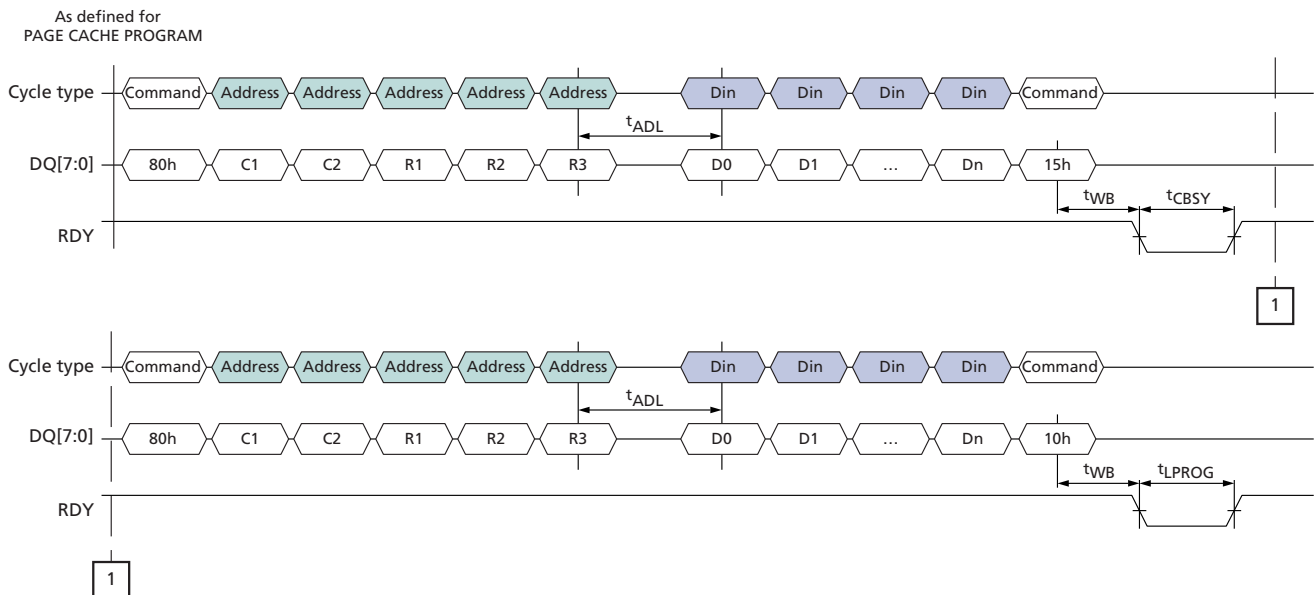


Figure 78: PROGRAM PAGE CACHE (80h–15h) Operation (End)





PROGRAM PAGE MULTI-PLANE (80h-11h)

The PROGRAM PAGE MULTI-PLANE (80h-11h) command enables the host to input data to the addressed plane's cache register and queue the cache register to ultimately be moved to the NAND Flash array. This command can be issued one or more times. Each time a new plane address is specified that plane is also queued for data transfer. To input data for the final plane and to begin the program operation for all previously queued planes, issue either the PROGRAM PAGE (80h-10h) command or the PROGRAM PAGE CACHE (80h-15h) command. All of the queued planes will move the data to the NAND Flash array. This command is accepted by the die (LUN) when it is ready (RDY = 1).

To input a page to the cache register and queue it to be moved to the NAND Flash array at the block and page address specified, write 80h to the command register. Unless this command has been preceded by an 11h command, issuing the 80h to the command register clears all of the cache registers' contents on the selected target. Write five address cycles containing the column address and row address; data input cycles follow. Serial data is input beginning at the column address specified. At any time during the data input cycle, the CHANGE WRITE COLUMN (85h) and CHANGE ROW ADDRESS (85h) commands can be issued. When data input is complete, write 11h to the command register. The selected die (LUN) will go busy (RDY = 0, ARDY = 0) for 'DBSY.

To determine the progress of 'DBSY, the host can monitor the target's R/B# signal or, alternatively, the status operations (70h, 78h) can be used. When the LUN's status shows that it is ready (RDY = 1), additional PROGRAM PAGE MULTI-PLANE (80h-11h) commands can be issued to queue additional planes for data transfer. Alternatively, the PROGRAM PAGE (80h-10h) or PROGRAM PAGE CACHE (80h-15h) commands can be issued.

When the PROGRAM PAGE (80h-10h) command is used as the final command of a multi-plane program operation, data is transferred from the cache registers to the NAND Flash array for all of the addressed planes during 'PROG. When the die (LUN) is ready (RDY = 1, ARDY = 1), the host should check the status of the FAIL bit for each of the planes to verify that programming completed successfully.

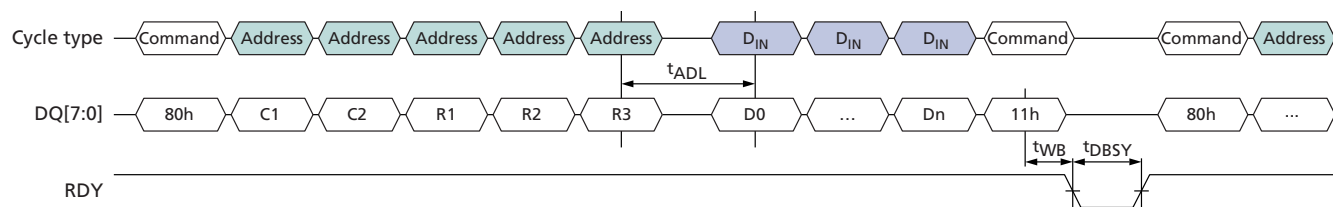
When the PROGRAM PAGE CACHE (80h-15h) command is used as the final command of a MULTI-PLANE PROGRAM CACHE operation, data is transferred from the cache registers to the data registers after the previous array operations finish. The data is then moved from the data registers to the NAND Flash array for all of the addressed planes. This occurs during 'CBSY. After 'CBSY, the host should check the status of the FAILC bit for each of the planes from the previous program cache operation, if any, to verify that programming completed successfully.

For the PROGRAM PAGE MULTI-PLANE (80h-11h), PROGRAM PAGE (80h-10h), and PROGRAM PAGE CACHE (80h-15h) commands, see Multi-Plane Operations for multi-plane addressing requirements.



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Figure 79: PROGRAM PAGE MULTI-PLANE (80h–11h) Operation



For JEDEC compliance the PROGRAM PAGE MULTI-PLANE (81h-11h) command is also supported. This would also include the last command 81h-10h to conclude a PROGRAM PAGE MULTI-PLANE sequence and 81h-15h to conclude a PROGRAM PAGE CACHE MULTI-PLANE sequence.



Erase Operations

Erase operations are used to clear the contents of a block in the NAND Flash array to prepare its pages for program operations.

Erase Operations

The ERASE BLOCK (60h-D0h) command, when not preceded by the ERASE BLOCK MULTI-PLANE (60h-D1h) command, erases one block in the NAND Flash array. When the die (LUN) is ready (RDY = 1, ARDY = 1), the host should check the FAIL bit to verify that this operation completed successfully.

MULTI-PLANE ERASE Operations

The ERASE BLOCK MULTI-PLANE (60h-D1h) command can be used to further system performance of erase operations by allowing more than one block to be erased in the NAND array. This is done by prepending one or more ERASE BLOCK MULTI-PLANE (60h-D1h) commands in front of the ERASE BLOCK (60h-D0h) command. See Multi-Plane Operations for details.

ERASE BLOCK (60h-D0h)

The ERASE BLOCK (60h-D0h) command erases the specified block in the NAND Flash array. This command is accepted by the die (LUN) when it is ready (RDY = 1, ARDY = 1).

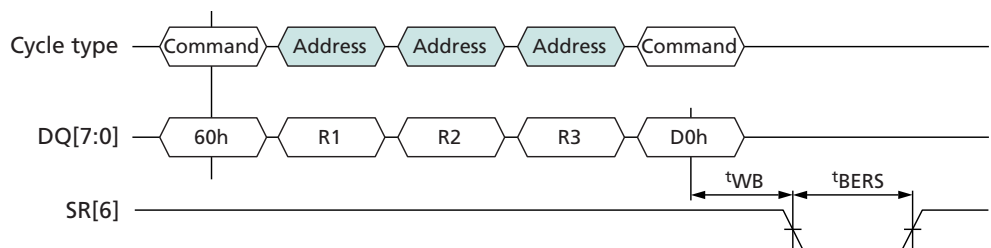
To erase a block, write 60h to the command register. Then write three address cycles containing the row address; the page address is ignored. Conclude by writing D0h to the command register. The selected die (LUN) will go busy (RDY = 0, ARDY = 0) for t_{BERS} while the block is erased.

To determine the progress of an ERASE operation, the host can monitor the target's R/B# signal, or alternatively, the status operations (70h, 78h) can be used. When the die (LUN) is ready (RDY = 1, ARDY = 1) the host should check the status of the FAIL bit.

In devices that have more than one die (LUN) per target, during and following interleaved die (multi-LUN) operations, the READ STATUS ENHANCED (78h) command must be used to select only one die (LUN) for status output. Use of the READ STATUS (70h) command could cause more than one die (LUN) to respond, resulting in bus contention.

The ERASE BLOCK (60h-D0h) command is used as the final command of a MULTI-PLANE ERASE operation. It is preceded by one or more ERASE BLOCK MULTI-PLANE (60h-D1h) commands. All of blocks in the addressed planes are erased. The host should check the status of the operation by using the status operations (70h, 78h). See Multi-Plane Operations for multi-plane addressing requirements.

Figure 80: ERASE BLOCK (60h-D0h) Operation





ERASE BLOCK MULTI-PLANE (60h-D1h)

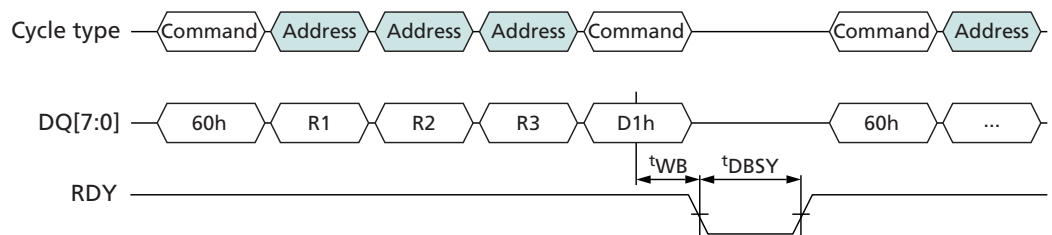
The ERASE BLOCK MULTI-PLANE (60h-D1h) command queues a block in the specified plane to be erased in the NAND Flash array. This command can be issued one or more times. Each time a new plane address is specified, that plane is also queued for a block to be erased. To specify the final block to be erased and to begin the ERASE operation for all previously queued planes, issue the ERASE BLOCK (60h-D0h) command. This command is accepted by the die (LUN) when it is ready (RDY = 1, ARDY = 1).

To queue a block to be erased, write 60h to the command register, then write three address cycles containing the row address; the page address is ignored. Conclude by writing D1h to the command register. The selected die (LUN) will go busy (RDY = 0, ARDY = 0) for t_{DBSY} .

To determine the progress of t_{DBSY} , the host can monitor the target's R/B# signal, or alternatively, the status operations (70h, 78h) can be used. When the LUN's status shows that it is ready (RDY = 1, ARDY = 1), additional ERASE BLOCK MULTI-PLANE (60h-D1h) commands can be issued to queue additional planes for erase. Alternatively, the ERASE BLOCK (60h-D0h) command can be issued to erase all of the queued blocks.

For multi-plane addressing requirements for the ERASE BLOCK MULTI-PLANE (60h-D1h) and ERASE BLOCK (60h-D0h) commands, see Multi-Plane Operations.

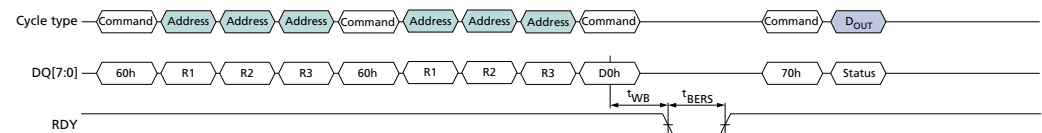
Figure 81: ERASE BLOCK MULTI-PLANE (60h-D1h) Operation



ERASE BLOCK MULTI-PLANE (60h-60h-D0h)

This operation behaves the same as the ERASE BLOCK MULTI-PLANE (60h-D1h) command followed by a ERASE BLOCK (60h-D0h) command.

Figure 82: ERASE BLOCK MULTI-PLANE (60h-60h-D0h) Operation





Copyback Operations

COPYBACK operations make it possible to transfer data within a plane from one page to another using the cache register. This is particularly useful for block management and wear leveling.

The COPYBACK operation is a two-step process consisting of a COPYBACK READ (00h-35h) and a COPYBACK PROGRAM (85h-10h) command. To move data from one page to another on the same plane, first issue the COPYBACK READ (00h-35h) command. When the die (LUN) is ready (RDY = 1, ARDY = 1), the host can transfer the data to a new page by issuing the COPYBACK PROGRAM (85h-10h) command. When the die (LUN) is again ready (RDY = 1, ARDY = 1), the host should check the FAIL bit to verify that this operation completed successfully.

To prevent bit errors from accumulating over multiple COPYBACK operations, it is recommended that the host read the data out of the cache register after the COPYBACK READ (00h-35h) completes prior to issuing the COPYBACK PROGRAM (85h-10h) command. The CHANGE READ COLUMN (05h-E0h) command can be used to change the column address. The host should check the data for ECC errors and correct them. When the COPYBACK PROGRAM (85h-10h) command is issued, any corrected data can be input. The CHANGE ROW ADDRESS (85h) command can be used to change the column address.

It is not possible to use the COPYBACK operation to move data from one plane to another or from one die (LUN) to another. Instead, use a READ PAGE (00h-30h) or COPYBACK READ (00h-35h) command to read the data out of the NAND, and then use a PROGRAM PAGE (80h-10h) command with data input to program the data to a new plane or die (LUN).

Between the COPYBACK READ (00h-35h) and COPYBACK PROGRAM (85h-10h) commands, the following commands are supported: status operations (70h, 78h), and column address operations (05h-E0h, 06h-E0h, 85h). Reset operations (FFh, FCh) can be issued after COPYBACK READ (00h-35h), but the contents of the cache registers on the target are not valid.

In devices which have more than one die (LUN) per target, once the COPYBACK READ (00h-35h) is issued, interleaved die (multi-LUN) operations are prohibited until after the COPYBACK PROGRAM (85h-10h) command is issued.

Multi-Plane Copyback Operations

Multi-plane copyback read operations improve read data throughput by copying data simultaneously from more than one plane to the specified cache registers. This is done by prepending one or more READ PAGE MULTI-PLANE (00h-32h) commands in front of the COPYBACK READ (00h-35h) command.

The COPYBACK PROGRAM MULTI-PLANE (85h-11h) command can be used to further system performance of COPYBACK PROGRAM operations by enabling movement of multiple pages from the cache registers to different planes of the NAND Flash array. This is done by prepending one or more COPYBACK PROGRAM (85h-11h) commands in front of the COPYBACK PROGRAM (85h-10h) command. See Multi-Plane Operations for details.



MLC 256Gb to 2Tb Async/Sync NAND Copyback Operations

COPYBACK READ (00h-35h)

The COPYBACK READ (00h-35h) command is functionally identical to the READ PAGE (00h-30h) command, except that 35h is written to the command register instead of 30h. See READ PAGE (00h-30h) (page 125) for further details.

Though it is not required, it is recommended that the host read the data out of the device to verify the data prior to issuing the COPYBACK PROGRAM (85h-10h) command to prevent the propagation of data errors.

Figure 83: COPYBACK READ (00h-35h) Operation

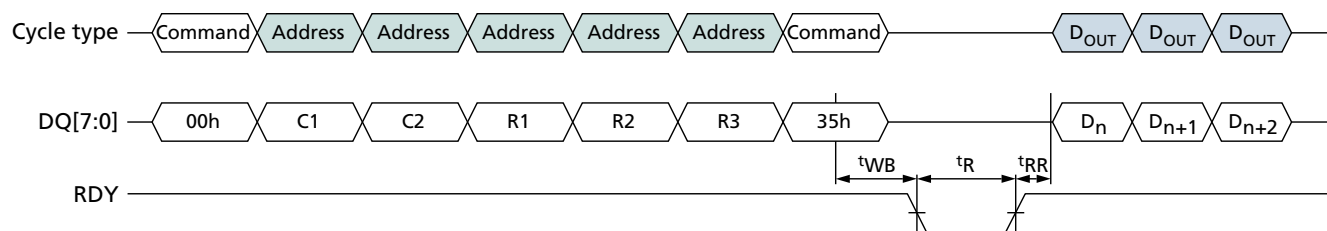
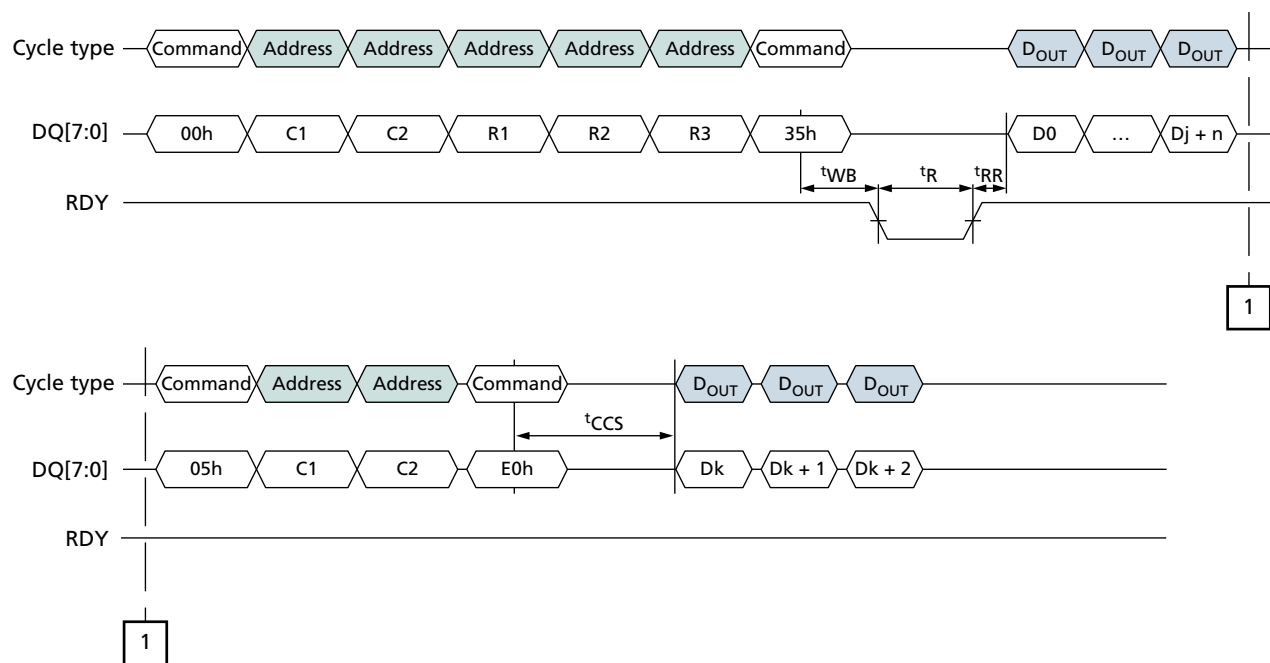


Figure 84: COPYBACK READ (00h-35h) with CHANGE READ COLUMN (05h-E0h) Operation





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COPYBACK PROGRAM (85h–10h)

The COPYBACK PROGRAM (85h-10h) command is functionally identical to the PROGRAM PAGE (80h-10h) command, except that when 85h is written to the command register, cache register contents are not cleared. See PROGRAM PAGE (80h-10h) for further details.

Figure 85: COPYBACK PROGRAM (85h–10h) Operation

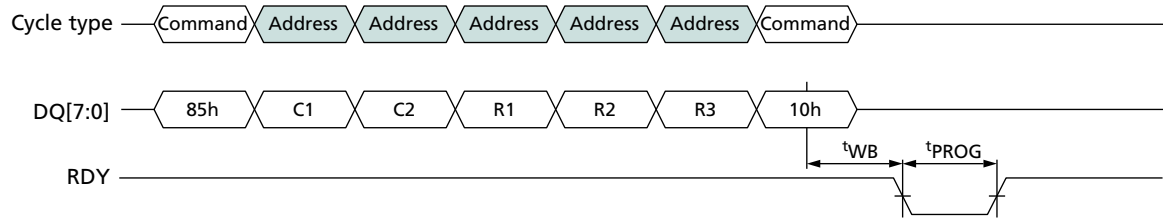
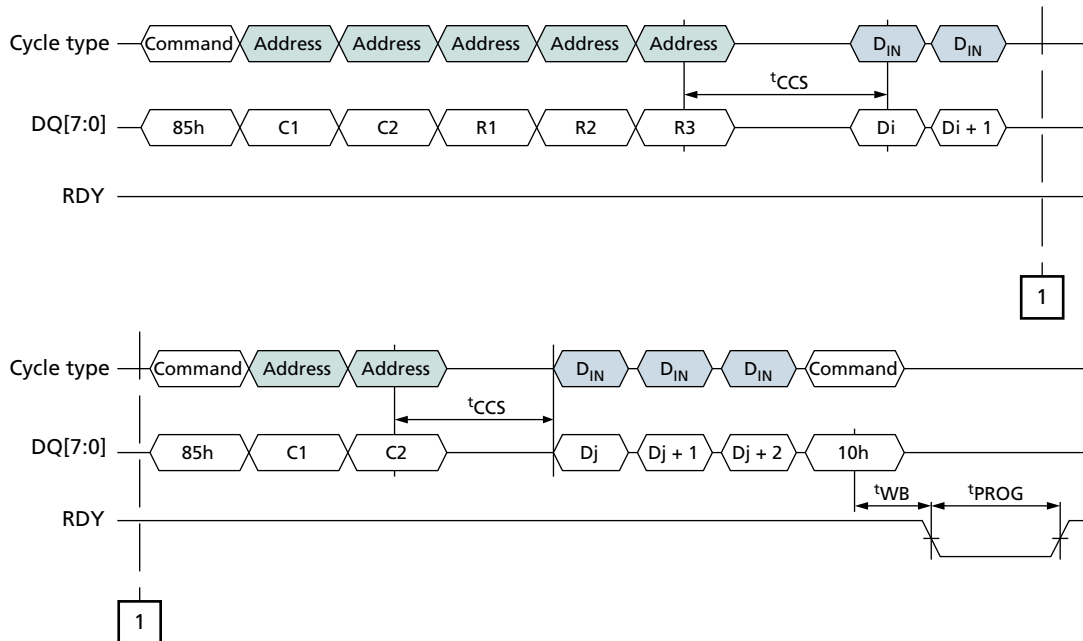


Figure 86: COPYBACK PROGRAM (85h-10h) with CHANGE WRITE COLUMN (85h) Operation



COPYBACK READ MULTI-PLANE (00h-32h)

The COPYBACK READ MULTI-PLANE (00h-32h) command is functionally identical to the READ PAGE MULTI-PLANE (00h-32h) command, except that the 35h command is written as the final command. The complete command sequence for the COPYBACK READ PAGE MULTI-PLANE is 00h-32h-00h-35h. See READ PAGE MULTI-PLANE (00h-32h) (page 130) for further details.

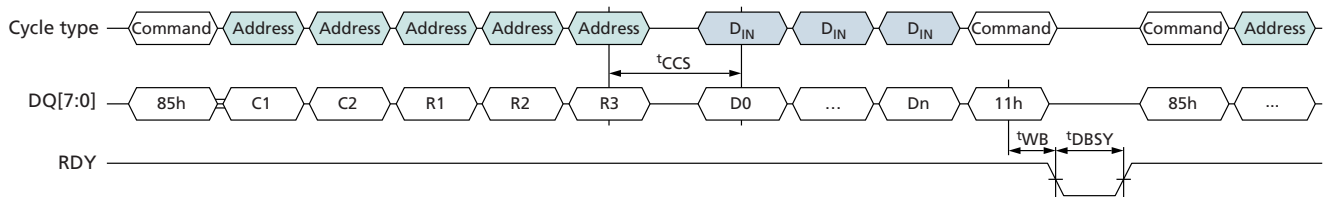


MLC 256Gb to 2Tb Async/Sync NAND One-Time Programmable (OTP) Operations

COPYBACK PROGRAM MULTI-PLANE (85h-11h)

The COPYBACK PROGRAM MULTI-PLANE (85h-11h) command is functionally identical to the PROGRAM PAGE MULTI-PLANE (80h-11h) command, except that when 85h is written to the command register, cache register contents are not cleared. See PROGRAM PAGE MULTI-PLANE (80h-11h) for further details.

Figure 87: COPYBACK PROGRAM MULTI-PLANE (85h-11h) Operation



One-Time Programmable (OTP) Operations

This Micron NAND Flash device offers a protected, one-time programmable NAND Flash memory area. Each target has a an OTP area with a range of OTP pages (see Table 30 (page 146)); the entire range is guaranteed to be good. Customers can use the OTP area in any way they desire; typical uses include programming serial numbers or other data for permanent storage.

The OTP area leaves the factory in an erased state (all bits are 1). Programming an OTP page changes bits that are 1 to 0, but cannot change bits that are 0 to 1. The OTP area cannot be erased, even if it is not protected. Protecting the OTP area prevents further programming of the pages in the OTP area.

Enabling the OTP Operation Mode

The OTP area is accessible while the OTP operation mode is enabled. To enable OTP operation mode, issue the SET FEATURES (EFh) command to feature address 90h and write 01h to P1, followed by three cycles of 00h to P2 through P4.

When the target is in OTP operation mode, all subsequent PAGE READ (00h-30h) and PROGRAM PAGE (80h-10h) commands are applied to the OTP area.

ERASE commands are not valid while the target is in OTP operation mode.

Programming OTP Pages

Each page in the OTP area is programming using the PROGRAM OTP PAGE (80h-10h) command. Each page can be programmed more than once, in sections, up to the maximum number allowed (see NOP in Table 30 (page 146)). The pages in the OTP area must be programmed in ascending order.

If the host issues a PAGE PROGRAM (80h-10h) command to an address beyond the maximum page-address range, the target will be busy for t_{OBSY} and the WP# status register bit will be 0, meaning that the page is write-protected.

Protecting the OTP Area



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To protect the OTP area, issue the OTP PROTECT (80h-10h) command to the OTP Protect Page. When the OTP area is protected it cannot be programmed further. It is not possible to unprotect the OTP area after it has been protected.

Reading OTP Pages

To read pages in the OTP area, whether the OTP area is protected or not, issue the PAGE READ (00h-30h) command.

If the host issues the PAGE READ (00h-30h) command to an address beyond the maximum page-address range, the data output will not be valid. To determine whether the target is busy during an OTP operation, either monitor R/B# or use the READ STATUS (70h) command. Use of the READ STATUS ENHANCED (78h) command is prohibited while the OTP operation is in progress.

Returning to Normal Array Operation Mode

To exit OTP operation mode and return to normal array operation mode, issue the SET FEATURES (EFh) command to feature address 90h and write 00h to P1 through P4.

If the RESET (FFh) command is issued while in OTP operation mode, the target will exit OTP operation mode and enter normal operating mode. If the synchronous interface is active, the target will exit OTP operation and enable the asynchronous interface.

If the SYNCHRONOUS RESET (FCh) command is issued while in the OTP operation mode, the target will exit OTP operation mode and the synchronous interface remains active.

Table 30: OTP Area Details

Description	Value
Number of OTP pages	27
OTP protect page address	01h
OTP start page address	04h
Number of partial page programs (NOP) to each OTP page	2

PROGRAM OTP PAGE (80h-10h)

The PROGRAM OTP PAGE (80h-10h) command is used to write data to the pages within the OTP area. To program data in the OTP area, the target must be in OTP operation mode.

To use the PROGRAM OTP PAGE (80h-10h) command, issue the 80h command. Issue five address cycles including the column address, the page address within the OTP page range, and a block address of 0. Next, write the data to the cache register using data input cycles. After data input is complete, issue the 10h command.

R/B# goes LOW for the duration of the array programming time, ^tPROG. The READ STATUS (70h) command is the only valid command for reading status in OTP operation mode. The RDY bit of the status register will reflect the state of R/B#. Use of the READ STATUS ENHANCED (78h) command is prohibited.

When the target is ready, read the FAIL bit of the status register to determine whether the operation passed or failed (see Status Operations).

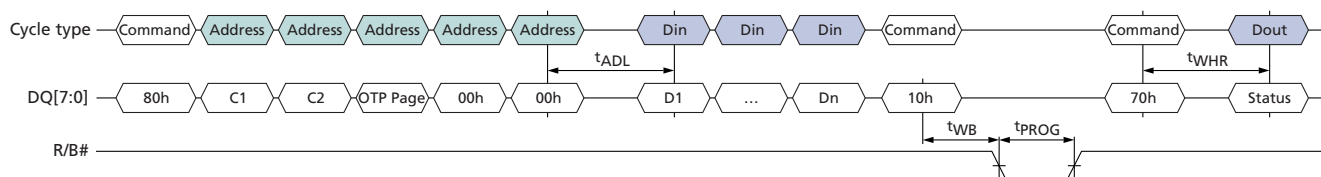


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The PROGRAM OTP PAGE (80h-10h) command also accepts the CHANGE WRITE COLUMN (85h) command during data input.

If a PROGRAM PAGE command is issued to the OTP area after the area has been protected, then R/B# goes LOW for t_{OBSY} . After t_{OBSY} , the status register is set to 60h.

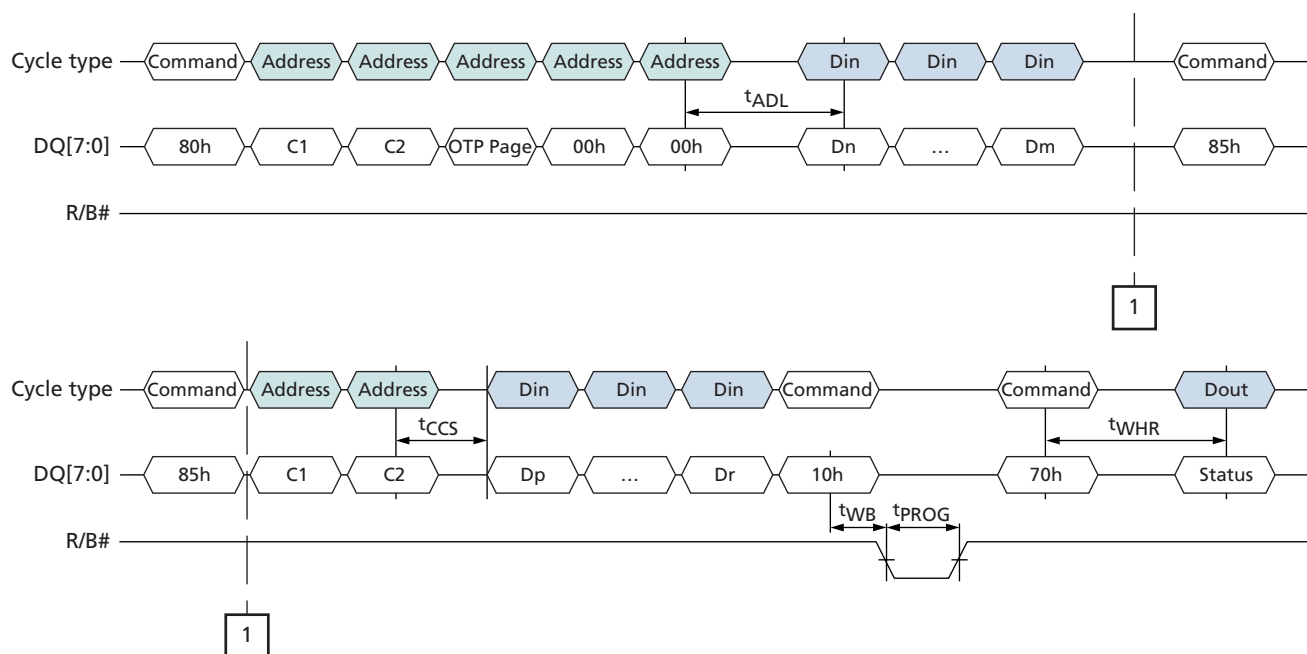
Figure 88: PROGRAM OTP PAGE (80h-10h) Operation





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Figure 89: PROGRAM OTP PAGE (80h-10h) with CHANGE WRITE COLUMN (85h) Operation



PROTECT OTP AREA (80h-10h)

The PROTECT OTP AREA (80h-10h) command is used to prevent further programming of the pages in the OTP area. The protect the OTP area, the target must be in OTP operation mode.

To protect all data in the OTP area, issue the 80h command. Issue five address cycles including the column address, OTP protect page address and block address; the column and block addresses are fixed to 0. Next, write 00h data for the first byte location and issue the 10h command.

R/B# goes LOW for the duration of the array programming time, t_{PROG} . The READ STATUS (70h) command is the only valid command for reading status in OTP operation mode. The RDY bit of the status register will reflect the state of R/B#. Use of the READ STATUS ENHANCED (78h) command is prohibited.

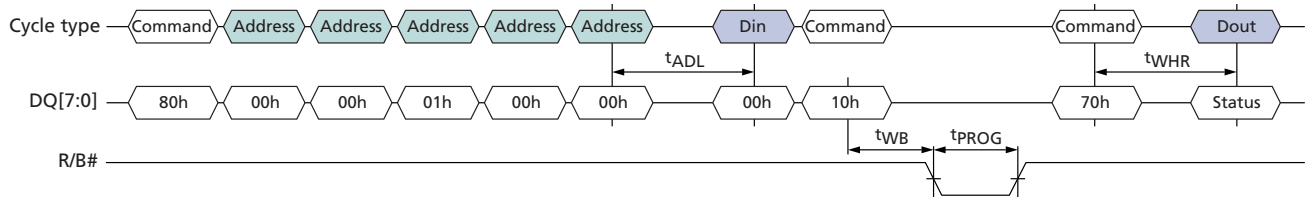
When the target is ready, read the FAIL bit of the status register to determine if the operation passed or failed (see Status Operations).

If the PROTECT OTP AREA (80h-10h) command is issued after the OTP area has already been protected, R/B# goes LOW for t_{OBSY} . After t_{OBSY} , the status register is set to 60h.

Figure 90: PROTECT OTP AREA (80h-10h) Operation



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Note: 1. OTP data is protected following a status confirmation.

READ OTP PAGE (00h-30h)

The READ OTP PAGE (00h-30h) command is used to read data from the pages in the OTP area. To read data in the OTP area, the target must be in OTP operation mode.

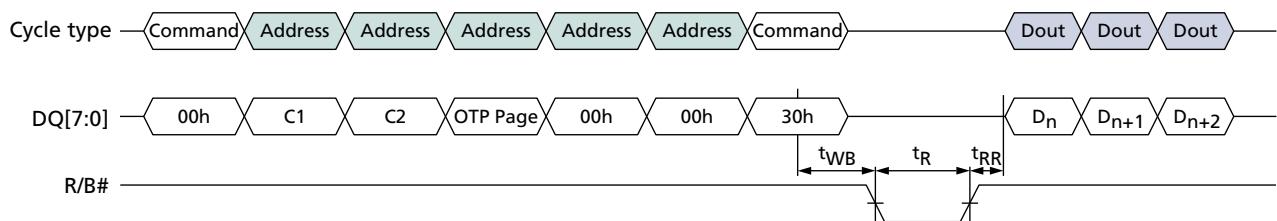
To use the READ OTP PAGE (00h-30h) command, issue the 00h command. Issue five address cycles including the column address, the page address within the OTP page range, and a block address of 0. Next, issue the 30h command. The selected die (LUN) will go busy (RDY = 0, ARDY = 0) for 'R as data is transferred.

To determine the progress of the data transfer, the host can monitor the target's R/B# signal, or alternatively the READ STATUS (70h) command can be used. If the status operations are used to monitor the die's (LUN's) status, when the die (LUN) is ready (RDY = 1, ARDY = 1) the host disables status output and enables data output by issuing the READ MODE (00h) command. When the host requests data output, it begins at the column address specified.

Additional pages within the OTP area can be read by repeating the READ OTP PAGE (00h-30h) command.

The READ OTP PAGE (00h-30h) command is compatible with the CHANGE READ COLUMN (05h-E0h) command. Use of the READ STATUS ENHANCED (78h) and CHANGE READ COLUMN ENHANCED (06h-E0h) commands are prohibited.

Figure 91: READ OTP PAGE (00h-30h) Operation





Multi-Plane Operations

Each NAND Flash logical unit (LUN) is divided into multiple physical planes. Each plane contains a cache register and a data register independent of the other planes. The planes are addressed via the low-order block address bits. Specific details are provided in Device and Array Organization.

Multi-plane operations make better use of the NAND Flash arrays on these physical planes by performing concurrent READ, PROGRAM, or ERASE operations on multiple planes, significantly improving system performance. Multi-plane operations must be of the same type across the planes; for example, it is not possible to perform a PROGRAM operation on one plane with an ERASE operation on another.

When issuing MULTI-PLANE PROGRAM or ERASE operations, use the READ STATUS (70h) command and check whether the previous operation(s) failed. If the READ STATUS (70h) command indicates that an error occurred (FAIL = 1 and/or FAILC = 1), use the READ STATUS ENHANCED (78h) command—time for each plane—to determine which plane operation failed.

Multi-Plane Addressing

Multi-plane commands require an address per operational plane. For a given multi-plane operation, these addresses are subject to the following requirements:

- The LUN address bit(s) must be identical for all of the issued addresses.
- The plane select bit, BA[9], must be different for each issued address.
- The page address bits, PA[8:0], must be identical for each issued address.

The READ STATUS (70h) command should be used following MULTI-PLANE PROGRAM PAGE and ERASE BLOCK operations on a single die (LUN).

Interleaved Die (Multi-LUN) Operations

In devices that have more than one die (LUN) per target, it is possible to improve performance by interleaving operations between the die (LUNs). An interleaved die (multi-LUN) operation is one that individual die (LUNs) involved may be in any combination of busy or ready status during operations.

Interleaved die (multi-LUN) operations are prohibited following RESET (FFh, FCh), identification (90h, ECh, EDh), and configuration (EEh, EFh) operations until ARDY = 1 for all of the die (LUNs) on the target.

During an interleaved die (multi-LUN) operation, there are two methods to determine operation completion. The R/B# signal indicates when all of the die (LUNs) have finished their operations. R/B# remains LOW while any die (LUN) is busy. When R/B# goes HIGH, all of the die (LUNs) are idle and the operations are complete. Alternatively, the READ STATUS ENHANCED (78h) command can report the status of each die (LUN) individually.

If a die (LUN) is performing a cache operation, like PROGRAM PAGE CACHE (80h-15h), then the die (LUN) is able to accept the data for another cache operation when status register bit 6 is 1. All operations, including cache operations, are complete on a die when status register bit 5 is 1.

Use the READ STATUS ENHANCED (78h) command to monitor status for the addressed die (LUN). When multi-plane commands are used with interleaved die (multi-LUN) op-



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erations, the multi-plane commands must also meet the requirements, see Multi-Plane Operations for details. After the READ STATUS ENHANCED (78h) command has been issued, the READ STATUS (70h) command may be issued for the previously addressed die (LUN).

See Command Definitions for the list of commands that can be issued while other die (LUNs) are busy.

During an interleaved die (multi-LUN) operation that involves a PROGRAM-series (80h-10h, 80h-15h, 80h-11h) operation and a READ operation, the PROGRAM-series operation must be issued before the READ-series operation. The data from the READ-series operation must be output to the host before the next PROGRAM-series operation is issued. This is because the 80h command clears the cache register contents of all cache registers on all planes.

During a interleaved die (multi-LUN) operation that involves PROGRAM-series (80h-10h, 80h-15h, 80h-11h) operations between multiple die (LUNs) on the same target, after data is inputted to the first die (LUN) addressed in that interleaved die (multi-LUN) sequence, before addressing the next die (LUN) with a PROGRAM-series (80h-10h, 80h-15h, 80h-11h) operation a program confirm command (via 80h-10h, 80-15h) must be issued to begin the array programming of the pervious die (LUN). If this is not done by the host prior to addressing the next die (LUN), data in all the cache registers of the previously die (LUNs) will be cleared by the PROGRAM-series (80h-10h, 80h-15h, 80h-11h) operation to the next die (LUN) in the interleaved die (multi-LUN) sequence. Utilizing the Program Clear functionality in feature address 01h can be utilized to avoid a PROGRAM-series (80h-10h, 80h-15h, 80h-11h) operation from clearing the contents of non-addressed NAND planes.

When issuing combinations of commands to multiple die (LUNs) (for example, Reads to one die (LUN) and Programs to another die (LUN)) or Reads to one die (LUN) and Reads to another die (LUN)), the host shall issue the READ STATUS ENHANCED (78h) command before reading data from any LUN. This ensures that only the LUN selected by the READ STATUS ENHANCED (78h) command responds to a data output cycle after being put into data output mode, and thus avoiding bus contention. After the READ STATUS ENHANCED (78h) command is issued to the selected die (LUN) a CHANGE READ COLUMN (05h-E0h) or CHANGE READ COLUMN ENHANCED (06h-E0h) command shall be issued prior to any data output from the selected die (LUN).



Error Management

Each NAND Flash die (LUN) is specified to have a minimum number of valid blocks (NVB) of the total available blocks. This means the die (LUNs) could have blocks that are invalid when shipped from the factory. An invalid block is one that contains at least one page that has more bad bits than can be corrected by the minimum required ECC. Additional blocks can develop with use. However, the total number of available blocks per die (LUN) will not fall below NVB during the endurance life of the product.

Although NAND Flash memory devices could contain bad blocks, they can be used quite reliably in systems that provide bad-block management and error-correction algorithms. This type of software environment ensures data integrity.

Internal circuitry isolates each block from other blocks, so the presence of a bad block does not affect the operation of the rest of the NAND Flash array.

NAND Flash devices are shipped from the factory erased. The factory identifies invalid blocks before shipping by attempting to program the bad-block mark into every location in the first page of each invalid block. It may not be possible to program every location with the bad-block mark. However, the first spare area location in each bad block is guaranteed to contain the bad-block mark. This method is compliant with ONFI Factory Defect Mapping requirements. See the following table for the first spare area location and the bad-block mark.

System software should check the first spare area location on the first page of each block prior to performing any PROGRAM or ERASE operations on the NAND Flash device. A bad block table can then be created, enabling system software to map around these areas. Factory testing is performed under worst-case conditions. Because invalid blocks could be marginal, it may not be possible to recover this information if the block is erased.

Over time, some memory locations may fail to program or erase properly. In order to ensure that data is stored properly over the life of the NAND Flash device, the following precautions are required:

- Always check status after a PROGRAM or ERASE operation
- Under typical conditions, use the minimum required ECC (see table below)
- Use bad-block management and wear-leveling algorithms

The first block (physical block address 00h) for each CE# is guaranteed to be valid with ECC when shipped from the factory.

Table 31: Error Management Details

Description	Requirement
Minimum number of valid blocks (NVB) per LUN	1998
Total available blocks per LUN	2096
First spare area location	Byte 16,384
Bad-block mark	00h
Minimum required ECC	60-bit ECC per 1141 bytes of data



Shared Pages

In MLC NAND Flash devices, each memory cell contains two bits of data. These bits are distributed across two NAND pages. Pages within a NAND block that share the same NAND memory cells are referred to as shared pages. If any program operation is interrupted (e.g. power loss or reset), data in previously programmed shared pages can also be corrupted.

The least significant numbered shared page must be programmed before the most significant numbered page of that pair can be programmed.

Table 32: Shared Pages

Shared Pages		Shared Pages	
0	–	1	6
2	–	3	–
4	–	5	9
7	12	8	13
10	16	11	17
14	20	15	21
18	24	19	25
22	28	23	29
26	32	27	33
30	36	31	37
34	40	35	41
38	44	39	45
42	48	43	49
46	52	47	53
50	56	51	57
54	60	55	61
58	64	59	65
62	68	63	69
66	72	67	73
70	76	71	77
74	80	75	81
78	84	79	85
82	88	83	89
86	92	87	93
90	96	91	97
94	100	95	101
98	104	99	105
102	108	103	109
106	112	107	113
110	116	111	117



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Table 32: Shared Pages (Continued)

Shared Pages		Shared Pages	
114	120	115	121
118	124	119	125
122	128	123	129
126	132	127	133
130	136	131	137
134	140	135	141
138	144	139	145
142	148	143	149
146	152	147	153
150	156	151	157
154	160	155	161
158	164	159	165
162	168	163	169
166	172	167	173
170	176	171	177
174	180	175	181
178	184	179	185
182	188	183	189
186	192	187	193
190	196	191	197
194	200	195	201
198	204	199	205
202	208	203	209
206	212	207	213
210	216	211	217
214	220	215	221
218	224	219	225
222	228	223	229
226	232	227	233
230	236	231	237
234	240	235	241
238	244	239	245
242	248	243	249
246	252	247	253
250	256	251	257
254	260	255	261
258	264	259	265
262	268	263	269



MLC 256Gb to 2Tb Async/Sync NAND Shared Pages

Table 32: Shared Pages (Continued)

Shared Pages		Shared Pages	
266	272	267	273
270	276	271	277
274	280	275	281
278	284	279	285
282	288	283	289
286	292	287	293
290	296	291	297
294	300	295	301
298	304	299	305
302	308	303	309
306	312	307	313
310	316	311	317
314	320	315	321
318	324	319	325
322	328	323	329
326	332	327	333
330	336	331	337
334	340	335	341
338	344	339	345
342	348	343	349
346	352	347	353
350	356	351	357
354	360	355	361
358	364	359	365
362	368	363	369
366	372	367	373
370	376	371	377
374	380	375	381
378	384	379	385
382	388	383	389
386	392	387	393
390	396	391	397
394	400	395	401
398	404	399	405
402	408	403	409
406	412	407	413
410	416	411	417
414	420	415	421



MLC 256Gb to 2Tb Async/Sync NAND Shared Pages

Table 32: Shared Pages (Continued)

Shared Pages		Shared Pages	
418	424	419	425
422	428	423	429
426	432	427	433
430	436	431	437
434	440	435	441
438	444	439	445
442	448	443	449
446	452	447	453
450	456	451	457
454	460	455	461
458	464	459	465
462	468	463	469
466	472	467	473
470	476	471	477
474	480	475	481
478	484	479	485
482	488	483	489
486	492	487	493
490	496	491	497
494	500	495	501
498	504	499	505
503	508	502	–
506	–	507	511
509	–	510	–



MLC 256Gb to 2Tb Async/Sync NAND Output Drive Impedance

Output Drive Impedance

Because NAND Flash is designed for use in systems that are typically point-to-point connections, an option to control the drive strength of the output buffers is provided. Drive strength should be selected based on the expected loading of the memory bus. There are four supported settings for the output drivers: 18 ohms, 25 ohms, 35 ohms, and 50 ohms.

The 35 ohm output drive strength setting is the power-on default value. The host can select a different drive strength setting using the SET FEATURES (EFh) command.

The output impedance range from minimum to maximum covers process, voltage, and temperature variations. Devices are not guaranteed to be at the nominal line.

Table 33: Output Drive Strength Conditions ($V_{CCQ} = 1.7\text{--}1.95V$)

Range	Process	Voltage	Temperature
Minimum	Fast-Fast	1.95V	T_A (MIN)
Nominal	Typical-Typical	1.8V	+25°C
Maximum	Slow-Slow	1.7V	T_A (MAX)

Table 34: Output Drive Strength Impedance Values ($V_{CCQ} = 1.7\text{--}1.95V$)

Output Strength	Rpd/Rpu	V_{OUT} to V_{SSQ}	Minimum	Nominal	Maximum	Unit
18 Ohms	Rpd	$V_{CCQ} \times 0.2$	8.2	14.0	23.8	ohms
		$V_{CCQ} \times 0.5$	11.7	18.0	24.3	ohms
		$V_{CCQ} \times 0.8$	14.0	23.0	34.5	ohms
	Rpu	$V_{CCQ} \times 0.2$	14.0	23.0	34.5	ohms
		$V_{CCQ} \times 0.5$	11.7	18.0	24.3	ohms
		$V_{CCQ} \times 0.8$	8.2	14.0	23.8	ohms
25 Ohms	Rpd	$V_{CCQ} \times 0.2$	11.4	20.0	32.0	ohms
		$V_{CCQ} \times 0.5$	16.3	25.0	33.7	ohms
		$V_{CCQ} \times 0.8$	20.0	31.0	49.0	ohms
	Rpu	$V_{CCQ} \times 0.2$	20.0	31.0	49.0	ohms
		$V_{CCQ} \times 0.5$	16.3	25.0	33.7	ohms
		$V_{CCQ} \times 0.8$	11.4	20.0	32.0	ohms
35 Ohms	Rpd	$V_{CCQ} \times 0.2$	16.0	27.5	44.5	ohms
		$V_{CCQ} \times 0.5$	22.7	35.0	47.3	ohms
		$V_{CCQ} \times 0.8$	28.0	45.0	69.0	ohms
	Rpu	$V_{CCQ} \times 0.2$	28.0	45.0	69.0	ohms
		$V_{CCQ} \times 0.5$	22.7	35.0	47.3	ohms
		$V_{CCQ} \times 0.8$	16.0	27.5	44.5	ohms



MLC 256Gb to 2Tb Async/Sync NAND Output Drive Impedance

Table 34: Output Drive Strength Impedance Values ($V_{CCQ} = 1.7\text{--}1.95\text{V}$) (Continued)

Output Strength	Rpd/Rpu	V_{OUT} to V_{SSQ}	Minimum	Nominal	Maximum	Unit
50 Ohms	Rpd	$V_{CCQ} \times 0.2$	24.0	40.0	63.0	ohms
		$V_{CCQ} \times 0.5$	32.5	50.0	67.5	ohms
		$V_{CCQ} \times 0.8$	40.0	64.0	98.0	ohms
	Rpu	$V_{CCQ} \times 0.2$	40.0	64.0	98.0	ohms
		$V_{CCQ} \times 0.5$	32.5	50.0	67.5	ohms
		$V_{CCQ} \times 0.8$	24.0	40.0	63.0	ohms

Table 35: Output Drive Strength Conditions ($V_{CCQ} = 2.7\text{--}3.6\text{V}$)

Range	Process	Voltage	Temperature
Minimum	Fast-Fast	3.6V	T_A (MIN)
Nominal	Typical-Typical	3.3V	+25°C
Maximum	Slow-Slow	2.7V	T_A (MAX)

Table 36: Output Drive Strength Impedance Values ($V_{CCQ} = 2.7\text{--}3.6\text{V}$)

Output Strength	Rpd/Rpu	V_{OUT} to V_{SSQ}	Minimum	Nominal	Maximum	Unit
18 Ohms	Rpd	$V_{CCQ} \times 0.2$	7.0	16.2	28.7	ohms
		$V_{CCQ} \times 0.5$	9.0	18.0	36.0	ohms
		$V_{CCQ} \times 0.8$	11.8	21.0	50.0	ohms
	Rpu	$V_{CCQ} \times 0.2$	11.8	21.0	50.0	ohms
		$V_{CCQ} \times 0.5$	9.0	18.0	36.0	ohms
		$V_{CCQ} \times 0.8$	7.0	14.0	28.7	ohms
25 Ohms	Rpd	$V_{CCQ} \times 0.2$	9.3	22.3	40.0	ohms
		$V_{CCQ} \times 0.5$	12.6	25.0	50.0	ohms
		$V_{CCQ} \times 0.8$	16.3	29.0	68.0	ohms
	Rpu	$V_{CCQ} \times 0.2$	16.3	29.0	68.0	ohms
		$V_{CCQ} \times 0.5$	12.6	25.0	50.0	ohms
		$V_{CCQ} \times 0.8$	9.3	19.0	40.0	ohms
35 Ohms	Rpd	$V_{CCQ} \times 0.2$	12.8	32.0	58.0	ohms
		$V_{CCQ} \times 0.5$	18.0	35.0	70.0	ohms
		$V_{CCQ} \times 0.8$	23.0	40.0	95.0	ohms
	Rpu	$V_{CCQ} \times 0.2$	23.0	40.0	95.0	ohms
		$V_{CCQ} \times 0.5$	18.0	35.0	70.0	ohms
		$V_{CCQ} \times 0.8$	12.8	32.0	58.0	ohms



MLC 256Gb to 2Tb Async/Sync NAND Output Drive Impedance

Table 36: Output Drive Strength Impedance Values ($V_{CCQ} = 2.7\text{--}3.6\text{V}$) (Continued)

Output Strength	Rpd/Rpu	V_{OUT} to V_{SSQ}	Minimum	Nominal	Maximum	Unit
50 Ohms	Rpd	$V_{CCQ} \times 0.2$	18.4	45.0	80.0	ohms
		$V_{CCQ} \times 0.5$	25.0	50.0	100.0	ohms
		$V_{CCQ} \times 0.8$	32.0	57.0	136.0	ohms
	Rpu	$V_{CCQ} \times 0.2$	32.0	57.0	136.0	ohms
		$V_{CCQ} \times 0.5$	25.0	50.0	100.0	ohms
		$V_{CCQ} \times 0.8$	18.4	45.0	80.0	ohms

Table 37: Pull-Up and Pull-Down Output Impedance Mismatch for Asynchronous and NV-DDR

Drive Strength	Minimum	Maximum	Unit	Notes
18 Ohms	0	6.3	ohms	1, 2
25 Ohms	0	8.8	ohms	1, 2
35 Ohms	0	12.3	ohms	1, 2
50 Ohms	0	17.5	ohms	1, 2

- Notes:
1. Mismatch is the absolute value between pull-up and pull-down impedances. Both are measured at the same temperature and voltage.
 2. Test conditions: $V_{CCQ} = V_{CCQ}(\text{MIN})$, $V_{OUT} = V_{CCQ} \times 0.5$, $T_A = T_{\text{OPER}}$.

Table 38: Pull-Up and Pull-Down Output Impedance Mismatch for NV-DDR2

Drive Strength	Minimum	Maximum	Unit	Notes
18 Ohms	0	3.2	ohms	1, 2
25 Ohms	0	4.4	ohms	1, 2
35 Ohms	0	6.2	ohms	1, 2
50 Ohms	0	8.8	ohms	1, 2

- Notes:
1. Mismatch is the absolute value between pull-up and pull-down impedances. Both are measured at the same temperature and voltage.
 2. Test conditions: $V_{CCQ} = V_{CCQ}(\text{MIN})$, $V_{OUT} = V_{CCQ} \times 0.5$, $T_A = T_{\text{OPER}}$.



MLC 256Gb to 2Tb Async/Sync NAND AC Overshoot/Undershoot Specifications

AC Overshoot/Undershoot Specifications

The supported AC overshoot and undershoot area depends on the timing mode selected by the host.

Table 39: Aynchronous Overshoot/Undershoot Parameters

Parameter	Timing Mode						Unit
	0	1	2	3	4	5	
Maximum peak amplitude provided for overshoot area	1	1	1	1	1	1	V
Maximum peak amplitude provided for undershoot area	1	1	1	1	1	1	V
Maximum overshoot area above V_{CCQ}	3	3	3	3	3	3	V-ns
Maximum undershoot area below V_{SSQ}	3	3	3	3	3	3	V-ns

Table 40: NV-DDR Overshoot/Undershoot Parameters

Parameter	Timing Mode						Unit
	0	1	2	3	4	5	
Maximum peak amplitude provided for overshoot area	1	1	1	1	1	1	V
Maximum peak amplitude provided for undershoot area	1	1	1	1	1	1	V
Maximum overshoot area above V_{CCQ}	3	3	3	2.25	1.8	1.5	V-ns
Maximum undershoot area below V_{SSQ}	3	3	3	2.25	1.8	1.5	V-ns

Table 41: NV-DDR2 Overshoot/Undershoot Parameters

Parameter	Signals	Timing Mode								Unit
		0	1	2	3	4	5	6	7	
Maximum peak amplitude provided for overshoot area	-	1	1	1	1	1	1	1	1	V
Maximum peak amplitude provided for undershoot area	-	1	1	1	1	1	1	1	1	V
Maximum overshoot area above V_{CCQ}	DQ[7:0], DQS, RE#	3	3	2.25	1.8	1.5	1.1	0.9	0.75	V-ns
	ALE, CLE, WE#	3	3	3	3	3	3	3	3	
Maximum undershoot area below V_{SSQ}	DQ[7:0], DQS, RE#	3	3	2.25	1.8	1.5	1.1	0.9	0.75	V-ns
	ALE, CLE, WE#	3	3	3	3	3	3	3	3	



MLC 256Gb to 2Tb Async/Sync NAND AC Overshoot/Undershoot Specifications

Figure 92: Overshoot

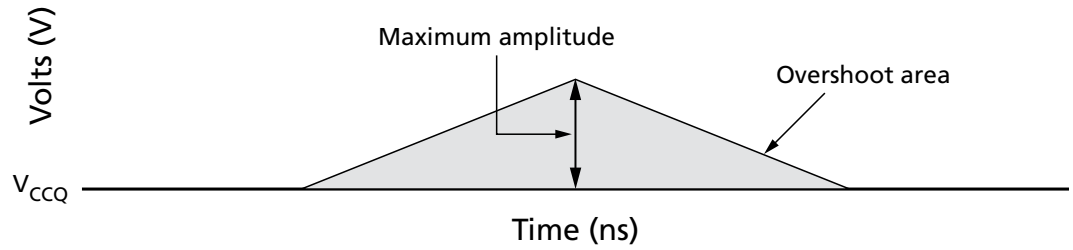
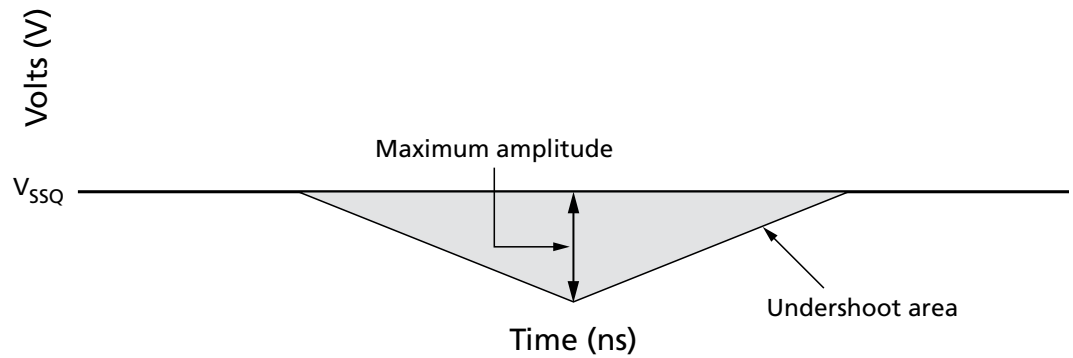


Figure 93: Undershoot





MLC 256Gb to 2Tb Async/Sync NAND Input Slew Rate

Input Slew Rate

Though all AC timing parameters are tested with a nominal input slew rate of 1 V/ns, it is possible to run the device at a slower slew rate. The input slew rates shown below are sampled, and not 100% tested. When using slew rates slower than the minimum values, timing must be derated by the host.

Table 42: Test Conditions for Input Slew Rate

Parameter	Value
Rising edge for setups	$V_{IL(DC)}$ To $V_{IH(AC)}$ for NV-DDR
	The last crossing of $V_{REFQ(DC)}$ and the first crossing of $V_{IH(AC)}$ min for NV-DDR2
Falling edge for setups	$V_{IH(DC)}$ To $V_{IL(AC)}$ for NV-DDR
	The last crossing of $V_{REFQ(DC)}$ and the first crossing of $V_{IL(AC)}$ max for NV-DDR2
Rising edge for holds	$V_{IL(DC)}$ To $V_{IH(AC)}$ for NV-DDR
	The first crossing of $V_{IL(DC)}$ max and the first crossing of $V_{REFQ(DC)}$ for NV-DDR2
Falling edge for holds	$V_{IH(DC)}$ To $V_{IL(AC)}$ for NV-DDR
	The first crossing of $V_{IH(DC)}$ min and the first crossing of $V_{REFQ(DC)}$ for NV-DDR2
Temperature range	T_A

The minimum and maximum input slew rate requirements that the device shall comply with below for NV-DDR operations. If the input slew rate falls below the minimum value, then derating shall be applied.

Table 43: NV-DDR Maximum and Minimum Input Slew Rate

Description	Timing Modes 0-5	Unit
Input slew rate (min)	0.5	V/ns
Input slew rate (max)	4.5	V/ns

Table 44: Input Slew Rate derating for NV-DDR ($V_{CCQ} = 1.7-1.95V$)

Command/ Address and DQ V/ns	CLK/DQS Slew Rate Derating $V_{IH(AC)}/V_{IL(AC)}= 540mV$, $V_{IH(DC)}/V_{IL(DC)}= 360mV$																Unit
	1		0.9		0.8		0.7		0.6		0.5		0.4		0.3		
	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	
1	0	0	0	0	-	-	-	-	-	-	-	-	-	-	-	-	ps
0.9	0	0	0	0	0	0	-	-	-	-	-	-	-	-	-	-	ps
0.8	-	-	0	0	0	0	0	0	-	-	-	-	-	-	-	-	ps
0.7	-	-	-	-	0	0	0	0	0	0	-	-	-	-	-	-	ps
0.6	-	-	-	-	-	-	0	0	0	0	0	0	-	-	-	-	ps
0.5	-	-	-	-	-	-	-	-	0	0	0	0	180	180	-	-	ps



MLC 256Gb to 2Tb Async/Sync NAND Input Slew Rate

Table 44: Input Slew Rate derating for NV-DDR ($V_{CCQ} = 1.7\text{--}1.95\text{V}$) (Continued)

Command/ Address and DQ V/ns	CLK/DQS Slew Rate Derating $V_{IH(AC)}/V_{IL(AC)}= 540mV$, $V_{IH(DC)}/V_{IL(DC)}= 360mV$																Unit
	1		0.9		0.8		0.7		0.6		0.5		0.4		0.3		
	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	
0.4	-	-	-	-	-	-	-	-	-	-	180	180	360	360	660	660	ps
0.3	-	-	-	-	-	-	-	-	-	-	-	-	660	660	920	920	ps

Table 45: Input Slew Rate derating for NV-DDR/NV-DDR2 ($V_{CCQ} = 2.7\text{--}3.6\text{V}$)

Command/ Address and DQ V/ns	CLK/DQS Slew Rate Derating $V_{IH(AC)}/V_{IL(AC)}= 990mV$, $V_{IH(DC)}/V_{IL(DC)}= 660mV$																Unit
	1		0.9		0.8		0.7		0.6		0.5		0.4		0.3		
	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	
1	0	0	0	0	-	-	-	-	-	-	-	-	-	-	-	-	ps
0.9	0	0	0	0	0	0	-	-	-	-	-	-	-	-	-	-	ps
0.8	-	-	0	0	0	0	0	0	-	-	-	-	-	-	-	-	ps
0.7	-	-	-	-	0	0	0	0	0	0	-	-	-	-	-	-	ps
0.6	-	-	-	-	-	-	0	0	0	0	0	0	-	-	-	-	ps
0.5	-	-	-	-	-	-	-	-	0	0	0	0	330	330	-	-	ps
0.4	-	-	-	-	-	-	-	-	-	-	330	330	660	660	1210	1210	ps
0.3	-	-	-	-	-	-	-	-	-	-	-	-	1210	1210	1760	1760	ps

The minimum and maximum input slew rate requirements that the device shall comply with below for NV-DDR2 operations. If the input slew rate falls below the minimum value, then derating shall be applied.

Table 46: NV-DDR2 Maximum and Minimum Input Slew Rate

Description	Single Ended	Differential	Unit
	Timing Modes 0-7	Timing Modes 0-7	
Input slew rate (min)	1.0	2.0	V/ns
Input slew rate (max)	4.5	9.0	V/ns

For DQ signals when used for input, the total data setup time (t_{DS}) and data hold time (t_{DH}) required is calculated by adding a derating value to the t_{DS} and t_{DH} values indicated for the timing mode. To calculate the total data setup time, t_{DS} is incremented by the appropriate Δ_{set} derating value. To calculate the total data hold time, t_{DH} is incremented by the appropriate Δ_{hold} derating value. Table 47 (page 164) provides the derating values when single-ended DQS is used. Table 48 (page 164) provides the derating values when differential DQS (DQS_t/DQS_c) is used.

The setup nominal slew rate for a rising signal is defined as the slew rate between the last crossing of $V_{REFQ(DC)}$ and the first crossing of $V_{IH(AC)}$ min. The setup nominal slew rate for a falling signal is defined as the slew rate between the last crossing of $V_{REFQ(DC)}$ and the first crossing of $V_{IL(AC)}$ max. If the actual signal is always earlier than the nominal slew rate line between the shaded ' $V_{REFQ(DC)}$ to AC region', then the derating value



MLC 256Gb to 2Tb Async/Sync NAND Input Slew Rate

uses the nominal slew rate shown in Figure 94 (page 166). If the actual signal is later than the nominal slew rate line anywhere between shaded ' $V_{REFQ(DC)}$ to AC region', then the derating value uses the slew rate of a tangent line to the actual signal from the AC level to the DC level shown in Figure 95 (page 167).

The hold nominal slew rate for a rising signal is defined as the slew rate between the last crossing of $V_{IL(DC)}$ max and the first crossing of $V_{REFQ(DC)}$. The hold nominal slew rate for a falling signal is defined as the slew rate between the last crossing of $V_{IH(DC)}$ min and the first crossing of $V_{REFQ(DC)}$. If the actual signal is always later than the nominal slew rate line between shaded 'DC to $V_{REFQ(DC)}$ region', then the derating value uses the nominal slew rate shown in Figure 96 (page 168). If the actual signal is earlier than the nominal slew rate line anywhere between the shaded 'DC to $V_{REFQ(DC)}$ region', then the derating value uses the slew rate of a tangent line to the actual signal from the DC level to the $V_{REFQ(DC)}$ level shown in Figure 97 (page 169).

If the tangent line is used for derating, the setup and hold values shall be derated from where the tangent line crosses $V_{REFQ(DC)}$, not the actual signal (refer to Figure 95 (page 167) and Figure 97 (page 169)).

For slew rates not explicitly listed in Table 47 (page 164) and Table 48 (page 164), the derating values should be obtained by linear interpolation. These values are typically not subject to production test; the values are verified by design and characterization.

Table 47: Input Slew Rate derating for NV-DDR2 single-ended ($V_{CCQ} = 1.7-1.95V$)

DQ	DQS Slew Rate Derating $V_{IH(AC)}/V_{IL(AC)}= 250mV$, $V_{IH(DC)}/V_{IL(DC)}= 125mV$																				
	2		1.5		1		0.9		0.8		0.7		0.6		0.5		0.4		0.3		
	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	
V/ns	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	Unit
2	0	0	0	0	0	0	–	–	–	–	–	–	–	–	–	–	–	–	–	–	ps
1.5	0	0	0	0	0	0	14	0	–	–	–	–	–	–	–	–	–	–	–	–	ps
1	0	0	0	0	0	0	14	0	31	0	–	–	–	–	–	–	–	–	–	–	ps
0.9	–	–	14	0	14	0	28	0	45	0	67	0	–	–	–	–	–	–	–	–	ps
0.8	–	–	–	–	31	0	45	0	63	0	85	0	115	0	–	–	–	–	–	–	ps
0.7	–	–	–	–	–	–	67	0	85	0	107	0	137	0	179	0	–	–	–	–	ps
0.6	–	–	–	–	–	–	–	–	115	0	137	0	167	0	208	0	271	0	–	–	ps
0.5	–	–	–	–	–	–	–	–	–	–	179	0	208	0	250	0	313	0	418	0	ps
0.4	–	–	–	–	–	–	–	–	–	–	–	–	271	0	313	0	375	0	480	0	ps
0.3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	418	0	480	0	594	0	ps

Table 48: Input Slew Rate derating for NV-DDR2 differential ($V_{CCQ} = 1.7-1.95V$)

DQ V/ns	DQS_t/DQS_c Slew Rate Derating V _{IH(AC)} /V _{IL(AC)} = 250mV, V _{IH(DC)} /V _{IL(DC)} = 125mV																Unit
	2		1.8		1.6		1.4		1.2		1		0.8		0.6		
	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	
2	0	0	–	–	–	–	–	–	–	–	–	–	–	–	–	–	ps
1.5	0	0	7	0	–	–	–	–	–	–	–	–	–	–	–	–	ps
1	0	0	7	0	16	0	–	–	–	–	–	–	–	–	–	–	ps
0.9	14	0	21	0	30	0	41	0	–	–	–	–	–	–	–	–	ps



MLC 256Gb to 2Tb Async/Sync NAND Input Slew Rate

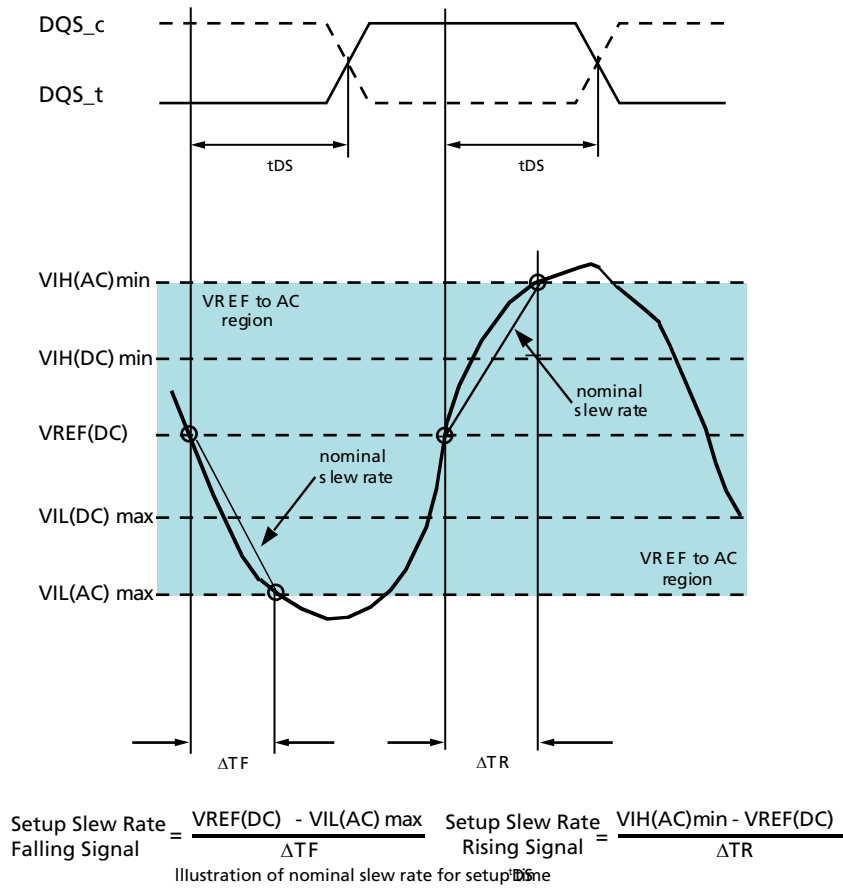
Table 48: Input Slew Rate derating for NV-DDR2 differential ($V_{CCQ} = 1.7\text{--}1.95\text{V}$) (Continued)

DQ V/ns	DQS_t/DQS_c Slew Rate Derating $V_{IH(AC)}/V_{IL(AC)}= 250mV$, $V_{IH(DC)}/V_{IL(DC)}= 125mV$																Unit
	2		1.8		1.6		1.4		1.2		1		0.8		0.6		
	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	set	hold	
0.8	31	0	38	0	47	0	58	0	73	0	–	–	–	–	–	–	ps
0.7	–	–	61	0	69	0	80	0	95	0	116	0	–	–	–	–	ps
0.6	–	–	–	–	99	0	110	0	125	0	146	0	176	0	–	–	ps
0.5	–	–	–	–	–	–	152	0	167	0	188	0	218	0	269	0	ps
0.4	–	–	–	–	–	–	–	–	229	0	250	0	282	0	333	0	ps
0.3	–	–	–	–	–	–	–	–	–	–	355	0	385	0	436	0	ps



MLC 256Gb to 2Tb Async/Sync NAND Input Slew Rate

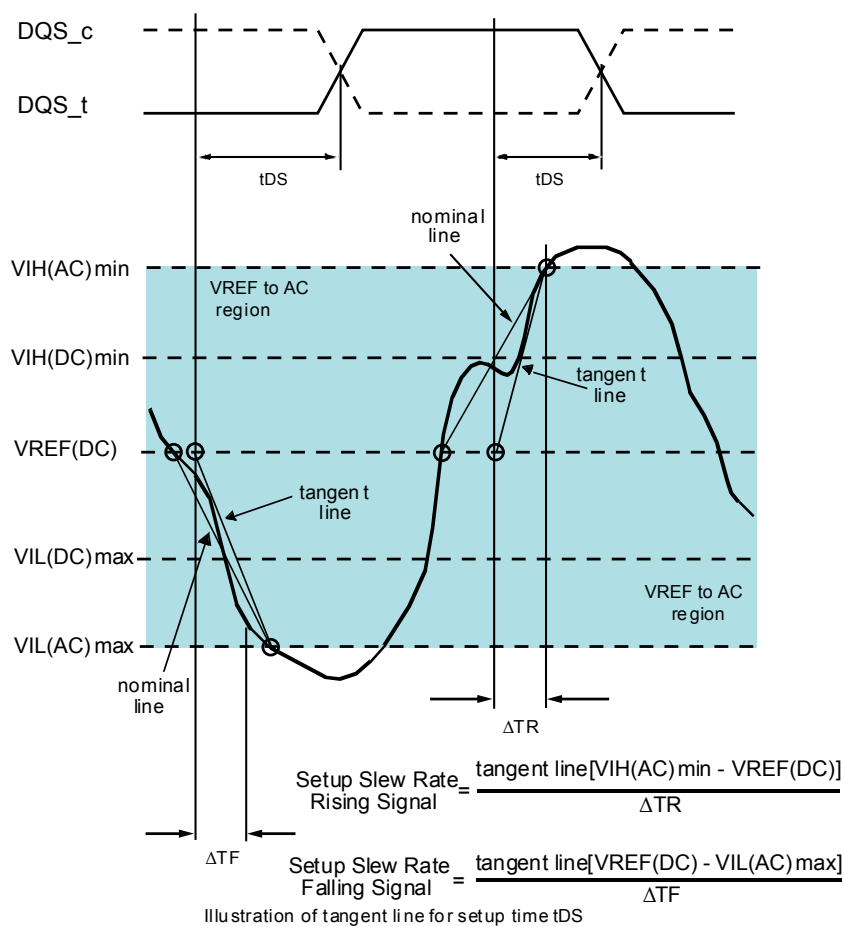
Figure 94: Nominal Slew Rate for Data Setup Time (t_{DS}), NV-DDR2 only





MLC 256Gb to 2Tb Async/Sync NAND Input Slew Rate

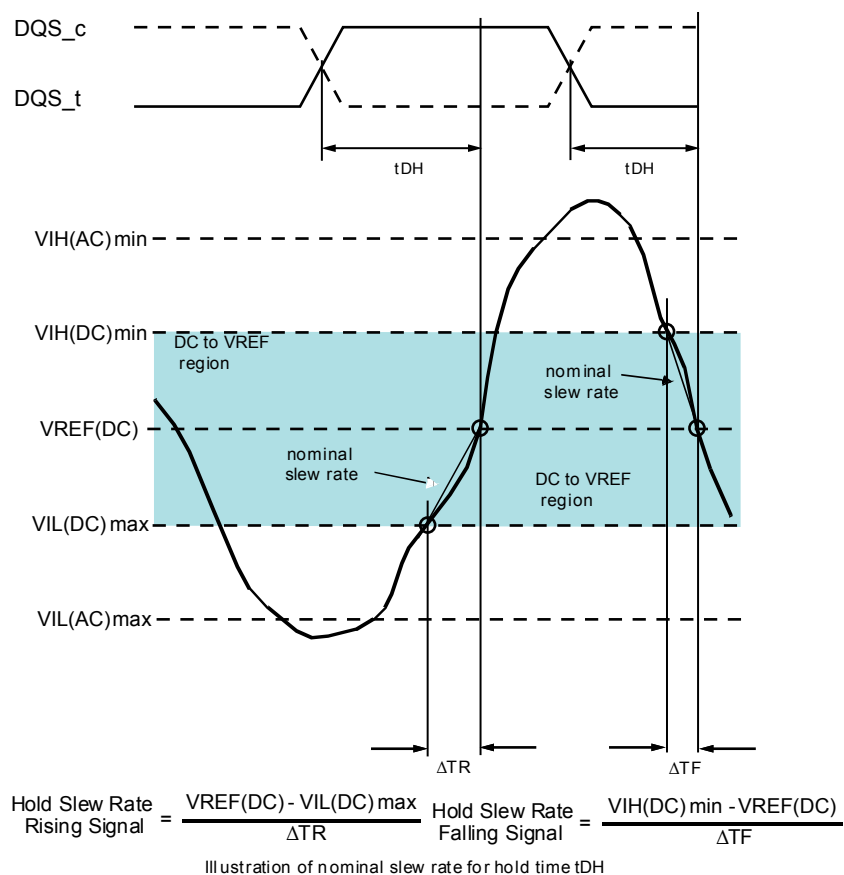
Figure 95: Tangent Line for Data Setup Time (t_{DS}), NV-DDR2 only





MLC 256Gb to 2Tb Async/Sync NAND Input Slew Rate

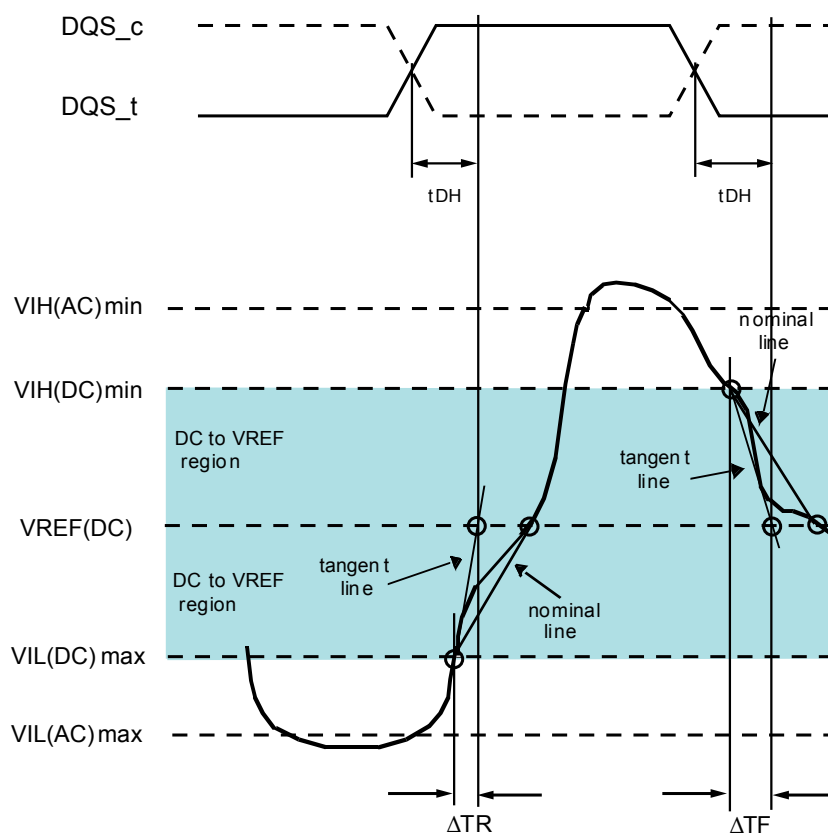
Figure 96: Nominal Slew Rate for Data Hold Time (t_{DH}), NV-DDR2 only





MLC 256Gb to 2Tb Async/Sync NAND Input Slew Rate

Figure 97: Tangent Line for Data Hold Time (t_{DH}), NV-DDR2 only



$$\text{Hold Slew Rate Rising Signal} = \frac{\text{tangent line [} VREF(DC) - VIL(DC)_{max} \text{]}}{\Delta TR}$$

$$\text{Hold Slew Rate Falling Signal} = \frac{\text{tangent line [} VIH(DC)_{min} - VREF(DC) \text{]}}{\Delta TF}$$

Illustration of tangent line for for hold time t_{DH}



MLC 256Gb to 2Tb Async/Sync NAND Output Slew Rate

Output Slew Rate

The output slew rate is tested using the following setup with only one die per DQ channel.

Table 49: Test Conditions for Output Slew Rate

Parameter	Asynchronous/NV-DDR Interface	NV-DDR2 Single-Ended	NV-DDR2 Differential
$V_{OL(DC)}$	$0.4 \times V_{CCQ}$	-	-
$V_{OH(DC)}$	$0.6 \times V_{CCQ}$	-	-
$V_{OL(AC)}^1$	$0.3 \times V_{CCQ}$	$V_{TT} - (V_{CCQ} \times 0.15)$	-
$V_{OH(AC)}^1$	$0.7 \times V_{CCQ}$	$V_{TT} + (V_{CCQ} \times 0.15)$	-
$V_{OLDiff(AC)}$	-	-	$-0.3 \times V_{CCQ}$
$V_{OHDiff(AC)}$	-	-	$0.3 \times V_{CCQ}$
Rising edge (t_{RISE})	$V_{OL(DC)}$ to $V_{OH(AC)}$	$V_{OL(AC)}$ to $V_{OH(AC)}$	-
Falling edge (t_{FALL})	$V_{OH(DC)}$ to $V_{OL(AC)}$	$V_{OH(AC)}$ to $V_{OL(AC)}$	-
Differential rising edge ($t_{RISEdiff}$)	-	-	$V_{OLDiff(AC)}$ to $V_{OHDiff(AC)}$
Differential falling edge ($t_{FALLdiff}$)	-	-	$V_{OHDiff(AC)}$ to $V_{OLDiff(AC)}$
Output slew rate rising edge	$[V_{OH(AC)} - V_{OL(DC)}] / t_{RISE}$	$[V_{OH(AC)} - V_{OL(AC)}] / t_{RISE}$	$[V_{OHDiff(AC)} - V_{OLDiff(AC)}] / t_{RISEdiff}$
Output slew rate falling edge	$[V_{OH(DC)} - V_{OL(AC)}] / t_{FALL}$	$[V_{OH(AC)} - V_{OL(AC)}] / t_{FALL}$	$[V_{OHDiff(AC)} - V_{OLDiff(AC)}] / t_{FALLdiff}$
Output reference load ¹	50 Ohm to V_{TT}	50 Ohm to V_{TT}	50 Ohm to V_{TT}
Temperature range	T_A	T_A	T_A

Note: 1. V_{TT} is $0.5 \times V_{CCQ}$.

Table 50: Output Slew Rate for Asynchronous and NV-DDR ($V_{CCQ} = 2.7\text{--}3.6V$)

Output Drive Strength	Min	Max	Unit
18 Ohms	1.5	10	V/ns
25 Ohms	1.5	9	V/ns
35 Ohms	1.2	7	V/ns
50 Ohms	1	5.5	V/ns

Table 51: Output Slew Rate for Single-Ended NV-DDR or NV-DDR2 ($V_{CCQ} = 1.7\text{--}1.95V$)

Output Drive Strength	Min	Max	Unit
18 Ohms	1	5.5	V/ns
25 Ohms	0.85	5	V/ns
35 Ohms	0.75	4	V/ns



MLC 256Gb to 2Tb Async/Sync NAND Output Slew Rate

Table 51: Output Slew Rate for Single-Ended NV-DDR or NV-DDR2 ($V_{CCQ} = 1.7-1.95V$) (Continued)

Output Drive Strength	Min	Max	Unit
50 Ohms	0.6	4	V/ns

Table 52: Output Slew Rate for Differential NV-DDR2 ($V_{CCQ} = 1.7-1.95$)

Output Drive Strength	Min	Max	Unit
18 Ohms	2.0	11.0	V/ns
25 Ohms	1.7	10.0	V/ns
35 Ohms	1.5	8.0	V/ns
50 Ohms	1.2	8.0	V/ns

Table 53: Output Slew Rate Matching Ratio for NV-DDR2

Drive Strength	Minimum	Maximum
Output Slew Rate matching ratio (pull-up to pull-down)	0.7	1.4

- Notes:
1. The output slew rate mismatch is determined by the ratio of fast slew rate and slow slew rate. If the rising edge is faster than the falling edge, then divide the rising slew rate by the falling slew rate. If the falling edge is faster than the rising edge, then divide the falling slew rate by the rising slew rate.
 2. The output slew rate mismatch is verified by design and characterization; it may not be subject to production testing.



MLC 256Gb to 2Tb Async/Sync NAND Power Cycle Requirements

Power Cycle Requirements

Upon power-down the NAND device requires a maximum voltage and minimum time that the host must hold V_{CC} and V_{CCQ} below the voltage prior to power-on.

Table 54: Power Cycle Requirements

Parameter	Value	Unit
Maximum V_{CC}/V_{CCQ}	100	mV
Minimum time below maximum voltage	100	ns



Electrical Specifications

Stresses greater than those listed can cause permanent damage to the device. This is a stress rating only, and functional operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not guaranteed. Exposure to absolute maximum rating conditions for extended periods can affect reliability.

Table 55: Absolute Maximum Ratings by Device

Parameter	Symbol	Min ¹	Max ¹	Unit
Voltage input	V _{IN}	-0.6	4.6	V
V _{CC} supply voltage	V _{CC}	-0.6	4.6	V
V _{CCQ} supply voltage	V _{CCQ}	-0.6	4.6	V
V _{PP} supply voltage	V _{PP}	-0.6	16	V
V _{REFQ} supply voltage	V _{REFQ}	-0.6	4.6	V
Storage temperature	T _{STG}	-65	+150	°C

Note: 1. Voltage on any pin relative to V_{SS}.

Table 56: Recommended Operating Conditions

Parameter		Symbol	Min	Typ	Max	Unit
Operating temperature	Commercial	T _A	0	–	70	°C
	Industrial		-40	–	+85	
V _{CC} supply voltage		V _{CC}	2.7	3.3	3.6	V
V _{CCQ} supply voltage		V _{CCQ}	2.7	3.3	3.6	V
			1.7	1.8	1.95	
V _{PP} supply voltage		V _{PP}	10.8	12.0	13.2	V
V _{REFQ} supply voltage		V _{REFQ}	0.49 x V _{CCQ}	0.5 x V _{CCQ}	0.51 x V _{CCQ}	V
V _{SS} ground voltage		V _{SS}	0	0	0	V

Table 57: Valid Blocks per LUN

Parameter	Symbol	Min	Max	Unit	Notes
Valid block number	NVB	1998	2096	Blocks	1

Note: 1. Invalid blocks are blocks that contain one or more bad bits beyond ECC. The device may contain bad blocks upon shipment. Additional bad blocks may develop over time; however, the total number of available blocks will not drop below NVB during the endurance life of the device. Do not erase or program blocks marked invalid from the factory.



MLC 256Gb to 2Tb Async/Sync NAND Electrical Specifications

Table 58: Capacitance: 152-Ball BGA Package

Description	Sym- bol	Single Die/Dual Die Package			Quad Die Package			Octal Die Package			16 Die Package			Unit	Note s
		Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	Min	Typ	Max		
Input capacitance (CLK/WE#)	C _{CK}	3.0	4.0	5.0	4.9	6.4	7.9	9.4	11.4	13.4	15.2	19.2	23.2	pF	3
Input capacitance (ALE, CLE)	C _{IN}	3.0	4.0	5.0	4.9	6.4	7.9	9.4	11.4	13.4	15.1	19.1	23.1	pF	3
Input/output capacitance (DQ[7:0], DQS/DQS_t, DQS_c, RE#/RE_t, RE_c, W/R#)	C _{IO}	3.7	4.7	5.7	6.6	8.1	9.6	13.0	15.0	16.9	24.4	28.4	32.4	pF	3
Input capacitance (CE#, WP#)	C _{OTHER}	–	–	5.0	–	–	7.9	–	–	13.0	–	–	23.0	pF	
Delta input capacitance	DC _{IN}	–	–	1.4	–	–	1.7	–	–	2	–	–	4	pF	
Delta input/output capacitance	DC _{IO}	–	–	1.4	–	–	1.7	–	–	2	–	–	4	pF	

- Notes:
1. Verified in device characterization; not 100% tested.
 2. Test conditions: T_A = 25°C, f = 100 MHz, V_{IN} = 0V.
 3. Values for C_{CK}, C_{IN} and C_{IO} (TYP) are estimates.

Table 59: Test Conditions

Parameter	Asynchronous and NV-DDR	NV-DDR2 single-ended	NV-DDR2 differential	Notes
Rising input transition	V _{IL(DC)} to V _{IH(AC)}	V _{IL(DC)} to V _{IH(AC)}	V _{ILdiff(DC)} max to V _{IHdiff(AC)} min	1
Falling input transition	V _{IH(DC)} to V _{IL(AC)}	V _{IH(DC)} to V _{IL(AC)}	V _{IHdiff(DC)} max to V _{ILdiff(AC)} max	1
Input rise and fall slew rates	1 V/ns	1 V/ns	2 V/ns	–
Input timing levels	V _{CCQ} /2	V _{REFQ}	cross-point	–
Output timing levels	V _{CCQ} /2	V _{TT}	cross-point	
Output load: Nominal output drive strength	50 Ohm to V _{TT}	50 Ohm to V _{TT}	50 Ohm to V _{TT}	2, 3, 4

- Notes:
1. The receiver will effectively switch as a result of the signal crossing the AC input level; it will remain in that status as long as the signal does not ring back above (below) the DC input LOW (HIGH) level.
 2. Transmission line delay is assumed to be very small.
 3. This test setup applies to all package configurations.
 4. V_{TT} is 0.5 x V_{CCQ}.



MLC 256Gb to 2Tb Async/Sync NAND Electrical Specifications – DC Characteristics and Operating Conditions (Asynchronous)

Electrical Specifications – DC Characteristics and Operating Conditions (Asynchronous)

Table 60: DC Characteristics and Operating Conditions (Asynchronous Interface)

Parameter	Conditions	Symbol	Min ¹	Typ ¹	Max ¹	Unit
Array read current (active)	–	I _{CC1_A}	–	28	50	mA
Array program current (active)	–	I _{CC2_A}	–	37	50	mA
Erase current (active)	–	I _{CC3_A}	–	25	50	mA
I/O burst read current	t _{RC} = t _{RC} (MIN); I _{OUT} = 0mA	I _{CC4R_A}	–	9	12	mA
		I _{CCQ4R_A}	–	5	10	mA
I/O burst write current	t _{WC} = t _{WC} (MIN)	I _{CC4W_A}	–	10	13	mA
		I _{CCQ4W_A}	–	1	3	mA
Bus idle current	–	I _{CC5_A}	–	5	7	mA
Current during first RESET command after power-on	–	I _{CC6}	–	–	10	mA
V _{PP} current (active)	V _{PP} is supplied and enabled during I _{CC1_A} , I _{CC2_A} , or I _{CC3_A}	I _{PPA}	–	–	5	mA
V _{PP} current (idle)	V _{PP} is supplied and not enabled	I _{PPI}	–	–	10	μA
Standby current - V _{CC}	CE# = V _{CCQ} - 0.2V; WP# = 0V/V _{CCQ}	I _{SB}	–	20	75	μA
Standby current - V _{CCQ}	CE# = V _{CCQ} - 0.2V; WP# = 0V/V _{CCQ}	I _{SBQ}	–	3	10	μA
Staggered power-up current	t _{RISE} = 1ms; C _{LINE} = 0.1μF	I _{ST}	–	–	10	mA

Note: 1. All values are per die (LUN) unless otherwise specified.

Electrical Specifications – DC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 61: DC Characteristics and Operating Conditions (NV-DDR and NV-DDR2 Interface) 3.3V V_{CCQ}

Parameter	Conditions	Symbol	Min ¹	Typ ¹	Max ¹	Unit
Array read current (active)	CE# = V _{IL} ; t _{CK} = t _{CK} (MIN) NV-DDR	I _{CC1_S}	–	28	50	mA
Array program current (active)	t _{CK} = t _{CK} (MIN) NV-DDR	I _{CC2_S}	–	37	50	mA
Erase current (active)	t _{CK} = t _{CK} (MIN) NV-DDR	I _{CC3_S}	–	25	50	mA
I/O burst read current	t _{CK} = t _{CK} (MIN) NV-DDR; t _{RC} = t _{RC} (MIN) NV-DDR2; I _{OUT} = 0mA	I _{CC4R_S}	–	18 ²	23 ²	mA
		I _{CCQ4R_S}	–	25 ²	30 ²	mA
I/O burst write current	t _{CK} = t _{CK} (MIN) NV-DDR; t _{DSC} = t _{DSC} (MIN) NV-DDR2	I _{CC4W_S}	–	16 ²	20 ²	mA
		I _{CCQ4W_S}	–	1	3	mA
Bus idle current	t _{CK} = t _{CK} (MIN) NV-DDR	I _{CC5_S}	–	6	10	mA



MLC 256Gb to 2Tb Async/Sync NAND Electrical Specifications – DC Characteristics and Operating Conditions (V_{CCQ})

Table 61: DC Characteristics and Operating Conditions (NV-DDR and NV-DDR2 Interface) 3.3V V_{CCQ} (Continued)

Parameter	Conditions	Symbol	Min ¹	Typ ¹	Max ¹	Unit
V _{PP} current (active)	V _{PP} is supplied and enabled during I _{CC1_S} , I _{CC2_S} , or I _{CC3_S}	I _{PPA}	–	–	5	mA
V _{PP} current (idle)	V _{PP} is supplied and not enabled	I _{PPI}	–	–	10	μA
Standby current - V _{CC}	CE# = V _{CCQ} - 0.2V; WP# = 0V/V _{CCQ}	I _{SB}	–	20	75	μA
Standby Current - V _{CCQ}	CE# = V _{CCQ} - 0.2V; WP# = 0V/V _{CCQ}	I _{SBQ}	–	3	10	μA

Notes: 1. All values are per die (LUN) unless otherwise specified.
2. For speeds up to 166MT/s.

Table 62: DC Characteristics and Operating Conditions (NV-DDR and NV-DDR2 Interface) 1.8V V_{CCQ}

Parameter	Conditions	Symbol	Min ¹	Typ ¹	Max ¹	Unit
Array read current (active)	CE# = V _{IL} ; t _{CK} = t _{CK} (MIN) NV-DDR	I _{CC1_S}	–	28	50	mA
Array program current (active)	t _{CK} = t _{CK} (MIN) NV-DDR	I _{CC2_S}	–	37	50	mA
Erase current (active)	t _{CK} = t _{CK} (MIN) NV-DDR	I _{CC3_S}	–	25	50	mA
I/O burst read current	t _{CK} = t _{CK} (MIN) NV-DDR; t _{RC} = t _{RC} (MIN) NV-DDR2; I _{OUT} = 0mA	I _{CC4R_S}	–	18 ²	23 ²	mA
		I _{CCQ4R_S}	–	15 ²	19 ²	mA
I/O burst write current	t _{CK} = t _{CK} (MIN) NV-DDR; t _{DSC} = t _{DSC} (MIN) NV-DDR2	I _{CC4W_S}	–	16 ²	20 ²	mA
		I _{CCQ4W_S}	–	1	3	mA
Bus idle current	t _{CK} = t _{CK} (MIN) NV-DDR	I _{CC5_S}	–	6	10	mA
V _{PP} current (active)	V _{PP} is supplied and enabled during I _{CC1_S} , I _{CC2_S} , or I _{CC3_S}	I _{PPA}	–	–	5	mA
V _{PP} current (idle)	V _{PP} is supplied and not enabled	I _{PPI}	–	–	10	μA
Standby current - V _{CC}	CE# = V _{CCQ} - 0.2V; WP# = 0V/V _{CCQ}	I _{SB}	–	20	75	μA
Standby Current - V _{CCQ}	CE# = V _{CCQ} - 0.2V; WP# = 0V/V _{CCQ}	I _{SBQ}	–	3	10	μA

Notes: 1. All values are per die (LUN) unless otherwise specified.
2. For speeds up to 166MT/s.

Electrical Specifications – DC Characteristics and Operating Conditions (V_{CCQ})



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Electrical Specifications – DC Characteristics and Operating Conditions (V_{CCQ})

Table 63: Asynchronous/NV-DDR DC Characteristics and Operating Conditions (3.3V V_{CCQ})

Parameter	Condition	Symbol	Min	Typ	Max	Unit	Notes
AC input high voltage	CE#, DQ[7:0], DQS, ALE, CLE, CLK (WE#), W/R# (RE#), WP#	$V_{IH(AC)}$	$0.8 \times V_{CCQ}$	–	$V_{CCQ} + 0.3$	V	
AC input low voltage		$V_{IL(AC)}$	–0.3	–	$0.2 \times V_{CCQ}$	V	
DC input high voltage	DQ[7:0], DQS, ALE, CLE, CLK (WE#), W/R# (RE#)	$V_{IH(DC)}$	$0.7 \times V_{CCQ}$	–	$V_{CCQ} + 0.3$	V	
DC input low voltage		$V_{IL(DC)}$	–0.3	–	$0.3 \times V_{CCQ}$	V	
Input leakage current	Any input $V_{IN} = 0V$ to V_{CCQ} (all other pins under test = 0V)	I_{LI}	–	–	± 10	μA	1
Output leakage current	DQ are disabled; $V_{OUT} = 0V$ to V_{CCQ}	I_{LO}	–	–	± 10	μA	1
Output low current (R/B#)	$V_{OL} = 0.4V$	$I_{OL} (R/B\#)$	8	10	–	mA	2

- Notes:
1. All leakage currents are per die (LUN). For example, if a die (LUN) had a leakage current of $\pm 10\mu A$ that would mean two die (LUNs) have a maximum leakage current of $\pm 20\mu A$ and four die (LUNs) have a maximum leakage current of $\pm 40\mu A$.
 2. DC characteristics may need to be relaxed if R/B# pull-down strength is not set to full strength. See Table 24 (page 107) for additional details.

Table 64: Asynchronous/NV-DDR DC Characteristics and Operating Conditions (1.8V V_{CCQ})

Parameter	Condition	Symbol	Min	Typ	Max	Unit	Notes
AC input high voltage	CE#, DQ[7:0], DQS, ALE, CLE, CLK (WE#), W/R# (RE#), WP#	$V_{IH(AC)}$	$0.8 \times V_{CCQ}$	–	$V_{CCQ} + 0.3$	V	
AC input low voltage		$V_{IL(AC)}$	–0.3	–	$0.2 \times V_{CCQ}$	V	
DC input high voltage	CE#, DQ[7:0], DQS, ALE, CLE, CLK (WE#), W/R# (RE#), WP#	$V_{IH(DC)}$	$0.7 \times V_{CCQ}$	–	$V_{CCQ} + 0.3$	V	
DC input low voltage		$V_{IL(DC)}$	–0.3	–	$0.3 \times V_{CCQ}$	V	
Input leakage current	Any input $V_{IN} = 0V$ to V_{CCQ} (all other pins under test = 0V)	I_{LI}	–	–	± 10	μA	1
Output leakage current	DQ are disabled; $V_{out} = 0V$ to V_{CCQ}	I_{LO}	–	–	± 10	μA	1
Output low current (R/B#)	$V_{OL} = 0.2V$	$I_{OL} (R/B\#)$	3	4	–	mA	2

- Notes:
1. All leakage currents are per die (LUN). For example, if a die (LUN) had a leakage current of $\pm 10\mu A$ that would mean two die (LUNs) have a maximum leakage current of $\pm 20\mu A$ and four die (LUNs) have a maximum leakage current of $\pm 40\mu A$.
 2. DC characteristics may need to be relaxed if R/B# pull-down strength is not set to full strength. See Table 24 (page 107) for additional details.

Table 65: NV-DDR2 DC Characteristics and Operating Conditions for Single-Ended signals (1.8V V_{CCQ})

Parameter	Condition	Symbol	Min	Typ	Max	Unit	Notes
AC input high voltage	DQ[7:0], DQS, ALE, CLE, WE#, RE#	$V_{IH(AC)}$	$V_{REFQ} + 0.250$	–	–	V	
AC input low voltage		$V_{IL(AC)}$	–	–	$V_{REFQ} - 0.250$	V	



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Electrical Specifications – DC Characteristics and Operating Conditions (V_{CCQ})

Table 65: NV-DDR2 DC Characteristics and Operating Conditions for Single-Ended signals (1.8V V_{CCQ}) (Continued)

Parameter	Condition	Symbol	Min	Typ	Max	Unit	Notes
AC input high voltage	CE#, WP#	$V_{IH(AC)}$	$0.8 \times V_{CCQ}$	–	$V_{CCQ} + 0.3$	V	
AC input low voltage		$V_{IL(AC)}$	–0.3	–	$0.2 \times V_{CCQ}$	V	
DC input high voltage	DQ[7:0], DQS, ALE, CLE, WE#, RE#	$V_{IH(DC)}$	$V_{REFQ} + 0.125$	–	$V_{CCQ} + 0.3$	V	2
DC input low voltage		$V_{IL(DC)}$	–0.3	–	$V_{REFQ} - 0.125$	V	2
DC input high voltage	CE#, WP#	$V_{IH(DC)}$	$0.7 \times V_{CCQ}$	–	$V_{CCQ} + 0.3$	V	
DC input low voltage		$V_{IL(DC)}$	–0.3	–	$0.3 \times V_{CCQ}$	V	
Input leakage current	Any input $V_{IN} = 0V$ to V_{CCQ} (all other pins under test = 0V)	I_{LI}	–	–	± 10	μA	1
Output leakage current	DQ are disabled; $V_{out} = 0V$ to V_{CCQ}	I_{LO}	–	–	± 10	μA	1
Output low current (R/B#)	$V_{OL} = 0.2V$	$I_{OL} (R/B\#)$	3	4	–	mA	3
V_{REFQ} leakage current	$V_{REFQ} = V_{CCQ}/2$ (all other pins not under test = 0V)	I_{VREFQ}	–	–	± 60	nA	

- Notes:
1. All leakage currents are per die (LUN). For example, if a die (LUN) had a leakage current of $\pm 10\mu A$ that would mean two die (LUNs) have a maximum leakage current of $\pm 20\mu A$ and four die (LUNs) have a maximum leakage current of $\pm 40\mu A$.
 2. These values are not defined. However, the single-ended signals (RE_t, RE_c, DQS_t, and DQS_c) need to be within the respective limits [$V_{IH(DC)}$ Max, $V_{IL(DC)}$ Min] for single-ended signals as well as the limitations for overshoot and undershoot.
 3. DC characteristics may need to be relaxed if R/B# pull-down strength is not set to full strength. See Table 24 (page 107) for additional details.

Table 66: NV-DDR2 DC Characteristics and Operating Conditions for Differential signals(1.8V V_{CCQ})

Parameter	Condition	Symbol	Min	Typ	Max	Unit	Notes
Differential AC input high voltage	DQS_t, DSQ_c, RE_t, RE_c	$V_{IHdiff(AC)}$	$2 \times [V_{IH(DC)} - V_{REFQ}]$	–	See Note	V	2
Differential AC input low voltage		$V_{ILdiff(AC)}$	See Note	–	$2 \times [V_{IL(AC)} - V_{REFQ}]$	V	2
Differential DC input high voltage	DQS_t, DSQ_c, RE_t, RE_c	$V_{IHdiff(DC)}$	$2 \times [V_{IH(DC)} - V_{REFQ}]$	–	See Note	V	2
Differential DC input low voltage		$V_{ILdiff(DC)}$	See Note	–	$2 \times [V_{IL(DC)} - V_{REFQ}]$	V	2
Input leakage current	Any input $V_{IN} = 0V$ to V_{CCQ} (all other pins under test = 0V)	I_{LI}	–	–	± 10	μA	1
Output leakage current	DQ are disabled; $V_{out} = 0V$ to V_{CCQ}	I_{LO}	–	–	± 10	μA	1



MLC 256Gb to 2Tb Async/Sync NAND Electrical Specifications – DC Characteristics and Operating Conditions (V_{CCQ})

Table 66: NV-DDR2 DC Characteristics and Operating Conditions for Differential signals(1.8V V_{CCQ}) (Continued)

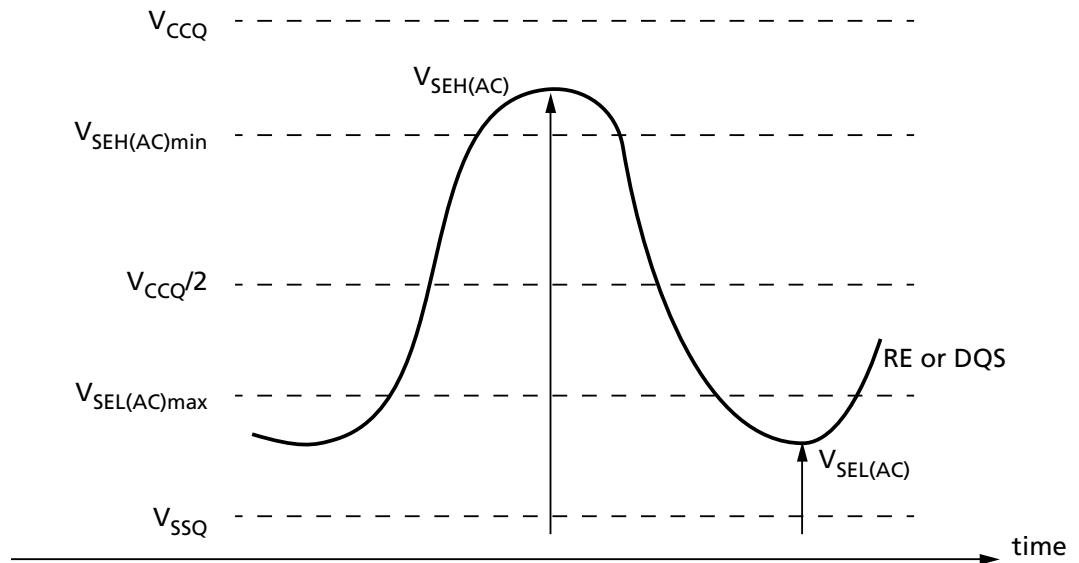
Parameter	Condition	Symbol	Min	Typ	Max	Unit	Notes
Output low current (R/B#)	$V_{OL} = 0.2V$	I_{OL} (R/B#)	3	4	–	mA	3
V_{REFQ} leakage current	$V_{REFQ} = V_{CCQ}/2$ (all other pins not under test = 0V)	I_{VREFQ}	–	–	± 60	nA	

- Notes:
1. All leakage currents are per die (LUN). For example, if a die (LUN) had a leakage current of $\pm 10\mu A$ that would mean two die (LUNs) have a maximum leakage current of $\pm 20\mu A$ and four die (LUNs) have a maximum leakage current of $\pm 40\mu A$.
 2. These values are not defined. However, the single-ended signals (RE_t, RE_c, DQS_t, and DQS_c) need to be within the respective limits [$V_{IH(DC)}$ Max, $V_{IL(DC)}$ Min] for single-ended signals as well as the limitations for overshoot and undershoot.
 3. DC characteristics may need to be relaxed if R/B# pull-down strength is not set to full strength. See Table 24 (page 107) for additional details.

Single-Ended Requirements for Differential signals

Each individual component of a differential signal (RE_t, RE_c, DQS_t, or DQS_c) shall comply with requirements for single-ended signals. RE_t and RE_c shall meet $V_{SEH(AC)}$ Min / $V_{SEL(AC)}$ Max in every half-cycle. DQS_t and DQS_c shall meet $V_{SEH(AC)}$ Min / $V_{SEL(AC)}$ Max in every half-cycle preceding and following a valid transition.

Figure 98: Single-Ended requirements for Differential Signals



While control (e.g., ALE, CLE) and DQ signal requirements are with respect to V_{REFQ} , the single-ended components of differential signals have a requirement with respect to $V_{CCQ}/2$; this is nominally the same. The transition of single-ended signals through the AC-levels is used to measure setup time. For single-ended components of differential signals the requirement to reach $V_{SEL(AC)}$ Max, $V_{SEH(AC)}$ Min has no bearing on timing, but adds a restriction on the common mode characteristics of these signals.



MLC 256Gb to 2Tb Async/Sync NAND Electrical Specifications – AC Characteristics and Operating Conditions (Asynchronous)

Table 67: Single-Ended Levels for RE_t, RE_c, DQS_t, DQS_c (1.8V V_{CCQ})

Parameter	Symbol	Min	Max	Unit	Notes
Single-Ended high level	V _{SEH(AC)}	V _{CCQ} /2 + 0.250	See note	V	1
Single-Ended low level	V _{SEL(AC)}	See note	V _{CCQ} /2 - 0.250	V	1

Note: 1. These values are not defined. However, the single-ended signals (RE_t, RE_c, DQS_t, and DQS_c) need to be within the respective limits [V_{IH(DC)} Max, V_{IL(DC)} Min] for single-ended signals as well as the limitations for overshoot and undershoot.

Table 68: Differential AC Input/Output Parameters

Parameter	Symbol	Min	Max	Unit	Notes
AC differential input cross-point voltage relative to V _{CCQ} / 2	V _{IX(AC)}	0.5 x V _{CCQ} - 0.175	0.5 x V _{CCQ} + 0.175	V	1
AC differential output cross-point voltage	V _{OX(AC)}	0.5 x V _{CCQ} - 0.2	0.5 x V _{CCQ} + 0.2	V	2, 3, 4

- Notes:
1. The typical value of V_{IX(AC)} is expected to be 0.5 x V_{CCQ} of the transmitting device. V_{IX(AC)} is expected to track variations in V_{CCQ}. V_{IX(AC)} indicates the voltage at which differential input signals shall cross.
 2. The typical value of V_{OX(AC)} is expected to be 0.5 x V_{CCQ} of the transmitting device. V_{OX(AC)} is expected to track variations in V_{CCQ}. V_{OX(AC)} indicates the voltage at which differential input signals shall cross.
 3. V_{OX(AC)} is measured with ½ DQ signals per data byte driving logic HIGH and ½ DQ signals per data byte driving logic LOW.
 4. V_{OX(AC)} is verified by design and characterization; it may not be subject to production testing.

Electrical Specifications – AC Characteristics and Operating Conditions (Asynchronous)

Table 69: AC Characteristics: Asynchronous Command, Address, and Data

Parameter	Symbol	Mode 0		Mode 1		Mode 2		Mode 3		Mode 4		Mode 5		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Clock period		100		50		35		30		25		20		ns	
Frequency		≈10		≈20		≈28		≈33		≈40		≈50		MHz	
ALE to data start	^t ADL	200	–	100	–	100	–	100	–	70	–	70	–	ns	1
ALE hold time	^t ALH	20	–	10	–	10	–	5	–	5	–	5	–	ns	
ALE setup time	^t ALS	50	–	25	–	15	–	10	–	10	–	10	–	ns	
ALE to RE# delay	^t AR	25	–	10	–	10	–	10	–	10	–	10	–	ns	
CE# access time	^t CEA	–	100	–	45	–	30	–	25	–	25	–	25	ns	
CE# HIGH hold time prior to VOLUME SELECT (E1h)	^t CEH	20	–	20	–	20	–	20	–	20	–	20	–	ns	
CE# hold time	^t CH	20	–	10	–	10	–	5	–	5	–	5	–	ns	



MLC 256Gb to 2Tb Async/Sync NAND

Electrical Specifications – AC Characteristics and Operating Conditions (Asynchronous)

Table 69: AC Characteristics: Asynchronous Command, Address, and Data (Continued)

Parameter	Symbol	Mode 0		Mode 1		Mode 2		Mode 3		Mode 4		Mode 5		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Delay before CE# HIGH for any Volume after a Volume is selected	t_{CEVDLY}	50	–	50	–	50	–	50	–	50	–	50	–	ns	
CE# HIGH to output High-Z	t_{CHZ}	–	100	–	50	–	50	–	50	–	30	–	30	ns	2
CLE hold time	t_{CLH}	20	–	10	–	10	–	5	–	5	–	5	–	ns	
CLE to RE# delay	t_{CLR}	20	–	10	–	10	–	10	–	10	–	10	–	ns	
CLE setup time	t_{CLS}	50	–	25	–	15	–	10	–	10	–	10	–	ns	
CE# HIGH to output hold	t_{COH}	0	–	15	–	15	–	15	–	15	–	15	–	ns	
CE# setup time	t_{CS}	70	–	35	–	25	–	25	–	20	–	15	–	ns	
CE# setup time for data input after CE# has been HIGH for > 1 μ s	t_{CS3}	100	–	100	–	100	–	100	–	100	–	100	–	ns	
Data hold time	t_{DH}	20	–	10	–	5	–	5	–	5	–	5	–	ns	
Data setup time	t_{DS}	40	–	20	–	15	–	10	–	10	–	7	–	ns	
ENi LOW until any issued command is ignored	t_{ENi}	–	15	–	15	–	15	–	15	–	15	–	15	ns	
CE# LOW until ENo LOW	t_{ENo}	–	50	–	50	–	50	–	50	–	50	–	50	ns	
Output High-Z to RE# LOW	t_{IR}	10	–	0	–	0	–	0	–	0	–	0	–	ns	
RE# cycle time	t_{RC}	100	–	50	–	35	–	30	–	25	–	20	–	ns	
RE# access time	t_{REA}	–	40	–	30	–	25	–	20	–	20	–	16	ns	3
RE# HIGH hold time	t_{REH}	30	–	15	–	15	–	10	–	10	–	7	–	ns	3
RE# HIGH to output hold	t_{RHOH}	0	–	15	–	15	–	15	–	15	–	15	–	ns	3
RE# HIGH to WE# LOW	t_{RHW}	200	–	100	–	100	–	100	–	100	–	100	–	ns	
RE# HIGH to output High-Z	t_{RHZ}	–	200	–	100	–	100	–	100	–	100	–	100	ns	2, 3
RE# LOW to output hold	t_{RLOH}	0	–	0	–	0	–	0	–	5	–	5	–	ns	3
RE# pulse width	t_{RP}	50	–	25	–	17	–	15	–	12	–	10	–	ns	
Ready to RE# LOW	t_{RR}	40	–	20	–	20	–	20	–	20	–	20	–	ns	
WE# HIGH to R/B# LOW	t_{WB}	–	200	–	100	–	100	–	100	–	100	–	100	ns	4
WE# cycle time	t_{WC}	100	–	45	–	35	–	30	–	25	–	20	–	ns	



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Electrical Specifications – AC Characteristics and Operating Conditions (Asynchronous)

Table 69: AC Characteristics: Asynchronous Command, Address, and Data (Continued)

Parameter	Symbol	Mode 0		Mode 1		Mode 2		Mode 3		Mode 4		Mode 5		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
WE# HIGH hold time	t_{WH}	30	–	15	–	15	–	10	–	10	–	7	–	ns	
WE# HIGH to RE# LOW	t_{WHR}	120	–	80	–	80	–	60	–	60	–	60	–	ns	
WE# pulse width	t_{WP}	50	–	25	–	17	–	15	–	12	–	10	–	ns	
WP# transition to WE# LOW	t_{WW}	100	–	100	–	100	–	100	–	100	–	100	–	ns	
Delay before next command after a Volume is selected	t_{VDLY}	50	–	50	–	50	–	50	–	50	–	50	–	ns	

- Notes:
1. Timing for t_{ADL} begins in the address cycle, on the final rising edge of WE# and ends with the first rising edge of WE# for data input.
 2. Data transition is measured $\pm 200\text{mV}$ from steady-state voltage with load. This parameter is sampled and not 100 percent tested.
 3. AC characteristics may need to be relaxed if output drive strength is not set to at least nominal.
 4. Do not issue a new command during t_{WB} , even if R/B# or RDY is ready.



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Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 70: AC Characteristics: NV-DDR Command, Address, and Data

Parameter	Symbol	Mode 0		Mode 1		Mode 2		Mode 3		Mode 4		Mode 5		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Clock period		50		30		20		15		12		10		ns	
Frequency		≈20		≈33		≈50		≈67		≈83		≈100		MHz	
Access window of DQ[7:0] from CLK	t_{AC}	3	20	3	20	3	20	3	20	3	20	3	20	ns	
ALE to data loading time	t_{ADL}	100	–	100	–	70	–	70	–	70	–	70	–	ns	
Command, address data delay	t_{CAD}	25	–	25	–	25	–	25	–	25	–	25	–	ns	1
DQ hold – command, address	t_{CAH}	10	–	5	–	4	–	3	–	2.5	–	2	–	ns	
ALE, CLE, W/R# hold	t_{CALH}	10	–	5	–	4	–	3	–	2.5	–	2	–	ns	
ALE, CLE, W/R# setup	t_{CALS}	10	–	5	–	4	–	3	–	2.5	–	2	–	ns	
DQ setup – command, address	t_{CAS}	10	–	5	–	4	–	3	–	2.5	–	2	–	ns	
CE# HIGH hold time prior to VOLUME SELECT (E1h)	t_{CEH}	20	–	20	–	20	–	20	–	20	–	20	–	ns	
Delay before CE# HIGH for any Volume after a Volume is selected	t_{CEVDLY}	50	–	50	–	50	–	50	–	50	–	50	–	ns	
CE# hold	t_{CH}	10	–	5	–	4	–	3	–	2.5	–	2	–	ns	
Average CLK cycle time	t_{CK} (avg)	50	100	30	50	20	30	15	20	12	15	10	12	ns	2
Absolute CLK cycle time, from rising edge to rising edge	t_{CK} (abs)	$t_{CK} \text{ (abs) MIN} = t_{CK} \text{ (avg)} + t_{JIT} \text{ (per) MIN}$ $t_{CK} \text{ (abs) MAX} = t_{CK} \text{ (avg)} + t_{JIT} \text{ (per) MAX}$												ns	
CLK cycle HIGH	t_{CKH} (abs)	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	t_{CK}	3
CLK cycle LOW	t_{CKL} (abs)	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	0.43	0.57	t_{CK}	3



MLC 256Gb to 2Tb Async/Sync NAND

Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 70: AC Characteristics: NV-DDR Command, Address, and Data (Continued)

Parameter	Symbol	Mode 0		Mode 1		Mode 2		Mode 3		Mode 4		Mode 5		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Data output end to W/R# HIGH	t_{CKWR}	$t_{CKWR}(\text{MIN}) = \text{RoundUp}[(t_{DQSCK}(\text{MAX}) + t_{CK})/t_{CK}]$												t_{CK}	
CE# setup	t_{CS}	35	–	25	–	15	–	15	–	15	–	15	–	ns	
CE# setup time for data input after CE# has been HIGH for > 1 μ s	t_{CS3}	75	–	75	–	75	–	75	–	75	–	75	–	ns	
Data In hold	t_{DH}	5	–	2.5	–	1.7	–	1.3	–	1.1	–	0.9	–	ns	
Access window of DQS from CLK	t_{DQSCK}	–	20	–	20	–	20	–	20	–	20	–	20	ns	
DQS, DQ[7:0] Driven by NAND	t_{DQSD}	–	18	–	18	–	18	–	18	–	18	–	18	ns	
DQS, DQ[7:0] to tri-state	t_{DQSHZ}	–	20	–	20	–	20	–	20	–	20	–	20	ns	4
DQS input high pulse width	t_{DQSH}	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	t_{CK}	
DQS input low pulse width	t_{DQSL}	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	t_{CK}	
DQS-DQ skew	t_{DQSQ}	–	5	–	2.5	–	1.7	–	1.3	–	1.0	–	0.85	ns	
Data input	t_{DQSS}	0.75	1.25	0.75	1.25	0.75	1.25	0.75	1.25	0.75	1.25	0.75	1.25	t_{CK}	
Data In setup	t_{DS}	5	–	3	–	2	–	1.5	–	1.1	–	0.9	–	ns	
DQS falling edge from CLK rising – hold	t_{DSH}	0.2	–	0.2	–	0.2	–	0.2	–	0.2	–	0.2	–	t_{CK}	
DQS falling to CLK rising – set-up	t_{DSS}	0.2	–	0.2	–	0.2	–	0.2	–	0.2	–	0.2	–	t_{CK}	
Data valid window	t_{DVW}	$t_{DVW} = t_{QH} - t_{DQSQ}$												ns	
ENi LOW until any issued command is ignored	t_{ENi}	–	15	–	15	–	15	–	15	–	15	–	15	ns	
CE# LOW until ENo LOW	t_{ENo}	–	50	–	50	–	50	–	50	–	50	–	50	ns	
Half clock period	t_{HP}	$t_{HP} = \text{Min}(t_{CKH}, t_{CKL})$												ns	



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Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 70: AC Characteristics: NV-DDR Command, Address, and Data (Continued)

Parameter	Symbol	Mode 0		Mode 1		Mode 2		Mode 3		Mode 4		Mode 5		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
The deviation of a given t_{CK} (abs) from a t_{CK} (avg)	t_{JIT} (per)	-0.7	0.7	-0.7	0.7	-0.7	0.7	-0.6	0.6	-0.6	0.6	-0.5	0.5	ns	
DQ-DQS hold, DQS to first DQ to go nonvalid, per access	t_{QH}	$t_{QH} = t_{HP} - t_{QHS}$												ns	
Data hold skew factor	t_{QHS}	-	6	-	3	-	2	-	1.5	-	1.2	-	1	ns	
Data output to command, address, or data input	t_{RHW}	100	-	100	-	100	-	100	-	100	-	100	-	ns	
Ready to data output	t_{RR}	20	-	20	-	20	-	20	-	20	-	20	-	ns	
CLK HIGH to R/B# LOW	t_{WB}	-	100	-	100	-	100	-	100	-	100	-	100	ns	
Command cycle to data output	t_{WHR}	80	-	80	-	80	-	80	-	80	-	80	-	ns	
DQS write pre-ambble	t_{WPRE}	1.5	-	1.5	-	1.5	-	1.5	-	1.5	-	1.5	-	t_{CK}	
DQS write post-ambble	t_{WPST}	1.5	-	1.5	-	1.5	-	1.5	-	1.5	-	1.5	-	t_{CK}	
W/R# LOW to data output cycle	t_{WRCK}	20	-	20	-	20	-	20	-	20	-	20	-	ns	
WP# transition to command cycle	t_{WW}	100	-	100	-	100	-	100	-	100	-	100	-	ns	
Delay before next command after a Volume is selected	t_{VDLY}	50	-	50	-	50	-	50	-	50	-	50	-	ns	

- Notes:
1. Delay is from start of command to next command, address, or data cycle; start of address to next command, address, or data cycle; and end of data to start of next command, address, or data cycle.
 2. $t_{CK}(\text{avg})$ is the average clock period over any consecutive 200-cycle window.
 3. $t_{CKH}(\text{abs})$ and $t_{CKL}(\text{abs})$ include static offset and duty cycle jitter.
 4. t_{DQSHZ} begins when W/R# is latched HIGH by CLK. This parameter is not referenced to a specific voltage level; it specifies when the device outputs are no longer driving.
 5. If RESET (FFh) is issued when the target is idle, the target goes busy for a maximum of 5 μ s.



MLC 256Gb to 2Tb Async/Sync NAND

Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 71: AC Characteristics: NV-DDR2 Command, Address, and Data for modes 0 - 4

Parameter	Symbol	Mode 0		Mode 1		Mode 2		Mode 3		Mode 4		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Clock period		30		25		15		12		10		ns	
Frequency		≈33		≈40		≈66		≈83		≈100		MHz	
Command and Address													
Access window of DQ[7:0] from RE# LOW or RE_t / RE_c	^t AC	3	25	3	25	3	25	3	25	3	25	ns	
ALE to data loading time	^t ADL	100	–	100	–	100	–	100	–	70	–	ns	
ALE to RE# LOW or RE_t / RE_c	^t AR	10	–	10	–	10	–	10	–	10	–	ns	
DQ hold – command, address	^t CAH	5	–	5	–	5	–	5	–	5	–	ns	
ALE, CLE hold	^t CALH	5	–	5	–	5	–	5	–	5	–	ns	
ALE, CLE setup with ODT disabled	^t CALS	15	–	15	–	15	–	15	–	15	–	ns	
ALE, CLE setup with ODT enabled	^t CALS2	25	–	25	–	25	–	25	–	25	–	ns	
DQ setup – command, address	^t CAS	5	–	5	–	5	–	5	–	5	–	ns	
CE# HIGH hold time prior to VOLUME SELECT (E1h)	^t CEH	20	–	20	–	20	–	20	–	20	–	ns	
Delay before CE# HIGH for any Volume after a Volume is selected	^t CEVDLY	50	–	50	–	50	–	50	–	50	–	ns	
CE# hold	^t CH	5	–	5	–	5	–	5	–	5	–	ns	
CE# HIGH to output Hi-Z	^t CHZ	–	30	–	30	–	30	–	30	–	30	ns	1
CLE HIGH to output Hi-Z	^t CLHZ	–	30	–	30	–	30	–	30	–	30		1
CLE to RE# LOW or RE_t / RE_c	^t CLR	10	–	10	–	10	–	10	–	10	–	ns	
CE# to RE# LOW or RE_t / RE_c	^t CR	10	–	10	–	10	–	10	–	10	–	ns	
CE# setup	^t CS	20	–	20	–	20	–	20	–	20	–	ns	
CE# setup for data output with ODT disabled	^t CS1	30	–	30	–	30	–	30	–	30	–	ns	



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Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 71: AC Characteristics: NV-DDR2 Command, Address, and Data for modes 0 - 4 (Continued)

Parameter	Symbol	Mode 0		Mode 1		Mode 2		Mode 3		Mode 4		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
CE# setup for DQS/DQ[7:0] with ODT enabled	t_{CS2}	40	–	40	–	40	–	40	–	40	–	ns	
CE# setup time to DQS (DQS _t) LOW after CE# has been HIGH for > 1 μ s	t_{CD}	100	–	100	–	100	–	100	–	100	–	ns	
ALE, CLE, WE#, hold time from CE# HIGH	t_{CSD}	10	–	10	–	10	–	10	–	10	–	ns	
ENi LOW until any issued command is ignored	t_{ENi}	–	15	–	15	–	15	–	15	–	15	ns	
CE# LOW until ENo LOW	t_{ENo}	–	50	–	50	–	50	–	50	–	50	ns	
Ready to data output	t_{RR}	20	–	20	–	20	–	20	–	20	–	ns	
CLK HIGH to R/B# LOW	t_{WB}	–	100	–	100	–	100	–	100	–	100	ns	
WE# cycle time	t_{WC}	25	–	25	–	25	–	25	–	25	–	ns	
WE# pulse width	t_{WH}	11	–	11	–	11	–	11	–	11	–	ns	
Command cycle to data output	t_{WHR}	80	–	80	–	80	–	80	–	80	–	ns	
WP# transition to command cycle	t_{WP}	11	–	11	–	11	–	11	–	11	–	ns	
WP# transition to command cycle	t_{WW}	100	–	100	–	100	–	100	–	100	–	ns	
Delay before next command after a Volume is selected	t_{VDLY}	50	–	50	–	50	–	50	–	50	–	ns	
Jitter													
The deviation of a given t_{DQS} (abs) / t_{DSC} (abs) from a t_{DQS} (avg) / t_{DSC} (avg)	t_{JITper} (DQS)	-2.4	2.4	-2.0	2.0	-1.2	1.2	-1.0	1.0	-0.8	0.8	ns	3, 5, 7
The deviation of a given t_{RC} (abs) / t_{DSC} (abs) from a t_{RC} (avg) / t_{DSC} (avg)	t_{JITper} (RE#)	-1.8	1.8	-1.5	1.5	-0.9	0.9	-0.75	0.75	-0.6	0.6	ns	3, 5, 7



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Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 71: AC Characteristics: NV-DDR2 Command, Address, and Data for modes 0 - 4 (Continued)

Parameter	Symbol	Mode 0		Mode 1		Mode 2		Mode 3		Mode 4		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Cycle to cycle jitter for DQS	$t_{JITcc}^{(DQS)}$	4.8	–	4.0	–	2.4	–	2.0	–	1.6	–	ns	3, 6
Cycle to cycle jitter for RE#	$t_{JITcc}^{(RE\#)}$	3.6	–	3.0	–	1.8	–	1.5	–	1.2	–	ns	3, 6
Data Input													
DQS setup time for data input start	t_{CDQSS}	30	–	30	–	30	–	30	–	30	–	ns	
DQS (DQS _t) HIGH and RE# (RE _t) HIGH setup to ALE, CLE and CE# LOW during data burst	t_{DBS}	5	–	5	–	5	–	5	–	5	–	ns	
Data In hold	t_{DH}	4.0	–	3.3	–	2.0	–	1.1	–	0.7	–	ns	10
Data In setup	t_{DS}	4.0	–	3.3	–	2.0	–	1.1	–	0.7	–	ns	10
DQS input high pulse width	t_{DQSH}	0.43	–	0.43	–	0.43	–	0.43	–	0.43	–	$t_{DSC}^{(avg)}$	
DQS input low pulse width	t_{DQSL}	0.43	–	0.43	–	0.43	–	0.43	–	0.43	–	$t_{DSC}^{(avg)}$	
Average DQS cycle time	$t_{DSC}^{(avg)}$ or t_{DSC}	30	–	25	–	15	–	12	–	10	–	ns	2
Absolute DQS cycle time, from rising edge to rising edge	$t_{DSC}^{(abs)}$	$t_{DSC}^{(abs)} \text{ MIN} = t_{DSC}^{(avg)} + t_{JITper}(DQS) \text{ MIN}$ $t_{DSC}^{(abs)} \text{ MAX} = t_{DSC}^{(avg)} + t_{JITper}(DQS) \text{ MAX}$										ns	
ENi LOW until any issued command is ignored	t_{ENi}	–	15	–	15	–	15	–	15	–	15	ns	
CE# LOW until ENo LOW	t_{ENo}	–	50	–	50	–	50	–	50	–	50	ns	
DQS write preamble with ODT disabled	t_{WPRE}	15	–	15	–	15	–	15	–	15	–	ns	
DQS write preamble with ODT enabled	t_{WPRE2}	25	–	25	–	25	–	25	–	25	–	ns	
DQS write postamble	t_{WPST}	6.5	–	6.5	–	6.5	–	6.5	–	6.5	–	ns	
DQS write postamble hold time	t_{WPSTH}	15	–	15	–	15	–	15	–	15	–	ns	



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Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 71: AC Characteristics: NV-DDR2 Command, Address, and Data for modes 0 - 4 (Continued)

Parameter	Symbol	Mode 0		Mode 1		Mode 2		Mode 3		Mode 4		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Data Output													
Access window of DQ[7:0] from CLK	^t AC	3	25	3	25	3	25	3	25	3	25	ns	
DQS (DQS _t) HIGH and RE# (RE _t) HIGH setup to ALE, CLE and CE# LOW during data burst	^t DBS	5	–	5	–	5	–	5	–	5	–	ns	
DQS-DQ skew	^t DQSQ	–	2.5	–	2.0	–	1.4	–	1.0	–	0.8	ns	
Access window of DQS from RE# or RE _t / RE _c	^t DQSRE	3	25	3	25	3	25	3	25	3	25	ns	
RE# LOW to DQS or DQ[7:0] driven	^t DQSD	5	18	5	18	5	18	5	18	5	18	ns	
DQS hold time after RE# LOW or RE _t /RE _c cross-point	^t DQSRH	5	–	5	–	5	–	5	–	5	–	ns	
Data valid window	^t DVW	^t DVW = ^t QH - ^t DQSQ										ns	
DQ-DQS hold, DQS to first DQ to go nonvalid, per access	^t QH	0.37	–	0.37	–	0.37	–	0.37	–	0.37	–	^t RC (avg)	9
DQS (DQS _t / DQS _c) output HIGH time	^t QSH	0.37	–	0.37	–	0.37	–	0.37	–	0.37	–	^t RC (avg)	9
DQS (DQS _t / DQS _c) output LOW time	^t QSL	0.37	–	0.37	–	0.37	–	0.37	–	0.37	–	^t RC (avg)	9
Average RE# cycle time	^t RC (avg) or ^t RC	30	–	25	–	15	–	12	–	10	–	ns	2
Absolute RE# cycle time	^t RC (abs)	^t RC (abs) MIN = ^t RC (avg) + ^t JITper(RE#) MIN ^t RC (abs) MAX = ^t RC (avg) + ^t JITper(RE#) MAX										ns	
Average RE# HIGH hold time	^t REH (avg)	0.45	0.55	0.45	0.55	0.45	0.55	0.45	0.55	0.45	0.55	^t RC (avg)	4
Absolute RE# HIGH hold time	^t REH (abs)	0.43	–	0.43	–	0.43	–	0.43	–	0.43	–	^t RC (avg)	
Data output to command, address, or data input	^t RHW	100	–	100	–	100	–	100	–	100	–	ns	



MLC 256Gb to 2Tb Async/Sync NAND

Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 71: AC Characteristics: NV-DDR2 Command, Address, and Data for modes 0 - 4 (Continued)

Parameter	Symbol	Mode 0		Mode 1		Mode 2		Mode 3		Mode 4		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Average RE# pulse width	$t_{RP} (avg)$	0.45	0.55	0.45	0.55	0.45	0.55	0.45	0.55	0.45	0.55	$t_{RC} (avg)$	4
Absolute RE# pulse width	$t_{RP} (abs)$	0.43	–	0.43	–	0.43	–	0.43	–	0.43	–	$t_{RC} (avg)$	
Read preamble with ODT disabled	t_{RPRE}	15	–	15	–	15	–	15	–	15	–	ns	
Read preamble with ODT enabled	t_{RPRE2}	25	–	25	–	25	–	25	–	25	–	ns	
Read postamble	t_{RPST}	$t_{DQSR} E + 0.5 * t_{RC}$	–	$t_{DQSR} E + 0.5 * t_{RC}$	–	$t_{DQSR} E + 0.5 * t_{RC}$	–	$t_{DQSR} E + 0.5 * t_{RC}$	–	$t_{DQSR} E + 0.5 * t_{RC}$	–	ns	
Read postamble hold time	t_{RPSTH}	15	–	15	–	15	–	15	–	15	–	ns	

- Notes:
- t_{CHZ} and t_{CLHZ} are not referenced to a specific voltage level, but specify when the device output is no longer driving.
 - The parameters $t_{RC}(avg)$ and $t_{DSC}(avg)$ are the average over any 200 consecutive periods and $t_{RC}(avg) / t_{DSC}(avg)$ min are the smallest rates allowed, with the exception of a deviation due to $t_{JIT} (per)$.
 - Input jitter is allowed provided it does not exceed values specified.
 - $t_{REH}(avg)$ and $t_{RP}(avg)$ are the average half clock period over any 200 consecutive clocks and is the smallest half period allowed, expect a deviation due to the allowed clock jitter. Input clock jitter is allowed provided it does not exceed values specified.
 - The period jitter $t_{JIT} (per)$ is the maximum deviation in the t_{RC} or t_{DSC} period from the average or nominal t_{RC} or t_{DSC} period. It is allowed in either the positive or negative direction.
 - The cycle-to-cycle jitter t_{JITcc} is the amount the clock period can deviate from one cycle to the next.
 - The duty cycle jitter applies to either the high pulse or low pulse; however, the two cumulatively cannot exceed t_{JITper} . As long as the absolute minimum half period ($t_{RP}(abs)$, $t_{REH}(abs)$, t_{DQSH} or t_{DQSL} is not less than 43 percent of the average cycle.
 - All timing parameter values assume differential signaling for RE# and DQS is used.
 - When the device is operated with input clock jitter, t_{QSL} , t_{QSH} , and t_{QH} need to be derated by the actual t_{JITper} in the input clock. (output deratings are relative to the NAND input RE pulse that generated the DQS pulse).
 - The t_{DS} and t_{DH} times listed are based on an input slew rate of 1 V/ns. If the input slew rate is not 1 V/ns, then the derating methodology shall be used.
 - Minimizing the amount of duty cycle jitter to more than 45 percent of the average cycle provides a larger t_{QH} , which in turn provides a larger data valid window. The device shall provide a minimum of 0.5 percent of larger data valid window for each 1% improvement in duty cycle jitter. For example, if the host provides a $t_{REH}(abs)$ of $0.49 * t_{RC}(avg)$ then the device shall provide at least a t_{QH} of $0.39 * t_{RC}(avg)$.



MLC 256Gb to 2Tb Async/Sync NAND

Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 72: AC Characteristics: NV-DDR2 Command, Address, and Data for timing modes 5 - 7

Parameter	Symbol	Mode 5		Mode 6		Mode 7		Unit	Notes
		Min	Max	Min	Max	Min	Max		
Clock period		7.5		6		5		ns	
Frequency		≈133		≈166		≈200		MHz	
Command and Address									
Access window of DQ[7:0] from RE# LOW or RE_t / RE_c	^t AC	3	25	3	25	3	25	ns	
ALE to data loading time	^t ADL	70	–	70	–	70	–	ns	
ALE to RE# LOW or RE_t / RE_c	^t AR	10	–	10	–	10	–	ns	
DQ hold – command, address	^t CAH	5	–	5	–	5	–	ns	
ALE, CLE hold	^t CALH	5	–	5	–	5	–	ns	
ALE, CLE setup with ODT disabled	^t CALS	15	–	15	–	15	–	ns	
ALE, CLE setup with ODT enabled	^t CALS2	25	–	25	–	25	–	ns	
DQ setup – command, address	^t CAS	5	–	5	–	5	–	ns	
CE# HIGH hold time	^t CEH	20	–	20	–	20	–	ns	
Delay before CE# HIGH for any Volume after a Volume is selected	^t CEVDLY	50	–	50	–	50	–	ns	
CE# hold	^t CH	5	–	5	–	5	–	ns	
CE# HIGH to output Hi-Z	^t CHZ	–	30	–	30	–	30	ns	1
CLE HIGH to output Hi-Z	^t CLHZ	–	30	–	30	–	30		1
CLE to RE# LOW or RE_t / RE_c	^t CLR	10	–	10	–	10	–	ns	
CE# to RE# LOW or RE_t / RE_c	^t CR	10	–	10	–	10	–	ns	
CE# setup	^t CS	20	–	20	–	20	–	ns	
CE# setup for data output with ODT disabled	^t CS1	30	–	30	–	30	–	ns	
CE# setup for DQS/ DQ[7:0] with ODT enabled	^t CS2	40	–	40	–	40	–	ns	



MLC 256Gb to 2Tb Async/Sync NAND

Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 72: AC Characteristics: NV-DDR2 Command, Address, and Data for timing modes 5 - 7 (Continued)

Parameter	Symbol	Mode 5		Mode 6		Mode 7		Unit	Notes
		Min	Max	Min	Max	Min	Max		
CE# setup time to DQS (DQS _t) LOW after CE# has been HIGH for > 1μs	^t CD	100	–	100	–	100	–	100	–
ALE, CLE, WE#, hold time from CE# HIGH	^t CSD	10	–	10	–	10	–	ns	
ENi LOW until any issued command is ignored	^t ENi	–	15	–	15	–	15	ns	
CE# LOW until ENo LOW	^t ENo	–	50	–	50	–	50	ns	
Ready to data output	^t RR	20	–	20	–	20	–	ns	
CLK HIGH to R/B# LOW	^t WB	–	100	–	100	–	100	ns	
WE# cycle time	^t WC	25	–	25	–	25	–	ns	
WE# pulse width	^t WH	11	–	11	–	11	–	ns	
Command cycle to data output	^t WHR	80	–	80	–	80	–	ns	
WP# transition to command cycle	^t WP	11	–	11	–	11	–	ns	
WP# transition to command cycle	^t WW	100	–	100	–	100	–	ns	
Delay before next command after a Volume is selected	^t VDLY	50	–	50	–	50	–	ns	
Jitter									
The deviation of a given ^t DQS (abs) / ^t DSC (abs) from a ^t DQS (avg) / ^t DSC (avg)	^t JITper (DQS)	-0.6	0.6	-0.48	0.48	-0.40	0.40	ns	3, 5, 7
The deviation of a given ^t RC (abs) / ^t DSC (abs) from a ^t RC (avg) / ^t DSC (avg)	^t JITper (RE#)	-0.45	0.45	-0.36	0.36	-0.30	0.30	ns	3, 5, 7
Cycle to cycle jitter for DQS	^t JITcc (DQS)	1.2	–	0.96	–	0.80	–	ns	3, 6
Cycle to cycle jitter for RE#	^t JITcc (RE#)	0.9	–	0.72	–	0.60	–	ns	3, 6
Data Input									
DQS setup time for data input start	^t CDQSS	30	–	30	–	30	–	ns	



MLC 256Gb to 2Tb Async/Sync NAND

Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 72: AC Characteristics: NV-DDR2 Command, Address, and Data for timing modes 5 - 7 (Continued)

Parameter	Symbol	Mode 5		Mode 6		Mode 7		Unit	Notes
		Min	Max	Min	Max	Min	Max		
DQS (DQS _t) HIGH and RE# (RE _t) HIGH setup to ALE, CLE and CE# LOW during data burst	^t DBS	5	–	5	–	5	–	ns	
Data In hold	^t DH	0.6	–	0.55	–	0.40	–	ns	10
Data In setup	^t DS	0.6	–	0.55	–	0.40	–	ns	10
DQS input high pulse width	^t DQSH	0.43	–	0.43	–	0.43	–	^t DSC (avg)	
DQS input low pulse width	^t DQSL	0.43	–	0.43	–	0.43	–	^t DSC (avg)	
Average DQS cycle time	^t DSC (avg) or ^t DSC	7.5	–	6	–	5	–	ns	2
Absolute DQS cycle time, from rising edge to rising edge	^t DSC (abs)	^t DSC (abs) MIN = ^t DSC (avg) + ^t JITper(DQS) MIN ^t DSC (abs) MAX = ^t DSC (avg) + ^t JITper(DQS) MAX						ns	
ENi LOW until any issued command is ignored	^t ENi	–	15	–	15	–	15	ns	
CE# LOW until ENO LOW	^t ENo	–	50	–	50	–	50	ns	
DQS write preamble with ODT disabled	^t WPRE	15	–	15	–	15	–	ns	
DQS write preamble with ODT enabled	^t WPRE2	25	–	25	–	25	–	ns	
DQS write postamble	^t WPST	6.5	–	6.5	–	6.5	–	ns	
DQS write postamble hold time	^t WPSTH	15	–	15	–	15	–	ns	
Data Output									
Access window of DQ[7:0] from CLK	^t AC	3	25	3	25	3	25	ns	
DQS (DQS _t) HIGH and RE# (RE _t) HIGH setup to ALE, CLE and CE# LOW during data burst	^t DBS	5	–	5	–	5	–	ns	
DQS-DQ skew	^t DQSQ	–	0.6	–	0.5	–	0.4	ns	
Access window of DQS from RE# or RE _t / RE _c	^t DQSRE	3	25	3	25	3	25	ns	
RE# LOW to DQS or DQ[7:0] driven	^t DQSD	5	18	5	18	5	18	ns	



MLC 256Gb to 2Tb Async/Sync NAND

Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

Table 72: AC Characteristics: NV-DDR2 Command, Address, and Data for timing modes 5 - 7 (Continued)

Parameter	Symbol	Mode 5		Mode 6		Mode 7		Unit	Notes
		Min	Max	Min	Max	Min	Max		
DQS hold time after RE# LOW or RE_t/RE_c crosspoint	t_{DQSRH}	5	–	5	–	5	–	ns	
Data valid window	t_{DVW}	$t_{DVW} = t_{QH} - t_{DQSQ}$						ns	
DQ-DQS hold, DQS to first DQ to go nonvalid, per access	t_{QH}	0.37	–	0.37	–	0.37	–	t_{RC} (avg)	9
DQS (DQS_t/DQS_c) output HIGH time	t_{QSH}	0.37	–	0.37	–	0.37	–	t_{RC} (avg)	9
DQS (DQS_t/DQS_c) output LOW time	t_{QSL}	0.37	–	0.37	–	0.37	–	t_{RC} (avg)	9
Average RE# cycle time	t_{RC} (avg) or t_{RC}	7.5	–	6	–	5	–	ns	2
Absolute RE# cycle time	t_{RC} (abs)	t_{RC} (abs) MIN = t_{RC} (avg) + $t_{JITper}(RE\#)$ MIN t_{RC} (abs) MAX = t_{RC} (avg) + $t_{JITper}(RE\#)$ MAX						ns	
Average RE# HIGH hold time	t_{REH} (avg)	0.45	0.55	0.45	0.55	0.45	0.55	t_{RC} (avg)	4
Absolute RE# HIGH hold time	t_{REH} (abs)	0.43	–	0.43	–	0.43	–	t_{RC} (avg)	
Data output to command, address, or data input	t_{RHW}	100	–	100	–	100	–	ns	
Average RE# pulse width	t_{RP} (avg)	0.45	0.55	0.45	0.55	0.45	0.55	t_{RC} (avg)	4
Absolute RE# pulse width	t_{RP} (abs)	0.43	–	0.43	–	0.43	–	t_{RC} (avg)	
Read preamble with ODT disabled	t_{RPRE}	15	–	15	–	15	–	ns	
Read preamble with ODT enabled	t_{RPRE2}	25	–	25	–	25	–	ns	
Read postamble	t_{RPST}	$t_{DQSRE} + 0.5 * t_{RC}$	–	$t_{DQSRE} + 0.5 * t_{RC}$	–	$t_{DQSRE} + 0.5 * t_{RC}$	–	ns	
Read postamble hold time	t_{RPSTH}	15	–	15	–	15	–	ns	

- Notes:
- t_{CHZ} and t_{CLHZ} are not referenced to a specific voltage level, but specify when the device output is no longer driving.
 - The parameters $t_{RC}(avg)$ and $t_{DSC}(avg)$ are the average over any 200 consecutive periods and $t_{RC}(avg) / t_{DSC}(avg)$ min are the smallest rates allowed, with the exception of a deviation due to t_{JIT} (per).
 - Input jitter is allowed provided it does not exceed values specified.



MLC 256Gb to 2Tb Async/Sync NAND Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2)

4. $t_{REH(avg)}$ and $t_{RP(avg)}$ are the average half clock period over any 200 consecutive clocks and is the smallest half period allowed, expect a deviation due to the allowed clock jitter. Input clock jitter is allowed provided it does not exceed values specified.
5. The period jitter $t_{JIT(per)}$ is the maximum deviation in the t_{RC} or t_{DSC} period from the average or nominal t_{RC} or t_{DSC} period. It is allowed in either the positive or negative direction.
6. The cycle-to-cycle jitter t_{JITcc} is the amount the clock period can deviate from one cycle to the next.
7. The duty cycle jitter applies to either the high pulse or low pulse; however, the two cumulatively cannot exceed t_{JITper} . As long as the absolute minimum half period ($t_{RP(abs)}$, $t_{REH(abs)}$, t_{DQSH} or t_{DQSL} is not less than 43 percent of the average cycle.
8. All timing parameter values assume differential signaling for RE# and DQS is used.
9. When the device is operated with input clock jitter, t_{QSL} , t_{QSH} , and t_{QH} need to be derated by the actual t_{JITper} in the input clock. (output deratings are relative to the NAND input RE pulse that generated the DQS pulse).
10. The t_{DS} and t_{DH} times listed are based on an input slew rate of 1 V/ns. If the input slew rate is not 1 V/ns, then the derating methodology shall be used.
11. Minimizing the amount of duty cycle jitter to more than 45 percent of the average cycle provides a larger t_{QH} , which in turn provides a larger data valid window. The device shall provide a minimum of 0.5 percent of larger data valid window for each 1% improvement in duty cycle jitter. For example, if the host provides a $t_{REH(abs)}$ of $0.49 * t_{RC(avg)}$ then the device shall provide at least a t_{QH} of $0.39 * t_{RC(avg)}$.



MLC 256Gb to 2Tb Async/Sync NAND Electrical Specifications – Array Characteristics

Electrical Specifications – Array Characteristics

Table 73: Array Characteristics

Parameter	Symbol	Typ	Max	Unit	Notes
Number of partial page programs	NOP	–	1	Cycles	1
ERASE BLOCK operation time	t_{BERS}	3	12	ms	8
Cache busy	t_{CBSY}	1100	3000	μs	3, 7
Change column setup time to data in/out or next command	t_{CCS}	–	300	ns	
Dummy busy time	t_{DBSY}	0.5	1	μs	
Busy time for SET FEATURES and GET FEATURES operations	t_{FEAT}	–	1	μs	
Busy time for interface change	t_{ITC}	–	1	μs	2
LAST PAGE PROGRAM operation time	t_{LPROG}	–	–	μs	3
Busy time for OTP DATA PROGRAM operation if OTP is protected	t_{OBSY}	–	50	μs	
Power-on reset time	t_{POR}	–	2	ms	
PROGRAM PAGE operation time	t_{PROG}	1600	3000	μs	7
READ PAGE operation time	t_R	–	115	μs	5
Cache read busy time	t_{RCBSY}	26	115	μs	6
Device reset time (Read/Program/Erase)	t_{RST}	–	5/10/500	μs	4

- Notes:
1. The pages in the OTP Block have an NOP of 2.
 2. t_{ITC} (MAX) is the busy time when the interface changes from asynchronous to synchronous using the SET FEATURES (EFh) command or synchronous to asynchronous using the RESET (FFh) command. During the t_{ITC} time, any command, including READ STATUS (70h) and READ STATUS ENHANCED (78h), is prohibited.
 3. t_{CBSY} and t_{LPROG} = t_{PROG} (last page) + t_{PROG} (last page - 1) - command load time (last page) - address load time (last page) - data load time (last page).
 4. If RESET command is issued when the target is READY, the target goes busy for a maximum of 5 μs .
 5. For Read Retry options 4 to 7, and 9, t_R Max maybe up to 350 μs .
 6. For Read Retry options 4 to 7, and 9, Cache read is not supported.
 7. In the case of a program operation that exceeds t_{PROG}/t_{CBSY} Max, that specific NAND block may be retired by the host system.
 8. In the case of an erase operation that exceeds t_{BERS} Max, that specific NAND block may be retired by the host system.



MLC 256Gb to 2Tb Async/Sync NAND Asynchronous Interface Timing Diagrams

Asynchronous Interface Timing Diagrams

Figure 99: RESET Operation

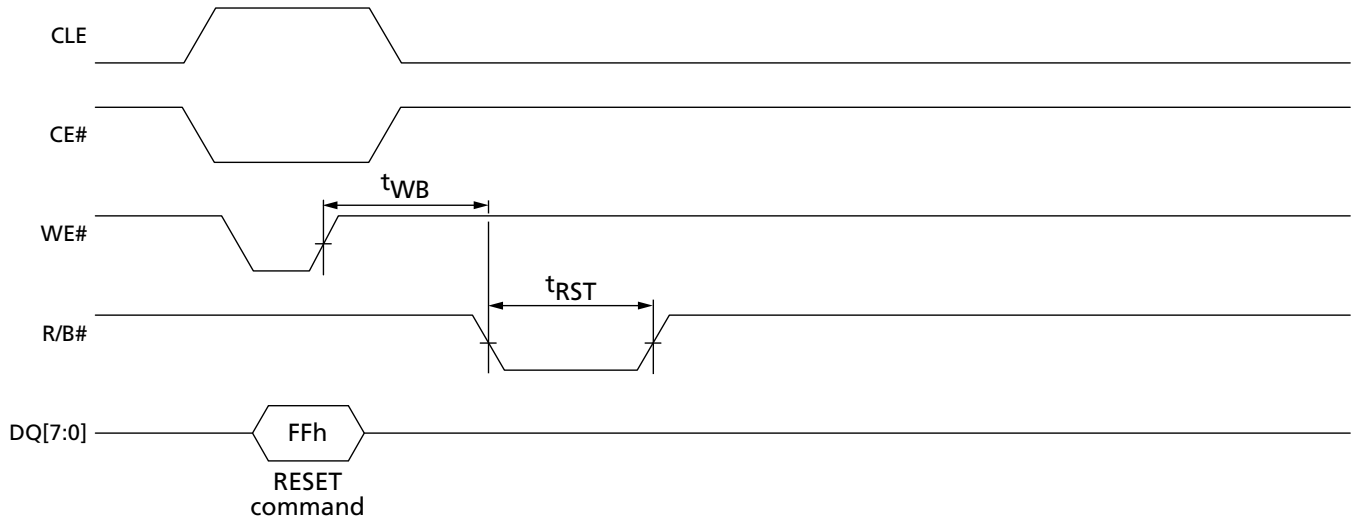
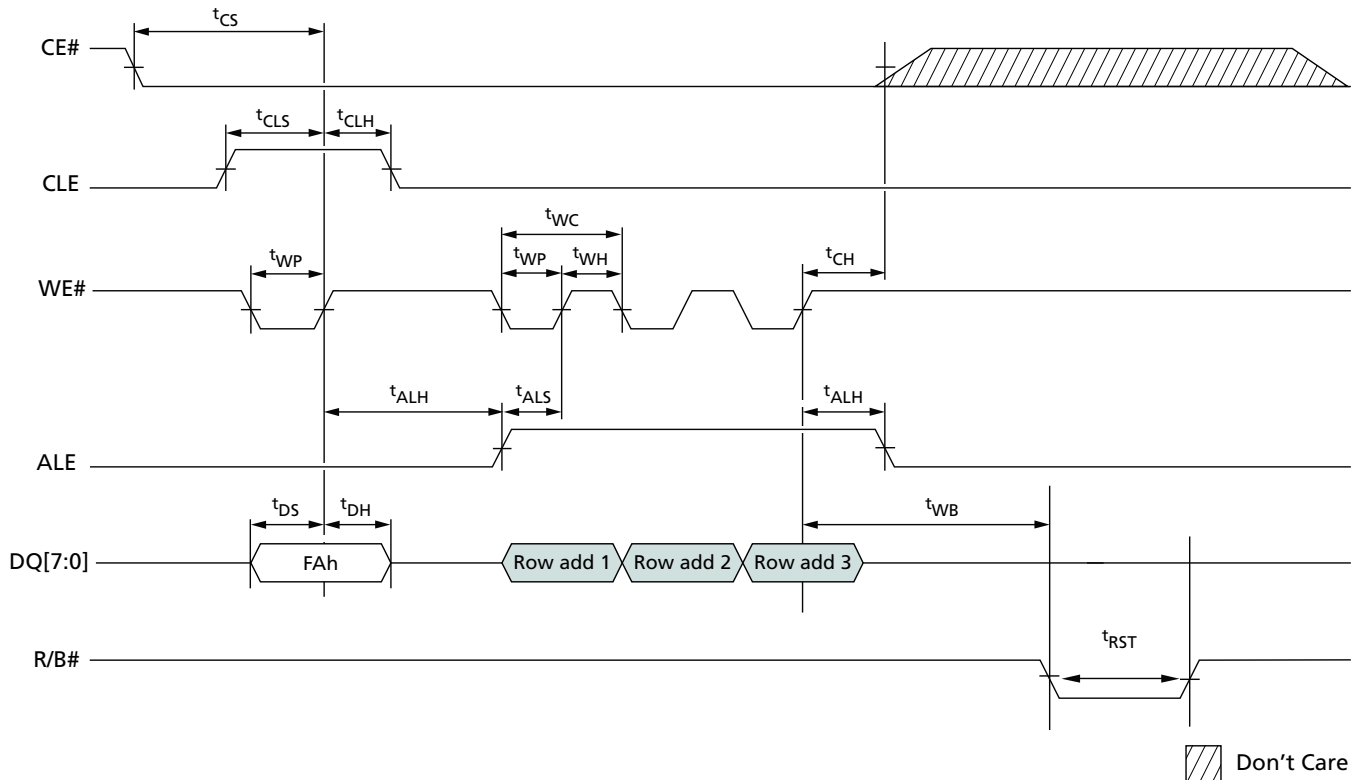


Figure 100: RESET LUN Operation





MLC 256Gb to 2Tb Async/Sync NAND Asynchronous Interface Timing Diagrams

Figure 101: READ STATUS Cycle

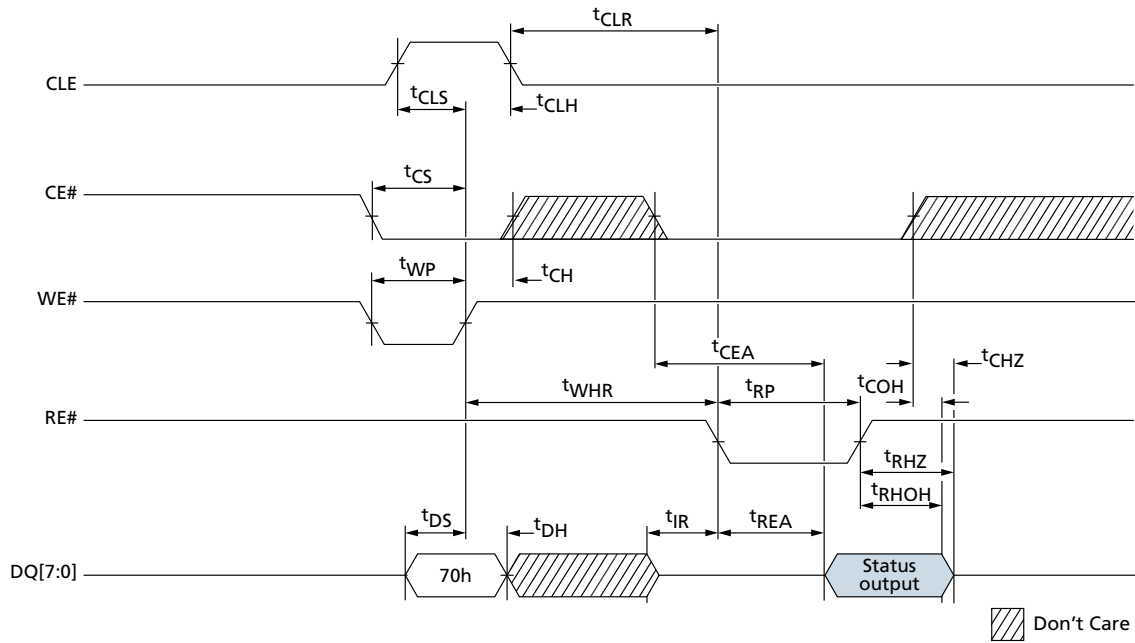
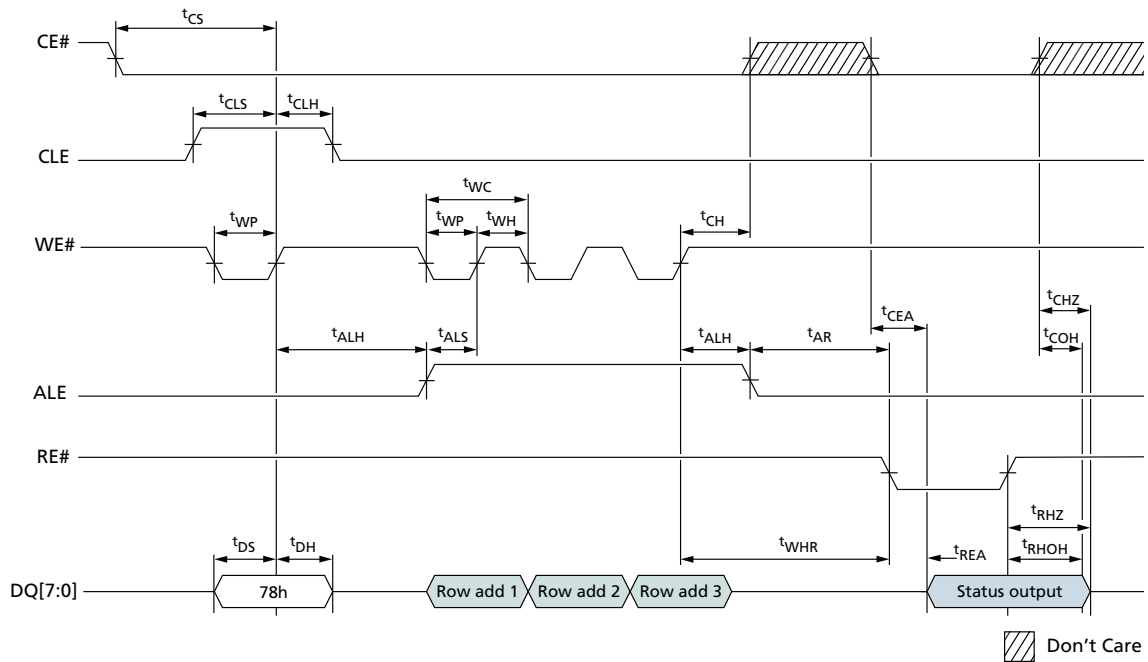


Figure 102: READ STATUS ENHANCED Cycle





MLC 256Gb to 2Tb Async/Sync NAND Asynchronous Interface Timing Diagrams

Figure 103: READ PARAMETER PAGE

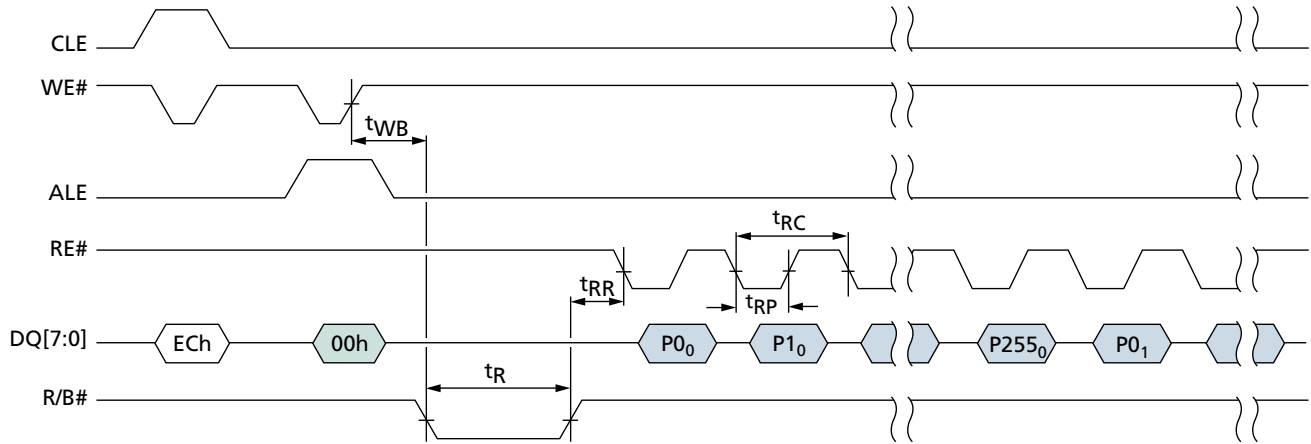
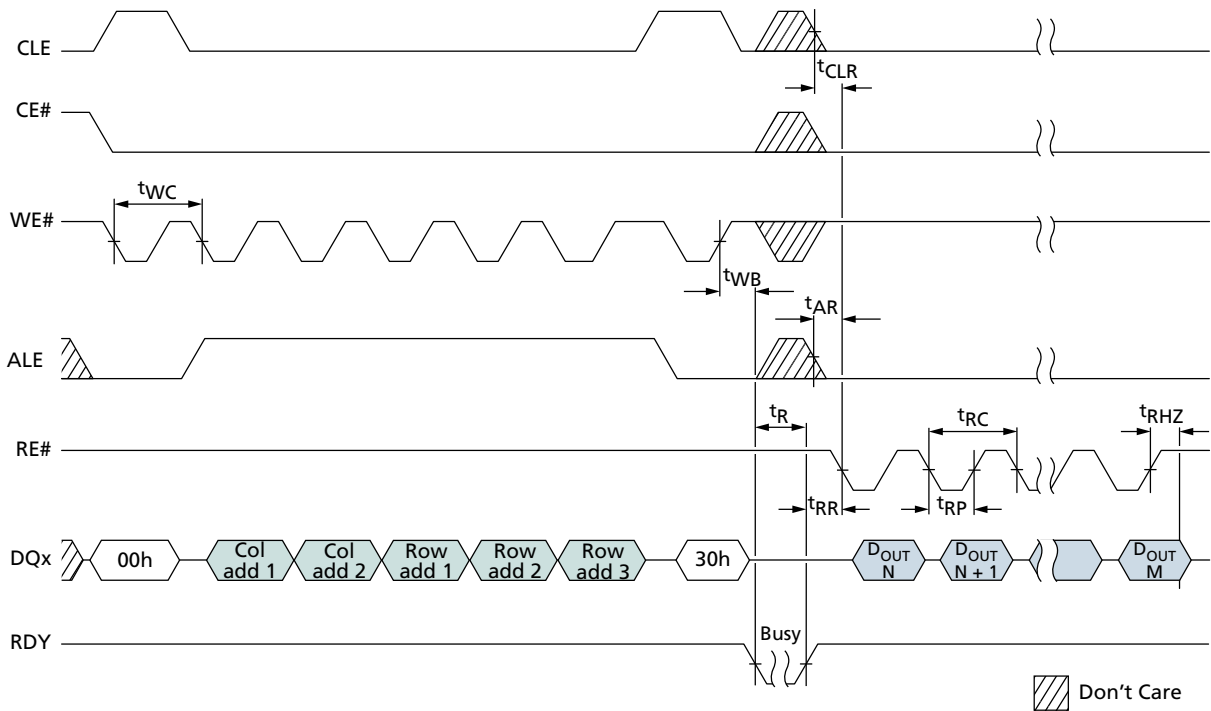


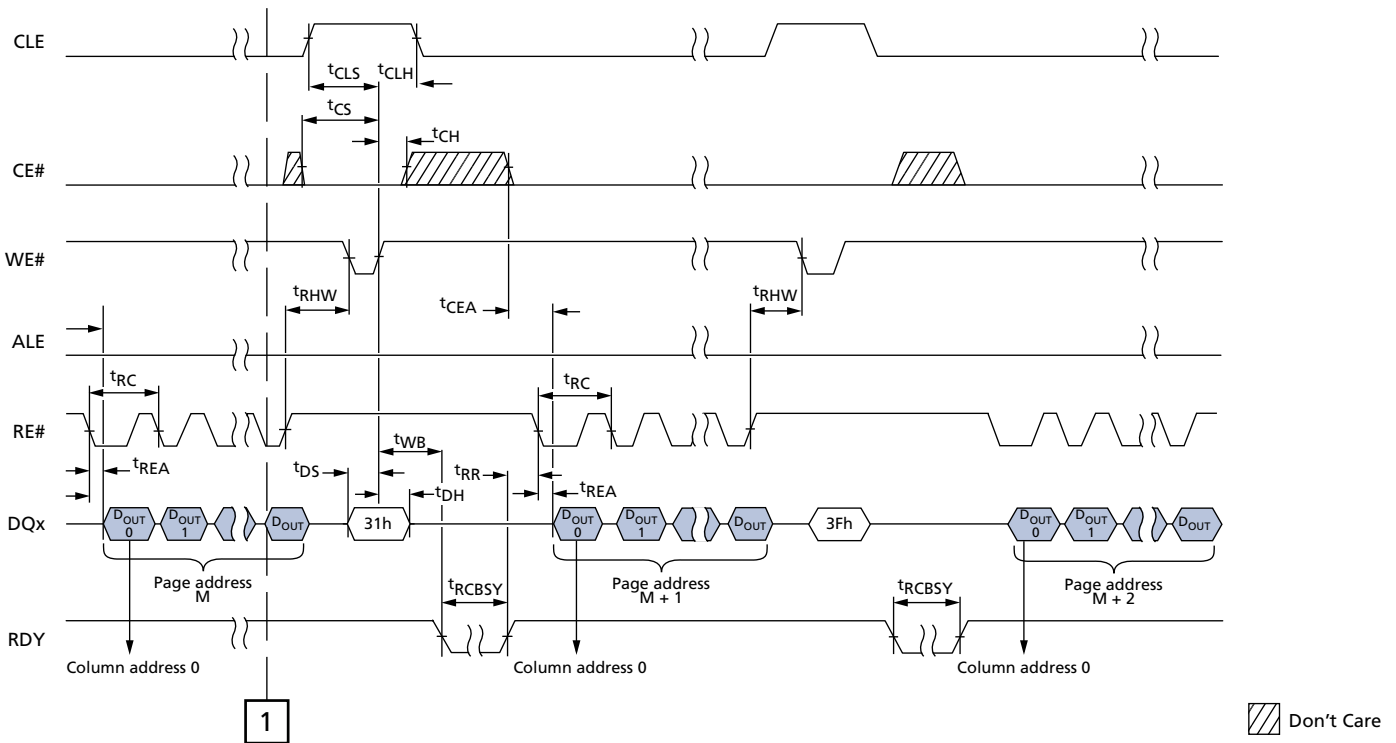
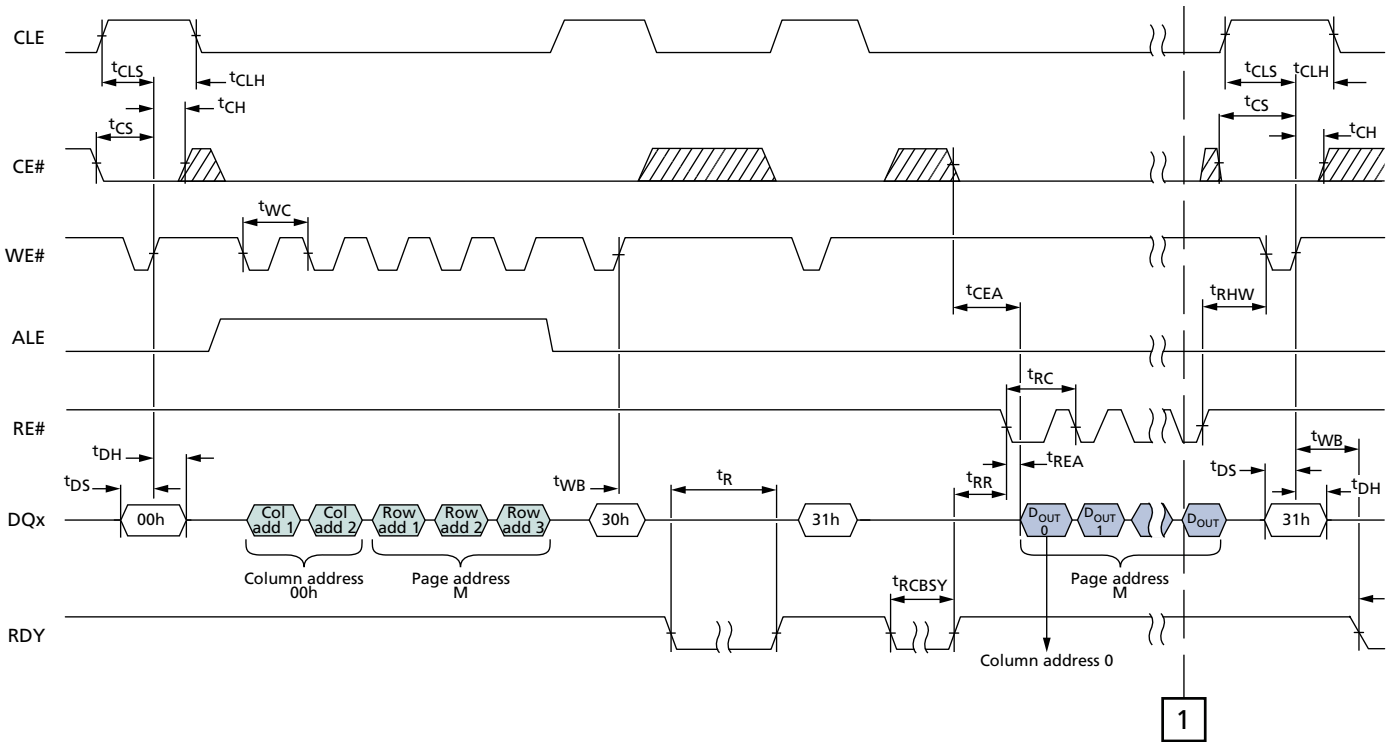
Figure 104: READ PAGE





MLC 256Gb to 2Tb Async/Sync NAND Asynchronous Interface Timing Diagrams

Figure 107: READ PAGE CACHE SEQUENTIAL

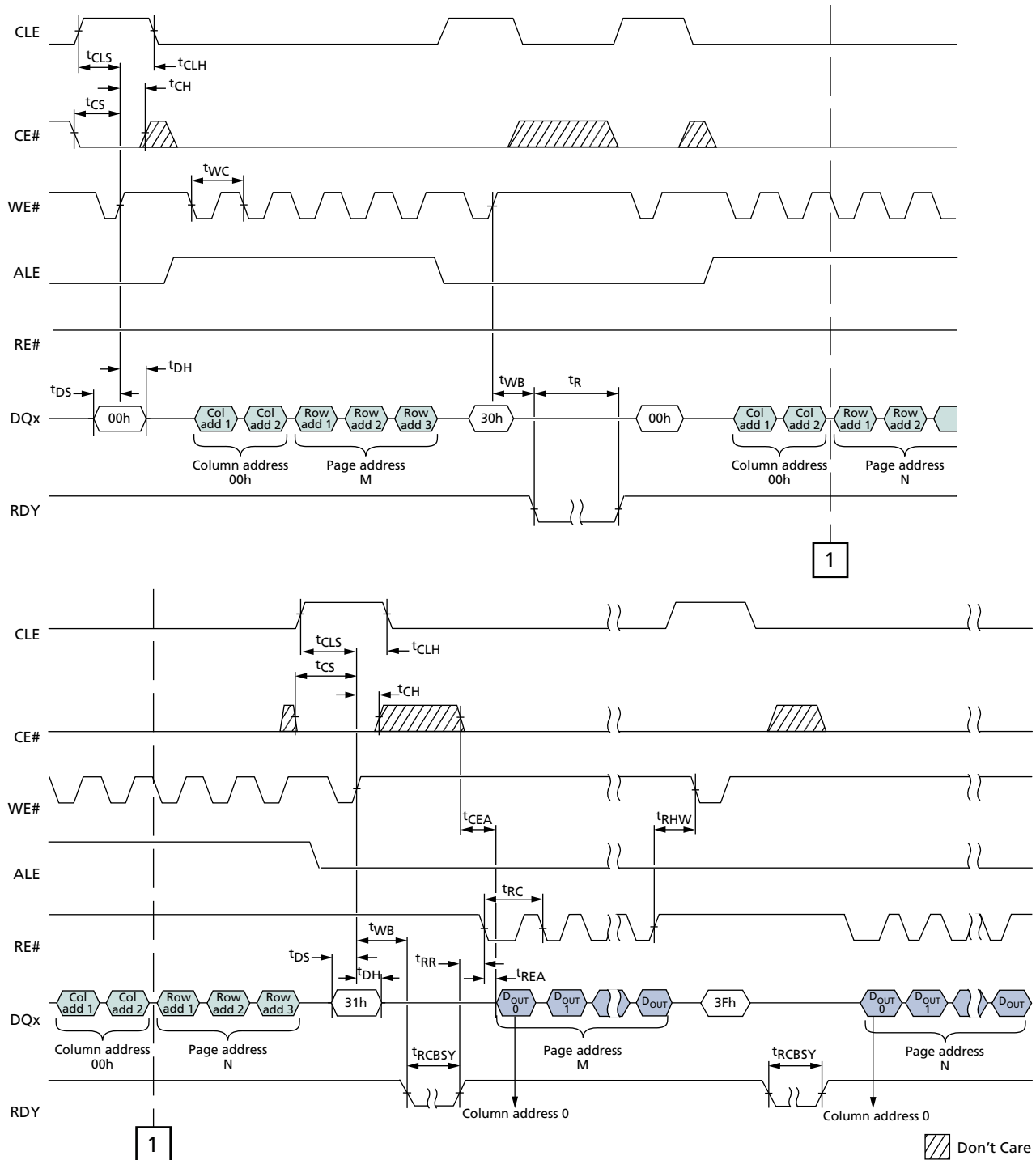


Don't Care



MLC 256Gb to 2Tb Async/Sync NAND Asynchronous Interface Timing Diagrams

Figure 108: READ PAGE CACHE RANDOM





MLC 256Gb to 2Tb Async/Sync NAND Asynchronous Interface Timing Diagrams

Figure 109: READ ID Operation

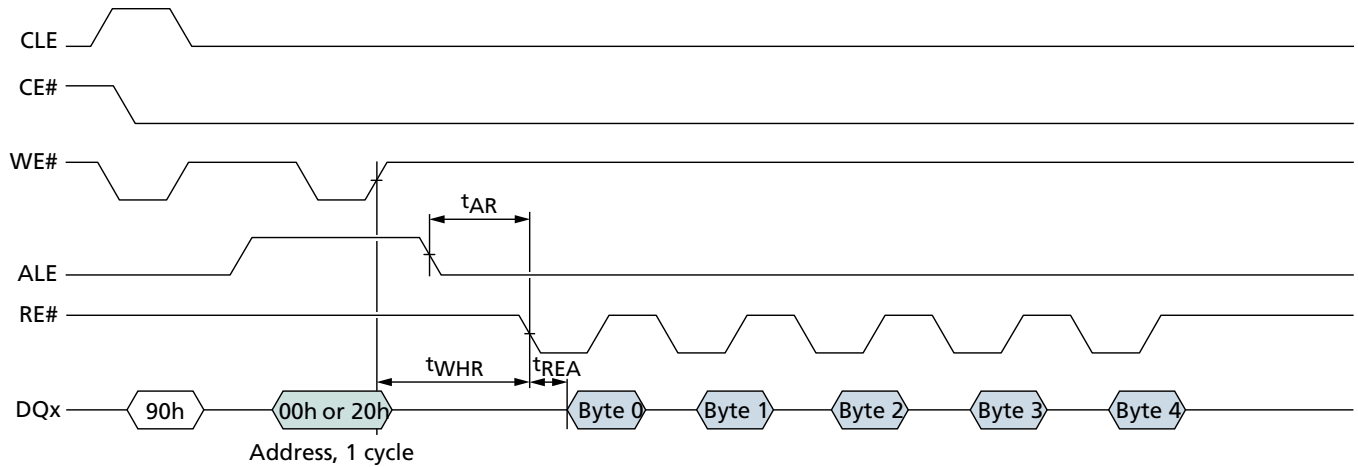
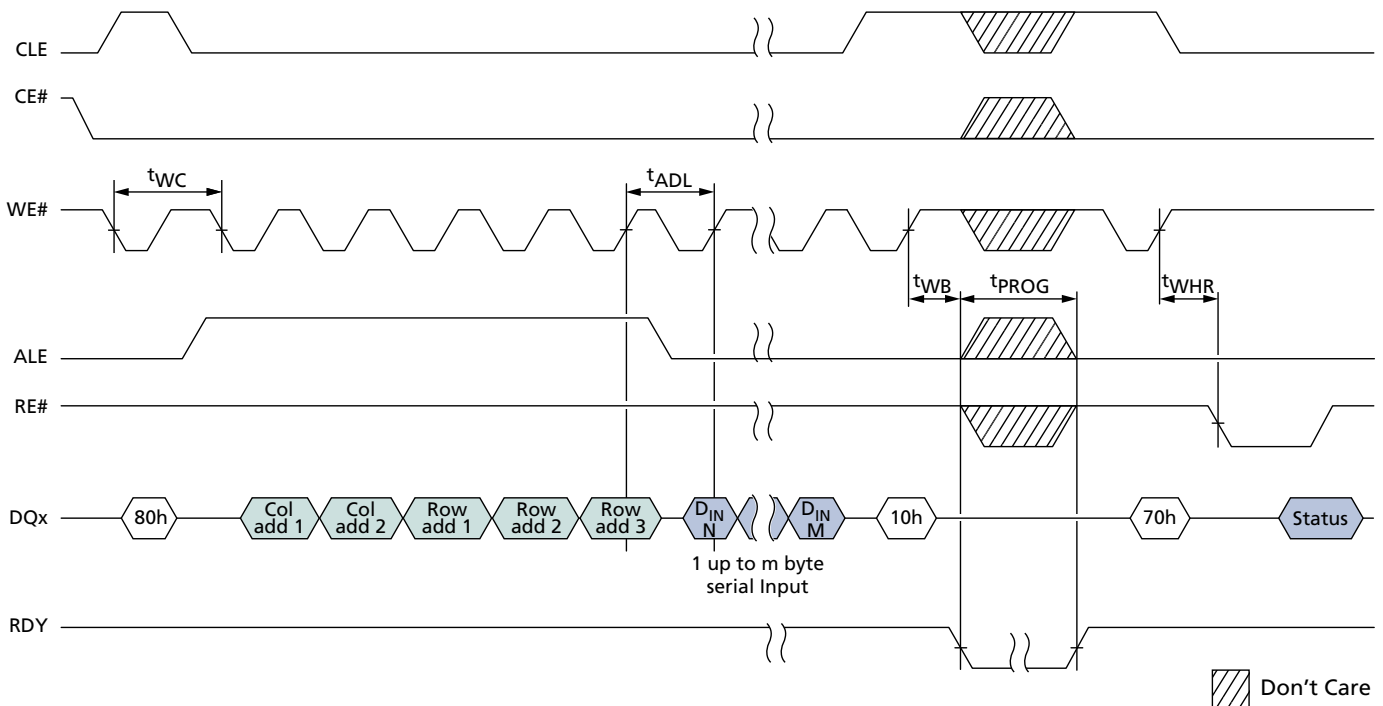


Figure 110: PROGRAM PAGE Operation





MLC 256Gb to 2Tb Async/Sync NAND Asynchronous Interface Timing Diagrams

Figure 111: PROGRAM PAGE Operation with CE# "Don't Care"

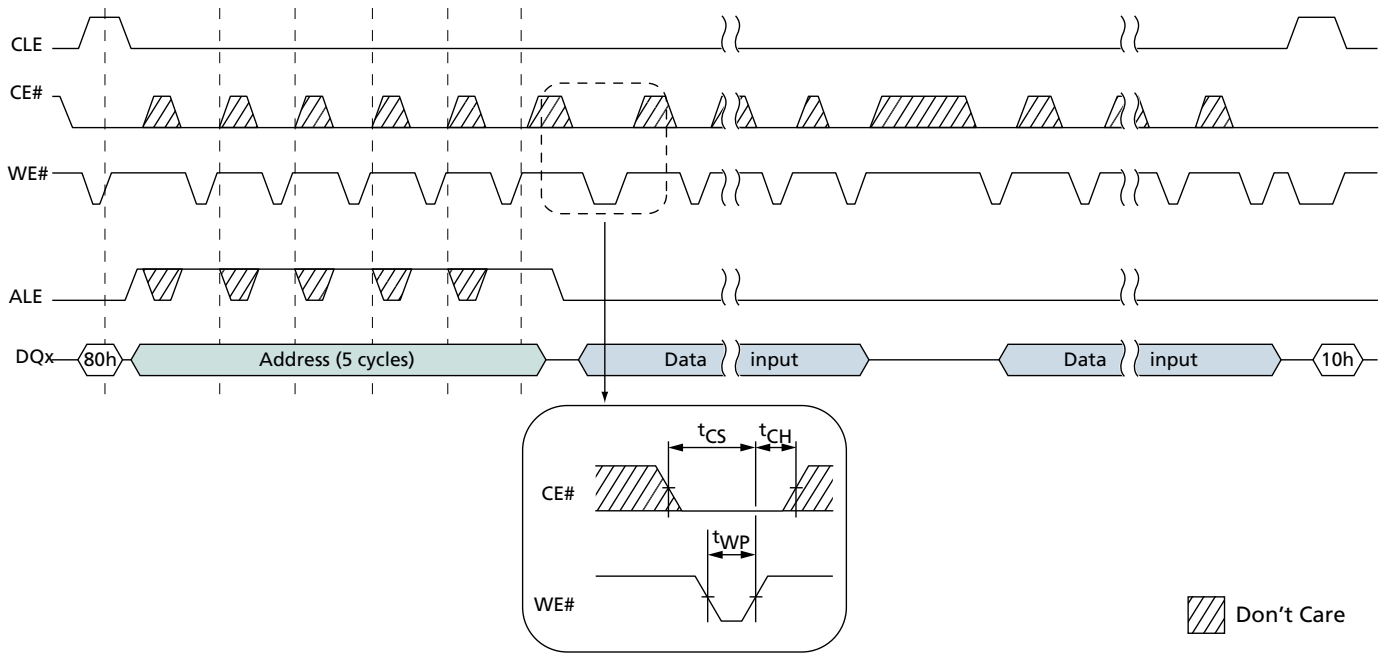
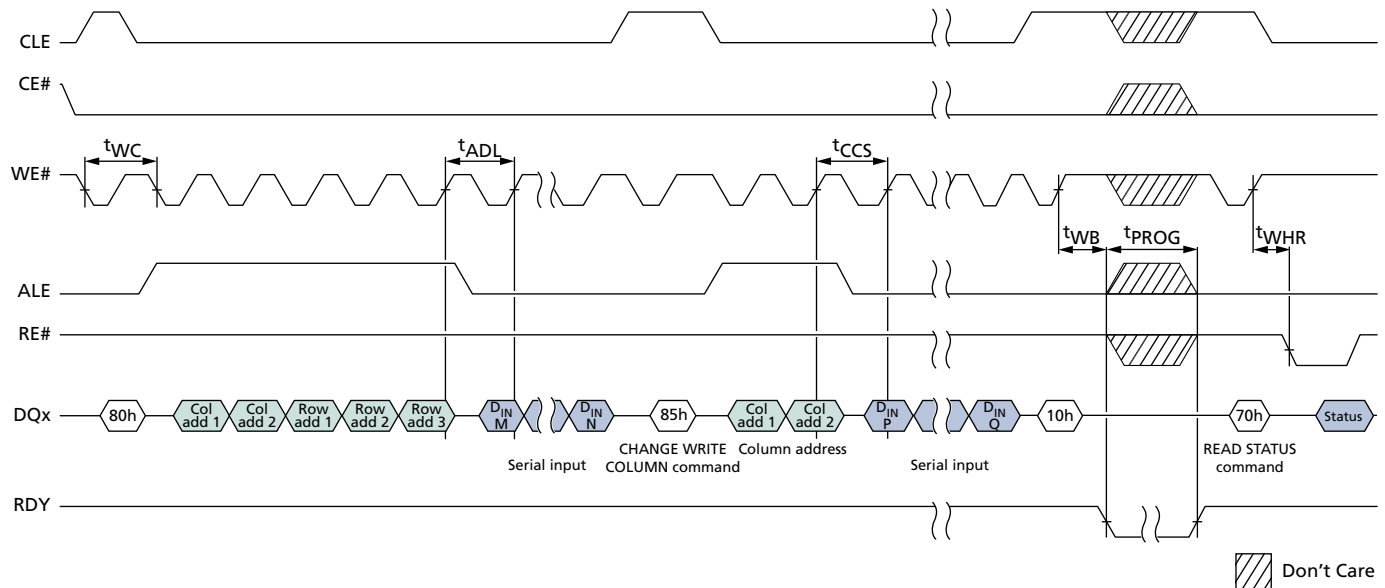


Figure 112: PROGRAM PAGE Operation with CHANGE WRITE COLUMN





MLC 256Gb to 2Tb Async/Sync NAND Asynchronous Interface Timing Diagrams

Figure 113: PROGRAM PAGE CACHE

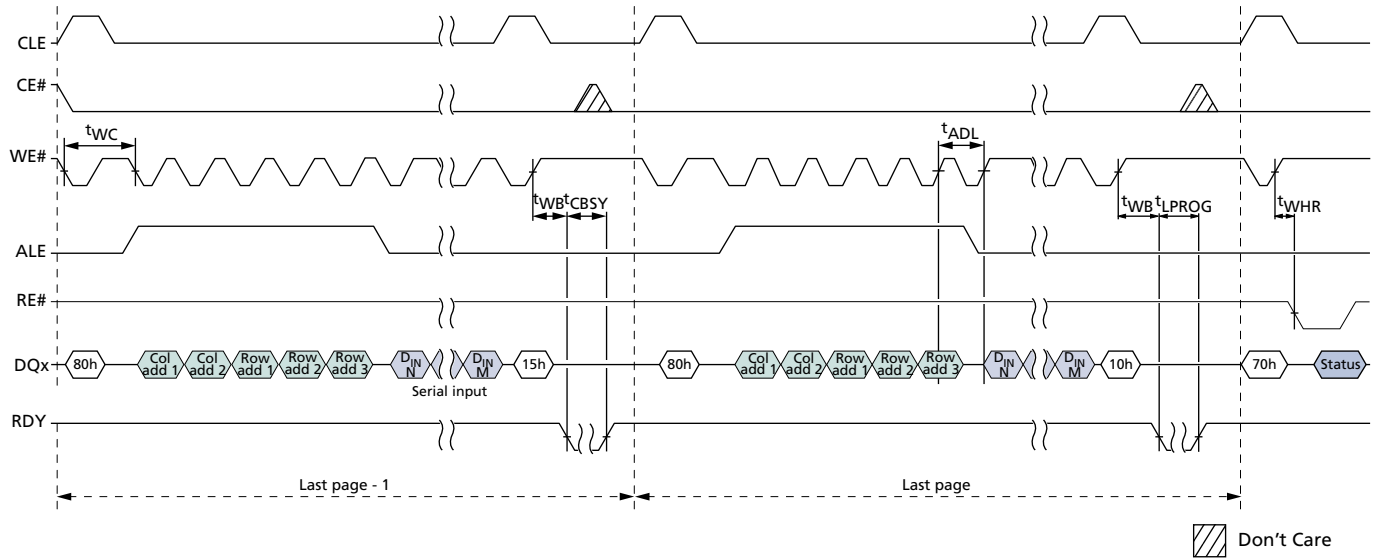
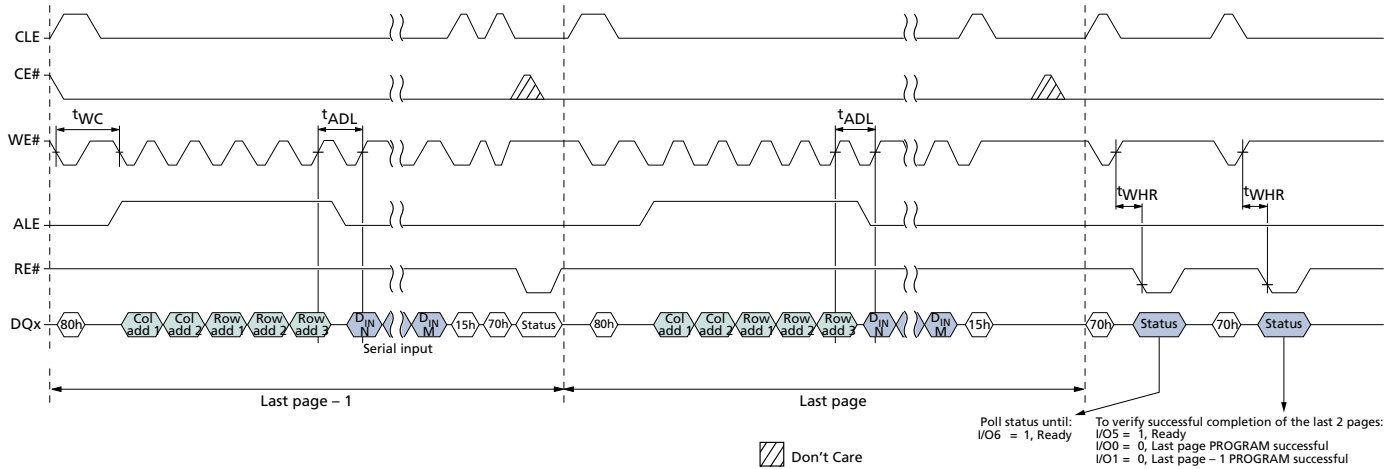


Figure 114: PROGRAM PAGE CACHE Ending on 15h





MLC 256Gb to 2Tb Async/Sync NAND Asynchronous Interface Timing Diagrams

Figure 115: COPYBACK

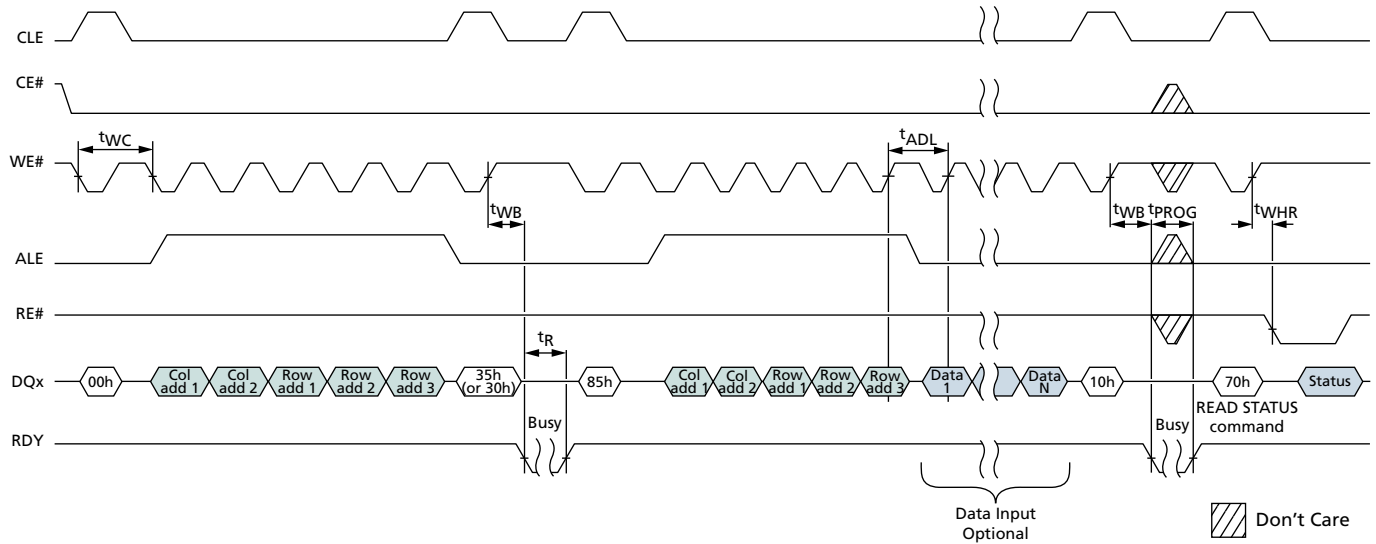
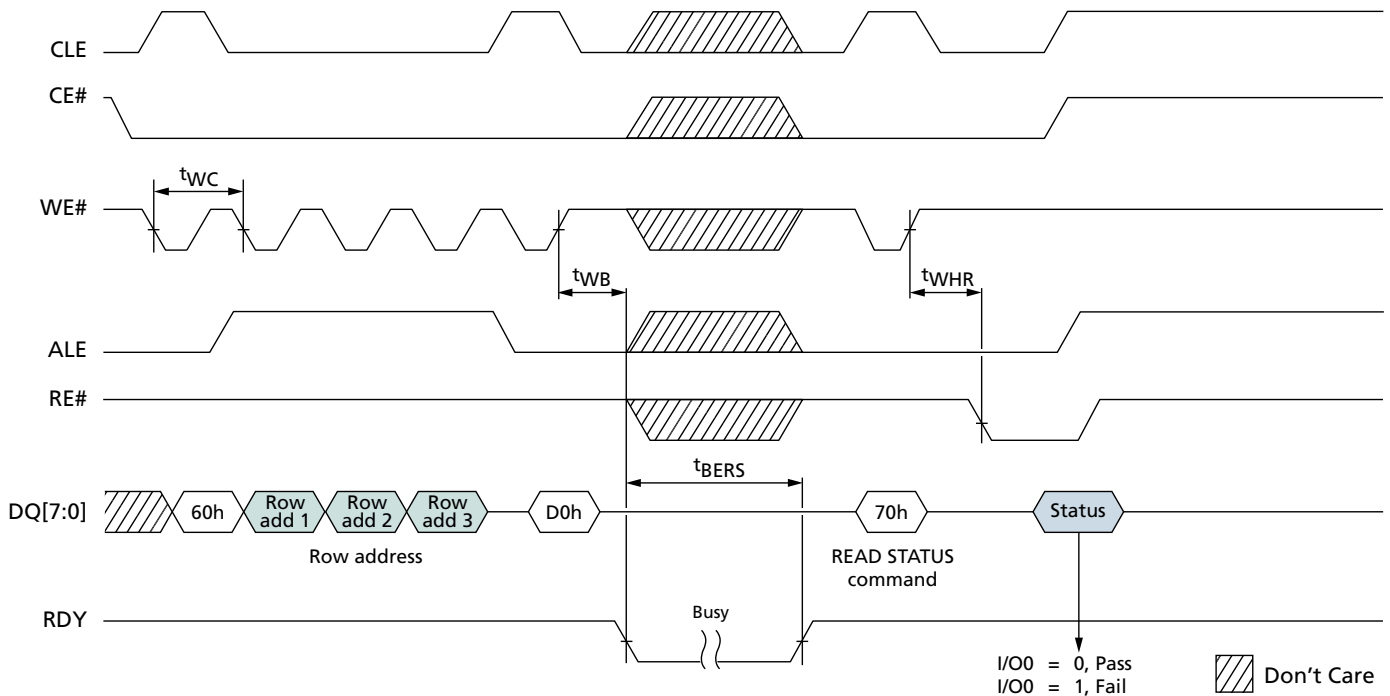
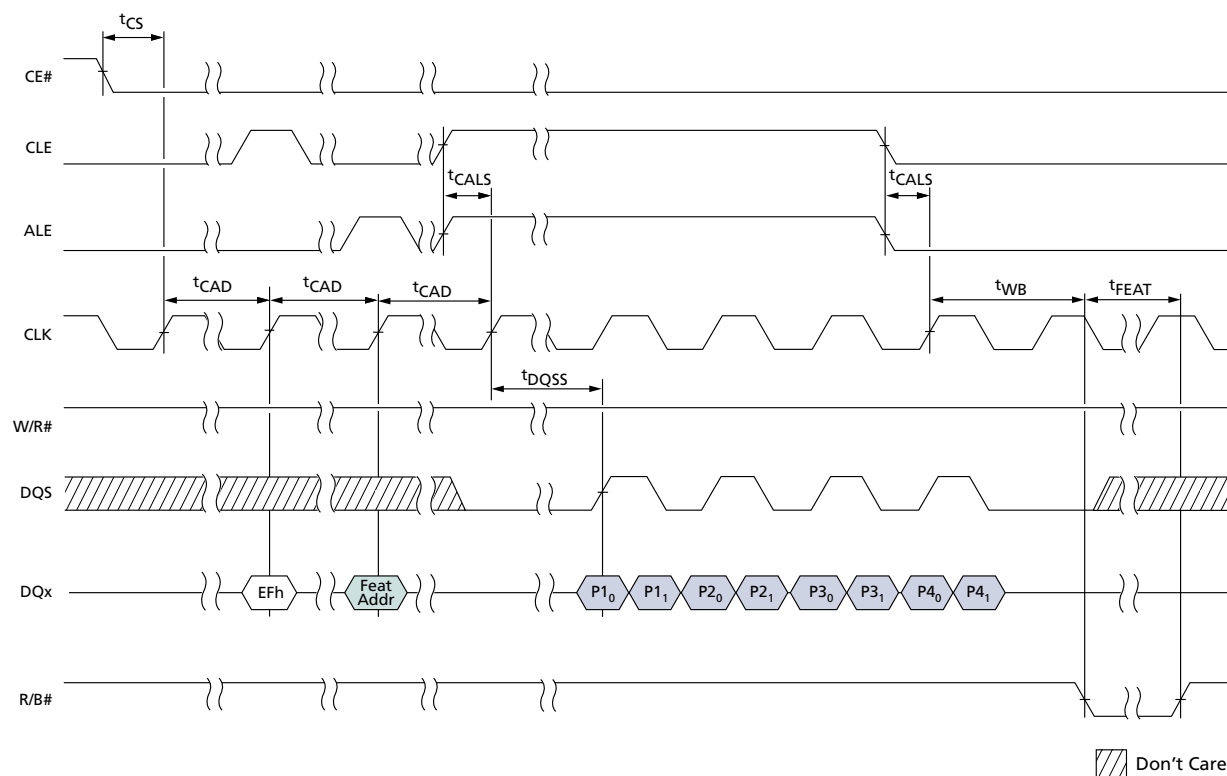


Figure 116: ERASE BLOCK Operation





NV-DDR Interface Timing Diagrams

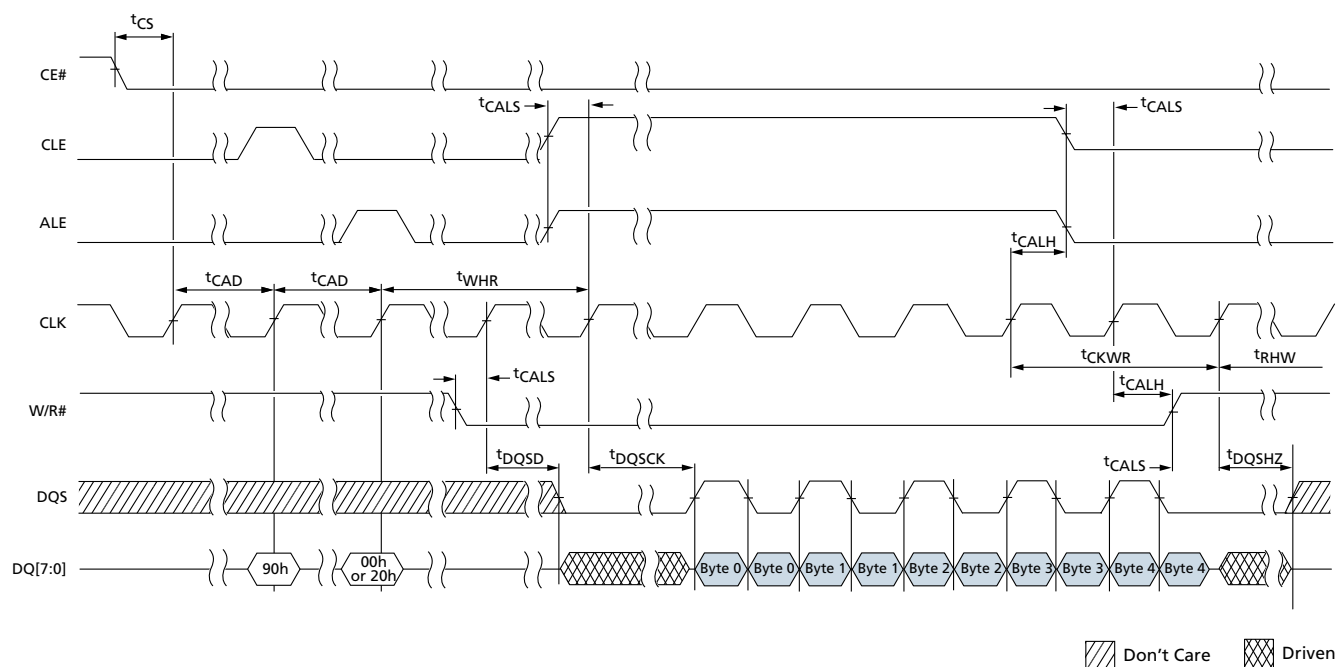
Figure 117: SET FEATURES Operation


- Notes:
1. When CE# remains LOW, t_{CAD} begins at the rising edge of the clock from which the last data byte is input for the subsequent command or data input cycle(s).
 2. t_{DSH} (MIN) generally occurs during t_{DQSS} (MIN).
 3. t_{DSS} (MIN) generally occurs during t_{DQSS} (MAX).
 4. The cycle that t_{CAD} is measured from may be an idle cycle (as shown), another command cycle, an address cycle, or a data cycle. The idle cycle is shown in this diagram for simplicity.



MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

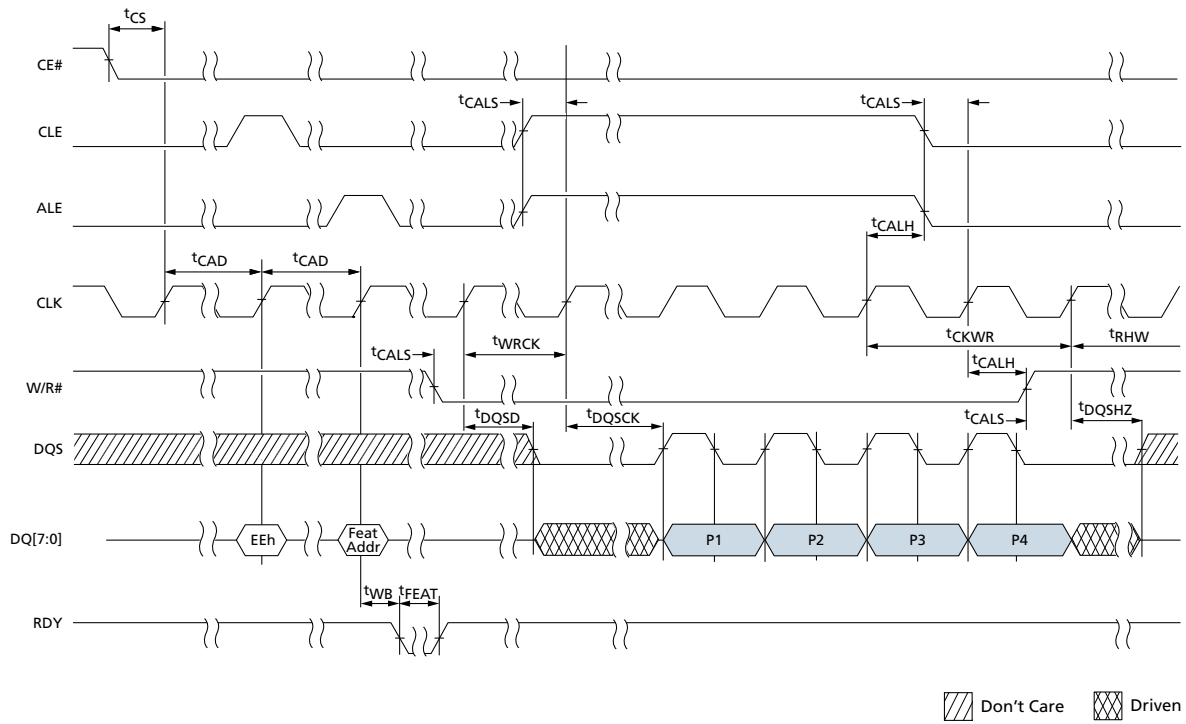
Figure 118: READ ID Operation





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

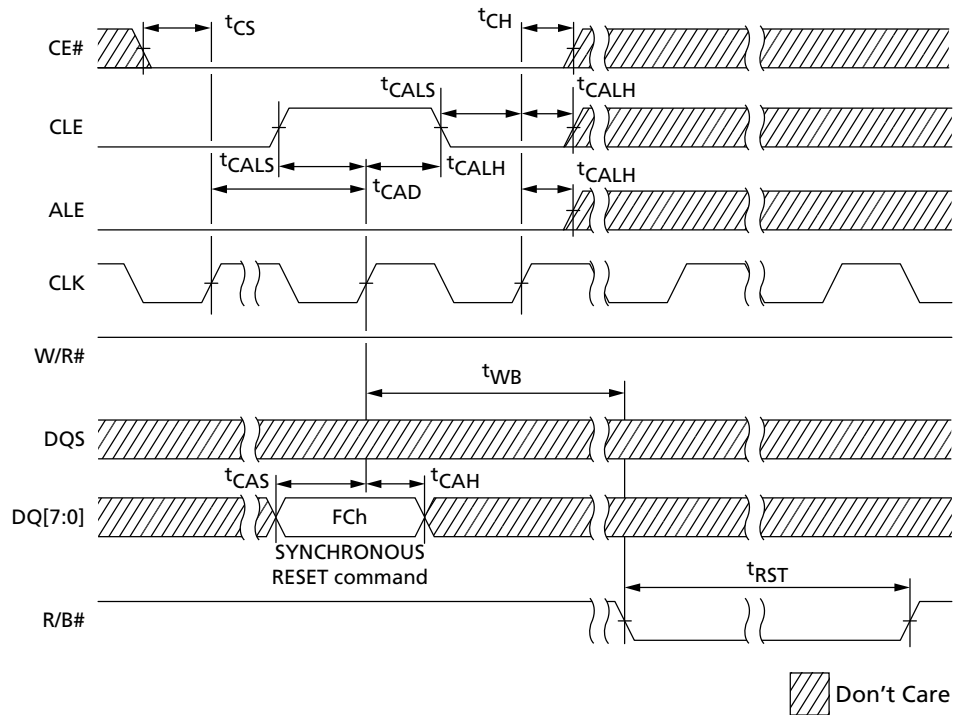
Figure 119: GET FEATURES Operation





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

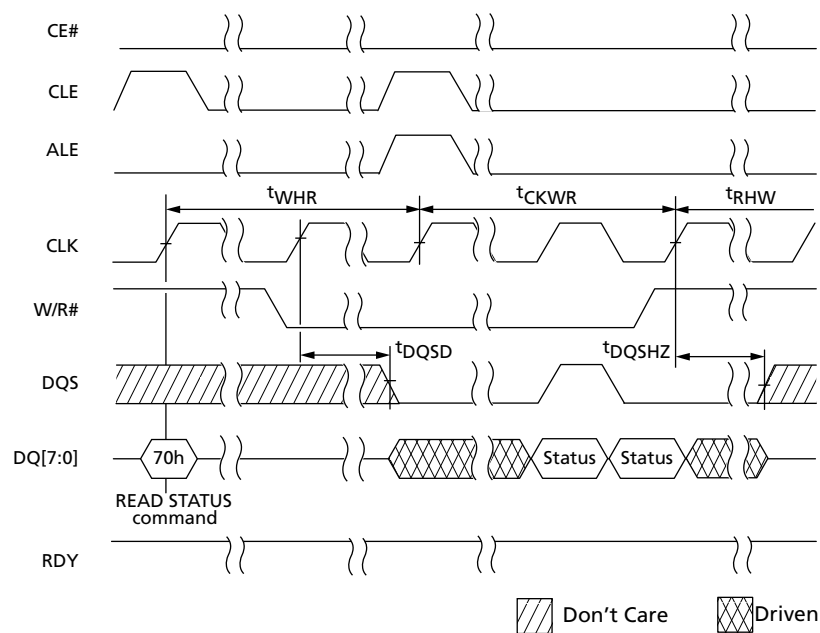
Figure 120: RESET (FCh) Operation





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

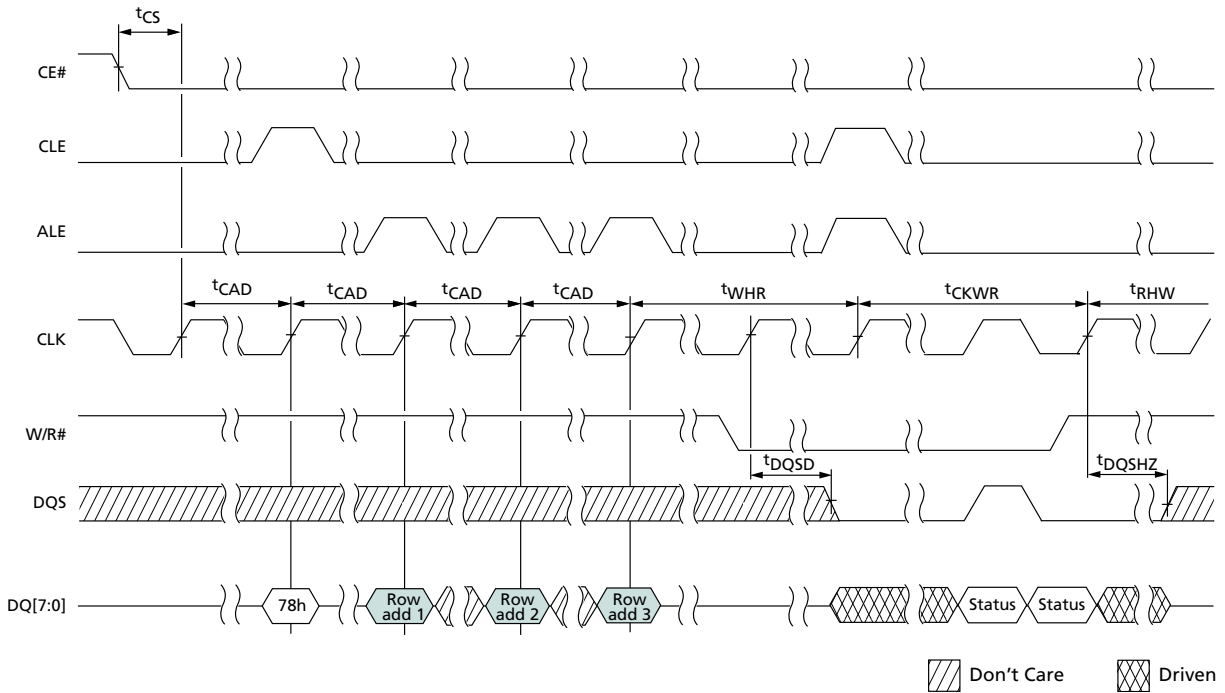
Figure 121: READ STATUS Cycle





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

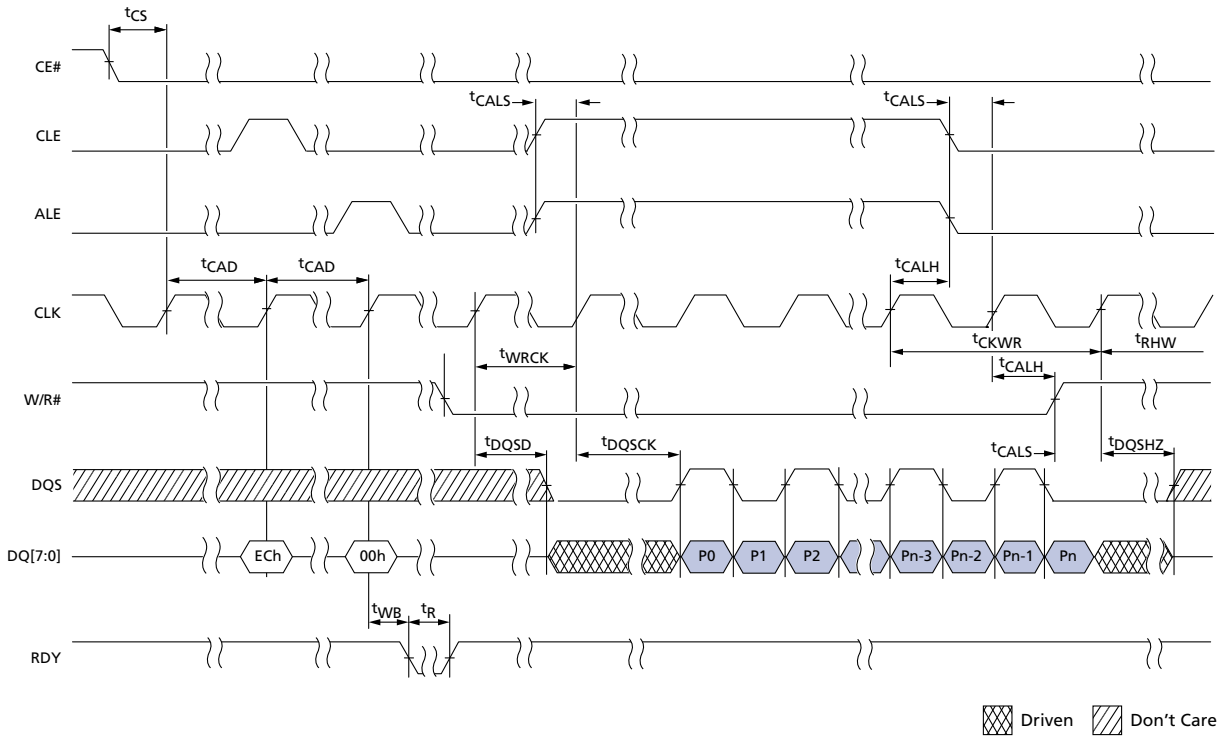
Figure 122: READ STATUS ENHANCED Operation





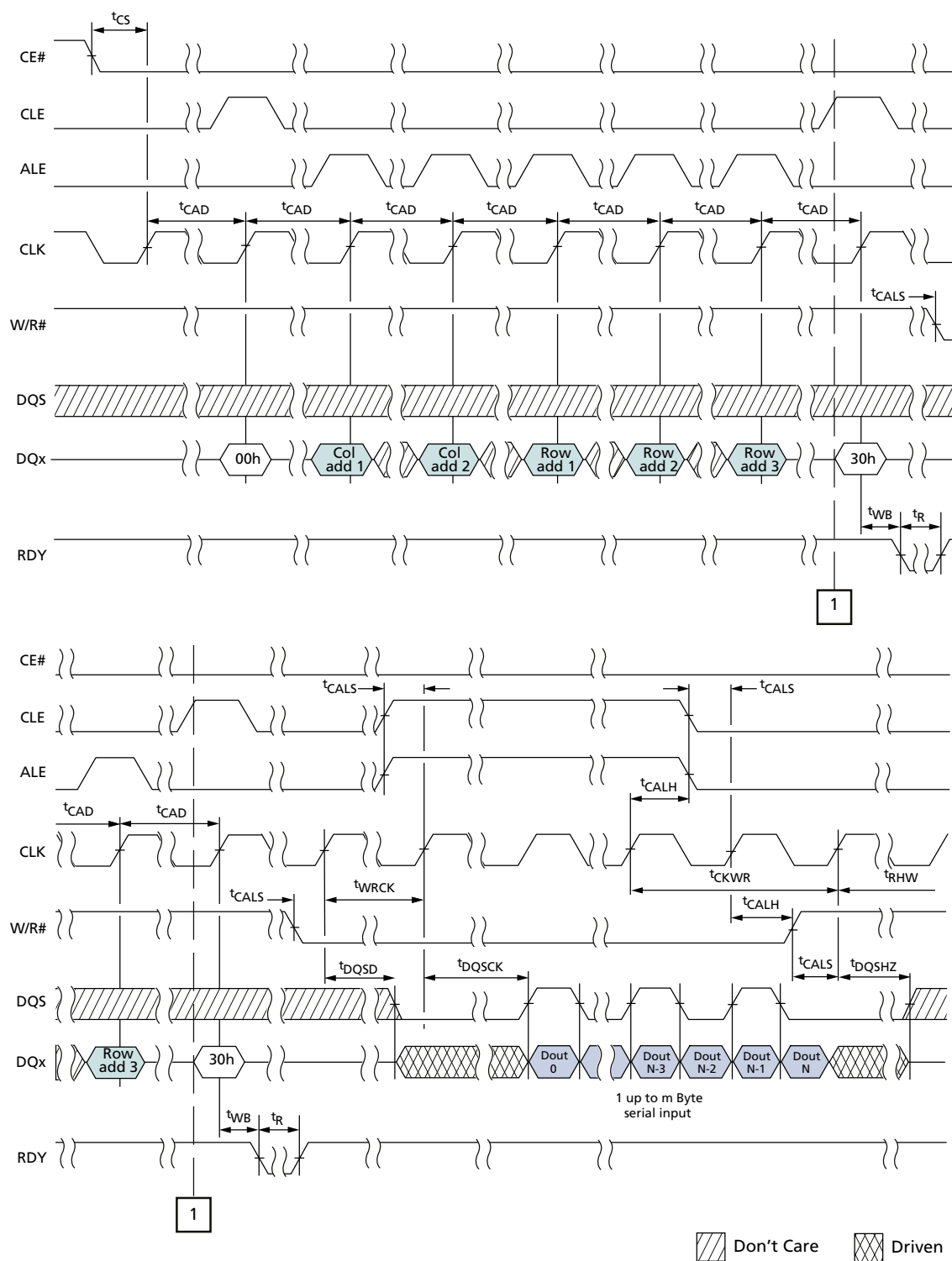
MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

Figure 123: READ PARAMETER PAGE Operation



MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

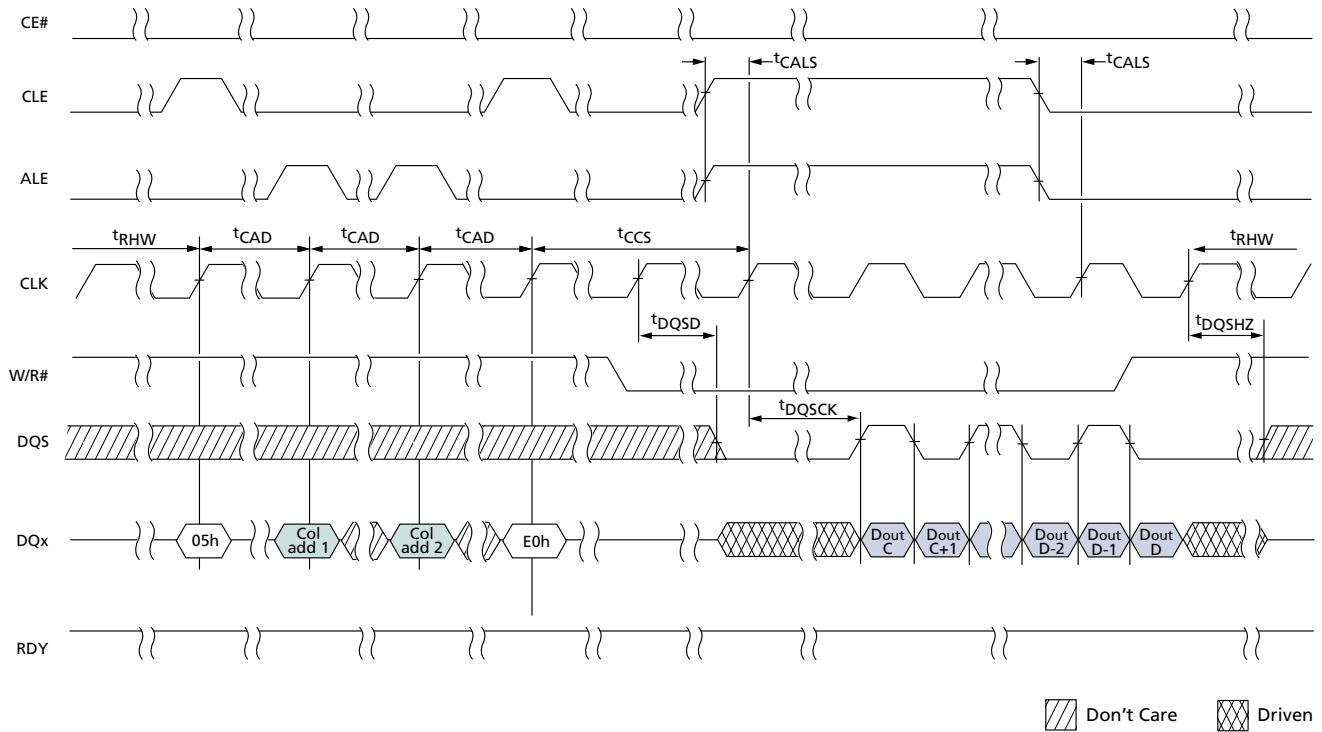
Figure 124: READ PAGE Operation





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

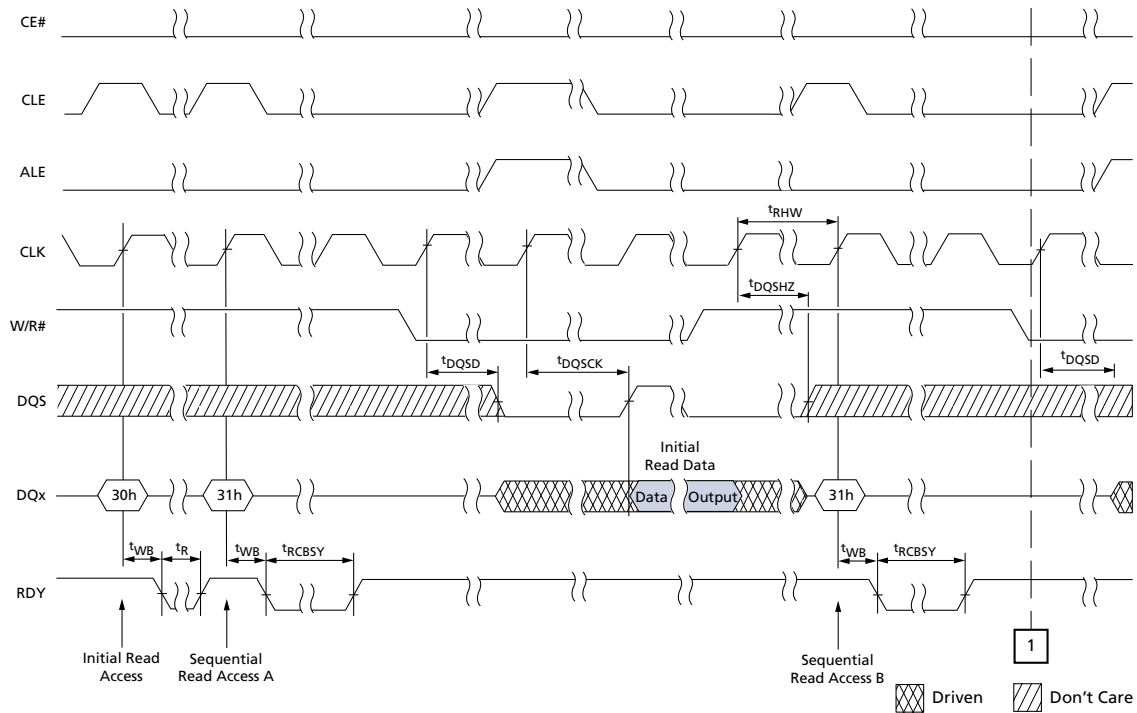
Figure 125: CHANGE READ COLUMN





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

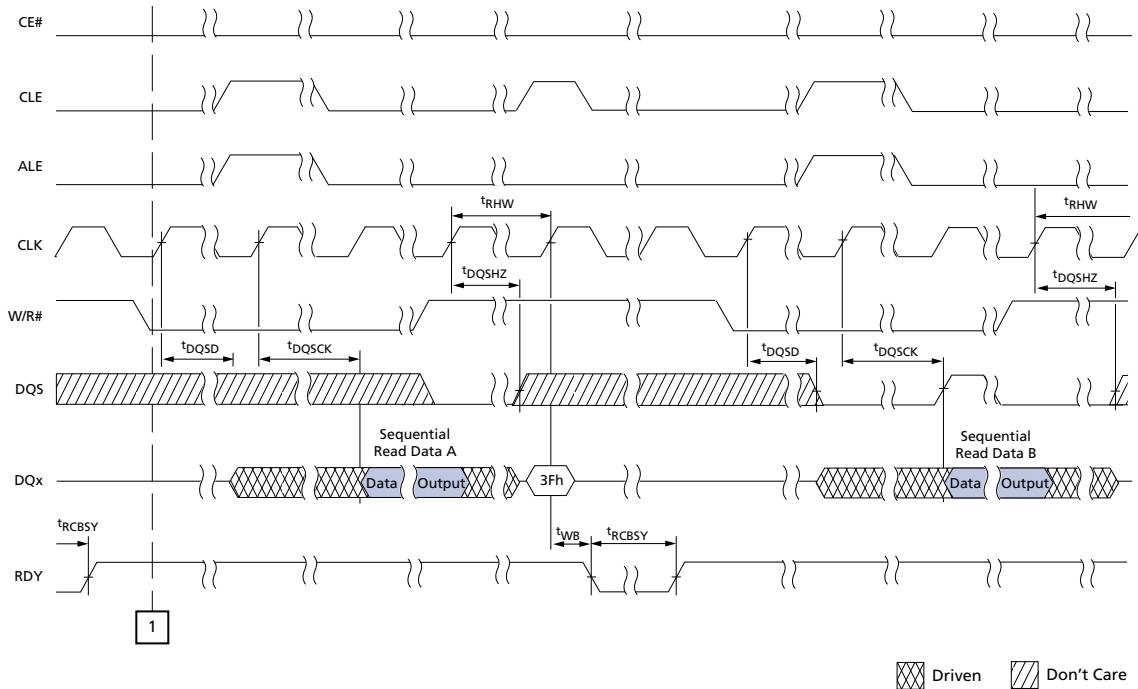
Figure 126: READ PAGE CACHE SEQUENTIAL (1 of 2)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

Figure 127: READ PAGE CACHE SEQUENTIAL (2 of 2)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

Figure 128: READ PAGE CACHE RANDOM (1 of 2)

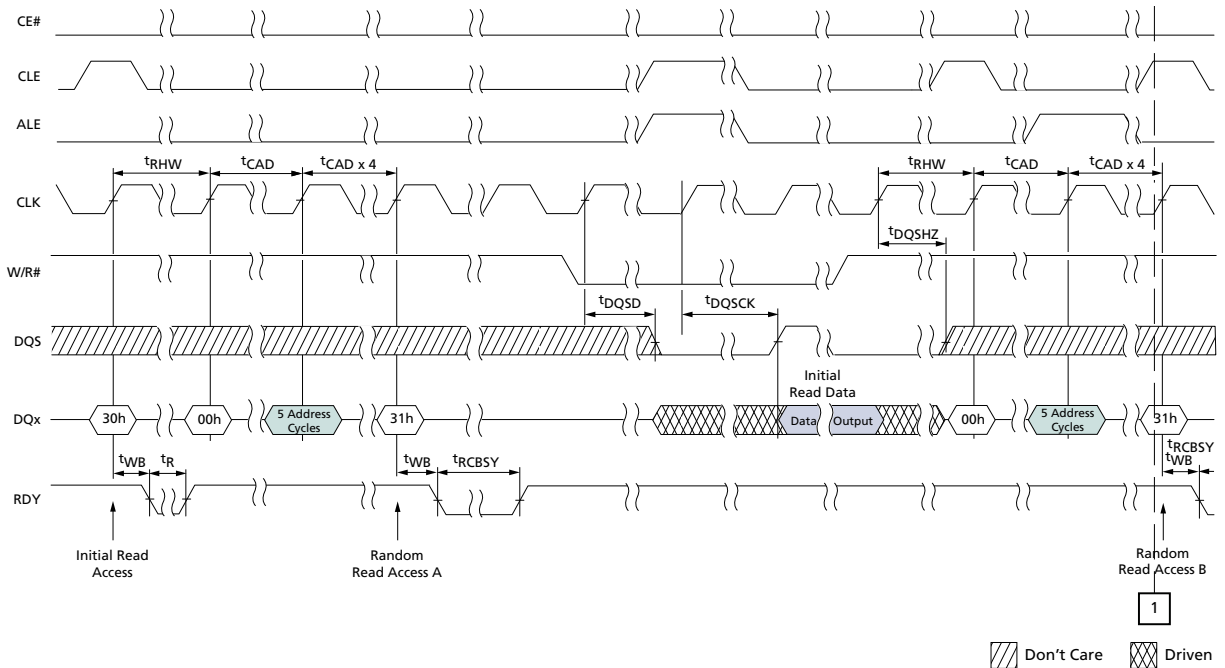
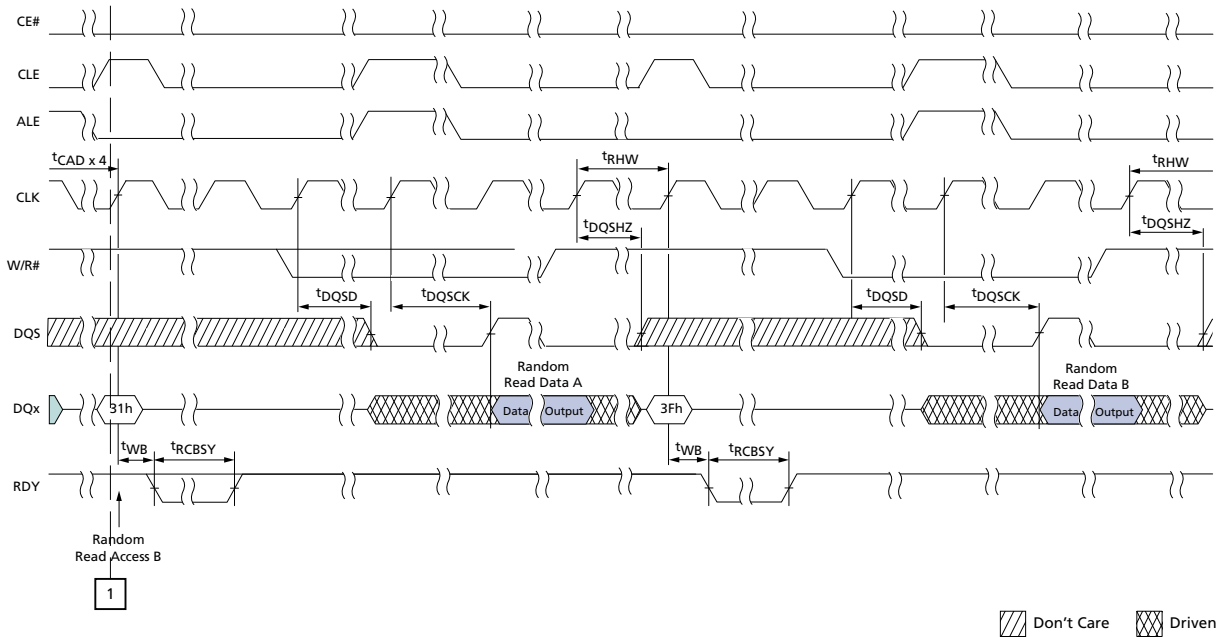


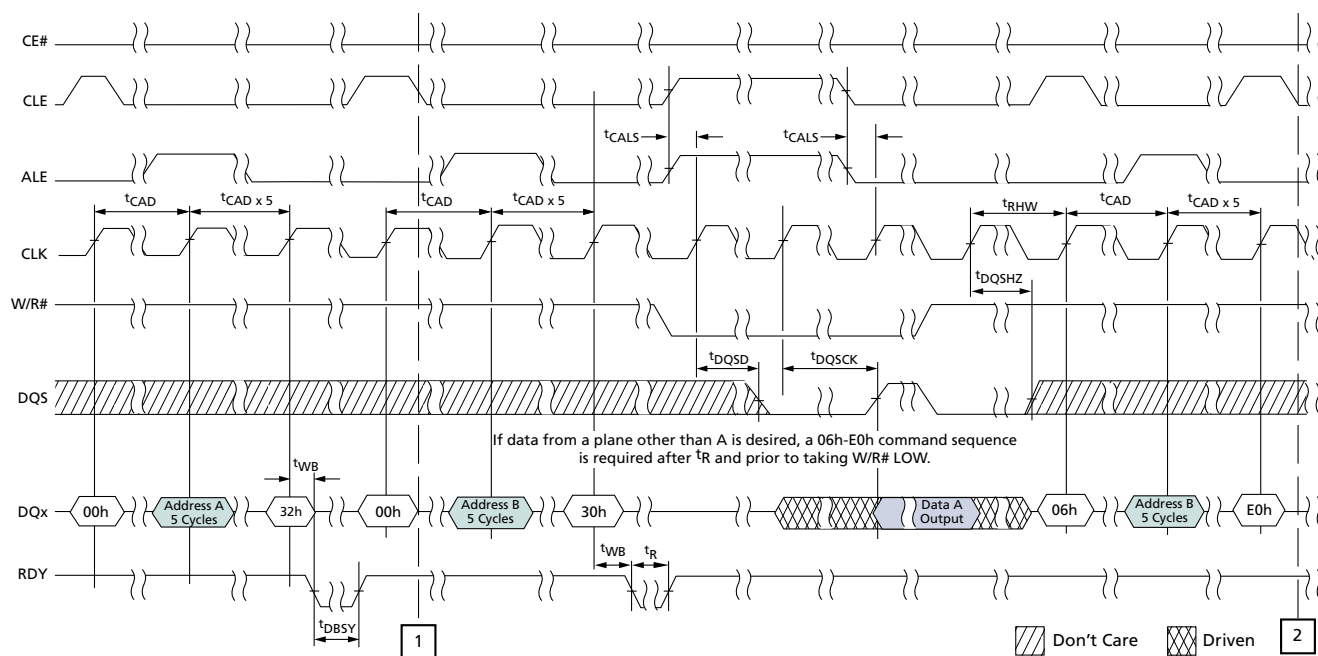
Figure 129: READ PAGE CACHE RANDOM (2 of 2)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

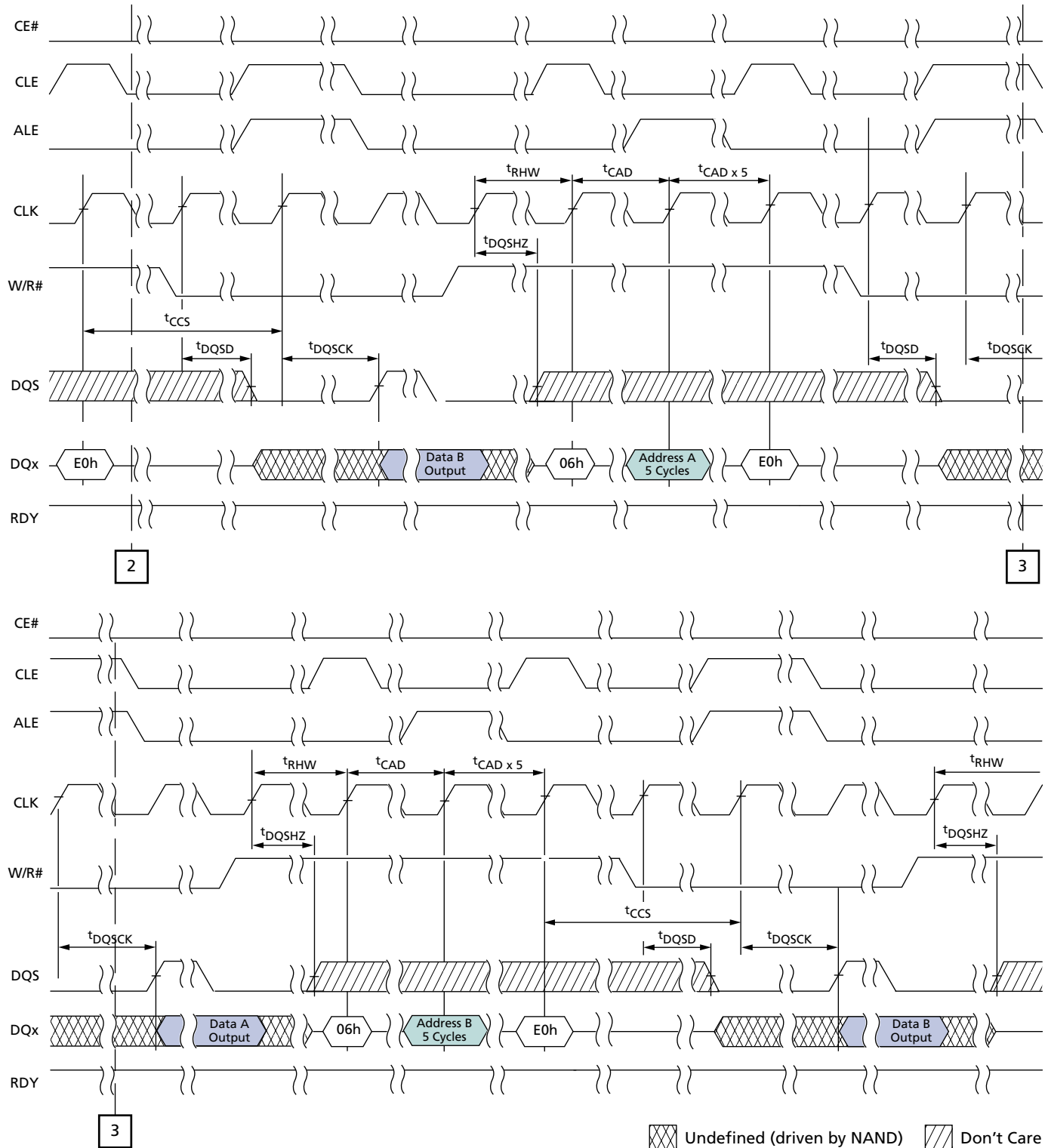
Figure 130: Multi-Plane Read Page (1 of 2)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

Figure 131: Multi-Plane Read Page (2 of 2)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

Figure 132: PROGRAM PAGE Operation (1 of 2)

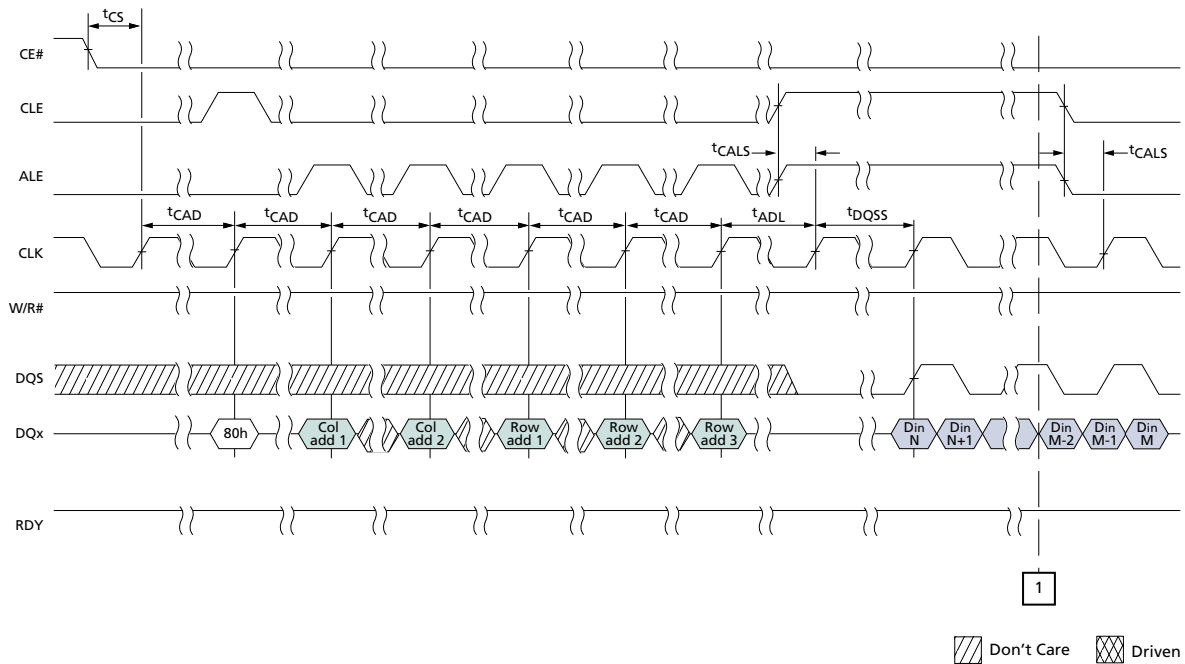
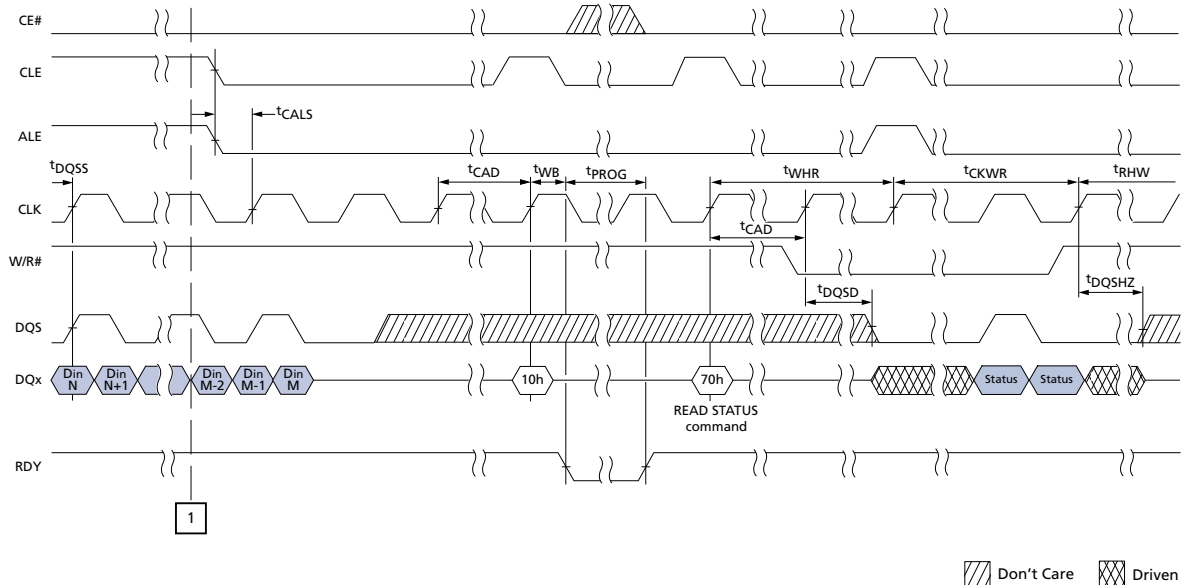


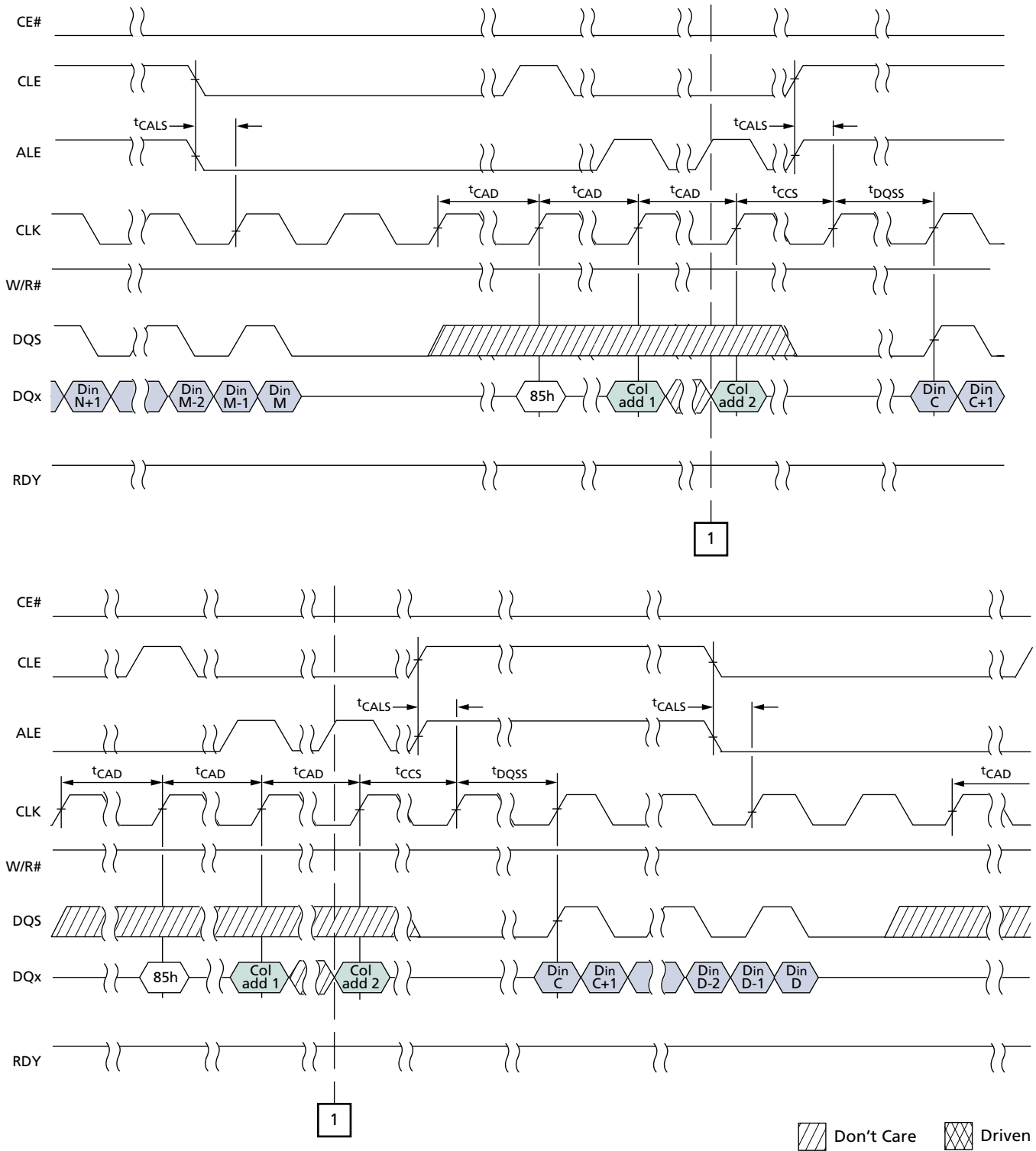
Figure 133: PROGRAM PAGE Operation (2 of 2)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

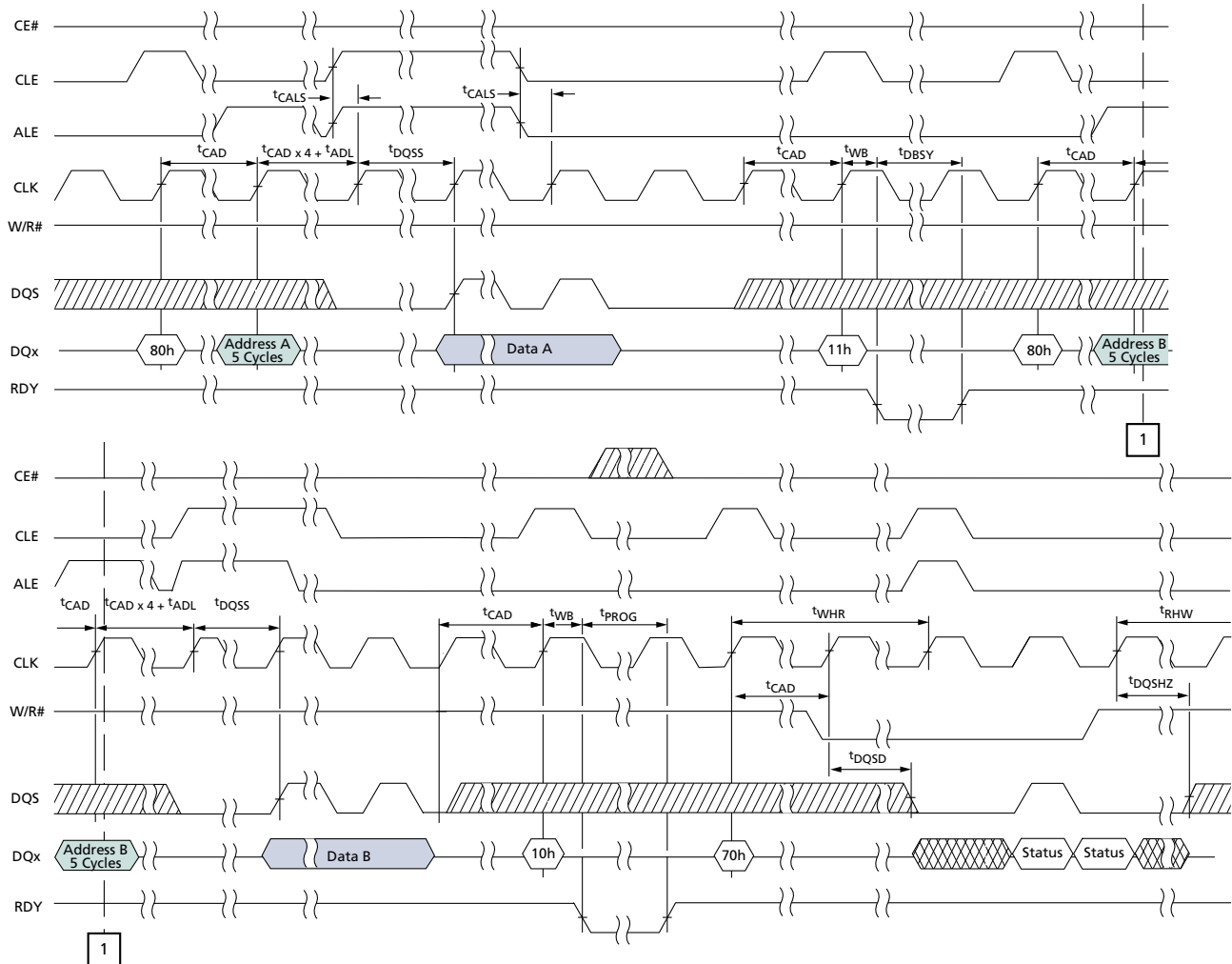
Figure 134: CHANGE WRITE COLUMN





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

Figure 135: Multi-Plane Program Page





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

Figure 136: ERASE BLOCK

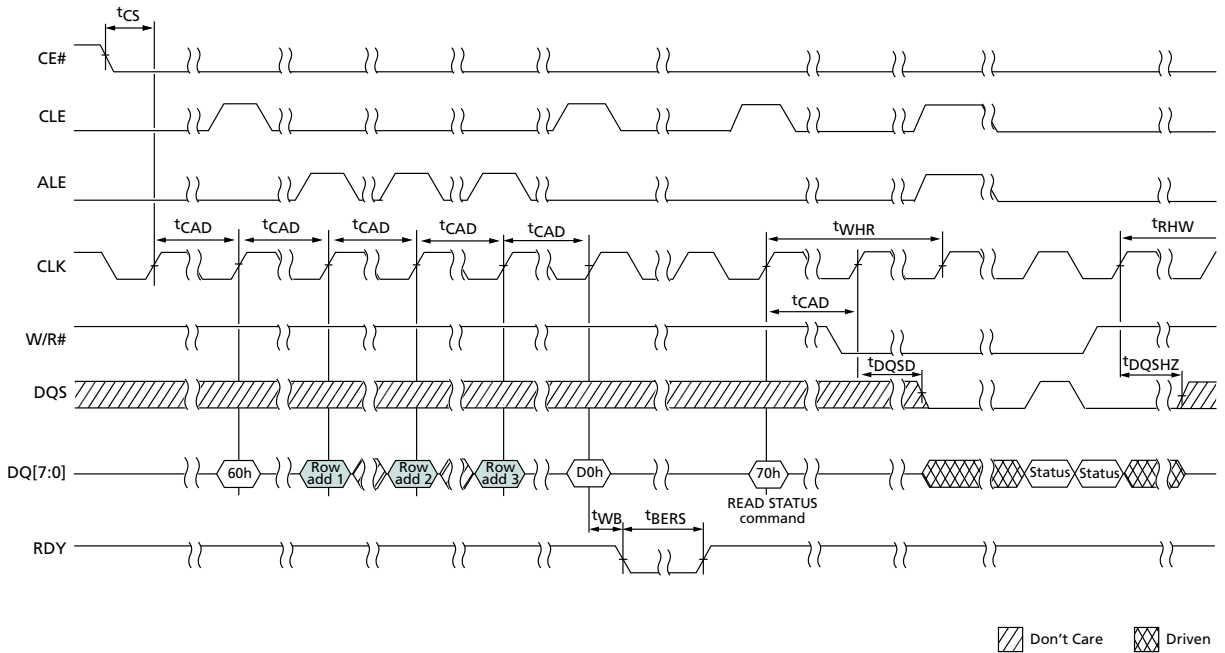
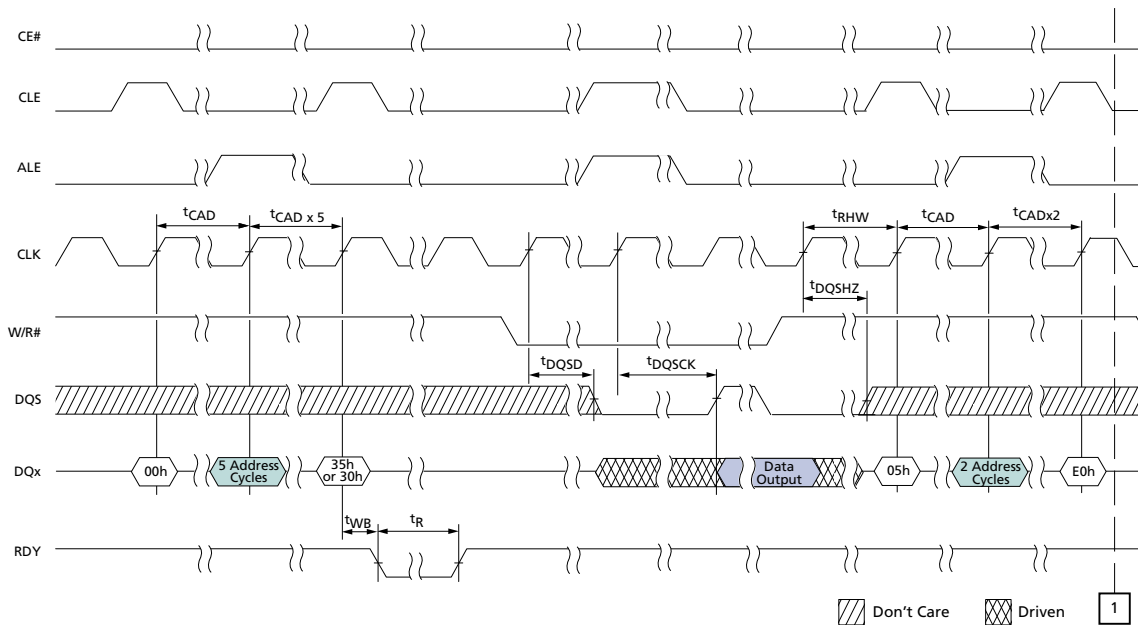


Figure 137: COPYBACK (1 of 3)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

Figure 138: COPYBACK (2 of 3)

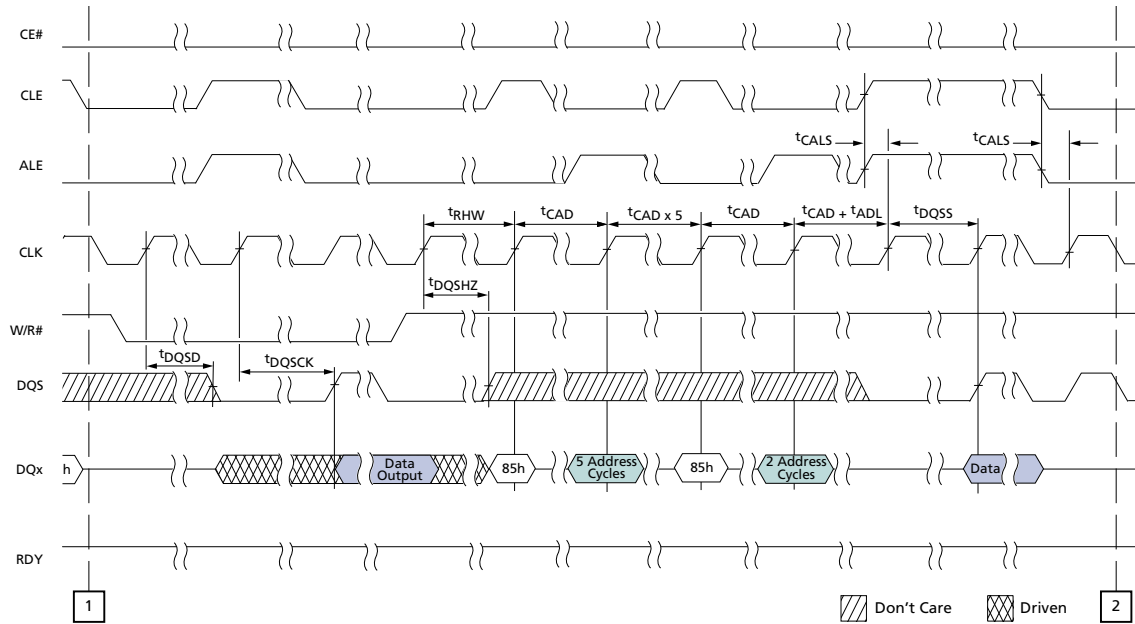
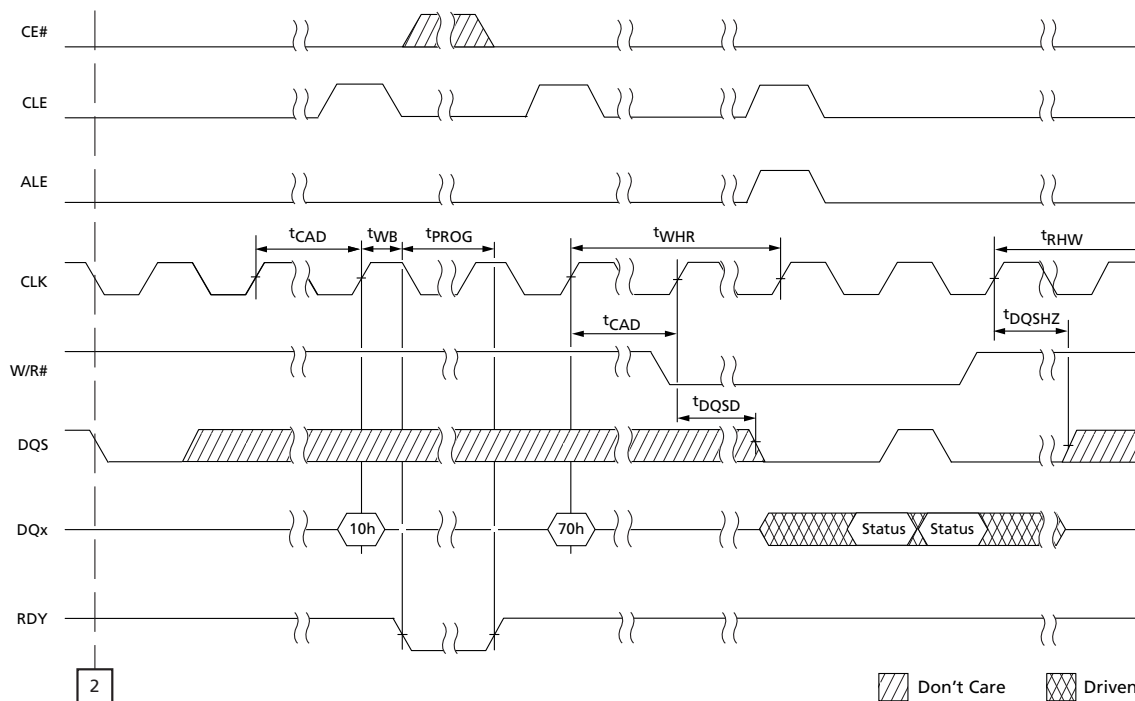


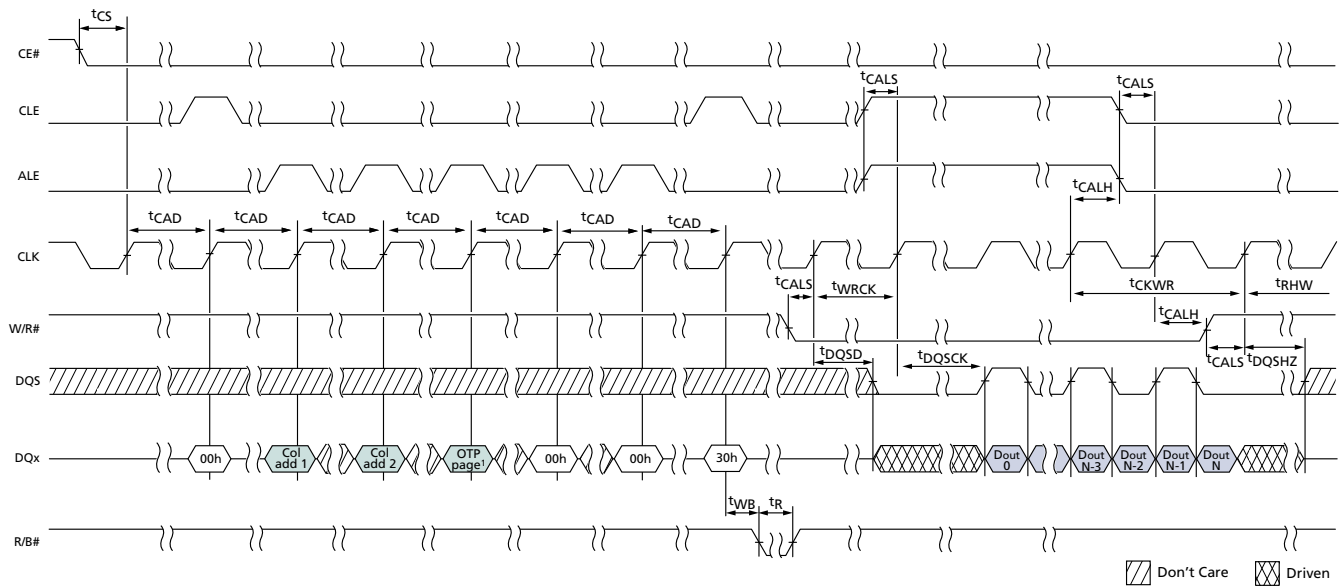
Figure 139: COPYBACK (3 of 3)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

Figure 140: READ OTP PAGE





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

Figure 141: PROGRAM OTP PAGE (1 of 2)

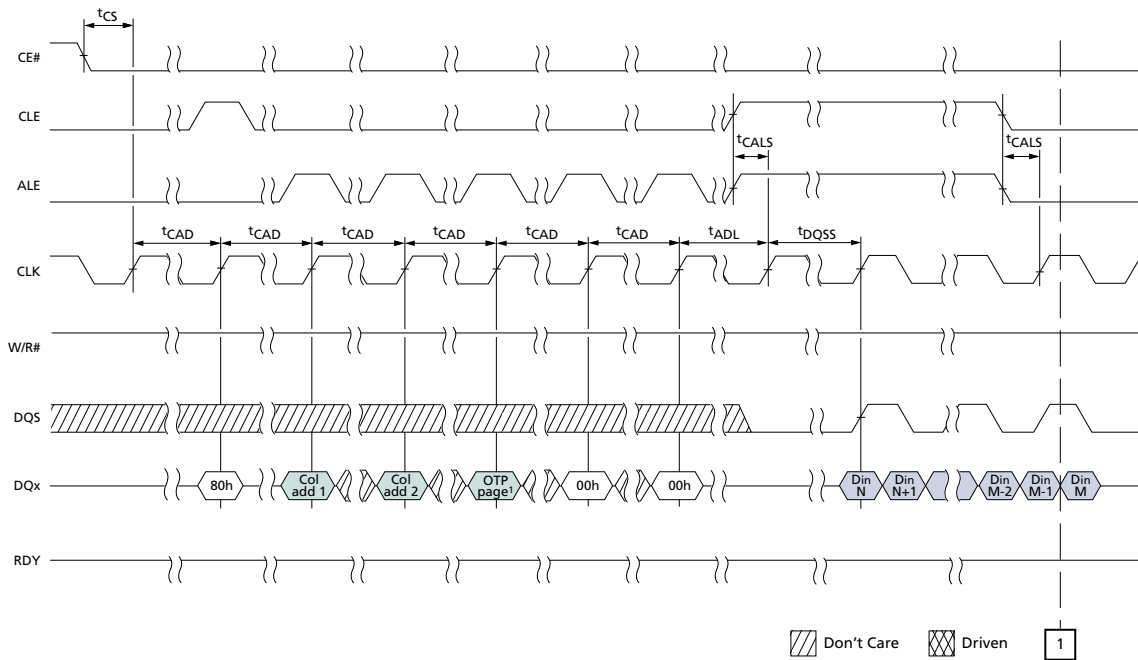
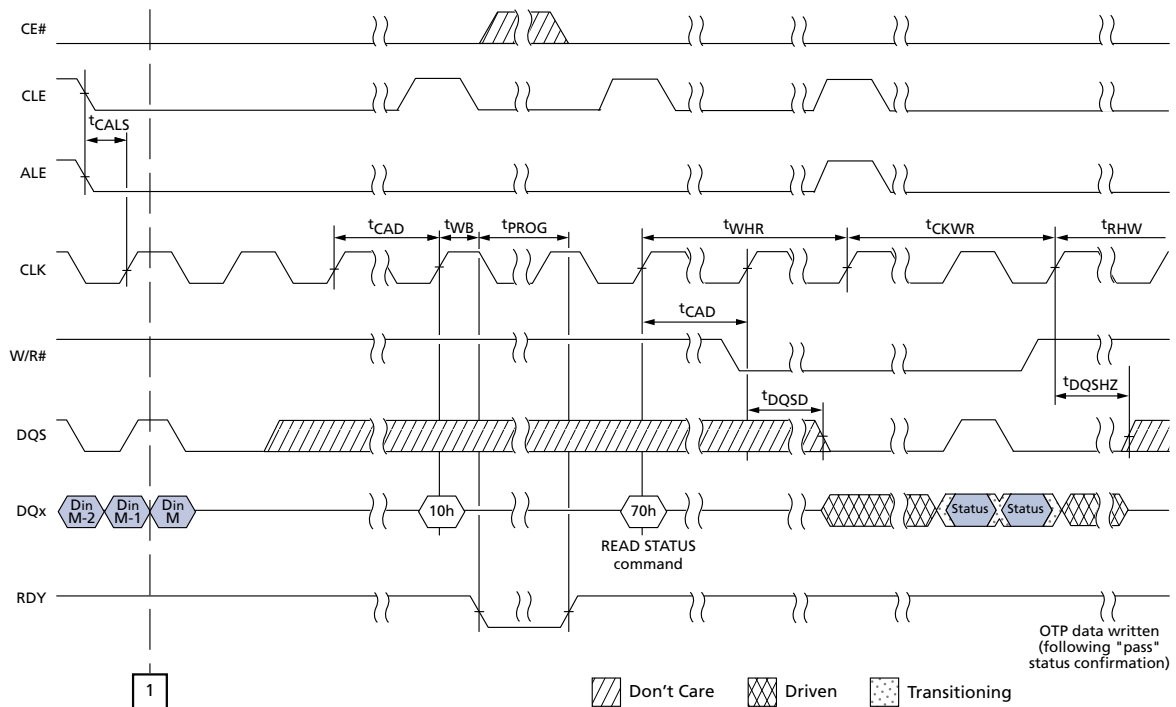


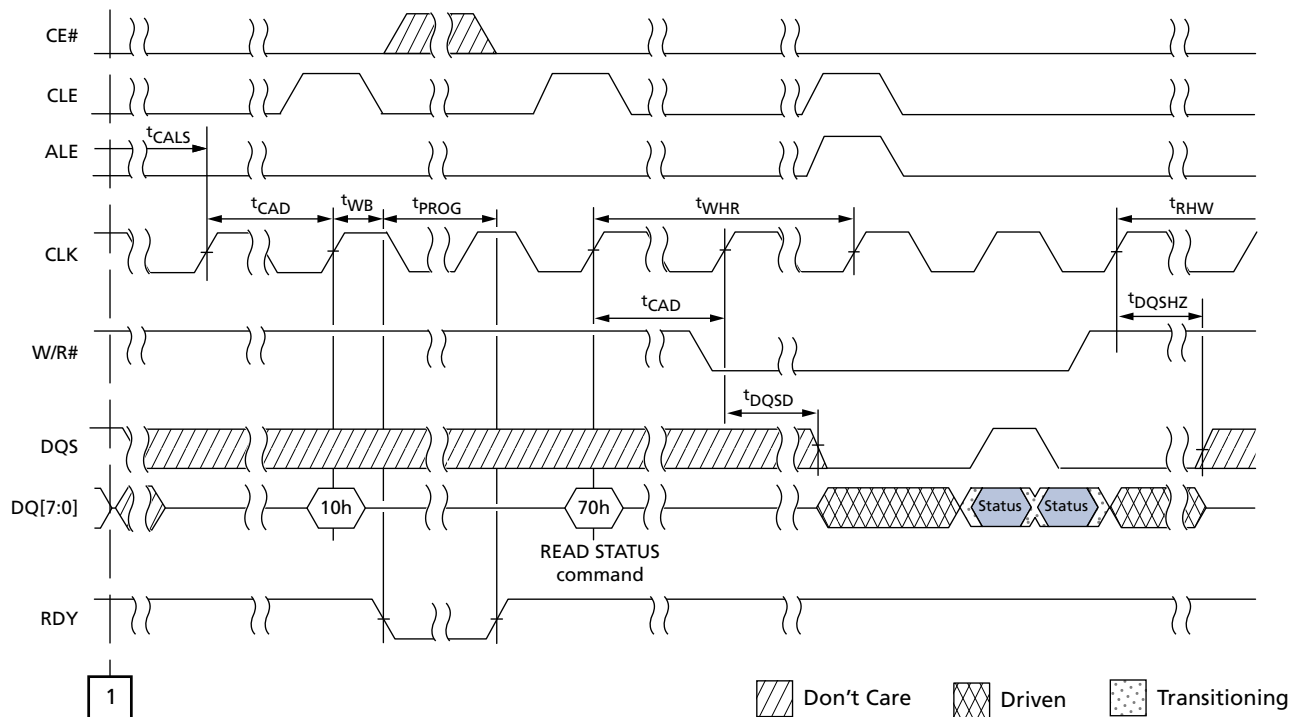
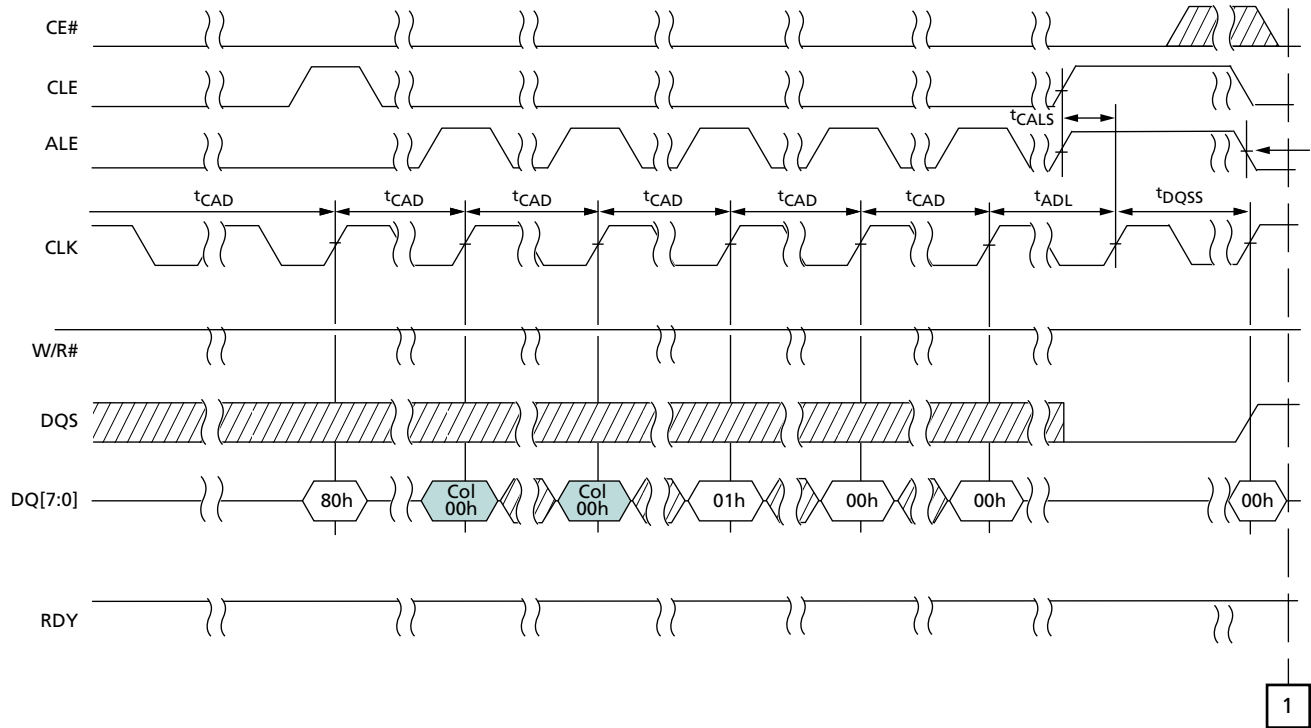
Figure 142: PROGRAM OTP PAGE (2 of 2)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR Interface Timing Diagrams

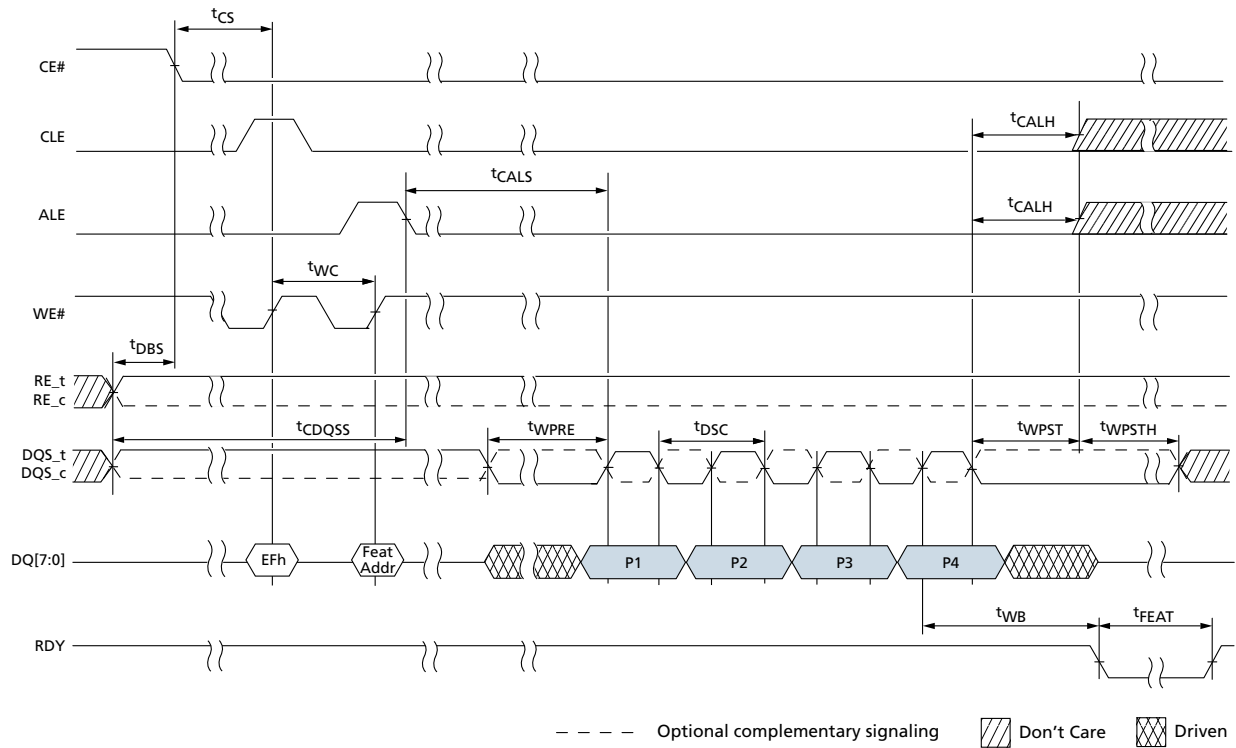
Figure 143: PROTECT OTP AREA





NV-DDR2 Interface Timing Diagrams

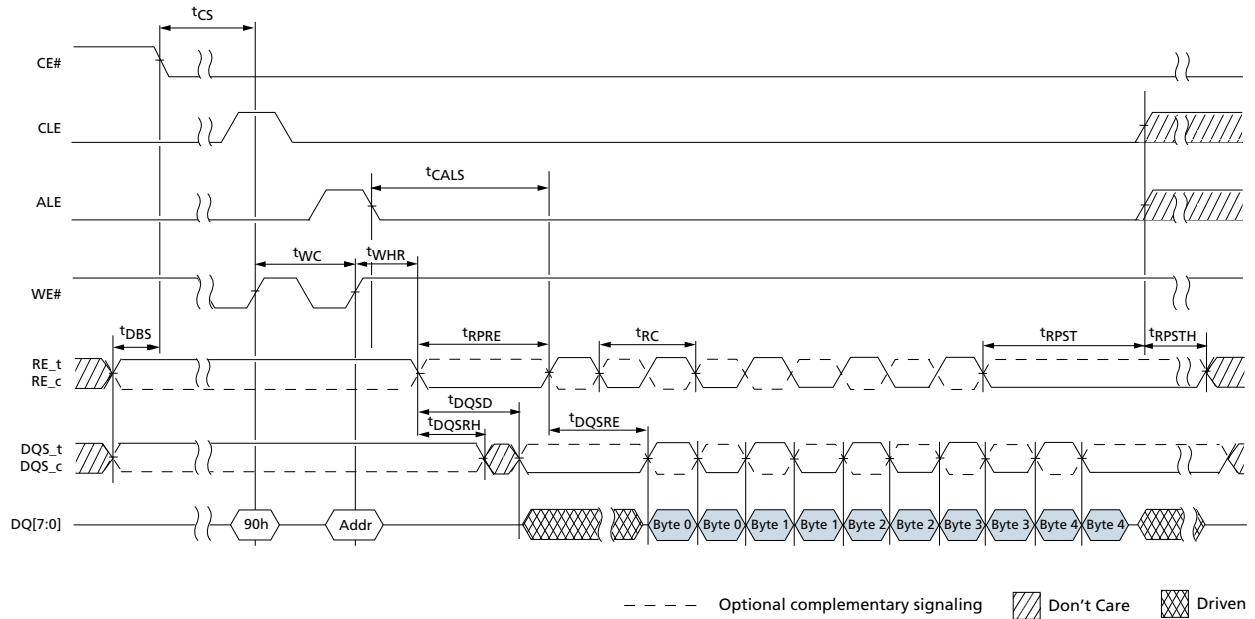
Figure 144: SET FEATURES Operation





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

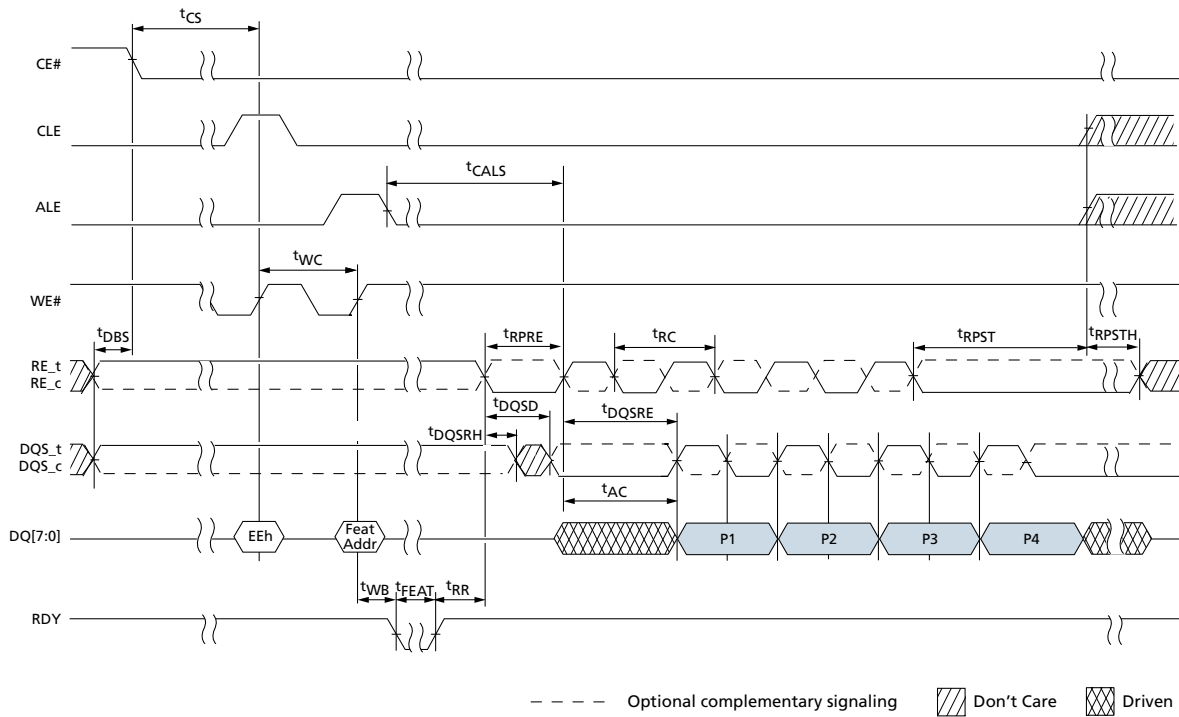
Figure 145: READ ID Operation





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

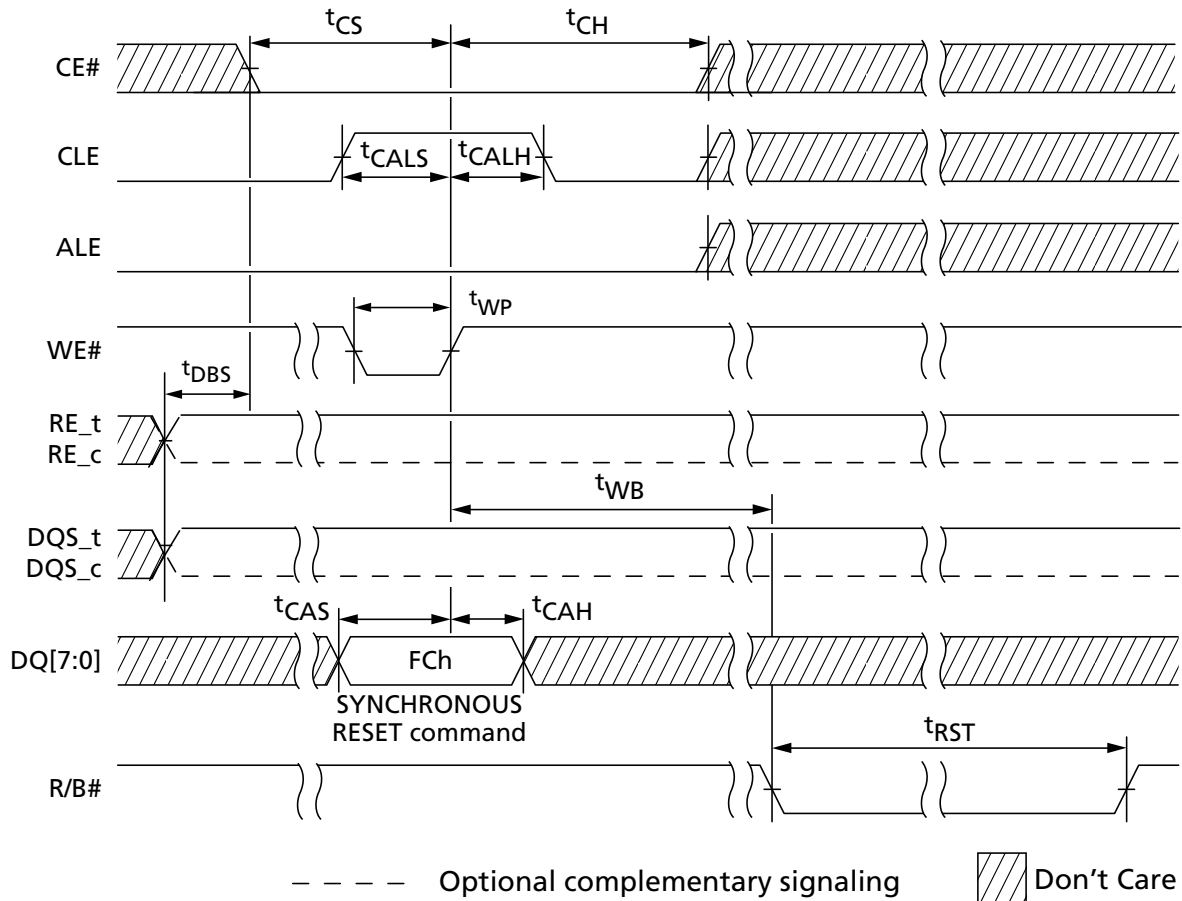
Figure 146: GET FEATURES Operation





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

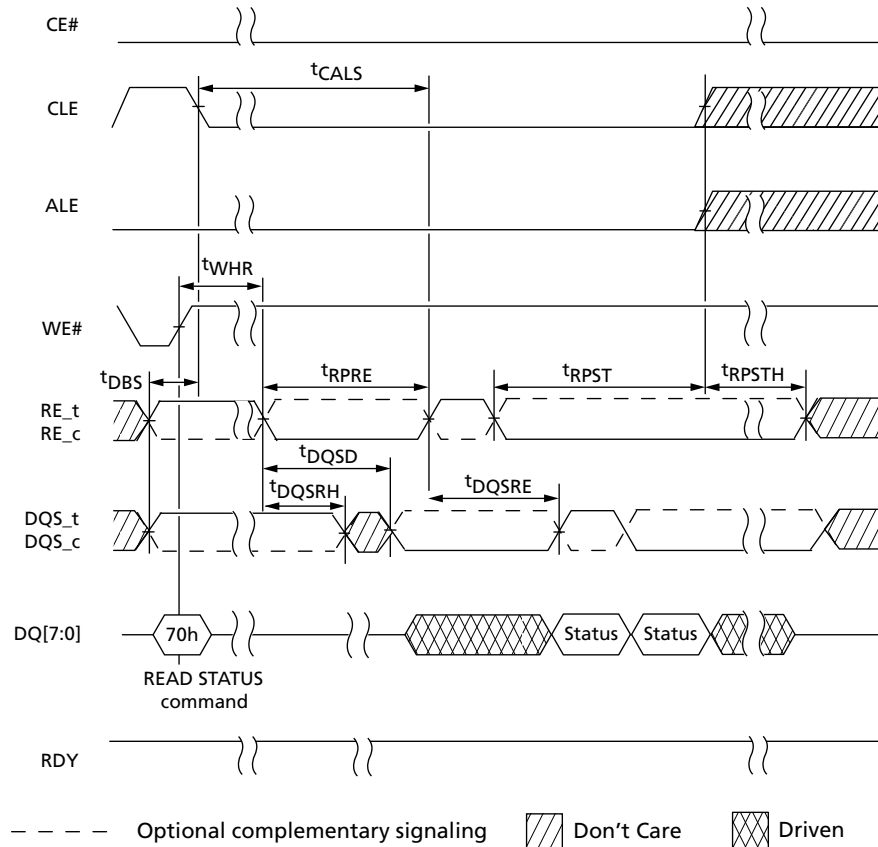
Figure 147: RESET (FCh) Operation





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

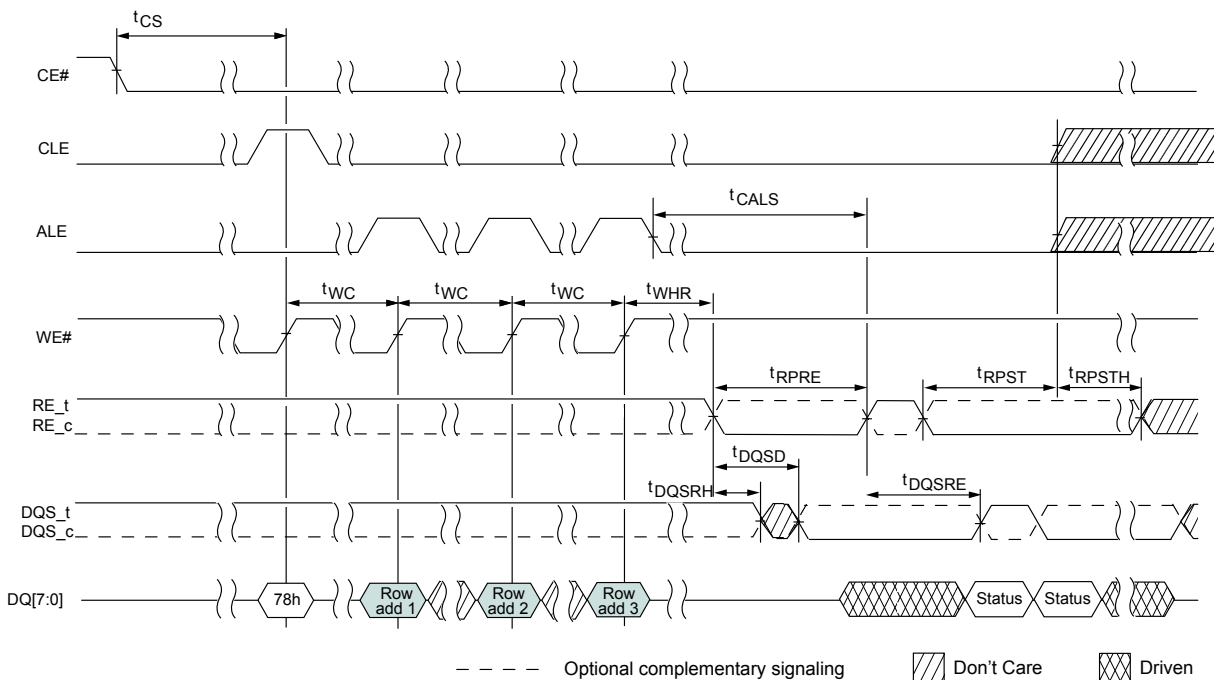
Figure 148: READ STATUS Cycle





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

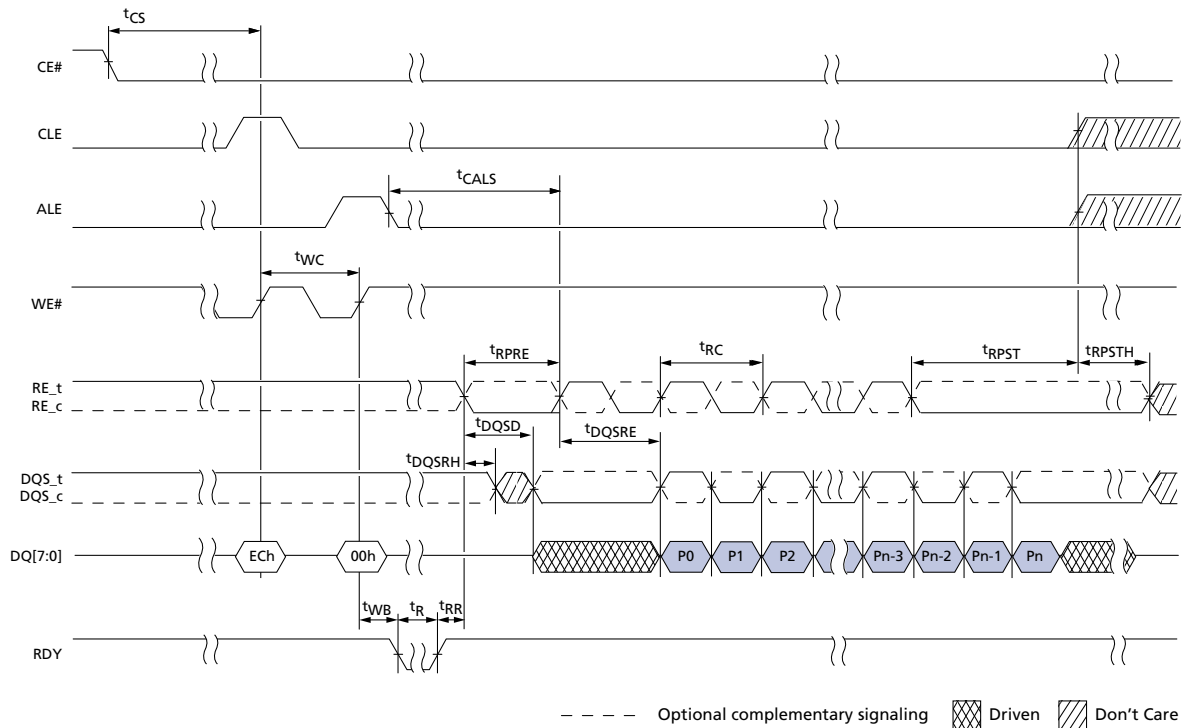
Figure 149: READ STATUS ENHANCED Operation





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

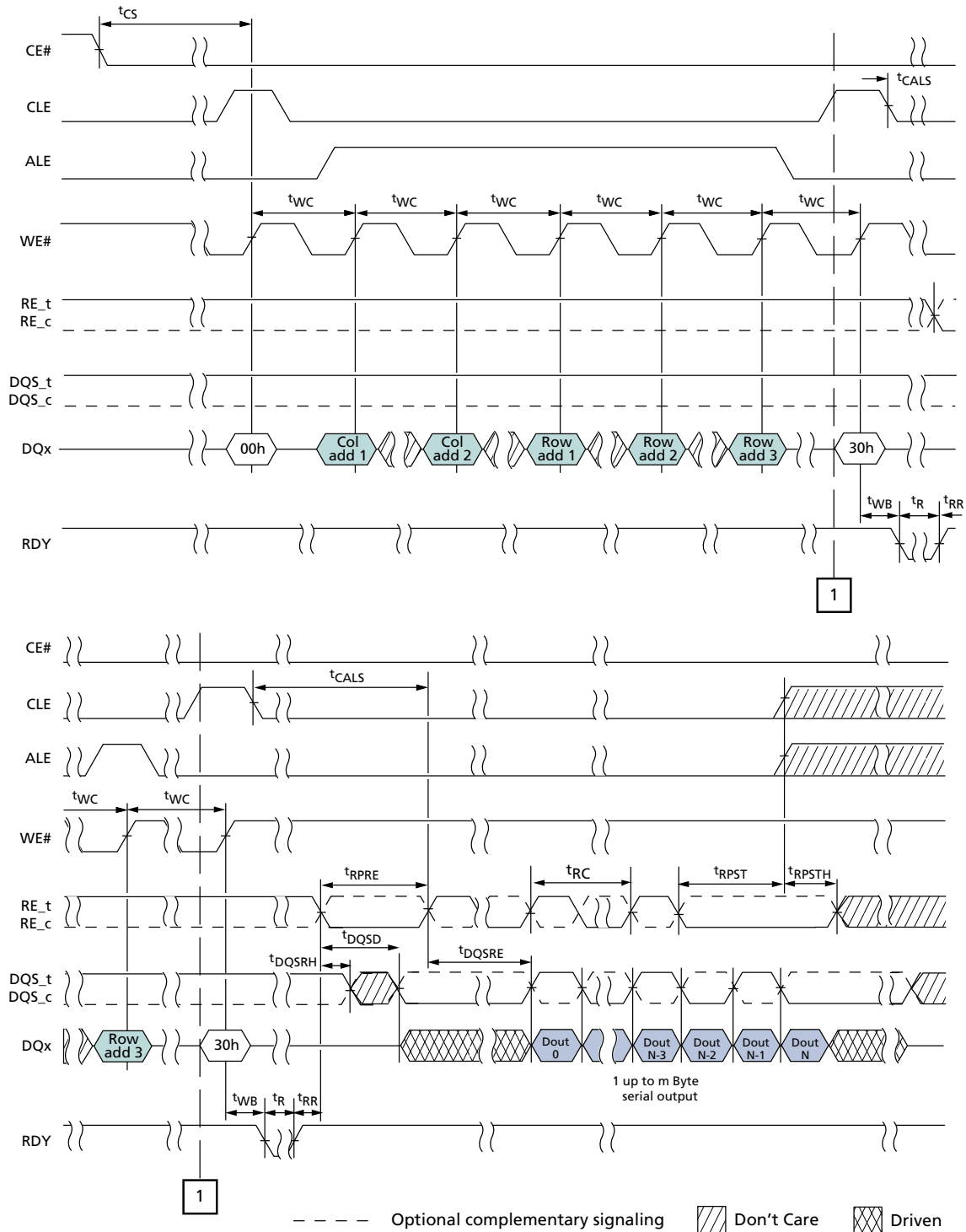
Figure 150: READ PARAMETER PAGE Operation





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

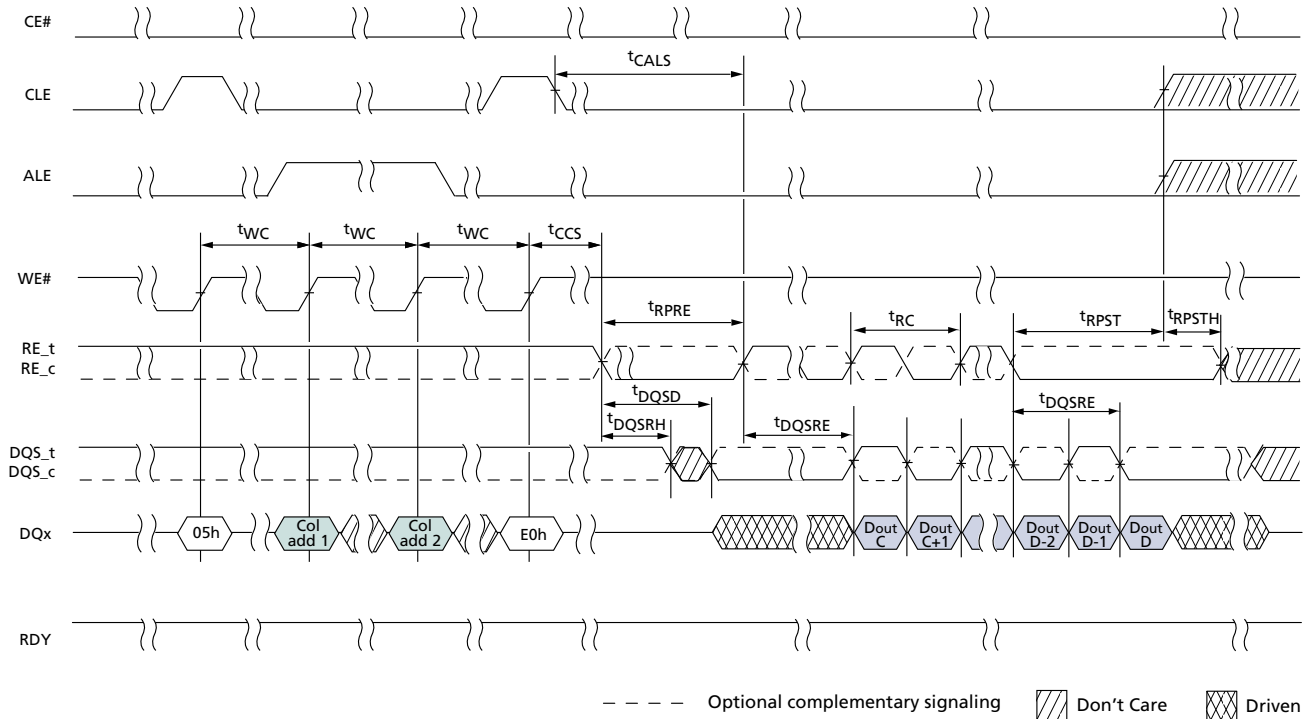
Figure 151: READ PAGE Operation





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

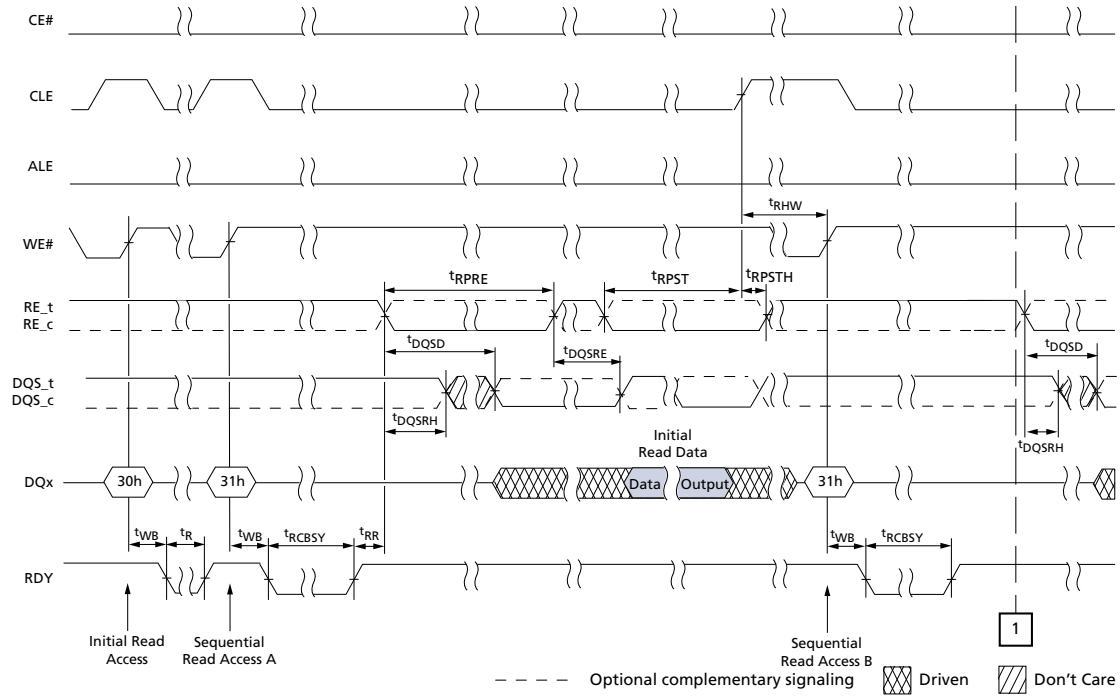
Figure 152: CHANGE READ COLUMN





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

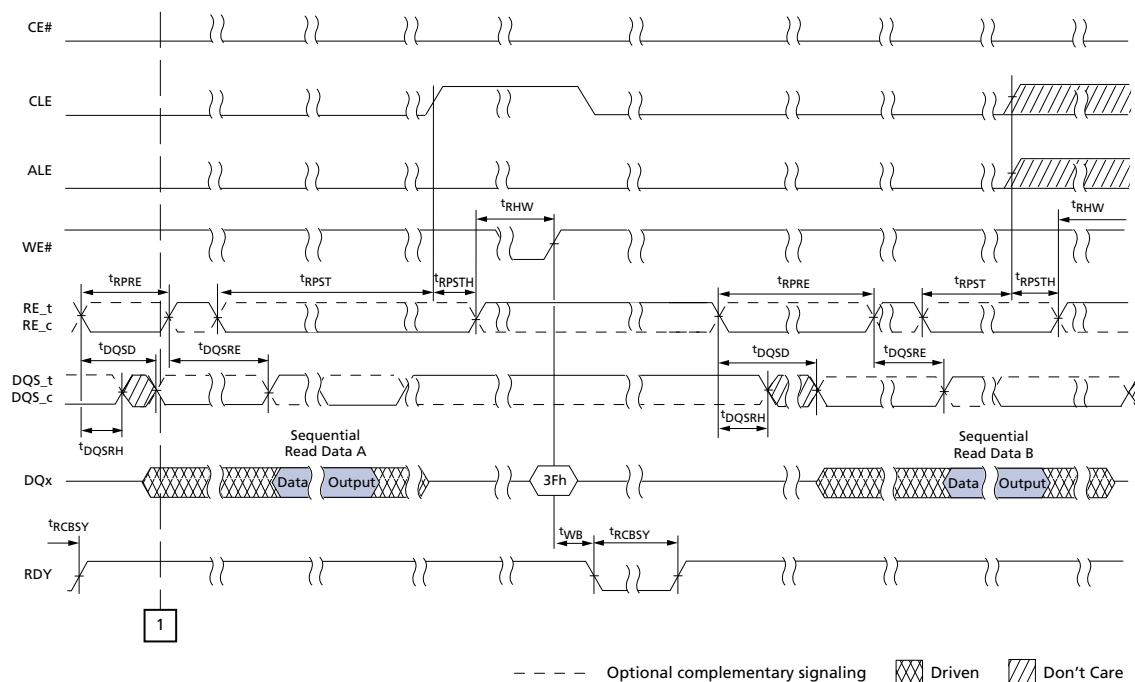
Figure 153: READ PAGE CACHE SEQUENTIAL (1 of 2)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

Figure 154: READ PAGE CACHE SEQUENTIAL (2 of 2)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

Figure 155: READ PAGE CACHE RANDOM (1 of 2)

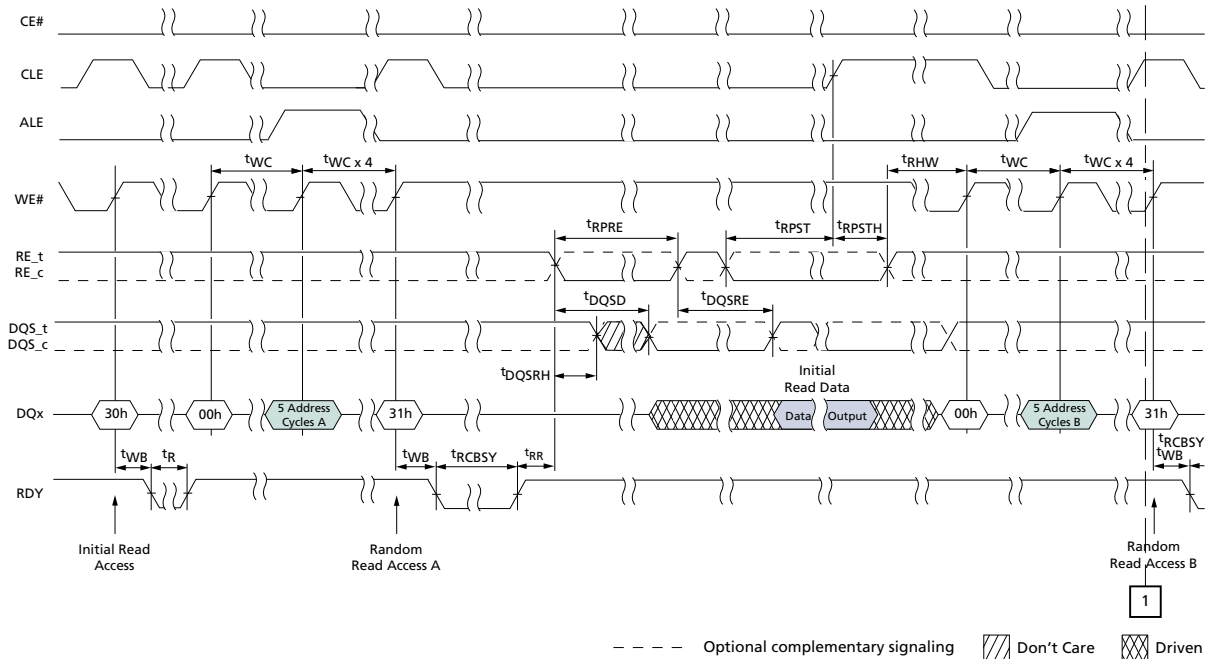
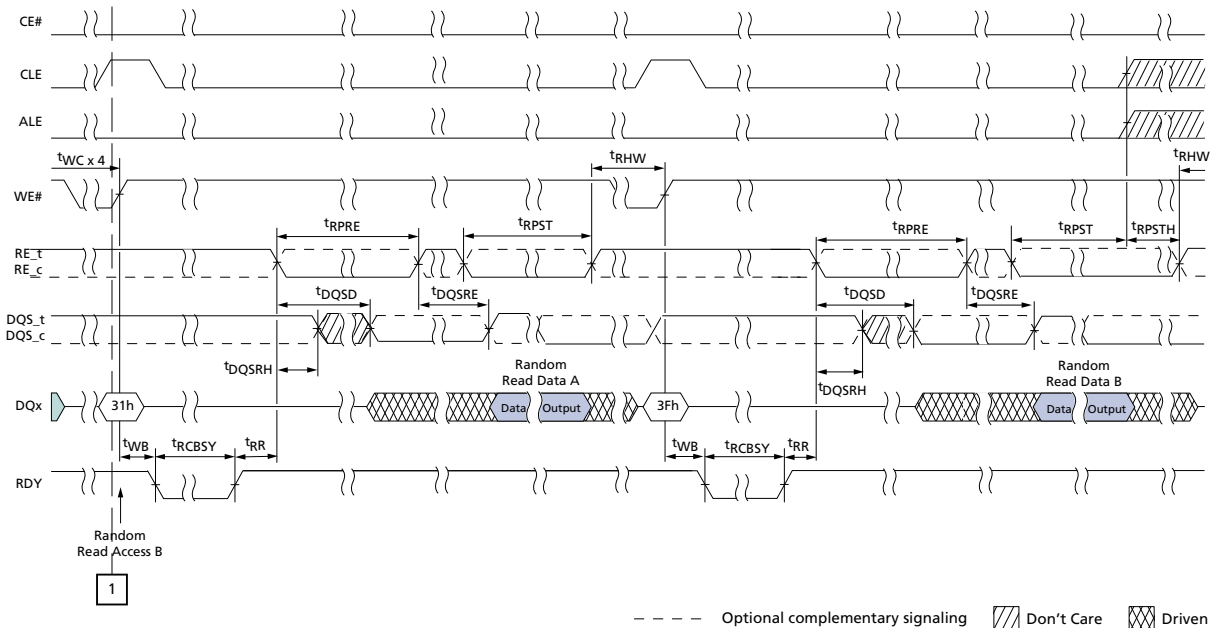


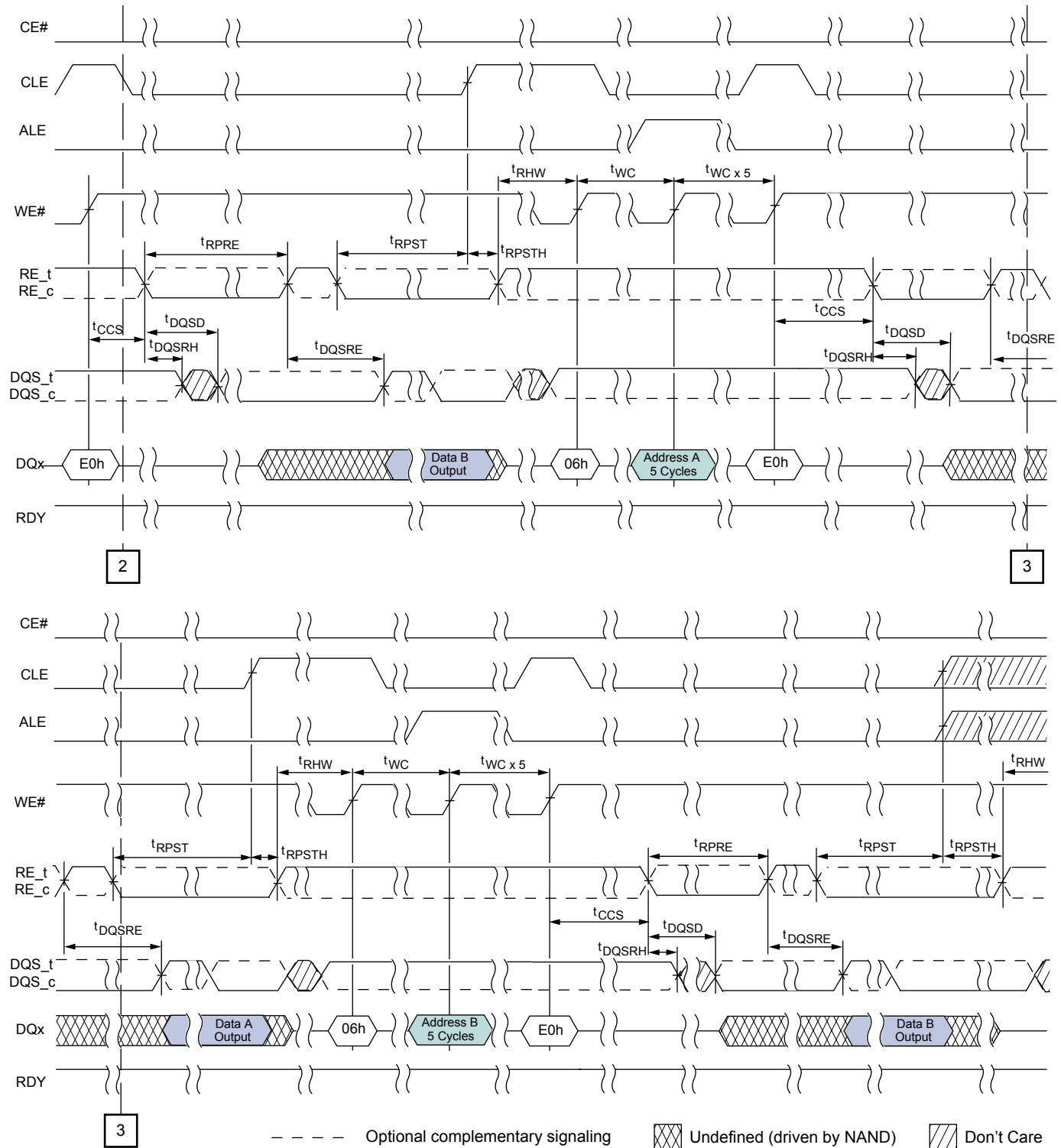
Figure 156: READ PAGE CACHE RANDOM (2 of 2)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

Figure 158: Multi-Plane Read Page (2 of 2)



MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

Figure 159: PROGRAM PAGE Operation (1 of 2)

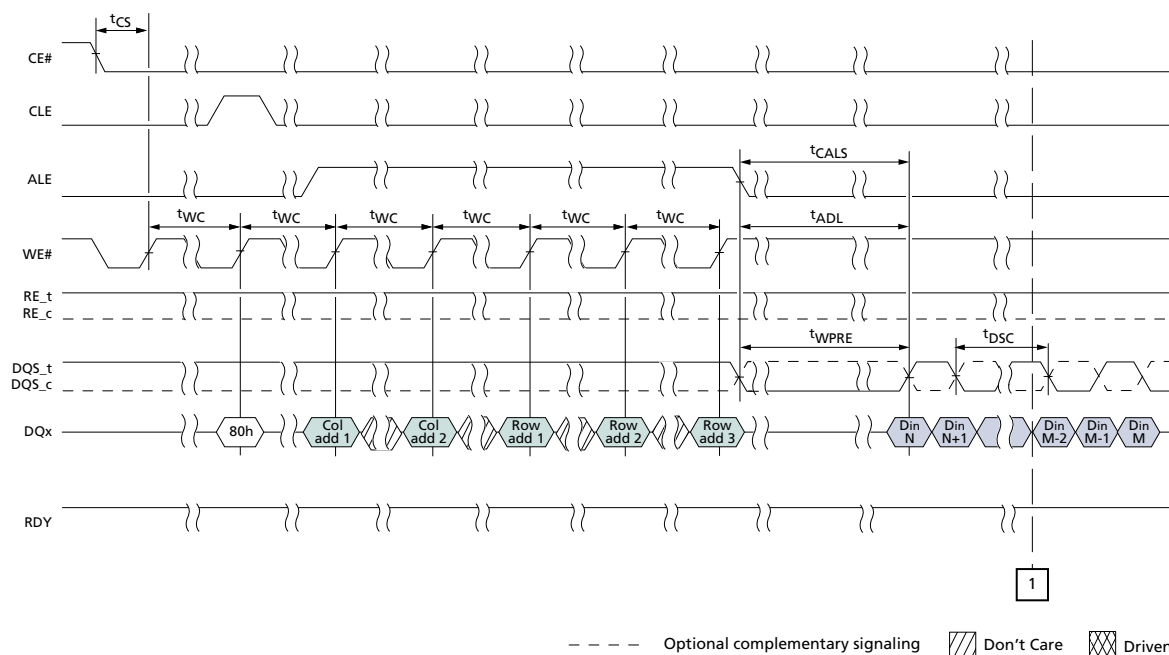
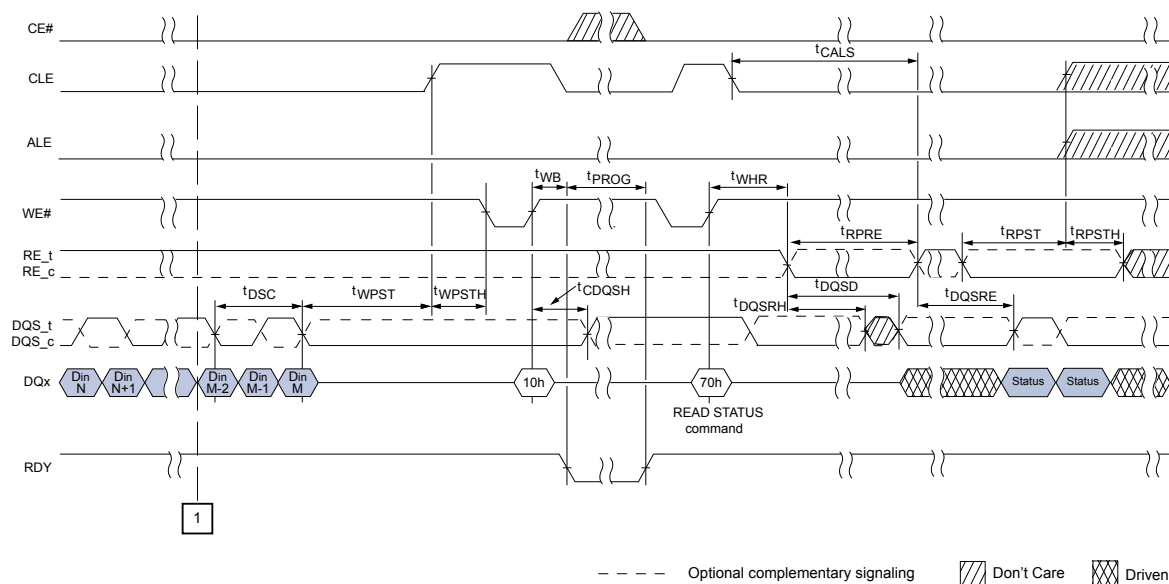


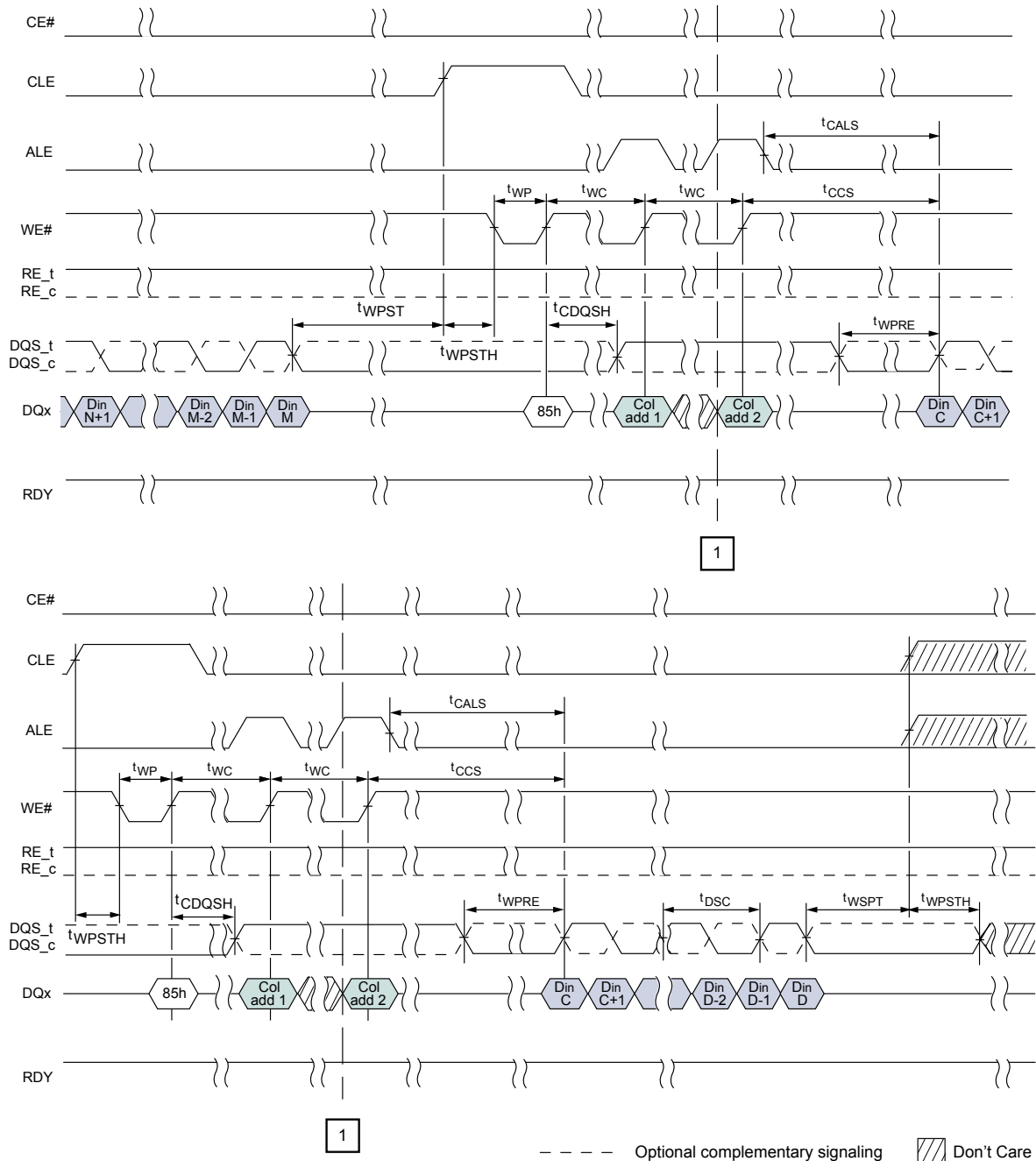
Figure 160: PROGRAM PAGE Operation (2 of 2)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

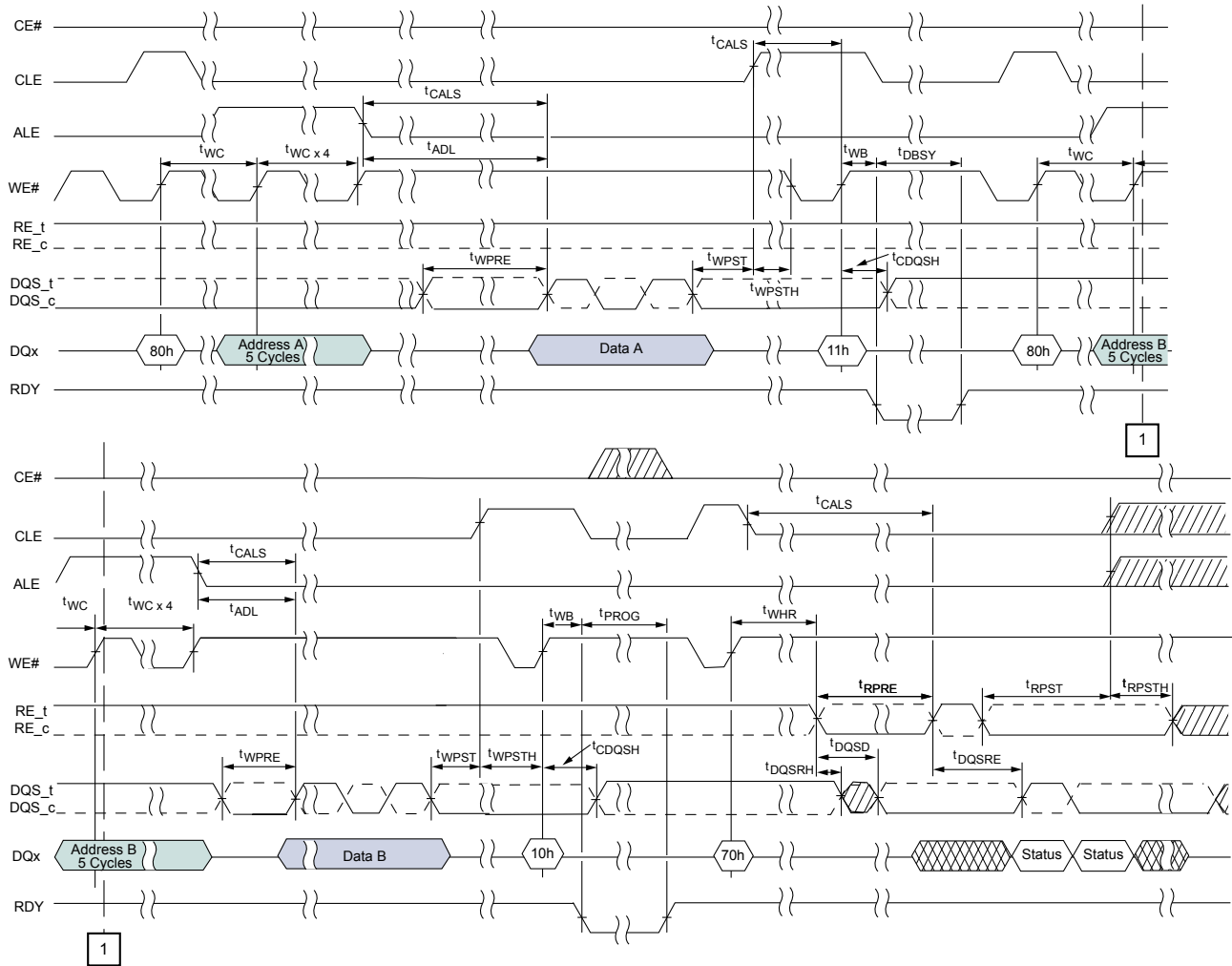
Figure 161: CHANGE WRITE COLUMN





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

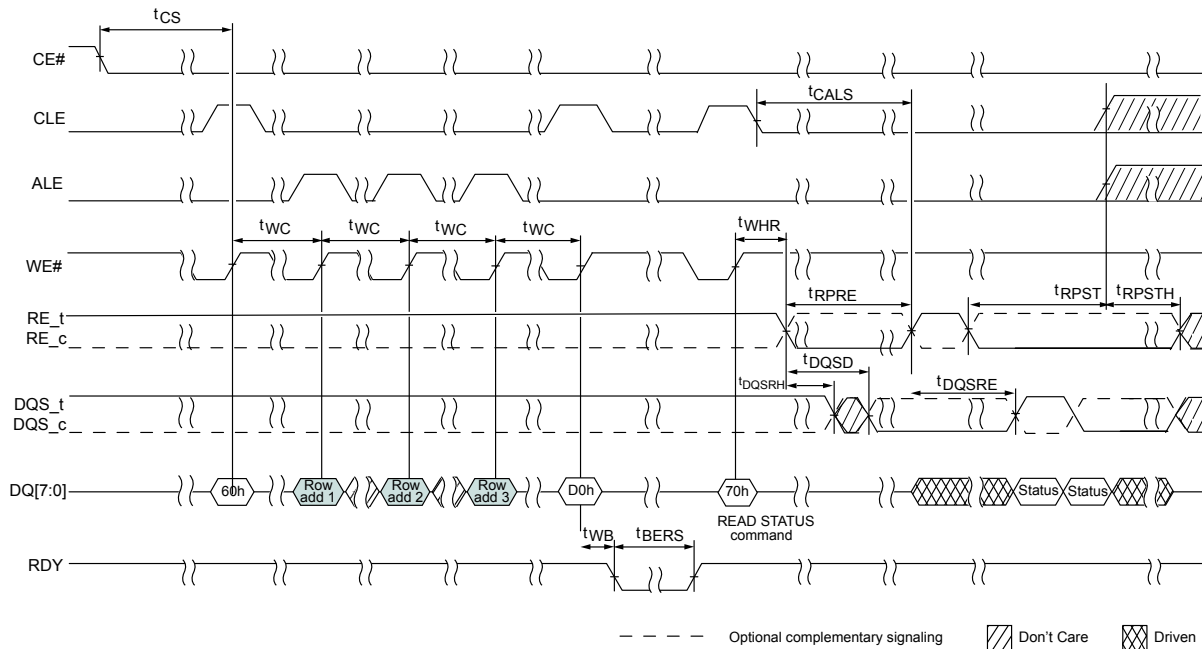
Figure 162: Multi-Plane Program Page





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

Figure 163: ERASE BLOCK





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

Figure 164: COPYBACK (1 of 3)

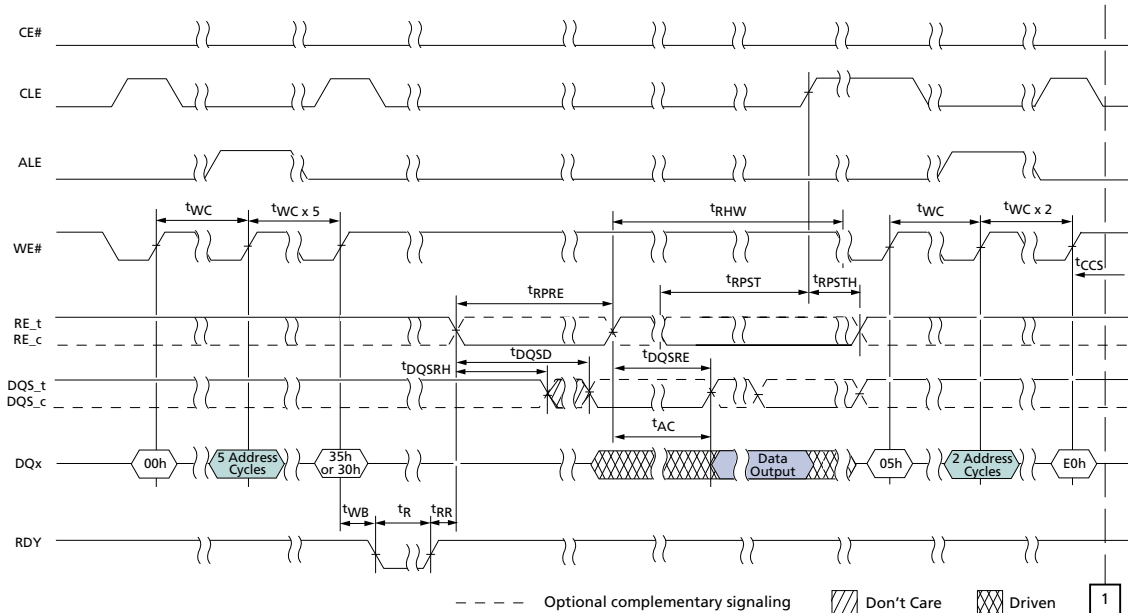
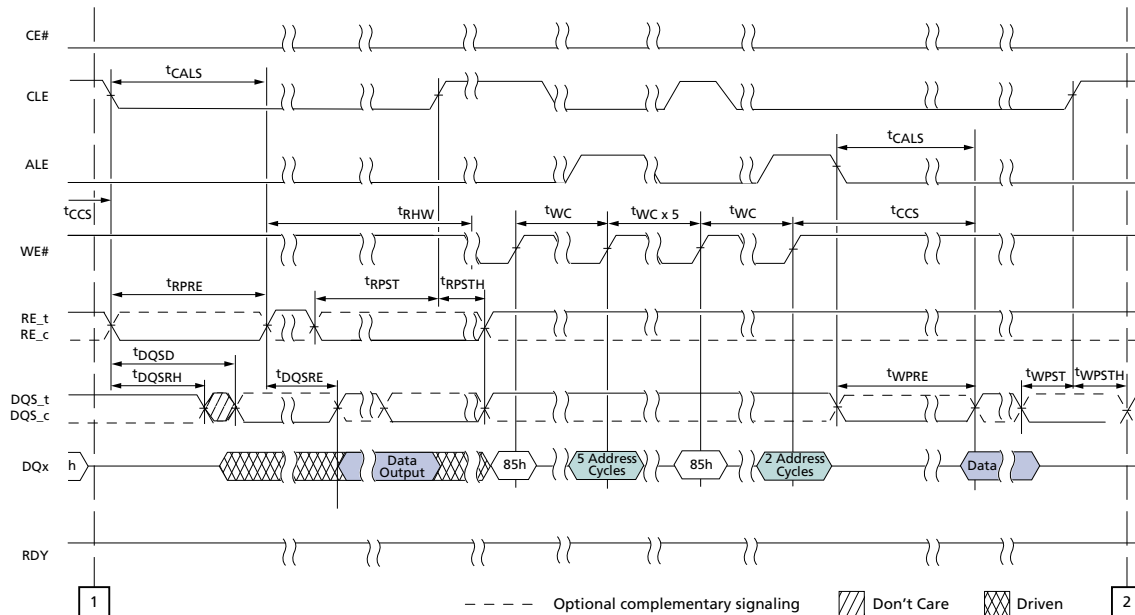


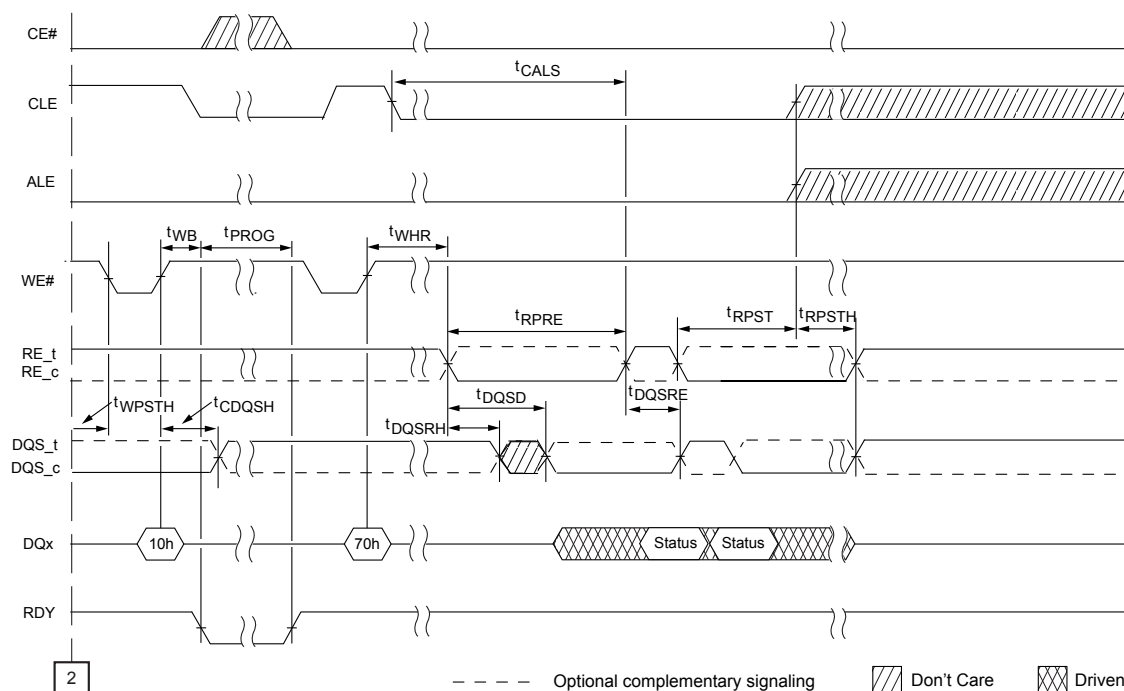
Figure 165: COPYBACK (2 of 3)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

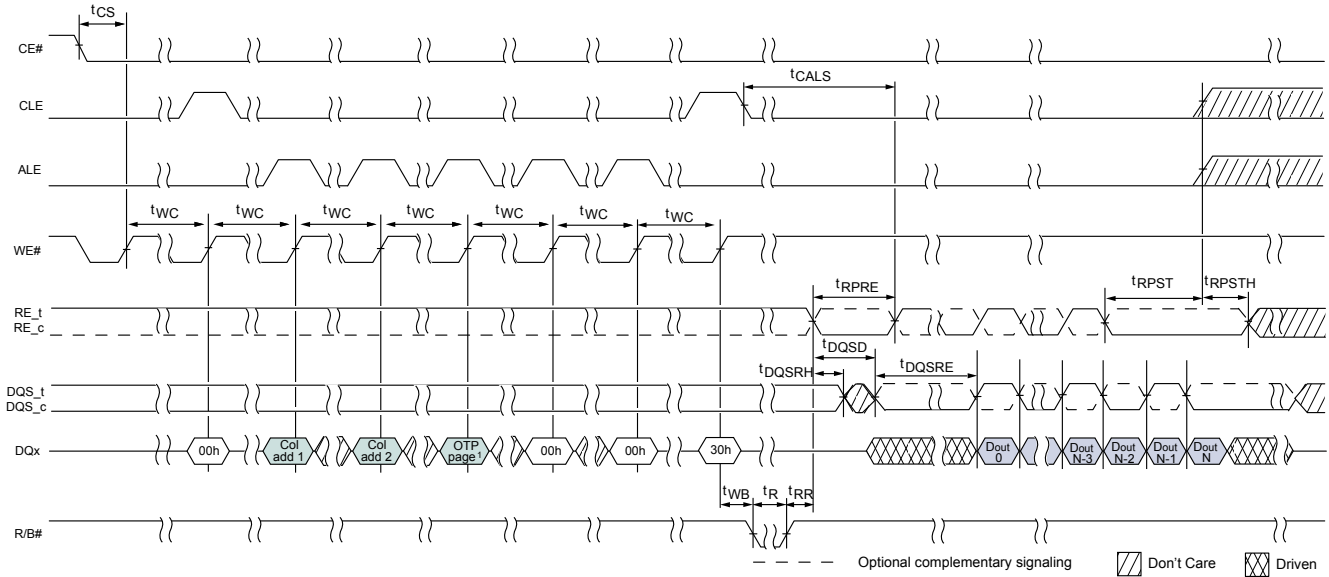
Figure 166: COPYBACK (3 of 3)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

Figure 167: READ OTP PAGE





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

Figure 168: PROGRAM OTP PAGE (1 of 2)

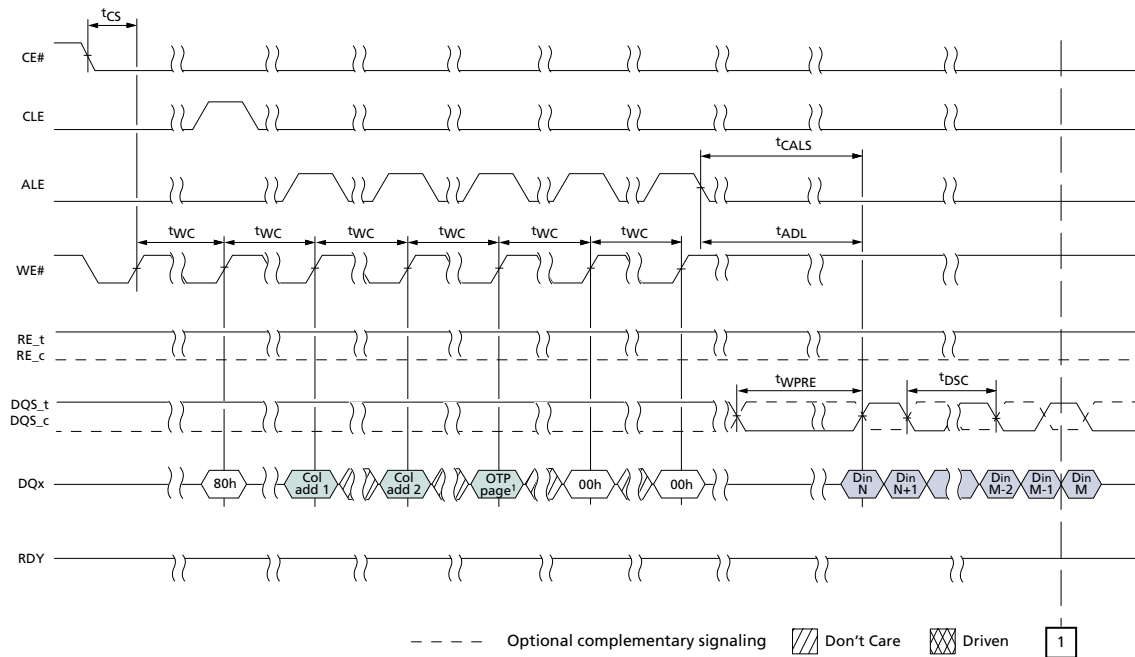
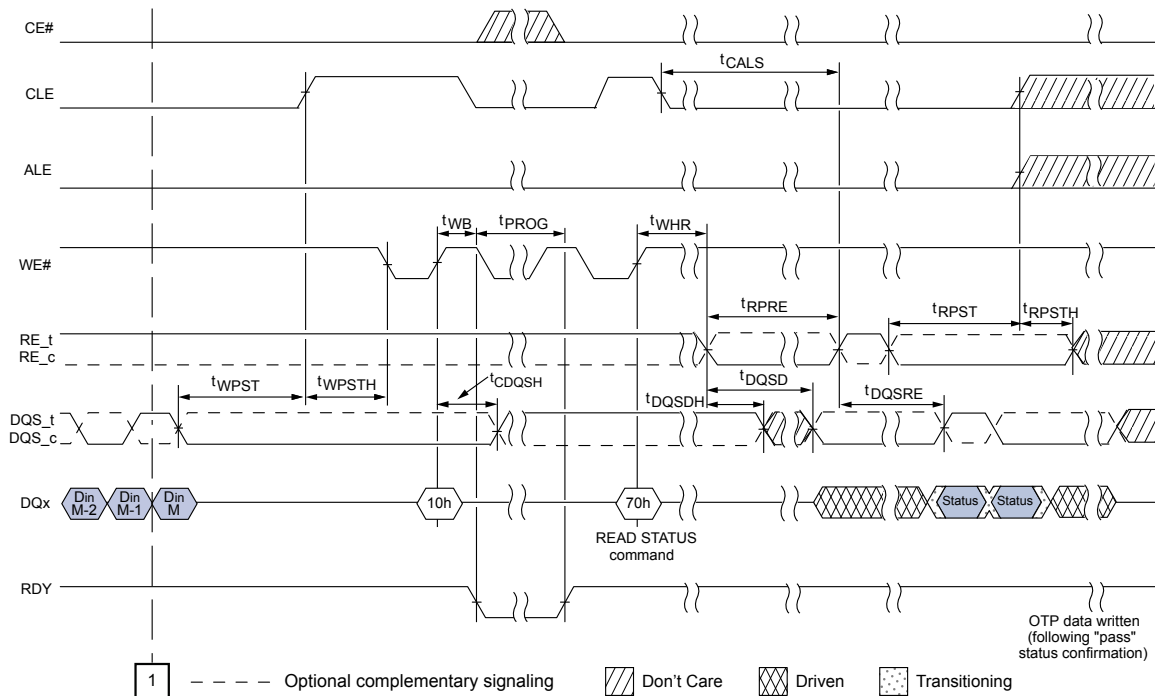


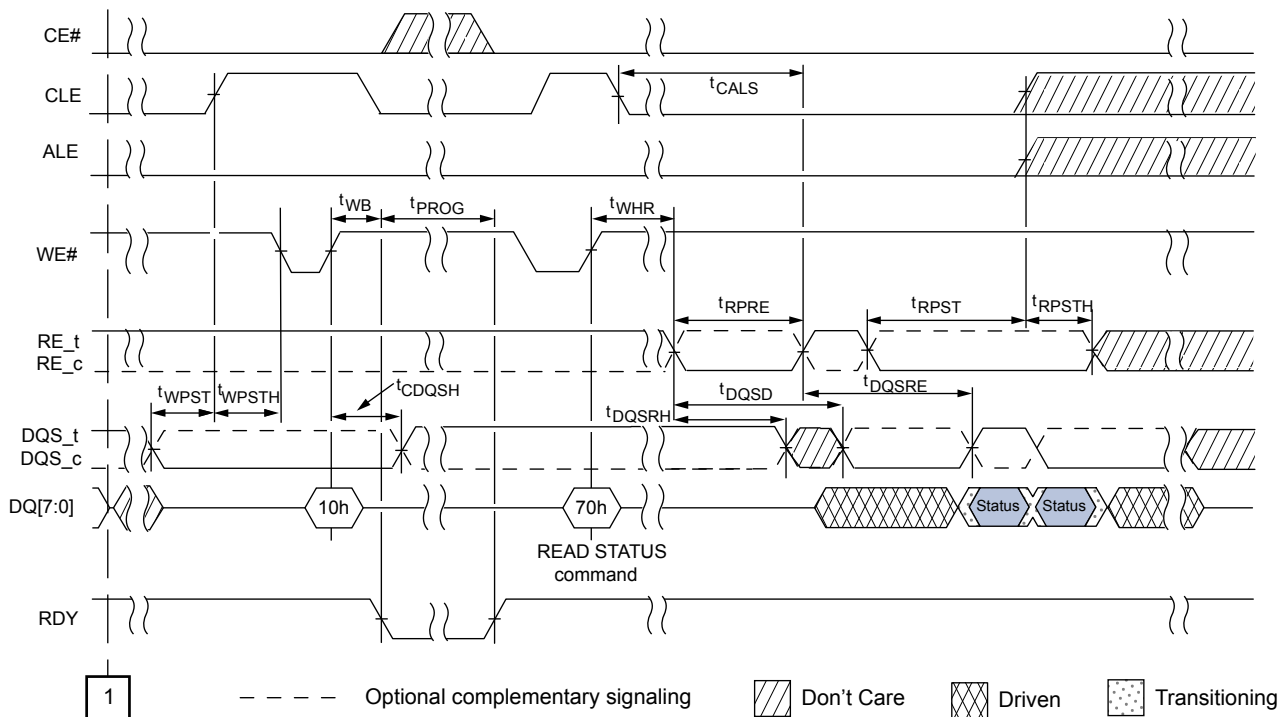
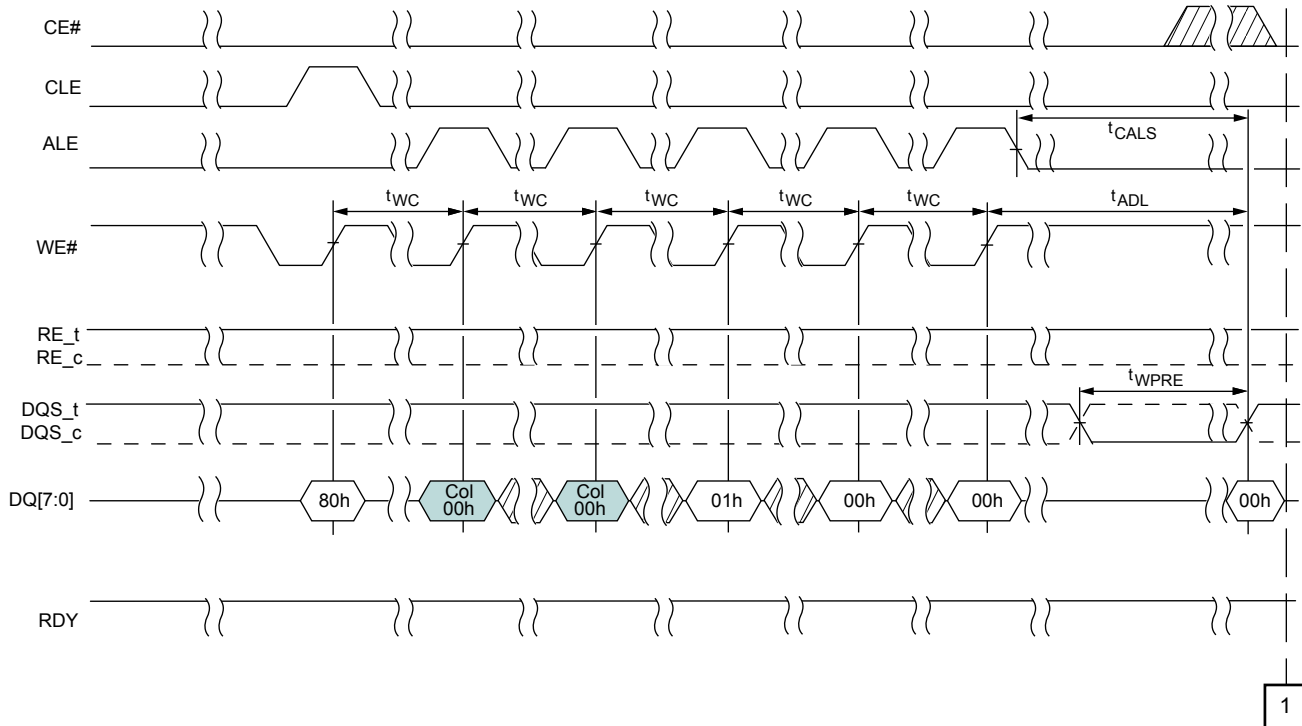
Figure 169: PROGRAM OTP PAGE (2 of 2)





MLC 256Gb to 2Tb Async/Sync NAND NV-DDR2 Interface Timing Diagrams

Figure 170: PROTECT OTP AREA





Revision History

Rev. G Production – 1/19/18

- Added "Important Notes and Warnings" section for further clarification aligning to industry standards
- Under Bus Operation – Asynchronous Interface section:
 - Under Asynchronous Commands subsection, corrected 'Asynchronous Command Latch Cycle' figure to show 'WP starting at the falling edge of CE#
 - Under Asynchronous Data Output subsection, added 'CS, 'REH, and 'CR parameters to the 'Asynchronous Data Output Cycles' figure after the falling edge of CE# for further clarification

Rev. F Production – 9/30/16

- Under Command Definitions section, added VOLUME SELECT (E1h) and ODT CONFIGURE (E2h) into the Command Set table
- Under Electrical Specifications section, under DC Characteristics and Operating Conditions (Asynchronous Interface) table:
 - Updated I_{CC4W_A} Typ to 10mA and Max to 13mA
 - Updated I_{SB} Typ to 20 μ A
- Under Electrical Specifications - DC Characteristics and Operating Conditions (NV-DDR and NV-DDR2) section, updated I_{SB} Typ to 20 μ A

Rev. E – 8/14

- Under ONFI Parameter Page Data Structure table for:
 - Updated bytes 128, 144, 146, 148, and 150 for capacitance for devices MT29F128G08CBECBH6, MT29F256G08CEECHB6, and MT29F2T08CTECBJ7 and bytes 144, 146, and 148 for capacitance for device MT29F1T08CUECBH8
 - Updated bytes 133–134 for 'PROG Max time for all devices
 - Updated byte 253 for all devices
 - Updated bytes 254–255 for all devices
- Under JEDEC Parameter Page Data Structure table:
 - Updated bytes 153–154 for 'PROG Max time for all devices
 - Updated bytes 163, 165, and 167 for capacitances for devices MT29F2T08CTECBJ7 and MT29F1T08CUECBH8
 - Updated byte 420 for all devices
 - Updated bytes 510–511 for all devices
- Under Input Slew Rate section, updated Input Slew Rate derating for NV-DDR2 single-ended and Input Slew Rate derating for NV-DDR2 differential tables per ONFI supported slew rates
- Under Electrical Specifications section, updated Recommended Operating Conditions table, V_{PP} Max to 13.2V
- Under Electrical Specifications section, updated Capacitance: 152-Ball BGA Package table for 16 die package
- Under Electrical Specifications – AC Characteristics and Operating Conditions (NV-DDR and NV-DDR2) section:



MLC 256Gb to 2Tb Async/Sync NAND Revision History

- Updated parameters t_{WPSTH} and t_{RPSTH} Min specification to 15ns
- Added the note, "Minimizing the amount of duty cycle jitter to more than 45 percent of the average cycle provides a larger t_{QH} , which in turn provides a larger data valid window. The device shall provide a minimum of 0.5 percent of larger data valid window for each 1% improvement in duty cycle jitter. For example, if the host provides a t_{REH} (abs) of $0.49 * t_{RC}(avg)$ then the device shall provide at least a t_{QH} of $0.39 * t_{RC}(avg)$."
- Under Electrical Characteristics – Array Characteristics section:
 - Updated t_{PROG}/t_{CBSY} Max to 3000 μ s
 - Added footnotes regarding block retirement when a program or erase operation exceeds t_{PROG} Max or t_{BERS} Max, respectively.

Rev. D – 3/14

- Under Device and Array Organization section added updated Device Organization for Sixteen-Die Package (152-ball BGA) figure for CE# assignments to match CE# and R/B# mapping relationships
- Under JEDEC Parameter Page Data Structure table updated byte 105 bit definition text to match JEDEC text
- Under Electrical Specifications – DC Characteristics and Operating Conditions (V_{CCQ}) section updated I_{VREFQ} Max to ± 60 nA

Rev. C – 2/14

- Under Features section:
 - Changed part number MT29F1T08CTCCB to MT29F2T08CTECB
 - Corrected typo in total page size from 18,526 bytes to 18,256 bytes
 - Included Read Retry under Advance Command Set
 - Added note on clarification of Read Retry operation
 - Changed references to 333MT/s NV-DDR2 interface speed to 166MT/s and 200MT/s NV-DDR interface speed to 166MT/s.
 - Removed reference to 272-ball BGA devices
- Under Part Numbering figure:
 - Under 'Classification', redefined T classification option
 - Under 'Speed grade', removed -6 reference
 - Under 'Package Code', added package code J7 and removed references to 272-ball BGA packages
- Under Signal Assignments section updated 152-Ball BGA (Ball-Down, Top View) figure to include signal assignments for 16DP package and removed reference to 272-ball BGA packages
- Under Package Dimensions section added 152-Ball LBGA – 14mm x 18mm (Package Code: J7) figure and 272-Ball VFPGA – 14mm x 18mm (Package Code: G1) figure and removed references to 272-ball BGA packages
- Under Device and Array Organization section added Device Organization for Sixteen-Die Package (152-ball BGA) figure and Device Organization for Four-Die Package (272-ball BGA) figure and removed references to 272-ball BGA packages
- Under Bus Operation - NV-DDR2 Interface, added note on ODT enabled during address input state under NV-DDR2 Interface Mode Selection table



- Under Identification Operations section:
 - Read ID Parameters for Address 00h table, changed part number MT29F1T08CTCCB to MT29F2T08CTECB and removed 272-ball BGA part numbers
 - Under ONFI Parameter Page Data Structure table:
 - Changed part number MT29F1T08CTCCB to MT29F2T08CTECB and removed 272-ball BGA part numbers
 - Updated bytes 128, 144, 146, 148, and 150 for capacitance
 - Updated byte 253
 - Updated bytes 254–255 for CRC
 - Under JEDEC Parameter Page Data Structure table:
 - Changed part number MT29F1T08CTCCB to MT29F2T08CTECB and removed 272-ball BGA part numbers
 - Updated bytes 163, 165, and 167 for capacitances
 - Updated byte 420
 - Updated bytes 510–511 for CRC
- Under Configuration Operations section, Feature Address 02h - updated the note to state a RESET (FFh) command will change the bit values back to default
- Under Status Operation section, updated reference to the NV-DDR and NV-DDR2 interfaces
- Under Read Operations section, added Read Retry Operations section for greater operational details
- Under Input Slew Rate section, updated Test Conditions for Input Slew Rate table
- Under Electrical Specifications section:
 - Updated Capacitance: 152-Ball BGA Package table for Single Die/Dual Die/Quad/Octal Die packages and added 16 die placeholder
 - Removed Capacitance: 272-Ball BGA Package table
- Under Electrical Specifications – DC Characteristics and Operating Conditions (Asynchronous):
 - Updated I_{CC1_A} TYP to 28mA
 - Updated I_{CC2_A} TYP to 37mA
 - Updated I_{CC4R_A} TYP to 9mA and Max to 12mA
 - Updated I_{CCQ4R_A} TYP to 5mA and Max to 10mA
 - Updated I_{CCQ4W_A} TYP to 1mA and Max to 3mA
- Under Electrical Specifications – DC Characteristics and Operating Conditions (Synchronous):
 - Updated I_{CC1_S} TYP to 28mA
 - Updated I_{CC2_S} TYP to 37mA
 - Updated I_{CC4R_S} TYP to 16mA and Max to 23mA for 3.3V V_{CCQ}
 - Updated I_{CCQ4R_S} TYP to 25mA and Max to 30mA for 3.3V V_{CCQ}
 - Updated I_{CC4R_S} TYP to 18mA and Max to 23mA for 1.8V V_{CCQ} for up to 166MT/s
 - Updated I_{CCQ4R_S} TYP to 15mA and Max to 19mA for 1.8V V_{CCQ} for up to 166MT/s
 - Updated I_{CCQ4W_S} TYP to 1mA and Max to 3mA for 3.3V V_{CCQ}
 - Updated I_{CCQ4W_S} TYP to 1mA and Max to 3mA for 1.8V V_{CCQ}



MLC 256Gb to 2Tb Async/Sync NAND Revision History

Rev. B – 8/13

- Added support for 1.8V/3.3V V_{CCQ} devices and related information
- Corrected spare area portion of device size to 936K under Features section
- Changed 272-ball BGA packages seating plane to 0.12mm
- Changed V_{PP} Max to 16V

Rev. A – 6/13

- Initial release

8000 S. Federal Way, P.O. Box 6, Boise, ID 83707-0006, Tel: 208-368-4000

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This data sheet contains minimum and maximum limits specified over the power supply and temperature range set forth herein. Although considered final, these specifications are subject to change, as further product development and data characterization sometimes occur.